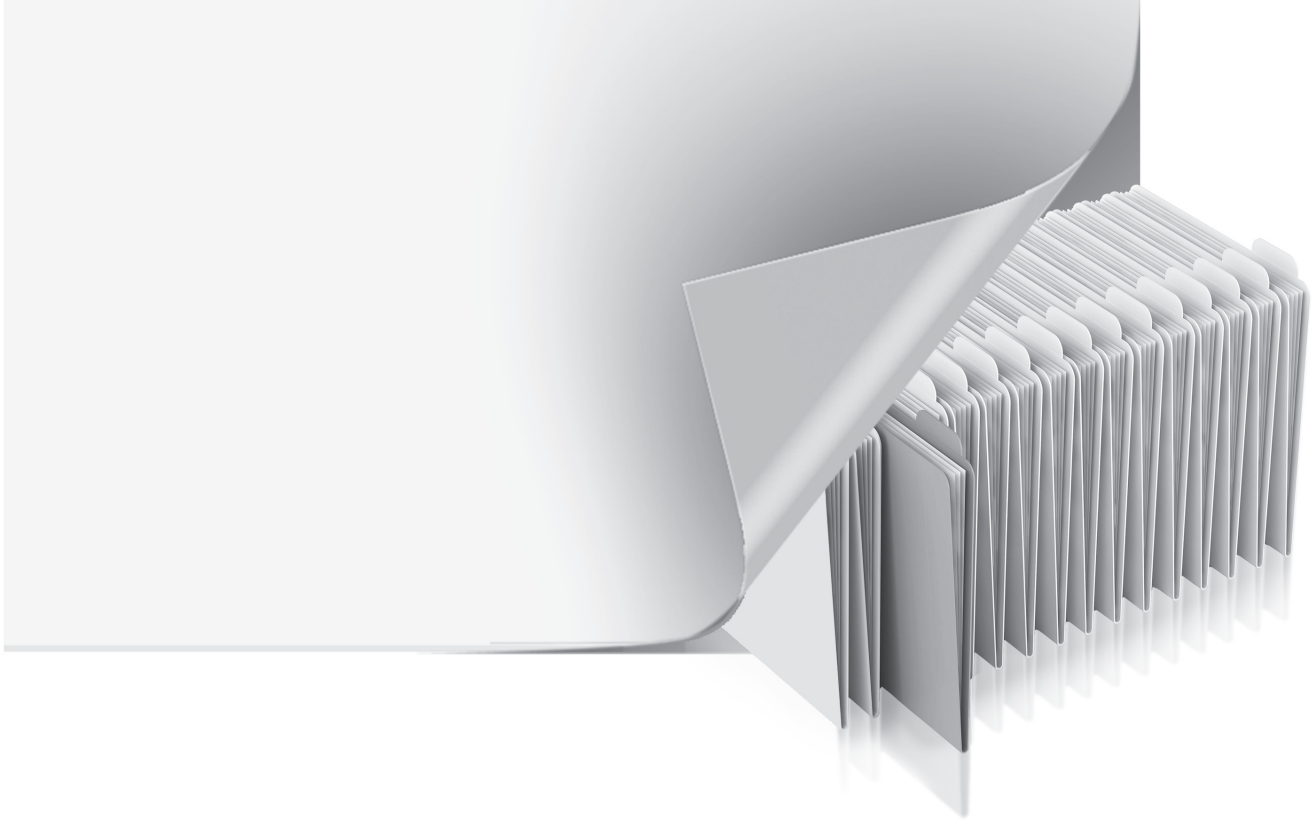
Administrative Healthcare Data

A Guide to Its Origin, Content, and Application Using SAS®

Craig Dickstein and Renu Gehring



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Contents

About This Book ix
About These Authors xi
Acknowledgments xiii
Chapter 1: The U.S. Healthcare System 1
Introduction 1
Data and Programming Used in This Book 2
Terminology 2
Flow of Administrative Healthcare Data 3
Key Players 4
Medical Claim Submission 4
Claim Processing 5
Recent Legislative Effects 6
HIPAA 6
Affordable Care Act 6
All Payer Claims Database 6
Continuing Enhancements 7
Conclusion 7
Chapter 2: Introduction to SAS Enterprise Guide and Sample Data 9
Introduction 9
Sample Data 9
What Is SAS Enterprise Guide? 11
SAS Libraries and Data Sets 12
Create a Permanent Library 12
View a SAS Data Set 13
SAS Data Types 13
Formats 14
DRG Format 15
Diagnosis Code Format 16
Applying Formats to SAS Variables 19
Formatting an Existing Variable 19
Placing Results of a Format into a New Variable 21
Conclusion 21
Chapter 3: The Payers 23
Introduction 23

Health Insurance 23
Medicare 24
Medicaid 25
Commercial Insurance 26
Others 26
TRICARE 26
CHAMPVA 26
FECA Black Lung 26
Conclusion 27
Chapter 4: The Providers 29
Introduction 29
Types of Providers 29
Facility 30
Professional 30
Pharmacy 31
Ancillary 31
National Provider Registry 32
NPI 32
Taxonomy 32
Other Provider Identifiers 33
Case Study: Standardizing Provider Names from the National Provider Registry 34
Case Study: Using Taxonomy Code to Identify Primary Care Physicians 36
Conclusion 38
Chapter 5: Facility Claims 39
Introduction 39
CMS-1450 Paper Claim Form 40
837I Electronic Claim Format 40
Data Elements Unique to Facilities 40
Type of Bill 40
Admission and Discharge Dates 42
Patient Discharge Status 43
Revenue Code 43
Diagnosis Codes 44
Present on Admission 45
Surgical Procedure Codes 45
DRG 45
Provider IDs 46
Others 46
Case Study: Calculating C-Section Rates among Hospitals 47
Create Summary Data Set for All Births 47
Create Summary Data Set for C-Section Births 49
Join Summary Data Sets 50
Create Bar Graphs 51
Case Study: Top Reasons for ER Utilization 53

Automating Reports with SAS Enterprise Guide 56
Creating Documentation in SAS Enterprise Guide 56
Conclusion 57
Chapter 6: Professional and Ancillary Claims 59
Introduction 59
Medical Claim Submission 60
CMS-1500 Claim Form 60
837P Electronic Claim Format 60
Data Elements Unique to CMS-1500/837P 60
Demographic Information 60
Diagnosis Codes 61
Diagnosis Pointer 61
Provider Identifiers 61
Procedure Codes and Modifiers 61
Place of Service 62
Provider Specialty 62
Payment Methodologies 62
Case Study: Identifying Children Who Miss Their Checkups 63
Identify Currently Enrolled Children under Six Years of Age 64
Import Excel Data 66
Identify Appropriate Professional Claims 68
Create Outreach Report 70
Create Internal Report 71
Case Study: Automating Reports with Macro Variables 73
Extract Code from Enterprise Guide Tasks 73
Import Data Code 74
Query Builder Code 75
Automate Code 76
Conclusion 81
Chapter 7: Pharmacy Claims 83
Introduction 83
NCPDP Claim Formats 84
Paper Claim Form 84
Electronic Format 84
Data Elements Unique to Pharmacy Data 86
Provider Identifiers 86
National Drug Code 86
Generic Product Identifier 88
Therapeutic Class Codes 88
Other Fields of Interest 89
Case Study: Computing Medication Adherence 90
PDC Computation 90
Data Expansion Using SAS Data Step 91
Create Study Period Data Set 92

Create Days Covered Data Set 94
Combine and Summarize Data 94
Graphing PDC 96
Automating PDC Graphs Generation 98
Conclusion 101
Chapter 8: Healthcare Claim Codes 103
Introduction 103
International Classification of Diseases 104
Diagnosis Codes 105
ICD-9-CM 105
ICD-10-CM 107
Surgical Procedure Codes 109
ICD-9-PCS 109
ICD-10-PCS 110
Current Procedural Terminology (CPT) 112
Category I 113
Category II 113
Category III 114
HCPCS 114
Level I 115
Level II 115
Level III 115
Modifiers 115
HIPPS 116
Other PPS Code Sets 117
NDC 117
LOINC 118
Case Study: Identifying a Patient with Complex Conditions 118
Code Simplification with SAS Array Processing 119
Identifying Members with Complex Conditions 120
Parameterizing Program 8.3 with Macro Variables 122
Case Study: Using Formats to Create Data Hygiene Routines 123
Conclusion 126
Chapter 9: The Members 127
Introduction 127
Member Demographics 128
Member Enrollment 129
Member Eligibility 130
Membership Issues of Interest 130
Membership Maintenance 130
Electronic Eligibility Inquiry 130
Changing Member ID 131
“Cross-Client” Projects 131
Householding 131

Member Months 131
Continuous Enrollment 132
Rate Setting and Risk Adjustment 132
Setting Rates 132
Adjusting Risk 134
Case Study: Creating Member Months Data 134
Creating a Callable Macro Program 135
Member Months Macro Program 137
Building a Member Months Table 139
Conclusion 141
Chapter 10: Computing and Tracking Financial Metrics 143
Introduction 143
Case Study: Bucketing Costs 143
Case Study: Calculating PMPM Costs 148
Case Study: Creating Reports 149
Conclusion 152
Chapter 11: HEDIS 153
Introduction 153
The Business Case 154
The Technical Challenges 154
Reporting System Components 155
Colorectal Cancer Screening 156
Definition 156
Eligible Population 156
Exclusions to Eligibility 157
Compliant Population 157
Hybrid Specification 158
Case Study: Developing a Rate for Colorectal Cancer Screening 158
Create a Driver Table 159
Clean Up of Membership Data 159
Check Continuous Enrollment 162
Identify the Denominator 165
Determine Compliant Population 172
Compute Rate 173
Conclusion 174
Chapter 12: Future Healthcare Data Issues 175
Introduction 175
Impact of the Affordable Care Act 175
Transparency in Pricing 176
ICD-10 176
Patient Centered Medical Home 176
Accountable Care Organization 177
Pharmacy Benefits Manager 177
Evolving Patient Medical Records 178

Electronic Medical Record 178
Electronic Health Record 178
Personal Health Record 178
Meaningful Use 179
Global Billing 179
All Payer Claims Database 179
Conclusion 180
Chapter 13: Extended Coding Examples 181
Introduction 181
Utility Macros 181
Age-as-of Calculation 181
Identifying Sparse Variables 182
Arrays of Detail Record Elements on the Header Record 187
Linking to the Diagnosis Pointer 190
Conclusion 193
Appendix 195
Glossary of Terms 195
CMS-1450 Claim Form 201
CMS-1500 Claim Form 202
Universal Claim Form for Prescription Drugs 203
Facility Type 205
Bill Sequence 206
Place of Service 207
Patient Status Code 210
Revenue Code 211
Index 225

About This Book

Purpose

This book explains the source and content of administrative healthcare data—the product of financial reimbursement for healthcare services. It integrates the business knowledge of healthcare data with practical and pertinent case studies as shown in SAS.

Progressing from simple to complex, the book provides an introduction to the key players in the healthcare industry, details the process and content of claims submissions, and showcases applications of the data in healthcare case studies. In the book you will find answers to the following questions and more.

- Who are the key players and how is data sourced, transmitted, and organized? How can this data be best managed in SAS to create actionable information for the health plan?
- What are the standard medical claim code sets and how does knowledge of them aid in their processing?
- How do you identify emergency room (ER) visits and the top reasons for ER utilization?
- How is pharmacy data different from medical claims data and what are the significant elements in each? Using SAS algorithms, how can you identify patients who are not in compliance with their medication regimen?
- What are utilization and financial metrics and how do you calculate them using SAS?
- What is HEDIS? What are the rules that define a performance measure? How do you calculate a HEDIS measure using SAS?

Is This Book for You?

This book is intended for healthcare analysts regardless of their level of proficiency or experience with SAS Enterprise Guide, SAS programming, or the healthcare industry. SAS solutions in the book progress from easy to complex. As a result, both novice and experienced programmers will learn some new and useful techniques. The book provides a concise, yet complete, foundational knowledge of the business of healthcare; it is certainly a must for analysts new to the industry and a great review for experienced healthcare analysts.

What's New in This Edition

This book is the much awaited update to a similar work by Marge Scerbo, Craig Dickstein, and Alan Wilson, *Health Care Data and SAS*® (Cary, NC: SAS Institute, 2001). Although this book covers some of the same material, changes in both the healthcare industry and SAS suggest the need for enhancing that important prior body of knowledge.

About the Examples

Software Used to Develop the Book's Content

Case studies and example code in this book were developed using SAS Enterprise Guide 5.1 and Base SAS version 9.4.

Example Code and Data

You can access the example code and data for this book by linking to its author page at http://support.sas.com/publishing/authors. Select the name of the author. Then, look for the cover thumbnail of this book, and select Example Code and Data to display the SAS programs that are included in this book.

For an alphabetical listing of all books for which example code and data is available, see http://support.sas.com/bookcode. Select a title to display the book's example code.

Keep in Touch

Given the rich and changing landscape of administrative healthcare data, this book cannot hope to have captured all possible business and data management concepts, despite our best efforts. It is with that acknowledgment that the authors encourage the reader to offer any thoughts on additional content that you feel would add to the usefulness of this book. We would gratefully accept any and all constructive input from our valued customers.

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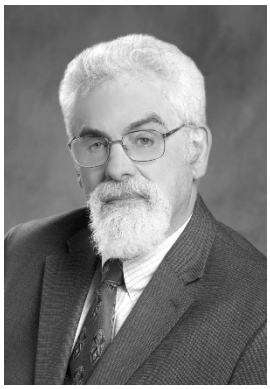
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Renu: I would like to thank my co-author, Craig, whose industry and technical knowledge continues to amaze me each time we interact. I also want to thank my family, especially my husband, John, without whose love and support this book would not be possible.

Craig: Renu is a pleasure to work with, and, without her patience, I would not have learned as much Enterprise Guide as this old "data step programmer" did. Without my wife Donna's support throughout, this endeavor would not have been possible. The healthcare industry, along with various colleagues, have been a strong influence in my professional life; there is always something to learn.

Chapter 1: The U.S. Healthcare System

Introduction 1

Data and Programming Used in This Book 2

Terminology 2

Flow of Administrative Healthcare Data 3

Key Players 4

Medical Claim Submission 4

Claim Processing 5

Recent Legislative Effects 6

- **HIPAA** 6
- **Affordable Care Act** 6
- **All Payer Claims Database** 6
- **Continuing Enhancements** 7

Conclusion 7

Introduction

The U.S. healthcare system is massive, multifaceted, and complex. So is the data that it produces. Your annual visit to the doctor generates data. If you are insured, a form of this data makes its way to your health insurance company, which reimburses your doctor for your care. When you pick up a prescription at your local pharmacy, another type of healthcare data is created. If you give birth at a hospital, the hospital produces yet more data. The insurer houses even more data—providers, benefit structures, and membership all contribute data content to the success of the total business operation.

This book focuses on healthcare data as experienced by a health insurance company. The data is the product of financial reimbursement for health care services. Commonly referred to as administrative healthcare data, it is the result of the relationships among providers, recipients, and payers of health care services. From birth to death, you are generating administrate healthcare data through your interactions with the provider community and your insurer. If you have ever been to a doctor's office, admitted to a hospital, or covered by an employer healthcare plan, then you already have a rudimentary understanding of the material addressed by this book. A comprehensive understanding of this data is a prerequisite for any analytics.

This book explains the source and content of administrative healthcare data and its related management. It illustrates concepts with actual healthcare case studies. Sample data is created in such a way that it closely simulates real healthcare data. All applications are created with SAS Enterprise Guide and Base SAS, which is further described in Chapter 2. They can be lifted straight from the book and put to use immediately.

This book is intended for the programmer/analyst charged with the analysis of administrative healthcare data. You will learn about how the data originates, what it contains, and best practices for programmatically managing this data. This book will give you the solid foundational knowledge to be a successful healthcare data analyst. This book is not intended to teach healthcare data analytics or analytical programming; that would be the next step in the readers' learning path.

Data and Programming Used in This Book

This book uses a fictitious insurance company, Healthy Living, Inc., to illustrate concepts of administrative healthcare data. The company's primary business is to pay medical claims to providers for services rendered to the company's insured members. As a result, Healthy Living, Inc., is the custodian of several large sources of post-adjudication data originating from institutional, professional, and ancillary providers.

Through the use of SAS Enterprise Guide, this book shows you how to build a number of analytical applications of Healthy Living, Inc.'s rich administrative healthcare data. Some key applications include:

- C-section rates across various hospitals
- Top reasons for emergency room (ER) utilization
- Outreach reports identifying children who miss their checkups
- Identifying patients who do not adhere to their medication regimes
- Reporting on key financial metrics

This book is intended for the healthcare analyst regardless of his or her level of proficiency with SAS Enterprise Guide or SAS programming. As a result, SAS code shown throughout the book is deliberately kept at an accessible level. This approach allows the healthcare analyst who is new to SAS to understand the programming techniques shown in the book. The advanced SAS programmer analyst also benefits from the simplified coding approach as they may add complexities and efficiencies to suit their purpose.

Terminology

Language is so important. It is difficult to run a data analysis project if the team members are not speaking a common language, defining terms in the same way, or deriving information with agreed upon algorithms. Terminology and language are of the utmost importance in the discussion and analysis of administrative healthcare data.

Every project should start at the whiteboard and not on a keyboard. Begin by defining common goals, terminology, and methodology. If the goal is to arrive at utilization metrics for office visits per member per year, how are we to define an office visit, a member, a year? You would be surprised at the variety of possible results when everyone is not on the same page.

The importance of getting the terminology on a common plane cannot be underestimated. It is okay to define an office visit by Place of Service *or* by CPT® code (Common Procedural Terminology). But it is not okay to define it both ways in the same project. Spend the time to get it right among the project team.

Table 1.1 defines some terms that will be used synonymously in this book to describe certain concepts.

Table 1.1: Synonymous Terminology

Concept	Synonymous term
Beneficiary of medical services	patient, member, recipient
Supplier of medical services	provider, practitioner
Reimburser of service cost	payer, insurer, managed care organization (MCO), health insurance plan
Medical claim	encounter, claim
Visit	episode of care, encounter

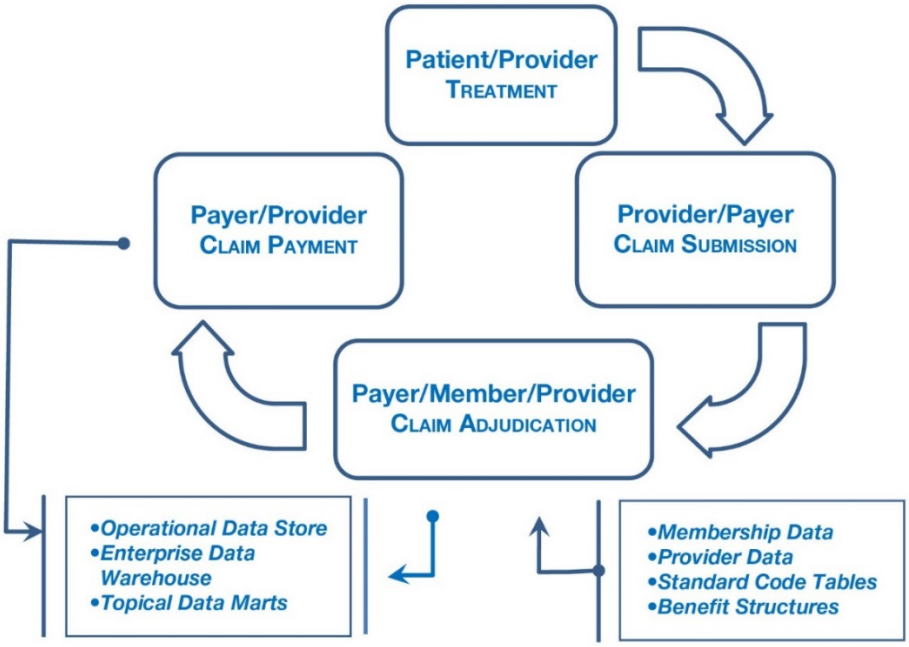
What is the difference between "health care" and "healthcare"? In this book, we will use the two-word phrase to describe the actions of the provider—a well-child checkup is health care. The phrase is an adjective modifying the noun. The single word we will use to describe the system as a whole—healthcare data, healthcare policy. It is generally used as an adjective.

Flow of Administrative Healthcare Data

The U.S. healthcare system is rife with stakeholders and unique relationships among them. To understand the flow of administrative healthcare data you need to understand those relationships and the supply chain that results in the data available to healthcare analysts. If this sounds simple, apologies; it is not!

Think about the flow of data from a provider perspective. The provider interacts with a patient (the insured member), initiating the gathering of information that is needed for the accurate and timely reimbursement by the payer for the services rendered. In a fee-for-service (FFS) model, the provider submits a claim to the payer for reimbursement. In a capitation model, a "medical claim" is still submitted, but only for the purpose of data collection, not actual payment. The payer then adjudicates the claim based on additional information about the member and the provider, resultant data is moved to an operational data store, and the member is notified of any out-of-pocket expenses for which they are responsible. Figure 1.1 graphically describes these important relationships. Reimbursement models will be discussed in Chapter 3.

Figure 1.1: Industry Relationships Drive the Movement of Administrative Healthcare Data



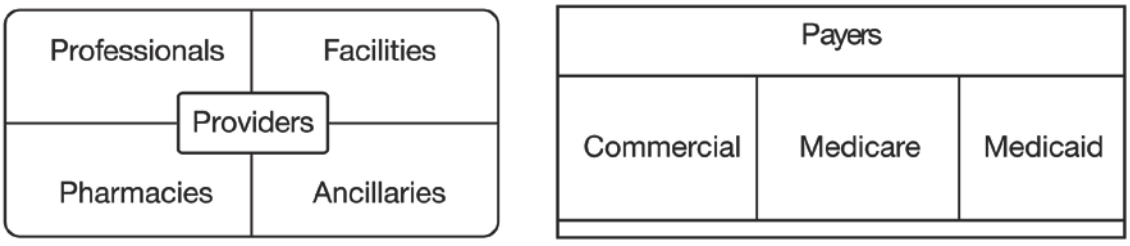
Key Players

One way to conceptualize the data origin is from a provider orientation. They initiate the data flow and, depending upon the provider type, use different claim submission mechanisms and provide different data elements. As Figure 1.2 illustrates, there are only four provider types—Professionals, Facilities, Pharmacies, and Ancillaries. These will be discussed in detail in Chapter 4.

There are three types of payers—Commercial, Medicare, and Medicaid. More on these in Chapter 3.

Policy makers, legislators, and regulators have a significant impact on the behavior of the above mentioned key players. Their role, while very important, will not be discussed in this book.

Figure 1.2: Industry Payers and Providers



Medical Claim Submission

The mechanism by which providers submit reimbursement information to payers has changed dramatically in the past few decades. Initially it was a paper-based system, with those forms and formats improving over time. Many commercial payers with tech-savvy decision makers then worked with their provider community to implement electronic data interchange (EDI) formats for the transmittal of medical claim information. These local initiatives to move away from paper-based instruments provided very efficient processes. With the

implementation of the Health Insurance Portability and Accountability Act of 1996 (HIPAA), yet greater strides were made in the efficient and effective submission of claims information. HIPAA mandated the use of clearly defined electronic formats under "Title II: Administrative Simplification," reducing over 400 EDI formats to a standard set of less than a dozen that are used by all providers and payers.

The form and format of the data transmittal differ slightly depending upon provider type. Professional and Ancillary providers use the CMS-1500 paper form or its electronic counterpart, the 837P format. Facilities use the CMS-1450 paper form or its electronic counterpart, the 837I format. Pharmacists, who have been electronic seemingly forever, use the National Council for Prescription Drug Programs (NCPDP) electronic format.

Despite the HIPAA mandate driven by compliance dates, paper forms are still being used in a limited way. We need not discuss the reasons for this here, but suffice it to say that you may see reference to paper forms for the foreseeable future.

Claim Processing

In Figure 1.1, consider the processes by which the payer receives, processes, and provisions administrative healthcare data. The figure suggests a complex relationship of data elements, both at the source and at the target. There are quite a few moving parts and pieces to consider. For analysts, the data source is typically the Enterprise Data Warehouse (EDW) and/or its many progeny, but for the payer enterprise it is a variety of claims and provider, member, and organizational policy information. Operational data sources reside (and are managed) in the EDW—membership, provider, and plan benefit data are all maintained as current by various departments within the health plan. It is this operational data that is necessary to accurately process inbound claims so that resultant adjudicated claim information is accurately stored in the EDW. The thoughtful analyst will understand how the data moves within the payer organization.

Adjudication, by definition, is "the act of pronouncing judgment based on the evidence presented." Medical claims adjudication is the process by which claims for reimbursement, as submitted by the provider (or patient), are processed into a payment transaction. The "evidence presented" comes in the form of membership, enrollment, and eligibility information; provider enrollment and contractual information; and the submitted claim describing the services rendered. The "judgment pronounced" is the payment made to the provider on behalf of the insured member. The complexities of an adjudication system are not within the scope of this book, but grasping the concept, with its input and outputs, is a key building block in our foundational knowledge.

Also not within the scope of this book is the notion that there is a strong relationship between a benefits structure, service costs, and insurance premiums. At the risk of defending insurance companies, it is fair to say that, in a simple fee-for-service model, insurance companies are a pass-thru facilitator for reimbursement, from member funds, of incurred provider service cost. Having said that, the insurer does have to make a profit on its business investment and, at the same time, be a contending player in the marketplace. A managed care, performance-based, or negotiated fee-for-service model allows for increased efficiencies, which leads to profitability and product sales. Health plans utilize many programs (e.g., disease management programs, provider profiling) to improve the efficiency of the delivery system. Measuring these precise dynamics is what health plan analysts do. Understanding this building block will direct you into the important world of healthcare policy, legislation, actuarial modeling, and sales/marketing.

Recent Legislative Effects

Much attention has been paid to the healthcare industry at the local, state, and federal levels. Several recent legislative efforts bear mentioning, although they will not be discussed further.

HIPAA

The implementation of HIPAA was the "Y2K" problem of the healthcare industry—a major transitional effort that was very resource intensive but necessary and productive for the industry.

From a data perspective, Title II: Administrative Simplification is the most interesting. This component of HIPAA requires the Department of Health and Human Services (HHS) to adopt national standards for electronic healthcare transactions and national identifiers for providers, health plans, employers, and individuals. To date, several of the most important gains from a data perspective are:

- 837 electronic claim format for Institutional, Professional, and Dental providers
- Unique national identifiers
  - National Provider Identifier (NPI)
  - Employer Identification Number (EIN)
- Codification of standard code sets such as diagnostic and procedure codes

We will learn more about these concepts in subsequent chapters.

Affordable Care Act

The passage of the Patient Protection and Affordable Care Act of 2010 (PPACA or commonly ACA) is a monumental change for the industry that is yet to be understood in its totality. While it is primarily "insurance reform," the effect on our data streams is not to be underestimated. At the time of writing this book not all provisions of the legislation have become effective. Political battles over its rollout are continuing while HHS continues to educate the public.

Provisions under the Affordable Care Act will further improve the issues around data that HIPAA initiated. These include requirements to adopt:

- operating rules for each transaction type
- a unique National Health Plan Identifier (HPID) and National Individual Identifier (NII)
- standard and operating rules for
  - electronic funds transfer (EFT)
  - electronic remittance advice (RA)
  - claims attachments

In addition, insurers will be required to certify their compliance with all rules and regulations. Substantial penalties for failures to certify or comply with the new standards and operating rules are provided for.

One concern to watch for as "improved" data streams and reimbursement models are defined is the possible loss of data granularity.

All Payer Claims Database

At the state level there has been much legislation surrounding the need for consolidated sources of claims information. At least twelve states have recently enacted legislation and/or started to collect healthcare claims data from commercial and public payers in an effort to establish an all-payer claims database (APCD).

While the contents of individual states' APCDs vary, they usually include data elements from member files, provider files, and claims files. The medical claims files include healthcare-related data elements such as diagnosis codes, procedure codes, pharmacy codes, insurance product type, facility type, cost amounts, and

provider information. In essence, the effort is to build a statewide or regional database that would mimic the structure and combine the information that MCOs have in-house.

Policy makers and legislators have been looking for a data source to begin to understand patterns and trends of healthcare utilization and costs. This should prove to be an excellent resource in the coming years. Keep an eye out for how you can play in this space.

Continuing Enhancements

While not directly related to legislative actions, it is important to note that the industry is frequently undergoing change as dictated, among other reasons, by changing business models and technology. Several examples will be discussed in later chapters—code sets are revised (e.g., moving from ICD-9 to ICD-10) and electronic transmission formats redefined (e.g., the 5010 version of the 837 electronic claim submission). It is mandatory that the analyst keep abreast of these changes and adjust business practices and programming as necessary.

Conclusion

Knowing the origin of every data element in any healthcare analytic project is of paramount importance. One cannot be the best analyst possible without an intimate knowledge of the data—from source to repository. From initiation, transformation, and relationship development to information and action, it is incumbent upon every analyst to understand the original source and content of administrative healthcare data.

Chapter 2: Introduction to SAS Enterprise Guide and Sample Data

Introduction 9

Sample Data 9

What Is SAS Enterprise Guide? 11

SAS Libraries and Data Sets 12

- **Create a Permanent Library** 12
- **View a SAS Data Set** 13
- **SAS Data Types** 13

Formats 14

- **DRG Format** 15
- **Diagnosis Code Format** 16

Applying Formats to SAS Variables 19

- **Formatting an Existing Variable** 19
- **Placing Results of a Format into a New Variable** 21

Conclusion 21

Introduction

This chapter shows the sample data used in the book and provides a quick introduction to SAS Enterprise Guide, the book's tool of choice for data management. We achieve these dual objectives through a seamless presentation of administrative healthcare data and SAS Enterprise Guide. You will view sample data and learn some basics of SAS Enterprise Guide at the same time. For instance, while becoming familiar with codes in healthcare data, you will understand how SAS formats can be used to categorize and interpret them.

In this chapter you will:

- Navigate in SAS Enterprise Guide
- Understand the structure of our sample claims data
- Learn about SAS data types
- Use some functions in creating new variables
- Use SAS formats effectively

Sample Data

Sample data for this book is created from the experience of a fictitious health insurance company, Healthy Living, Inc. The company's primary purpose is to adjudicate and pay claims for the healthcare services its members receive from providers. As a result, it receives and sends electronic data to a wide variety of sources. As shown in Table 2.1, sample data used in the book reflects the range of claims the company processes.

Table 2.1: Types of Claims Data

Type of claim	Examples of healthcare services
Facility	Inpatient hospital, outpatient surgery, emergency room (ER) visit, rehabilitation, home health, hospice, and so on
Professional/Ancillary	Routine office visits, specialist visit, durable medical equipment, transportation services, and so on
Pharmacy	Prescription drugs

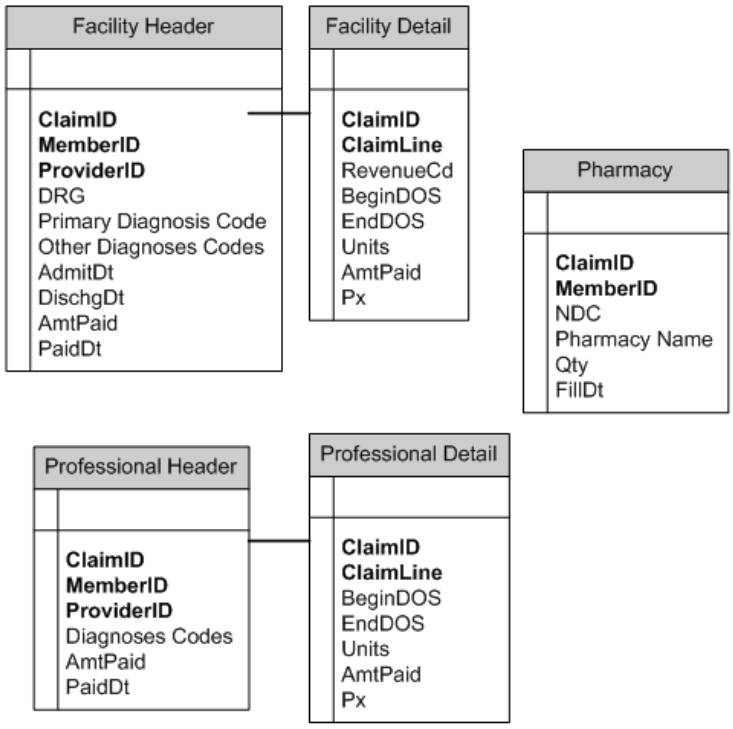
Data for the three types of claims is stored in five SAS data sets, which, along with sample code and supplemental data tables, are available for download from the authors' webpages. The key SAS data sets are listed below.

- Facility Claim Header
- Facility Claim Detail
- Professional Claim Header
- Professional Claim Detail
- Pharmacy

A simple way to distinguish between header and detail files is to remember that header files describe the patient and their medical condition, while detail files contain the treatments provided. Both sets of files are intended for payment purposes.

The header files are uniquely defined by ClaimID, whereas detail files are defined by ClaimID and ClaimLine. The join (match merge) relationship between header and detail tables is one to many as one claim from the header file may join with multiple claim lines from the detail file. Pharmacy claims are point-of-sale data and contain information about prescription drug fills. Line numbers do not exist in Pharmacy data, as each fill or refill is a unique claim. Figure 2.1 provides a conceptual representation of claims data that is worked with in this book. Further detail found in these possible files is discussed later in the book.

Figure 2.1: Conceptual Data Model



For the sake of simplicity only selected columns from tables and selected tables are displayed. For example, dimension tables containing healthcare codes and their nomenclature are not included in Figure 2.1. This omission is by design, as healthcare codes are discussed extensively in subsequent chapters.

A member's healthcare utilization history is complete only when all five tables in the data model in Figure 2.1 are queried. Visits to the ER and inpatient stays come from the two Facility files. Visits to professionals such as a primary care doctor or a specialist are found in the Professional files, as well as any ancillary services such as diabetic supplies. A member's prescription drug history is gathered from the Pharmacy file. In the next sections, you will learn about SAS Enterprise Guide while viewing one of the tables shown in Figure 2.1.

Figure 2.1 shows ProviderID on Facility and Professional header tables. The reality is somewhat different and more complex. A Professional claim may be billed under one provider. However, each claim line could be generated by a different practitioner. The Billing Provider is on the header table and the Servicing (Rendering) Provider is on the detail table for professional claims. A Facility claim may have several providers listed—Billing, Attending, Operating, and Other(s). These are found on the header table with no providers noted on the detail table.

What Is SAS Enterprise Guide?

SAS Enterprise Guide is a client-based application that provides point-and-click access to the power of SAS programming. If you have never programmed in SAS, you can point and click your way through tasks that generate code in the background. The application offers extensive help, including self-paced tutorials and task-specific help. These may be accessed from the **Help** menu. If you are an expert SAS programmer, you can use it to write code and check your syntax. Regardless of your level of SAS experience, you can use SAS Enterprise Guide to increase your knowledge of SAS software. Play with one of its numerous tasks and study the code it creates.

SAS Enterprise Guide contains a full host of data management, analytical, and even administrative capabilities. The application offers myriad wizards to guide you through these tasks. The **Tasks** menu in Enterprise Guide includes data management, manipulation, statistical analysis, and reporting abilities. Table 2.2 shows some of its capabilities.

Table 2.2: Selected Capabilities of Tasks Menu in Enterprise Guide

Data	Describe	Graph
Filter and Sort	List Data	Bar Chart
Append Table	Summary Statistics	Pie Chart
Create Format	Summary Tables	Line Plot
Transpose	List Report Wizard	Scatter Plot
Random Sample	One-way Frequencies	Bubble Plot
Rank	Table Analysis	Tile Chart

An especially useful feature of SAS Enterprise Guide is that it allows you to create, run, and schedule a process flow. A process flow is a collection of tasks and code nodes that can be connected with each other. For example, you import data via the **Import Data** task. You manipulate and transform this data using SAS code. Finally, you use the **Summary Tables** task to create a report. You can link the **Import Data** task with SAS code, which can be linked to the **Summary Tables** task. SAS Enterprise Guide allows you to run this process piecemeal or in its entirety. Chapter 5 shows you how to create and run a process flow.

SAS Libraries and Data Sets

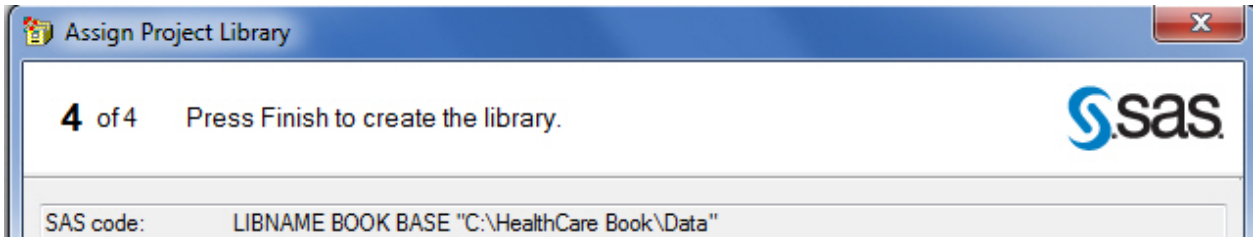
A SAS library is a collection of SAS data sets and is used to organize data. A SAS data set can be permanent or temporary. A permanent data set sits in a physical location, whereas a temporary data set, while also in a physical location, exists only for the duration of a SAS session. Temporary data sets are stored in a library referred to as WORK. Permanent data sets are stored in permanent libraries that can be predefined by SAS or defined by the user. SASUSER is an example of a library defined by SAS.

Create a Permanent Library

Use SAS Enterprise Guide to create a library to access the data sets used in the book.

1. Select **Assign Project Library** from **Tools** in SAS Enterprise Guide.
2. On page 1 of **Assign Project Library** task, enter BOOK in **Name**.
3. On page 2 of **Assign Project Library** task, enter a physical location in **Path**.
4. On page 4, verify the code created by the task. This is shown in Figure 2.2.

Figure 2.2: Create a Library in SAS Enterprise Guide



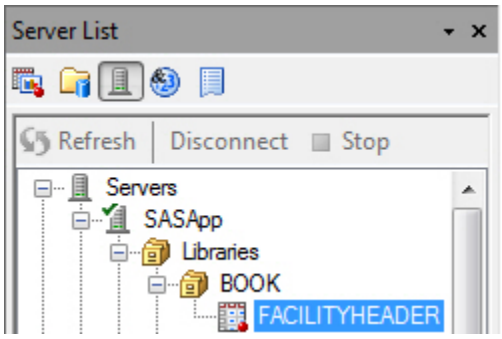
The **Assign Project Library** task generates the **LIBNAME** statement, which is a programmatic way to allocate a library as a resource to our session. The keyword BOOK is a shorthand reference to the physical folder location.

View a SAS Data Set

Open FacilityHeader data set.

1. Click on  to view data sets.
2. Navigate to BOOK library as shown in Figure 2.3.

Figure 2.3: Navigate through SAS Server Libraries



3. Double click on FacilityHeader to view data. (This is the default action; be aware that the default action may have been redefined by the system administrator or the user.)

SAS Enterprise Guide provides you with multiple ways to access its point-and-click tasks. In addition to the **Tasks** menu, tasks are also presented whenever a data set is viewed. As Figure 2.4 shows, myriad data management and analytical tasks are available from the task line located above column names of a data set.

Figure 2.4: FacilityHeader Data Set

FACILITYHEADER

Filter and Sort | Query Builder | Data | Describe | Graph | Analyze | Export | Send To

	ClaimID	MemberID	Sex	DOB	ProviderID	AdmitDt	DischgDt	DRG	AmtPaid	PaidDt	BillType	Pdx
1	C077627745	M474065868234...	F	1982-01-19	P623457604	2011-02-08	2011-02-11	766	7338.37	2011-03-27	111	65421
2	C095733386	M473079937226...	F	1971-02-28	P623457604	2011-01-25	2011-01-29	765	11111.70	2011-03-04	111	65421
3	C095744102	M475025611242...	F	1970-01-20	P623457601	2011-01-28	2011-01-31	766	8357.31	2011-03-04	111	64201
4	C095744832	M000887106044...	F	1985-05-19	P623459983	2011-01-25	2011-01-28	766	5421.20	2011-03-04	111	65981
5	C095745566	M001999791072...	F	1988-10-27	P623459983	2011-01-09	2011-01-11	767	5414.71	2011-03-04	111	66401
6	C095745613	M002889966095...	F	1980-03-07	P623457604	2011-01-23	2011-01-25	765	5414.71	2011-03-04	111	65981
7	C095745642	M003199877102...	F	1973-02-28	P623459983	2011-01-16	2011-01-19	767	5414.71	2011-03-04	111	64881
8	C095757137	M002046933073...	F	1980-04-12	P623459983	2011-01-29	2011-02-01	766	3287.93	2011-03-04	111	65421
9	C095761276	M000117744011...	F	1972-04-08	P623459983	2011-01-30	2011-02-03	765	6119.45	2011-03-04	111	65421
10	C095763270	M386943399153...	F	1978-04-02	P623459983	2011-02-01	2011-02-03	774	3324.54	2011-03-04	111	65421
11	C095768046	M001484017061...	F	1976-12-25	P623457601	2011-02-01	2011-02-04	765	6119.45	2011-03-04	111	65421
12	C095768066	M477083374262...	F	1986-01-06	P623457604	2011-01-31	2011-02-03	766	4873.05	2011-03-04	111	66331
13	C095771806	M000122750011...	F	1987-11-21	P623459983	2011-01-28	2011-02-08	774	3287.93	2011-03-04	111	64891

SAS Data Types

SAS has two types of variables: character and numeric.

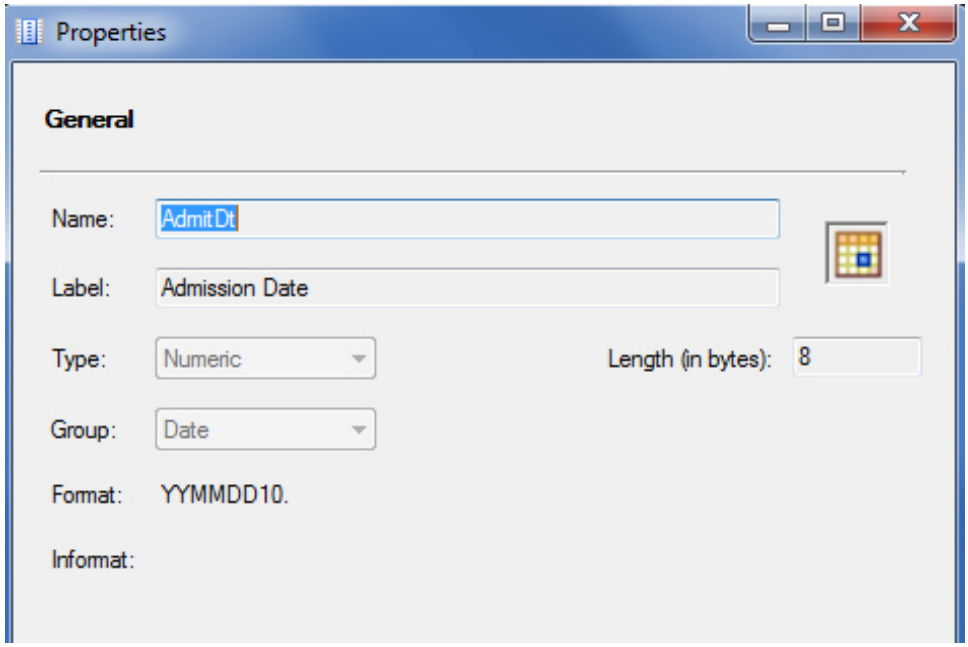
Character, or text, variables in SAS are denoted by  . As shown in Figure 2.4, ClaimID is a character variable.

Numeric variables are denoted by  . AmtPaid in the FacilityHeader data set is a numeric variable. A date variable is also numeric as SAS stores its value as the number of days since January 1, 1960.

Notice that the date variables such as AdmitDt in Figure 2.4 are marked by the calendar icon  . This is because it has been formatted as a date. A format changes the appearance of a variable's display, not its internal value.

To find out more about AdmitDt and its format, you can view the variable's properties. Right click on AdmitDt and choose **Properties**. As Figure 2.5 shows, AdmitDt is numeric, but it is formatted to appear as a calendar date. This makes it convenient for you to interpret dates.

Figure 2.5: View Properties of a Variable



The format *yymmdd* is supplied by SAS. The number 10 denotes its length. An example is 2012-12-01. To distinguish formats from SAS variables, a period is used at the end of a format name. In the next section, you will learn how to create your own formats.

Formats

Healthcare data is full of codes. As shown in Figure 2.4, both DRG (Diagnosis Related Group) and PDx (principle diagnosis code) columns contain codes. User-created formats can be used to create a crosswalk between codes and their nomenclature.

DRGs are used to classify patients with similar resource consumption for payment and statistical reasons. A claim's PDx and other information are used by grouper software to generate the DRG code. Please note that, for the moment, DRG is used in a generic way. There are many "flavors" of DRG, as will be discussed in Chapter 5. Other healthcare codes will be discussed extensively later in the book.

All healthcare codes are nominal and should be stored and used as character data. If you do not intend to do arithmetic on a variable, then store it as a character.

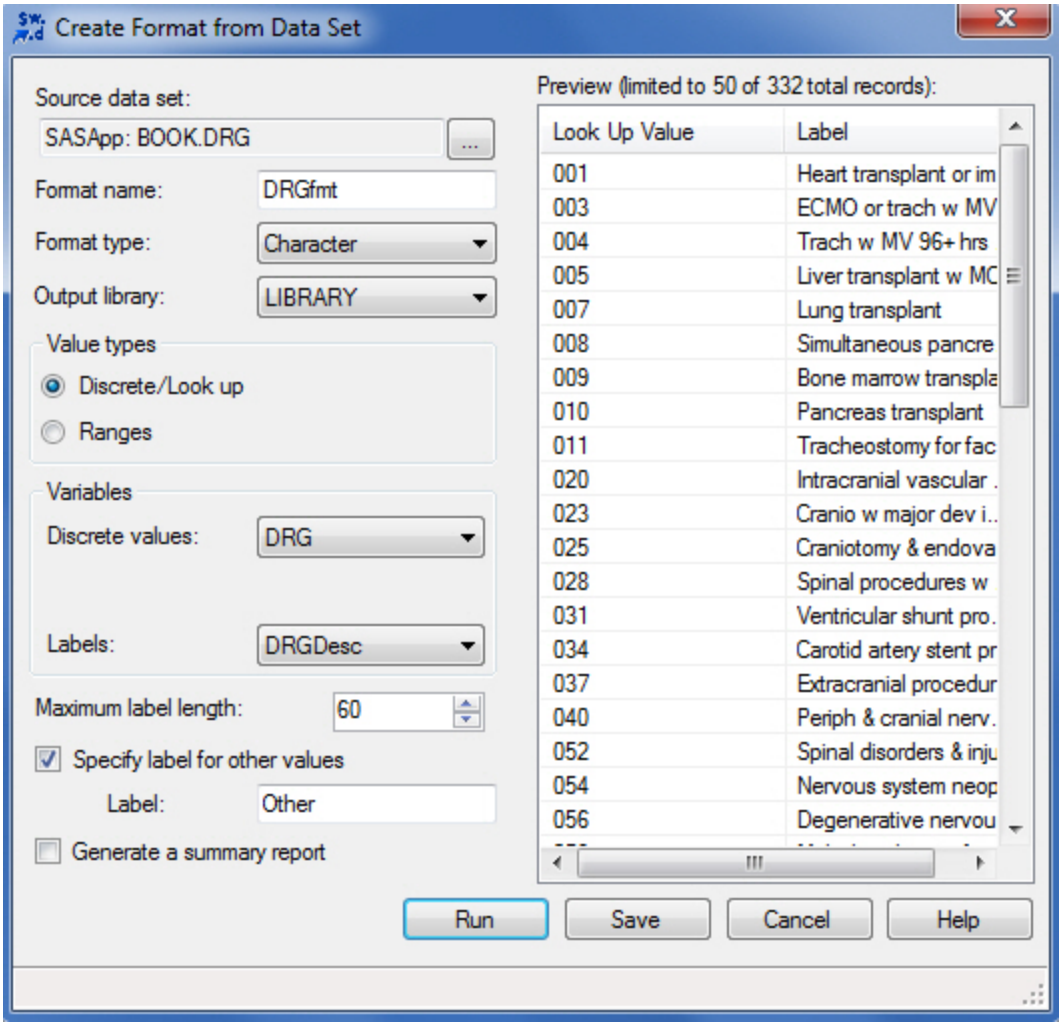
DRG Format

Let us begin by creating a format for DRG. DRG codes and their nomenclature are stored in a data set called DRG (a dimension table), which is located in the library referenced by BOOK.

Create a format from an existing data set.

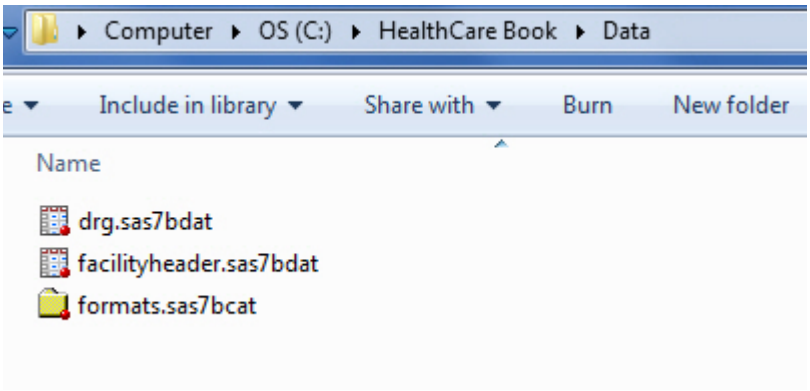
1. Select **Create Format from Data Set** from Tasks—Data.
2. In **Create Format from Data Set**, specify the following as shown in Figure 2.6.
   a. Select Book.DRG under **Source data set**.
   b. Key in DRGfmt under **Format name**. This is a user-chosen name.
   c. Select **Character** under **Format Type**. Formats can either be numeric or character. DRG is a character format because DRG code is stored as a character variable.
   d. Select **LIBRARY** as the **Output library**. This specifies that the format catalog will be located in a permanent library. A format catalog can be thought of as an object containing one or more formats.
   e. Select **Discrete/Look up** in **Value types**. This specifies that there is a one-to-one correspondence between DRG code and its description.
   f. In **Variables**, pick DRG. In **Labels**, pick DRGdesc.
   g. Choose 60 as the length of the label. This is to ensure that none of the DRG nomenclature will be truncated.
   h. Verify from **Preview** that Look up Value and Label appear correctly.

Figure 2.6: Create Format from Data Set



3. Click **Run** and verify with Windows Explorer that the formats catalog has been created (see Figure 2.7).

Figure 2.7: The Formats Catalog



By default, SAS searches for formats in two libraries, WORK and LIBRARY. You also have the option of adding other libraries to the search path with the FMTSEARCH option in SAS. For the purpose of simplicity, this book saves formats in LIBRARY, which has been assigned the same physical location as BOOK.

You will need to click **Run** at the end of each procedure in this book, unless a step explicitly tells you otherwise.

Diagnosis Code Format

Often data sets containing codes need to be manipulated before a format can be created and used appropriately. As Figure 2.8 shows, the look-up table Dx contains diagnosis codes and their nomenclature. Notice that the codes contain decimals, an industry standard.

Figure 2.8: The Dx Look up Table

	Dx	DxDesc
1	650	NORMAL DELIVERY
2	651.01	TWIN PREGNANCY, DELIVERED
3	V27.0	OUTCOME OF DELIVERY SINGLE LIVEBORN

The FacilityHeader table contains PDx and other diagnosis codes (Dx2–Dx8) columns that need to be formatted. As Figure 2.9 shows, these columns do not contain decimal places. The decimals have been stripped to save space, a customary practice in most databases.

Please note that the data format inconsistencies discussed in this section (diagnosis codes with or without decimals) are not uncommon in healthcare data storage and management. The reader is cautioned to be vigilant throughout any given project.

Figure 2.9: Diagnosis Code Columns in FacilityHeader

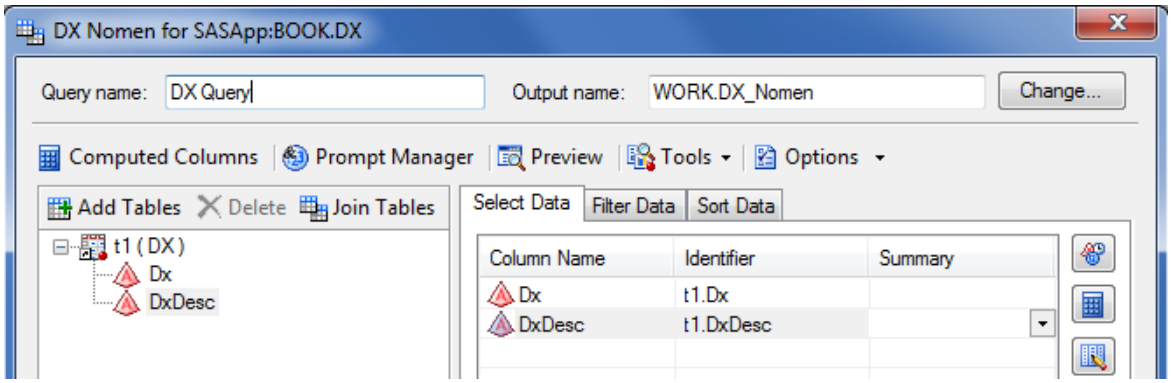
Pdx	Dx_2	Dx_3	Dx_4
650	V270		
650	V270		
650	V270		
650	V270		
65101	2851	64822	V272
65101	V272	65701	65951

To create a format that can be correctly applied to the diagnosis code columns in FacilityHeader, you must first remove decimals from the Dx columns in the Dx table.

This can be done using SAS Enterprise Guide's tasks.

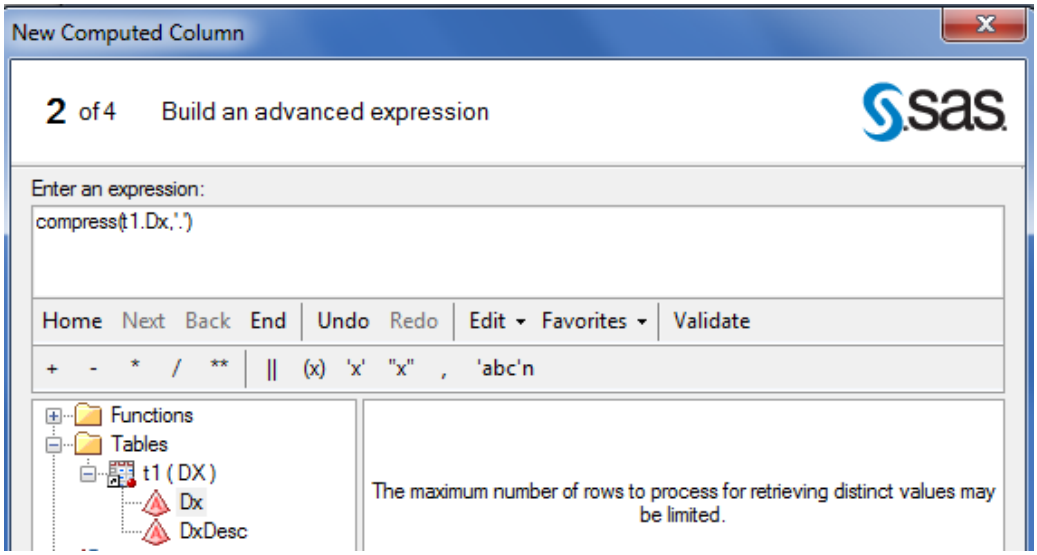
1. Select **Query Builder** from Tasks—Data.
2. In **Query Builder**, specify the following as shown in Figure 2.10.
   a. Type in Dx Query in **Query Name**.
   b. Type in Dx_Nomen in **Output Name**. The resultant data set will be called Dx_Nomen, and it will reside in the temporary WORK library.
   c. Select Dx and DxDesc columns. There are several ways to do this. One way is to right click on each column in the left-hand pane and choose **Select Column.** Another is to drag and drop a column to the right-hand pane. A third way is to double click the variable.

Figure 2.10: Query Builder Task



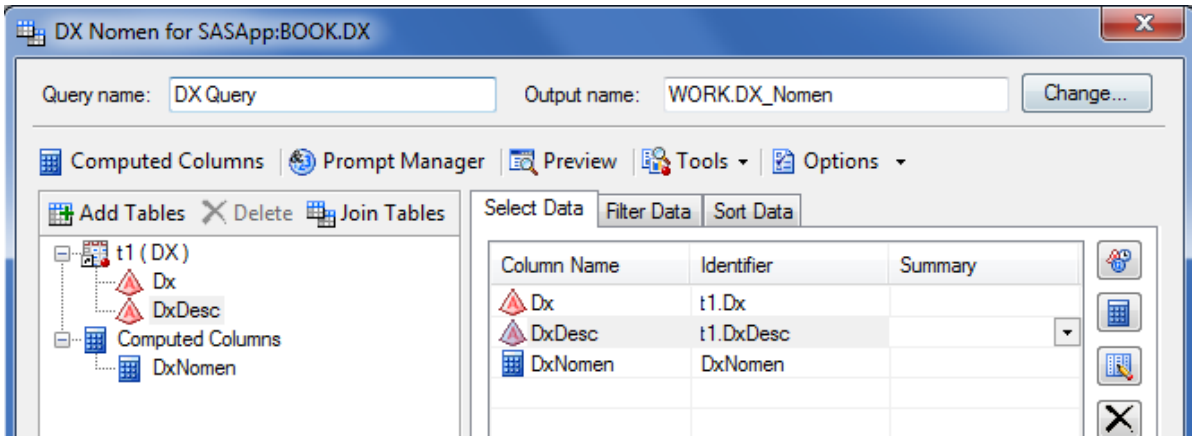
3. To create a computed column containing Dx codes without any decimals, click on **Computed Columns** located under **Query name**.
4. In the **Computed Columns** window, click **New**. This starts the **New Computed Column** wizard.
5. In Step 1, select radio button **Advanced expression** in **Select a type** window.
6. In Step 2, use the compress function to remove decimal places from Dx. This is done by entering the expression as shown in Figure 2.11.

Figure 2.11: Create a Computed Column



7. In Step 3, **Modify Additional Options**, enter DxNomen in the identifier and column name fields. Identifier is an element that is used by SAS Enterprise Guide to generate code. For the purpose of point-and-click tasks, the identifier can be the same as column name.
8. Confirm that a new column DxNomen is added (Figure 2.12).

Figure 2.12: Query Builder Task with Computed Column



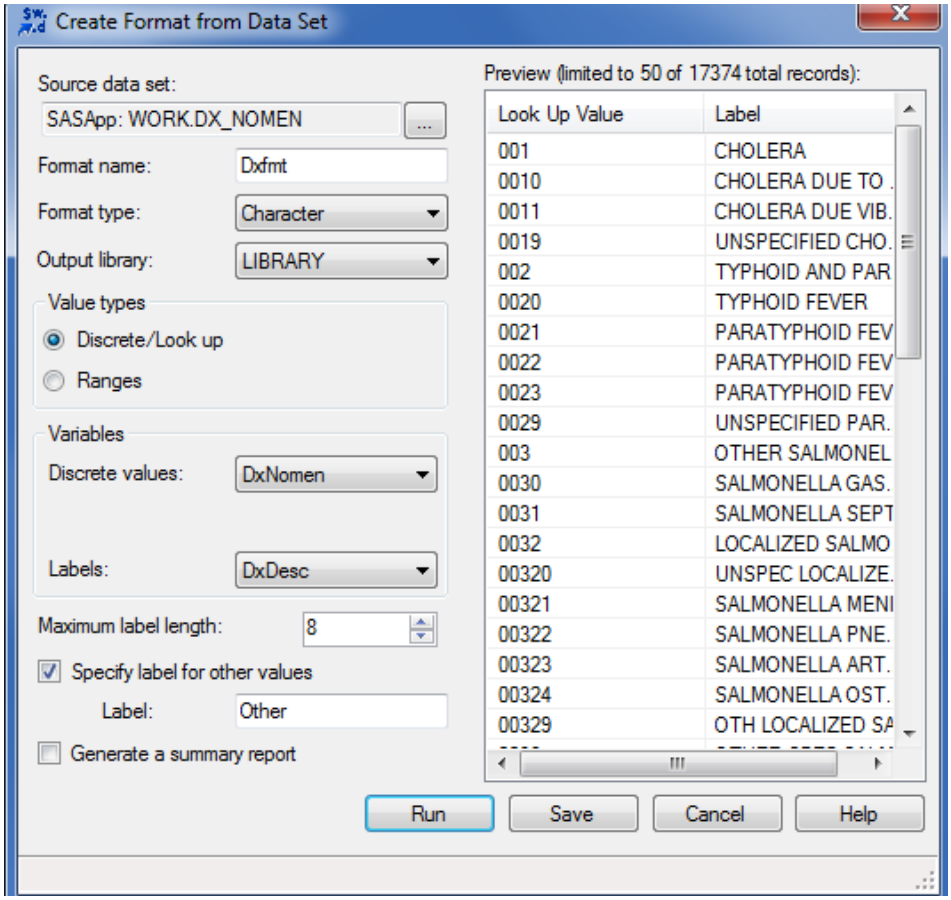
9. **Run** the query and confirm that the data set Dx_Nomen contains a column called DxNomen, which has been stripped of its decimals (Figure 2.13).

Figure 2.13: Data Set DxNomen

Dx	DxDesc	DxNomen
001	CHOLERA	001
001.0	CHOLERA DUE...	0010
001.1	CHOLERA DUE...	0011
001.9	UNSPECIFIED...	0019
002	TYPHOID AND...	002
002.0	TYPHOID FEVER	0020
002.1	PARATYPHOID...	0021
002.2	PARATYPHOID...	0022
002.3	PARATYPHOID...	0023

10. Create a format with data set Dx_Nomen following steps 1–3 from the **DRG Format** section. A completed **Create Format from Data Set** window is shown in Figure 2.14.

Figure 2.14: Diagnosis Format



SAS is full of functions. There are functions that operate on character variables, those that work on numeric variables, and even those that carry out date manipulations.

The COMPRESS function works on character variables, character strings, or expressions that resolve to either of the aforementioned. Its most common usage is to remove a character from a variable value. Without its second argument, it can be used to remove spaces from a variable.

Example 1: COMPRESS(‘99.0’,’.’)

Example 2: COMPRESS(‘99 0’)

The result of both examples is a character string equaling 990.

Applying Formats to SAS Variables

This section demonstrates how to apply formats to data set variables so that the descriptors for healthcare codes are displayed. This can be done in two ways. The first is to change the appearance of an existing variable. The second is to create a new variable containing the format.

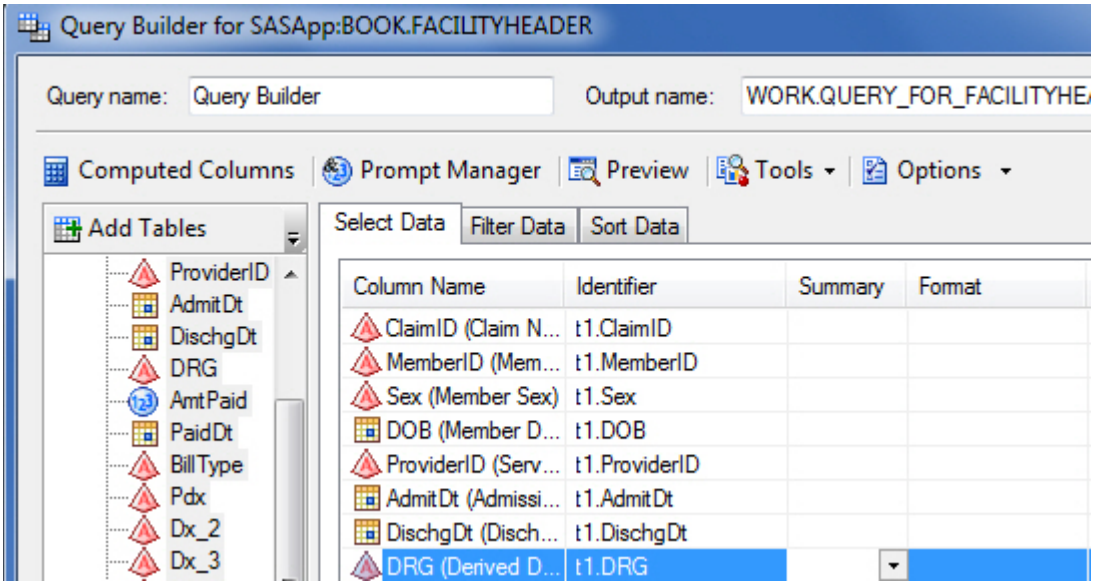
Formatting an Existing Variable

Use Query Builder to apply the DRGFMT. format to the existing DRG column in FacilityHeader.

1. Select **Query Builder** from **Tasks—Data**.
2. In **Query Builder**, select all columns from FacilityHeader.

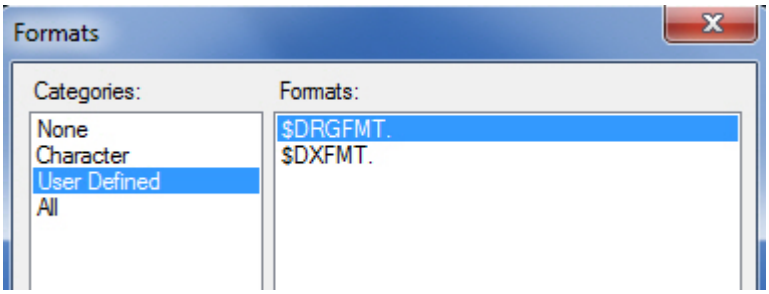
3. Double click on the cell corresponding to the DRG row and the Formats column as shown in Figure 2.15.

Figure 2.15: Format an Existing Variable



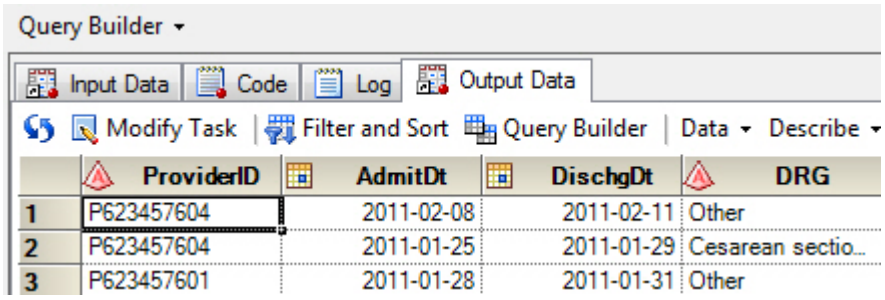
4. Click **Change** in the Properties for DRG window.
5. Select **User Defined** under **Categories** and then choose $DRGFMT (see Figure 2.16).

Figure 2.16: Use a User-Defined Format



6. Confirm that the variable DRG has the format $DRGFMT. and now shows the description. Remember that DRG is still stored as a nominal code; it only appears as the description because of the format (see Figure 2.17).

Figure 2.17: Formatted DRG



Placing Results of a Format into a New Variable

Codes are so universal in healthcare data that analysts routinely like to see both the code and its nomenclature in the same table. To see diagnosis code and its nomenclature as two separate variables, you will need to apply the format and place the result in a new column.

The previous sections have shown you how to use Query Builder to create a new variable. This section focuses on the PUT function, which allows you to apply the $DRGFMT. format.

The PUT function has two arguments and is generally used to convert numeric data to character. Its result is always character. The second argument of the PUT function is a format. Consider the following two uses:

```
DischgDtChar=put(DischgDt,mmddyy10.);
```

DischgDtChar, a character variable, is produced using the *mmddyy10* format.

```
DRGDesc=put(DRG,$DRGfmt.);
```

DRGDesc, a character variable, is created by first taking the value of DRG and applying the format $DRGfmt. to it. The $ sign signifies that the format is a character format, as it was created for a character variable.

For the sake of simplicity, formats are applied and added as new variables to data sets used in the book. This approach will enable you to access codes and their nomenclature in the sample tables, saving you the trouble of downloading format catalogs.

> Using formats to look up codes and their nomenclature is an extremely powerful technique. It avoids time and resource-intensive joins between tables.
>
> However, format catalogs are often specific to a particular SAS version. Older format catalogs may require conversion to be used in newer versions of SAS. This is the key reason that this book saves nomenclature as an additional field in the claims data tables. This is not an efficient technique, but it serves the purpose of making the data widely available.

Conclusion

This chapter has shown that Healthy Living, Inc.'s data model is composed of claims data from a variety of sources. It has also covered the fundamentals of SAS Enterprise Guide as applied to a key claims table, the FacilityHeader. Specifically, it showed you how to:

- Navigate in SAS Enterprise Guide
- Use some functions in creating new variables
- Use SAS formats effectively

Chapter 3: The Payers

Introduction 23
Health Insurance 23
Medicare 24
Medicaid 25
Commercial Insurance 26
Others 26
TRICARE 26
CHAMPVA 26
FECA Black Lung 26
Conclusion 27

Introduction

The "payer" of healthcare claims is an important player in the U.S. healthcare system. Also known as an insurer or managed care organization, the payer plays a key role in the financial interaction between providers and recipients of healthcare services. An insured recipient of service is also known as a "member" of a payer organization.

In this chapter you will:

- Learn about models of health insurance, including the rudiments of various reimbursement mechanisms.
- Identify and differentiate between major payers.

Health Insurance

Many different business models exist in today's marketplace and many more will come and go with evolving policy and legislative changes. This is a healthy process and serves to benefit the consumer of healthcare services. This section highlights some common models. You should explore the insurance model(s) used in your organization and understand that each model impacts health care delivery and utilization differently.

From a health plan's member perspective several major benefit delivery models can be defined as follows. The industry is rife with many variations of these models. The definitions are intended only to provide a framework from which to think about the workings of health insurance. You should understand your own organization's benefit model.

- **Health Maintenance Organization (HMO)**: HMO is a delivery model that requires members to choose an in-network primary care physician (PCP). This professional provider is the gatekeeper for the member's primary care and referrals to specialists. The insurer pays a fixed monthly fee (*capitation*) to the member's PCP, regardless of how much health care is needed or delivered. This was one of the original "managed care" models rolled out in the early 1980s. HMOs provide a wide variety of medical services, from office visits to hospitalization and surgery. There are exceptions, but most

HMO members must receive their medical treatment from providers within the insurer's provider network.

- **Preferred Provider Organization (PPO)**: In this model, there is no requirement for the choice of a PCP (gatekeeper). Rather than a capitated arrangement, services are reimbursed as they are incurred. Members are encouraged to use the insurer's provider network with financial incentives such as variable co-pays and coinsurance. Services rendered by healthcare providers who are not part of the PPO network incur more out-of-pocket expense for the member.
- **Point of Service (POS)**: A hybrid of HMO and PPO, this healthcare option allows members to choose, at the time medical services are needed, whether they will go to a provider within the plan's network or seek medical care outside the network. The member pays no deductible and only a small co-payment when the healthcare provider is within the network of the managed care organization (MCO). A PCP is usually chosen and is responsible for all referrals within the network. If the member goes outside of the network for health care, a deductible and co-payment are a percentage of the provider's charges.

From a provider's perspective two major reimbursement models can be defined as follows:

- **Fee-for-Service (FFS):** Providers are paid a specified amount for each service provided. FFS is also referred to as an "indemnity" arrangement. The fee schedule is arrived at by contractual agreement and is generally less than the provider charges; FFS is sometimes called a "discounted" fee-for-service arrangement. The procedure-specific fees may be arrived at as a percent of charge, a flat rate, a per diem, a case rate, or any number of other inventive arrangements. If the provider is within the payer's network, they cannot balance bill the patient, which provides a strong incentive for a member to use network providers.
- **Capitation**: A capitated arrangement between the payer and providers pays for services based on the number of members who are covered for specific services over a specified period of time rather than the cost of or number of services that are actually provided.

A healthcare claim is simply a request for reimbursement of services rendered. One of the primary functions of the health insurer is to adjudicate claims submitted by the service providers on behalf of the insured members. Also known as "claims processing," this is the practice of bringing together requisite information to decide what payment amount is to be sent to the provider. Residual amounts may accrue to the member as out-of-pocket expenses. Since the payer is the financial link between the member/patient and the provider, there is a provider component (the contract) and a member component (a benefit structure) that feeds the adjudication process. Two products result from adjudication: the provider paycheck and administrative healthcare data. The latter is the subject of this book.

Medicare

Medicare is the largest of all the payer organizations within the industry. This health plan is administered by the Centers for Medicare & Medicaid Services (CMS), one of eleven operating divisions within the Department of Health and Human Services (www.hhs.gov). According to the CMS website (www.cms.gov), in 2012 the program covered in excess of 50 million individuals at a cost of $536 billion—16 percent of the federal budget. CMS contracts with nongovernment carriers to enroll members and reimburse providers. To become a Medicare beneficiary (an eligible member), one must be over the age of sixty-five, be disabled, or have end-stage renal disease (ESRD). Table 3.1 details the major components of Medicare coverage:

Table 3.1: Elements of Medicare

Component	Coverage	Cost
Part A	▪ Inpatient hospital care ▪ Hospice care ▪ Some home health care ▪ Limited skilled nursing home care	None direct to the beneficiary—covered by employment taxation

Component	Coverage	Cost
Part B	▪ Outpatient hospital care ▪ Physician and other professional services ▪ Occupational and physical therapy ▪ Other home health care	Annual premium and deductibles set by Social Security income dependencies
Part C	▪ A package of Parts A, B, and D coverage constructed and offered by commercial insurers ▪ Replaces "traditional Medicare" for those with special needs	Premiums and out-of-pocket expenses are set by the insurer based on a capitation rate received from CMS
Part D	▪ Prescription drugs ▪ CMS approved plans are provided by commercial insurers	Premium set by the commercial carrier

Medigap Supplemental Insurance is privately purchased coverage for services not covered by Parts A or Part B. Deductibles and coinsurance apply. Ten standardized plans are sold by insurance companies and regulated by CMS.

Medicare is so dominant in the U.S. healthcare system that few providers and no insurance companies can ignore it. Medicare payments are either made directly to providers or indirectly to insurance companies. Medicare payment and condition classification methodologies are industry standards, used by providers, insurance companies, and state governments. For example, hospitals leverage Medicare methodologies with their non-Medicare business models. State governments routinely implement Medicare methodologies or a variant of Medicare methodologies for their Medicaid programs.

Medicare's payment methodologies are categorized by type of health service. It pays for hospital services on a prospective or predetermined basis. Its inpatient prospective payment system (IPPS) is based on the Diagnosis Related Groups (DRG) originally developed by Yale University. DRGs classify all possible diagnostic and procedural codes from the International Classification of Diseases, 9th Revision (ICD-9), into meaningful groupings. Medicare's outpatient prospective payment system (OPPS) is similar to its inpatient prospective methodology—outpatient services are classified under groups called Ambulatory Payment Classifications (APCs). Long-term acute care and psychiatric acute care facilities use similar DRG systems. Skilled Nursing Facilities (SNF) use Resource Utilization Groups (RUGs) and rehabilitation facilities use Case Mix Groups (CMGs).

Medicare uses a resource-based relative value scale (RBRVS) to reimburse physicians. Each service is given a relative value unit (RVU). The physician's fee is calculated by multiplying a conversion factor to the RVUs, adjusting for geographical cost variations.

Medicaid

Medicaid is a health insurance program jointly funded by state and federal governments but administered by individual states. The Kaiser Family Foundation (www.kff.org) reported that "in June 2012, Medicaid enrollment reached 54.1 million as high unemployment and falling incomes led many families to turn to Medicaid for coverage." While the number of Medicaid-insured lives is greater than that of Medicare, the plans themselves are local to states and therefore thought of as separate health plans.

Eligibility for Medicaid benefits is based on family income as it relates to the federal poverty level (FPL). With the Affordable Care Act the default income threshold is 133% of FPL. By waiver from CMS, revenue-poor states may lower this threshold. A generous state legislature may raise the bar for its constituents.

Benefit structures vary among the states as well. CMS mandates a minimum set of benefits to include inpatient, outpatient, and professional medical services. States may, at their discretion, add benefits such as vision and dental care and prescription drugs. Medicaid is particularly attentive to preventive care services for mothers,

babies, and adolescents, requiring that these services be provided for the eligible population. Nursing home services are also paid for by state Medicaid programs.

"Dual eligibles" are individuals who are entitled to traditional Medicare (Parts A and/or B) coverage and are also eligible for some Medicaid benefits. Individuals with Medicare and limited income may receive help from a state Medicaid program to pay for out-of-pocket medical expenses. These benefits are sometimes called Medicare Savings Programs (MSP). Services that are covered by both programs are *paid first by Medicare* and the difference by Medicaid, up to the state's payment limit.

Commercial Insurance

For those people who do not qualify for one of the government health plans, commercial insurers offer a full range of benefit structures and reimbursement models. These companies may operate as for-profit or not-for-profit entities but are always regulated by the insurance department of the state in which they do business.

The Affordable Care Act has had a major impact on this segment of the industry. Rules have been set forward regarding minimum eligibility/enrollment criteria and out-of-pocket cost for preventive care services.

Some of the better known companies in this category are United Healthcare, CIGNA, and the Blue Cross and Blue Shield plans. There are also many small regional carriers that specialize in segments of the market.

> Commercial insurance is typically thought of as the employer group market, but the insurer will also offer coverage plans for the individual market. This is occasionally referred to as the self-pay or direct-pay population.

Others

Several lesser known payers bear mentioning. You will see these are listed on the CMS-1500 claim form as a potential source for reimbursement. They also pay facility claims as appropriate.

TRICARE

TRICARE, formerly known as the Civilian Health and Medical Program of the Uniformed Services (CHAMPUS), is the health care program for approximately 9.6 million active-duty service members, National Guard and Reserve members, retirees, their families, survivors, and certain former spouses worldwide. The program is managed by the Defense Health Agency (DHA). As a major component of the Military Health System, the TRICARE health program combines the health care resources at military hospitals and clinics (or direct care) with networks of civilian health care professionals, institutions, pharmacies, and suppliers to provide access to high-quality health care.

CHAMPVA

Civilian Health and Medical Program of the Department of Veterans Affairs (CHAMPVA) covers veterans' families not eligible for TRICARE who are dependents of veterans with a 100 percent permanent disability received while on active duty or who died from a disability received while on active duty.

> TRICARE and CHAMPVA are often confused with each other, but they are completely separate programs. The Department of Veterans Affairs administers CHAMPVA.

FECA Black Lung

FECA Black Lung is a program that provides medical services to eligible beneficiaries for diagnostic and therapeutic services related to black lung disease as defined under the Black Lung Benefits Act of 2000 (BLBA). The Department of Labor's Office of Workers' Compensation Programs (OWCP) is responsible for

payment of all reasonable charges stemming from covered medical services provided to claimants eligible under the Federal Employees' Compensation Act (FECA).

Conclusion

Payers compensate providers for healthcare services received by their members. You now understand the basics of:

- Who is providing health insurance to whom
- The various basic delivery models that are being employed.

Chapter 4: The Providers

Introduction 29

Types of Providers 29

- **Facility** 30
- **Professional** 30
- **Pharmacy** 31
- **Ancillary** 31

National Provider Registry 32

- **NPI** 32
- **Taxonomy** 32
- **Other Provider Identifiers** 33

Case Study: Standardizing Provider Names from the National Provider Registry 34

Case Study: Using Taxonomy Code to Identify Primary Care Physicians 36

Conclusion 38

Introduction

Providers of health care are the initiators in the flow of administrative healthcare data. They provide a variety of services for the members of a health plan and therefore seek reimbursement by that health plan. This claim for service reimbursement, and its subsequent payment, generates much of the administrative healthcare data that we analyze. It therefore makes sense to view our data in the context of who initiated it, why, and how.

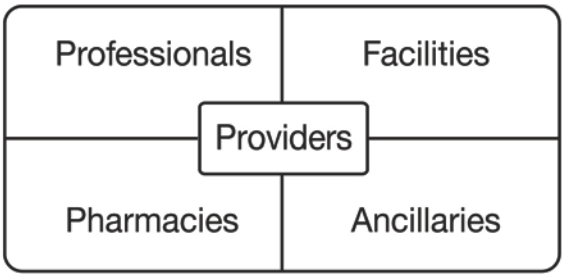
In this chapter you will:

- Differentiate between four provider types.
- Understand how providers submit claims.
- Learn about a national provider database, the National Provider Registry.
- Use character functions in SAS Enterprise Guide to standardize provider names.
- Leverage data elements within the National Provider Identifier to identify primary care physicians in SAS Enterprise Guide.

Types of Providers

There are four categories of healthcare providers: Facilities, Professional, Ancillaries, and Pharmacies. In many payer organizations data is stored and organized around this theme. Orienting our thinking about administrative healthcare data around these four provider types will serve us well. But, at the risk of oversimplifying, it must be understood that there are several subcategories and nuances about the data they generate. We will explore the providers of healthcare services in this chapter.

Figure 4.1: Four categories of healthcare providers



Facility

Facilities, or institutions, are the "bricks and mortar" of the healthcare industry. While we commonly think of hospitals when we think of facilities, there are many other types of facilities: Skilled Nursing Facility (SNF), Home Health Agency (HHA), and Intermediate Care Facility (ICF), to name a few. Within these are many subcategories that more fully describe the facility type. Much of what we do with facility data will rely on our ability to further understand and categorize on type of facility. Please see Table A.1 in the Appendix for a complete listing as defined by the industry.

Facilities have only two mechanisms by which to submit claims to a payer: the CMS-1450 paper form and the 837I electronic format. All payers are obligated to receive these formats. Two key data elements on the claim identify the type of facility and type of service and frequently serve analytic projects for data filtering and categorization. A facsimile of the CMS-1450 form is reproduced as Figure A.1 in the Appendix, and its content and other useful data elements will be discussed in detail in Chapter 5. Table 4.1 describes the two key data elements that allow us to tease apart the claim to determine more detail about the submitting provider.

Table 4.1: Key data elements on Facility claims

Code	Description
Type of Bill	Components of this data element describe the type of facility (e.g., hospital) and the type of care (e.g., inpatient, outpatient).
Revenue Code	This code describes the type of service on the service line detail, of which there will be many on a facility claim. This data element is analogous to a facility's cost centers (e.g., laboratory, emergency room, room and board, or radiology) and is a possible basis for submitted charges.

Professional

Professional providers come in all shapes and sizes and can be either organizations (i.e., group practices or billing organizations) or individuals. Physicians, nurse practitioners, therapists, and dentists are but a few of the types of professional providers. Billing organizations will often bill on behalf of their "associates," but for our purposes they are considered individual or professional providers.

This provider type has two mechanisms by which to submit claims to all payers: the CMS-1500 paper form and the 837P electronic format. A facsimile of this form is reproduced as Figure A.2 in the Appendix, and its content will be discussed in detail in Chapter 6.

A distinguishing feature of this claim type is that the provider of clinical interest, the Rendering Provider, is listed on the service level detail line and not in the claim header information. The Billing Provider is at the header level. This is to say that we may find multiple professional providers rendering services on a single medical claim.

Another data element that bears mentioning at this juncture is the Place of Service (POS) field. It is a required field for professional claim submissions. POS is at the service line detail and defines the physical location, or setting, in which the service was performed. It is analogous to, but not the same as, the Type of Bill (aka, Bill Type) seen on facility claims. Since professional providers have feet (facilities do not), they are often moving about providing services in multiple settings. This concept affords us a mechanism to filter or group data for reporting or comparative purposes.

For a list of the POS codes, see www.cms.gov/Medicare/Coding/place-of-service-codes/Place_of_Service_Code_Set.html. A list current as of this writing is in Table A.3.

For many facility medical services (e.g., caesarian section, hospital radiology) there will be a professional component to the data. While we think of an inpatient surgery as a facility claim, we must not overlook the charges that the professional provider (the surgeon in this case) submits in order to get paid for her time. Many projects require that we continue to think about all the providers that may have played a part in the delivery of healthcare services for a given patient and the data sources that this thought may suggest.

Pharmacy

This category of provider includes the street-corner pharmacies (e.g., RiteAid, Walmart), mail order pharmacies, and, in some cases, specialty pharmacies. Pharmacy Benefit Manager (PBM) companies are often contracted by the payers to administer these claims. The PBM data is sometimes housed separately. Of particular note is that it does not include hospital formularies; these are cost centers within a facility and are billed out on facility claims under the appropriate revenue codes.

A PBM is a third-party administrator (TPA) of prescription drug programs for a health plan. They may also be a wholly owned service within an integrated healthcare system. They are responsible for processing and paying prescription drug claims, developing and maintaining the formulary, contracting with pharmacies, and negotiating discounts and rebates with drug manufacturers.

The National Council for Prescription Drug Programs (NCPDP [http://www.ncpdp.org]) is a not-for-profit membership organization that creates and maintains the national standards for electronic pharmacy services. While paper claim forms exist for pharmacy providers (see Figures A.3a and A.3b in the Appendix for a facsimile), they typically are not used. The current electronic format is the NCPDP Telecommunication Standard Version D.0. As a provider group, pharmacists were at the forefront of electronic claim submission. Pharmacy claims are further discussed in Chapter 7.

Ancillary

Ancillary providers can be thought of as an "other" category of provider. They provide services such as laboratory tests, radiology tests, ambulance services, and physical, speech, and occupational therapies. Durable medical equipment (DME) and durable medical supply (DMS) vendors also fall into this category. By claim volume and cost measures, ancillary providers are a very large component of the healthcare system and are often overlooked in data analytic projects.

Due, in part, to the fact that these services are ancillary to professional medical services, this group of providers also submits claims with the CMS-1500 and 837P formats. They use the same billable codes and POS as professional providers.

Not all ancillary is created equal. For example, laboratory results data does not come in via a CMS-1500/837P. Only the billable laboratory encounter (CPT code) comes in via a claim/encounter. Laboratory results rank as highly as claim/encounter data in reporting, providing detail on the outcome of the claim/encounter.

National Provider Registry

The administrative simplification component of the Health Insurance Portability and Accountability Act of 1996 (HIPAA) required the adoption of a standard unique identifier for healthcare providers. The stated purpose was to improve the efficiency and effectiveness of electronic transmission of health information. The Centers for Medicare & Medicaid Services (CMS) developed the National Plan and Provider Enumeration System (NPPES) to assign the unique National Provider Identifier (NPI). The Registry, as NPPES is more generally known, contains considerable information about all providers and therefore can be thought of as a national provider database.

The Registry may be downloaded at no cost from the NPPES website (nppes.viva-it.com/NPI_Files.html). This version has more than 300 disclosable fields of information, the kinds of which are found in Table 4.2.

Table 4.2: Disclosable data content of the National Provider Registry

Item	Content
National Provider Identifier (NPI)	The unique and essential field on the file
Entity Type Code	1—Individual, 2—Organization
Identification Numbers	Employer ID Number (EIN), Tax ID Number (TIN), state license, legacy provider numbers
Names	Legal, legal business, others
Addresses	Mailing and physical
Taxonomy Codes	Primary and fourteen others
Dates	Activation and deactivation

NPI

NPI is a unique ten-digit identification number issued to healthcare providers in the United States by CMS. It has no built-in intelligence; that is to say, there is no embedded meaning to this nominal piece of data.

All providers within the healthcare system must have an NPI; the HIPAA final rule is in effect. All Medicare claims must use the NPI, and commercial payers should be requiring its use as well. The advent of this identifier is a godsend for the healthcare industry. We are now able to uniquely identify providers in our administrative data regardless of the source. If you are not seeing this field fully populated in your data, you should be asking some questions.

Taxonomy

The Healthcare Provider Taxonomy Code (HPTC) is unique, alphanumeric, and ten characters in length. The code is structured into three components: Provider Type, Classification, and Area of Specialization. A few examples can be seen in Table 4.3. Taxonomy code values are updated twice a year and are publicly available at http://www.wpc-edi.com/reference/. When a provider (individual, group, or institution) signs up via NPPES for an NPI they choose a primary taxonomy (specialty category) and up to fourteen others. It is important to note that taxonomy is self-reported. CMS does not verify that the provider is board certified in any taxonomy reported. CMS merely validates that the taxonomy reported exists in the available list of taxonomies. Also, depending upon Entity Type code (individual or organization), there is a very different domain of values from which to choose.

Table 4.3: Taxonomy code structure

Taxonomy code	Provider type (position 1–2)	Classification (position 3–4)	Specialization (position 5–10)
103TC0700X	Behavioral Health & Social Service Providers	Psychologist	Clinical
207NP0225X	Allopathic & Osteopathic Physicians	Dermatology	Pediatric Dermatology
282NC2000X	Hospitals	General Acute Care Hospital	Children
3416S0300X	Transportation Services	Ambulance	Water Transport

It should be evident from these few examples that the perceptive programmer/analyst can quickly parse the taxonomy code into its component parts and use the embedded intelligence to a project's advantage. It may also be a useful task to determine if local definitions of provider type and specialty may be preferable to the self-reported national provider registry data.

Other Provider Identifiers

Not only does the Registry list other identifiers associated with a provider, but you will see a variety of identifiers in a payer's data warehouse. We list such possibilities in Table 4.4 but encourage the reader to consider their veracity. When at all possible the industry should be using NPI as a best practice.

Table 4.4: Other provider IDs

Identifier	Description
TIN	Tax ID Number (for IRS purposes)
EIN	Employer ID Number (other government reporting purposes)
SSN	Social Security Number (for Social Security Administration [SSA] business)
UPIN	Universal Physician ID Number (an older attempt at an NPI)
DEA	Drug Enforcement Agency identifier (not all providers need to have one)
Commercial Insurance	Legacy identifiers (assigned to a payer's network of providers for internal use)
Medicare	Several have been in use over the years; NPI should be a replacement. The Medicare ID may synonymously be called CMS Certification Number (CCN), CMS Provider Number, or CMS ID, among others.
Medicaid	Several in use over the years; NPI should be a replacement.

In several cases CMS (and other payers) require Medicare IDs in order to use their pricing methodologies. This six-digit ID cannot be replaced by NPI. Health plans that pay on a Prospective Payment System basis (e.g., Diagnosis Related Group [DRG], Ambulatory Payment Classifications [APC], Resource Utilization Group [RUG]) *must* store the Medicare ID, as that is what drives the reimbursement function. The CCN continues to serve a critical role in verifying that a provider has been Medicare certified and for what type of services.

Provider data is housed and maintained wherever reimbursement is performed—the payer organizations. A variety of information is managed for the purpose of adjudicating and paying claims, building provider networks, credentialing, and reporting. Because of the complexity of the content and its management, it is incumbent upon the analyst to be vigilant regarding its accuracy.

Case Study: Standardizing Provider Names from the National Provider Registry

The Provider Services department at Healthy Living, Inc., has two primary objectives. The first is to negotiate contracts with providers. The second is to maintain an accurate and up-to-date provider database. This ensures that claims are processed without any glitches. The department understands that hand-entered data is difficult to maintain, so it utilizes the national Registry to standardize provider data elements such as name, address, phone and fax numbers, and provider specialty.

Figure 4.2 shows a sample of data from the Registry. Please note that only pertinent data elements are shown.

Figure 4.2: Sample Registry Records

NPI	Entity_Type_CD	Legal_Bus_Name	Prov_Last_Name	Prov_1st_Name	Prov_Middle_Name	Prov_Name_Prefix_Txt	Prov_Taxon_CD_1
1932102134	1		RAMZY	JOSEPH	I	DR.	207Y00000X
1699778761	1		CATTS	RICHARD	BRYAN	MR.	363LF0000X
1427051226	1		TOOZE	STEVEN	E	DR.	207X00000X
1083617047	1		SORDI	MARK	A	DR.	207Q00000X
1205839164	2	MID-ATLANTIC FAM...					207Q00000X
1013910926	2	CHRISTIANA CARE...					332B00000X

- Legal_Bus_Name is populated for organizations (Entity_Type_CD=2).
- Prov_Last_Name and Prov_1st_Name are populated for individuals (Entity_Type_CD=1).

This section shows you how to use SAS functions to create a standardized Provider Name data element. This is done using two Query Builder tasks in SAS Enterprise Guide.

The two Query Builder tasks may be collapsed into one. However, the two-step solution demonstrated below is easier to follow. Analysts should weigh programmatic efficiency and compactness with ease of understanding and maintenance.

- The first Query Builder task uses SAS functions to create two temporary variables, one each for individuals and organizations within the Registry.
- The second Query Builder task uses results from the first to create a standard Provider Name element that is populated for all NPIs.

Let us create the two Query Builder tasks.

1. Open BOOK.NPI. For review on opening a SAS data set, see Chapter 2.
2. Select **Query Builder** from **Tasks—Data**.
3. In **Query Builder**, enter the following information. Type in "NPI Based Provider Name Step 1" in **Query Name**. Enter WORK.NPI_ProvNameTemp in **Output Name**. Select all columns from NPI data set.
4. Create two computed columns using the expressions shown below. Name these computed columns ProvNameTemp1 and ProvNameTemp2, respectively. For a review on creating computed columns, see Chapter 2. Confirm results in Figure 4.3.

```
propcase(t1.Legal_Bus_Name)
```

The PROPCASE function capitalizes the first letter of each word, thus making names more reader friendly.

```
catx(", ",propcase(t1.Prov_Last_Name),propcase(t1.Prov_1st_Name))
```

The CATX function concatenates two or more variables, using a separator specified in the first argument. Its subsequent arguments list variables to be concatenated. A nice feature of the function is that it also removes trailing and leading blanks from each of the variables being concatenated.

Figure 4.3: Data set WORK.NPI_ProvNameTemp

NPI	ProvNameTemp1	ProvNameTemp2	Entity_ Type_ CD	Legal_Bus_Name	Prov_Last_Name	Prov_1st_Name
1932102134		Ramzy,Joseph	1		RAMZY	JOSEPH
1699778761		Catts,Richard	1		CATTS	RICHARD
1427051226		Tooze,Steven	1		TOOZE	STEVEN
1083617047		Sordi,Mark	1		SORDI	MARK
1205839164	Mid-Atlantic Family Prac...		2	MID-ATLANTIC FAM...		
1013910926	Christiana Care Health...		2	CHRISTIANA CARE...		

5. While data set NPI_ProvNameTemp is open, select **Query Builder** from **Tasks—Data**. This will create a Query Builder task based on the data set created by the first Query Builder task.
6. In **Query Builder**, type "NPI Based Provider Name Step 2" in **Query Name**. Enter WORK.NPI_ProvName in **Output Name**. Select all columns from the NPI data set except the two temporary variables.
7. Create a computed column, ProvName, using the following expression:

```
coalesce(t1.ProvNameTemp1,t1.ProvNameTemp2)
```

The COALESCE function returns the first non-missing value from a list of variables. The function serves to select a value from the two temporary variables. If ProvNameTemp1 is missing, then the value from ProvNameTemp2 is used.

8. Confirm results in Figure 4.4.

Figure 4.4: Standardized Provider Name from the Registry

NPI	Entity_Type_CD	Legal_Bus_Name	Prov_Last_Name	Prov_1st_Name	ProvName
1932102134	1		RAMZY	JOSEPH	Ramzy,Joseph
1699778761	1		CATTS	RICHARD	Catts,Richard
1427051226	1		TOOZE	STEVEN	Tooze,Steven
1083617047	1		SORDI	MARK	Sordi,Mark
1205839164	2	MID-ATLANTIC F...			Mid-Atlantic Family Prac...
1013910926	2	CHRISTIANA CA...			Christiana Care Health...

The CATS function is closely related to the CATX function shown above. While the CATX function inserts a separator string, the CATS function does not. For example, CATS("Inpatient","Hospital") results in a value of "InpatientHospital." The SCAN function may be used to parse the result of the CATX function. For example, scan(ProvNameTemp1,1,",") extracts the first word (last name) of the individual providers, whereas scan(ProvNameTemp1,2,",") results in the second word (first name) of the providers. The COALESCE function can take more than two variables as arguments, creating an intelligent business hierarchy. This may be useful with contact phone numbers. If the daytime phone number is missing, then the mobile phone number should be used. If the mobile phone number is missing, then the emergency contact phone number should be used and so on.

Case Study: Using Taxonomy Code to Identify Primary Care Physicians

Because one of the insurance products offered by Healthy Living, Inc., does not employ the gatekeeper approach (an HMO), they would like to encourage members to seek care from primary care physicians rather than specialists. The company's analysts in the Provider Services department utilize the taxonomy codes found within the Registry to identify primary care physicians.

Figure 4.5 shows select taxonomy codes and their associated nomenclature variables.

Figure 4.5: Select taxonomy codes and associated descriptors

Code	Type	Classification	Specialization
207Q00000X	Allopathic & Osteopathic Physicians	Family Medicine	
207QG0300X	Allopathic & Osteopathic Physicians	Family Medicine	Geriatric Medicine
207R00000X	Allopathic & Osteopathic Physicians	Internal Medicine	
207RG0300X	Allopathic & Osteopathic Physicians	Internal Medicine	Geriatric Medicine
207V00000X	Allopathic & Osteopathic Physicians	Obstetrics & Gyn...	
208000000X	Allopathic & Osteopathic Physicians	Pediatrics	
208D00000X	Allopathic & Osteopathic Physicians	General Practice	
207ZB0001X	Allopathic & Osteopathic Physicians	Pathology	Blood Banking &...
207ZC0006X	Allopathic & Osteopathic Physicians	Pathology	Clinical Pathology

This section uses taxonomy codes to create a basic and expanded definition of primary care physicians. The basic definition contains the following codes:

- 207Q00000X: Physicians, Family Medicine, no specialization
- 207R00000X: Physicians, Internal Medicine, no specialization
- 208D00000X: Physicians, General Practice, no specialization

The expanded definition of primary care physicians is built by adding the following codes to the basic definition:

- 207QG0300X: Physicians, Family Medicine, Geriatric Medicine
- 207RG0300X: Physicians, Internal Medicine, Geriatric Medicine
- 207V00000X: Physicians, Obstetrics & Gynecology, no specialization
- 208000000X: Physicians, Pediatrics, no specialization

A taxonomy code value of "207Q00000X" may be broken up into three distinct pieces for easier identification of physician type, classification, and specialty. The first two position values, "20," are described by physician type. The next two indicate classification. The remaining six characters indicate a specialty.

An industry implementation of taxonomy codes should leverage SAS formats to break out taxonomy codes into different hierarchies. Depending on project needs, one or more hierarchies may be used. For example, a project involving utilization of an older patient population may leverage a SAS format created from *Specialization* to query all professionals specializing in Geriatric Medicine.

As SAS formats have been discussed extensively in Chapter 2, this section focuses on program code to create basic formula to identify primary care physicians.

Let us work with the WORK.NPI_ProvName data set to identify primary care physicians.

1. Open WORK.NPI_ProvName. Recall that this is the data set created in the previous case study.
2. Select **Query Builder** from **Tasks—Data**.
3. In **Query Builder**, specify the following. Type in "Identify PCP" in **Query Name**. Enter "WORK.NPI" in **Output Name**. Select NPI, Entity_Type_CD, ProvName, and Prov_Taxon_CD_1.
4. Create a computed column, PCP, using the following expression.

```
(t1.Prov_Taxon_CD_1 in ("207Q00000X","207R00000X","208D00000X"))
```

There are two sets of parenthesis in the above expression.

The inner parenthesis is used with the IN operator, which lists a number of values. This operator may also be used with a numeric variable.

The outer set of parentheses illustrates Boolean logic, which results in a numeric value of 1 (True) or 0 (False). If the taxonomy code of the current observation being processed matches any one of the values listed, then PCP evaluates to 1. Otherwise, PCP resolves to 0.

5. Confirm results in Figure 4.6.

Figure 4.6: Basic PCP definition

NPI	Entity_Type_CD	ProvName	Prov_Taxon_CD_1	PCP
1932102134	1	Ramzy,Joseph	207Y00000X	0
1699778761	1	Catts,Richard	363LF0000X	0
1427051226	1	Tooze,Steven	207X00000X	0
1083617047	1	Sordi,Mark	207Q00000X	1
1205839164	2	Mid-Atlantic Fam...	207Q00000X	1
1013910926	2	Christiana Care...	332B00000X	0
1245233105	1	Prigg,J	363A00000X	0
1295738250	1	Bahtiarian,Grego...	207Q00000X	1
1437152337	1	Lucca,Joseph	2251G0304X	0
1356344816	1	Tanchanco,Rodri...	207R00000X	1

6. Create an Expanded PCP Flag by repeating Step 4 and using the following expression:

```
(t1.Prov_Taxon_CD_1 in ("207Q00000X","207R00000X","208D00000X",
   "207QG0300X","207RG0300X","207V00000X",
   "208000000X"))
```

The IN operator in SAS can be used with the colon modifier. To identify values that begin with either "10" or "20", the following expression may be used: (t1.Prov_Taxon_CD_1 in: ("20", "10")). While very powerful and useful, caution should be exercised with this functionality; knowing the complete domain of values being addressed will prevent unexpected results.

Boolean logic can be used with some creativity to result in values other than 1 or 0. The following code creates age categories (1, 2, 3) based on numeric value of age.

(t1.Age < 2) + (2 * (t1.Age between 2 and 18)) + (3 * (t1.Age >= 19))

If age is less than 2 then the computed expression resolves to 1. If age is between 2 and 18, the result is multiplied by 2 to yield a value of 2. If age is 19 or greater, the result is multiplied by 3. Notice the use of extra parenthesis to guarantee the order of evaluation. While not needed everywhere in this example, it is a good work habit and also improves readability.

Conclusion

This chapter compared and contrasted provider types and their claim submission mechanisms. It introduced you to a national provider database, the National Provider Registry. The chapter has also shown you how to leverage character functions and Boolean logic in SAS Enterprise Guide to effectively utilize the Registry data elements.

Chapter 5: Facility Claims

Introduction 39

CMS-1450 Paper Claim Form 40

837I Electronic Claim Format 40

Data Elements Unique to Facilities 40

- Type of Bill 40
- Admission and Discharge Dates 42
- Patient Discharge Status 43
- Revenue Code 43
- Diagnosis Codes 44
- Present on Admission 45
- Surgical Procedure Codes 45
- DRG 45
- Provider IDs 46
- Others 46

Case Study: Calculating C-Section Rates among Hospitals 47

- Create Summary Data Set for All Births 47
- Create Summary Data Set for C-Section Births 49
- Join Summary Data Sets 50
- Create Bar Graphs 51

Case Study: Top Reasons for ER Utilization 53

Automating Reports with SAS Enterprise Guide 56

Creating Documentation in SAS Enterprise Guide 56

Conclusion 57

Introduction

Facilities are different from other provider types. They provide dissimilar services and are reimbursed based on different types of information. Successful healthcare analysts must tailor their analytics to the specifics of facility claims data.

In this chapter you will:

- Understand the claim submission mechanism used by facilities.
- Learn about data elements specific to facilities.
- Analyze facility header data in SAS Enterprise Guide to create bar graphs illustrating different Cesarean section (C-section) rates among hospitals.
- Combine facility header and detail data in SAS Enterprise Guide to identify top reasons for emergency room utilization.

CMS-1450 Paper Claim Form

The CMS-1450 form is the standard paper claim form used by institutional providers. It may be referred to as a UB-92 or UB-04 form, as the prior naming convention was UB-xx, where xx was the two-digit year of its release. UB stands for Uniform Billing. HIPAA-related waivers still allow the use of this paper format, but we hope it will disappear over time.

The National Uniform Billing Committee (NUBC [http://www.nubc.org]) is responsible for the design and maintenance of the CMS-1450 form. NUBC was formed in 1975 by the American Hospital Association (AHA) for the purpose of developing a uniform data collection format for all institutional healthcare services.

837I Electronic Claim Format

The 837I electronic claim format is used to submit healthcare claim billing information from facilities to payers, either directly or via intermediary billing services and claims clearinghouses. The 837I transaction provides all the necessary information to allow the payer to begin to adjudicate the claim. As of March 31, 2012, healthcare providers must be compliant with version 5010 of the HIPAA EDI standards. The 5010 standards divide the 837 transaction set into three groups, as follows: 837P for professionals, 837I for institutions, and 837D for dental practices.

As with the professional provider counterpart, the 837I provides significant improvement over the paper form—more detail and more value-add. The format continues to undergo change as both providers and payers find room for improvement. Documentation of the ANSI (American National Standards Institute) 837 formats may be obtained from the Washington Publishing Company (www.wpc-edi.com).

Data Elements Unique to Facilities

This section focuses on data elements found only on the facility's claim submission. These are not found elsewhere in administrative healthcare data, as they are either specific to facility claims and/or specific to a facility type.

Type of Bill

Type of Bill (TOB), Form Locator (FL) 4, describes three distinct pieces of information: type of facility, type of care, and sequence of the bill. It is a required field. For the electronic submission (837I), it is a three-character field with each character describing distinct pieces of information. For the CMS-1450 paper submission, a four-character field is required, with the first position being a zero. The leading zero should be stripped for consistency. Type of Bill is extremely important. You can utilize its built-in intelligence to filter or categorize large volumes of data to retain claims of interest. As with many of the healthcare codes, Type of Bill should be stored, managed, and used as a character type. It is nominal-level data.

Some documentation suggests that the first two characters denote facility type (see Table A.1 in the Appendix). Tables 5.1–5.5 show the domains of each component of TOB.

Table 5.1: First position of TOB (-xx)

Code	Nomenclature
1	Hospital
2	Skilled Nursing
3	Home Health
4	Religious Non-Medical Health Care Institution

Code	Nomenclature
6	Intermediate Care
7	Clinic
8	Special Facility

Table 5.2: Second position of TOB when first is not 7 or 8 (x-x)

Code	Nomenclature
1	Inpatient
2	Inpatient (Medicare Part B only)
3	Outpatient
4	Other
5	Level I
6	Level II
8	Swing Bed

Table 5.3: Second position of TOB for clinics (7-x)

Code	Nomenclature
1	Rural Health
2	Renal Dialysis Center
3	Freestanding
4	Outpatient Rehabilitation Facility (ORF)
5	Comprehensive Outpatient Rehab Facility (CORF)
6	Community Mental Health Center
7	Federally Qualified Health Center (FQHC)
8	Licensed Freestanding Emergency Medical Facility
9	Other

Table 5.4: Second position of TOB for special facilities (8-x)

Code	Nomenclature
1	Hospice (not hospital based)
2	Hospice (hospital based)
3	Ambulatory Surgical Center
4	Freestanding Birthing Center
5	Critical Access Hospital
6	Residential Facility
9	Other

Table 5.5: Third position of TOB—bill sequence (xx-)

Code	Nomenclature
0	Non-Payment/Zero Claim
1	Admit thru Discharge Claim
2	Interim—First Claim
3	Interim—Continuing Claim
4	Interim—Last Claim
5	Late Charge Only
7	Replacement of Prior Claim
8	Void/Cancel of Prior Claim
Others	Less common values exist and may be found in Table A.2 in the Appendix.

The information given in Tables 5.1–5.5, A.1, and A.2 is not meant as a perfect substitute for documentation from the NUBC (http://www.nubc.org/) and the AHA (http://www.aha.org). While we believe the information here is accurate at the time of this writing, the nuance of its use is best guided by vendor documentation. In addition, the documentation is updated quarterly and can change frequently.

Admission and Discharge Dates

Admit date and hour are required fields for inpatient stays, and discharge hour is required for the final inpatient claim. Of particular note is that an inpatient discharge date is not submitted on the claim form. It is derived by the adjudication system or the warehouse acquisition process. The question then is “how?”

Typically the Statement Through Date (FL 6) is taken to be the discharge date if the Patient Discharge Status (FL 17) indicates that the patient indeed left the premises. Otherwise, the Statement Through Date is the end of the billing cycle. Other information on the claim form situationally contributes to the determination of the date we see in our data source. The takeaway here is that the analyst may need to review local operational documentation if the concept is important to the project at hand.

Patient Discharge Status

Patient Discharge Status (FL 17), synonymously known as patient disposition or patient status, is a two-character code that indicates the disposition of the patient at the end of the period covered by the claim (Statement From and Through Dates). It is a required field for the current claim submission formats. In the past it was "situational"—required only for inpatient confinement. "Discharge" may be a misleading term as it is required for all facility encounters, not just inpatient stays. See Table A.4 in the Appendix for a list of codes current as of October 1, 2013.

Patient Discharge Status becomes a key piece of information when, among other tasks, you wish to combine claims into a complete episode of care with a set of business rules. Specifically, prolonged stays in a Skilled Nursing Facility (SNF) can generate monthly interim bills with a discharge status of "30—still a patient." These are not separate encounters. They are one encounter but are split-billed and need to be combined into one encounter. This requires the use of several fields to determine when the encounter begins and when it ends, and specifically where the patient went, or not. Other analytic needs will suggest the importance of knowing the patient's disposition as of the Statement Through Date.

Revenue Code

Revenue codes are required and are found on the detail service lines (FL 42) of the CMS-1450 claim form and, of course, on the 837I electronic format. The revenue code identifies specific accommodations, ancillary services, or special arrangements. In combination with a line-level HCPCS (Healthcare Common Procedure Coding System), Rate, or HIPPS (Health Insurance Prospective Payment System) code, it provides the basis for the line-level service charge. Rather than finding multiple occurrences of the same revenue code on a claim, you will find that the "units" field serves as a multiplier for the unit charge.

The architecture of this nominal-level code set is four digits, with the first three representing a category and the last a subcategory. Please note that in older claims you may find a three-digit code; simply prefix that representation with a zero to arrive at some consistency. Again, do not lose sight of proper variable type definitions in project work. If working with nominal- or ordinal-level data, regardless of content, store and manage as character type. Many healthcare claim codes look like numbers but are not numeric data.

Examples of revenue code categories are in Table 5.6. Note that the "x" will take on values of 0–9 and represents a subcategory. A complete list of revenue codes may be obtained by subscription to the NUBC (www.nubc.org). Code set subscription services can be very useful, as updates may be as frequent as quarterly. Table A.5 in the Appendix lists the codes valid for the 2014 federal fiscal year.

Table 5.6: Revenue code category examples

Code	Description
010x–021x	Room and Board Services
025x	Pharmacy
029x	Durable Medical Equipment (DME)
036x	Operating Room Services
045x	Emergency Department
050x	Outpatient Services

Understanding the content of this code will provide the definition for many business rules and the formatting of reports and queries. While use of this code can be very helpful, it is submitted by the facility, is highly subjective, and is prone to gaming by the facility depending on reimbursement schemes. For example: If a facility is paid a per diem with a pass through on high-cost drugs, the facility may put all its drugs under revenue code 636. However, 636 requires the drug to be a specialty drug and a supporting HCPCS code must be present.

Diagnosis Codes

International Classification of Diseases, Nth Revision, Clinical Modification (ICD CM) codes are commonly called diagnosis codes. Unlike the professional claim, facilities claims submission allows for many different diagnosis codes. Table 5.7 describes these. Generally, the electronic format provides for more code values to be entered than does the paper format. Facility claims analysis and reporting should consider these alone or in combination as the project requires.

Table 5.7: CMS-1450/837I diagnosis code (DX) types

Type	Description
Principal	The condition chiefly responsible for the facility episode determined after study and work up by the physician. Also known as Primary or Discharge. Required for all facility claims.
Other	Conditions that coexist during the episode. Situational for facility claims.
Admission	The reason for the admission. It is often different from the Principal diagnosis. Given that the Principal diagnosis is the diagnosis after study, if the patient is admitted for a symptom, for instance, shortness of breath (SOB), a Principal diagnosis of congestive heart failure (CHF) could result. Required for all inpatient claims.
Patient Reason	The patient reason for an outpatient visit. Required for all outpatient visits; optional otherwise.
External Cause of Injury (ECI)	Required when an injury, poisoning, or adverse effect is the cause for seeking medical treatment. It describes the mechanism of injury. Depending on the provider and the data warehouse, the E-code may be stored as “other diagnoses.”

Regardless of any field naming convention that you may run into, the only ordinality in these diagnosis code fields is Principal and Other. No concept of secondary, tertiary, and so on exists when referring to diagnosis codes.

Present on Admission

All claims involving inpatient admissions (Bill Type= "11x") must include a Present on Admission (POA) code value for Principal, Other, and ECI diagnosis. In many adjudication scenarios this field will affect the claim disposition and payment.

For discharges occurring on or after October 1, 2008 (the inception of MS-DRG [Medicare Severity DRG]), CMS will not pay hospitals for Hospital Acquired Conditions (HAC), cases where one of the selected conditions was not present on admission. That is, the case would be paid as though the diagnosis were not present. Most commercial insurers have followed CMS's lead. As you can imagine, this has become a real incentive for hospitals to carefully monitor and correct the causes of these conditions.

Table 5.8 shows the complete domain of values for POA.

Table 5.8: Present on Admission code values

Code Value	Nomenclature
Y	Yes—present at the time of inpatient admission.
N	No—not present at the time of inpatient admission.
U	Documentation is insufficient to determine if the condition is present on admission.
W	Clinically undetermined.
Blank or 1	Exempt from POA reporting (ICD documentation lists the diagnosis codes that are exempt).

Surgical Procedure Codes

International Classification of Diseases, Nth Revision, Procedure Coding System (ICD PCS) codes, most commonly called Surgical Procedure codes, are header-level information (as are diagnosis codes) on facility claims and are submitted only for inpatient claims. As with diagnosis codes, there will be a Principal code and potentially multiple Others. The codes are documented in Volume 3 of the ICD and are maintained by the World Health Organization.

The electronic claim submission format (837I) allows for twelve such codes to be submitted and also has been designed for handling the next version of ICD surgical procedure codes (ICD-10-PCS). When one of these codes is submitted, Operating Physician information is required. The Operating Physician is the individual with primary responsibility for performing the surgical procedure(s). The Other Operating Physician is the individual performing a secondary surgical procedure or assisting the Operating Physician. This Other Operating Physician segment can only be submitted when Operating Physician information is also sent on the claim**.**

DRG

Diagnostic Related Group (DRG) is required by Medicare to be submitted in FL 71 on the CMS-1450 and in the 837I format. Many payers will derive this code to ensure accurate payment. DRG applies to acute care inpatient hospitals only. The DRG and DRG-based reimbursement systems (All Patient DRG [AP-DRG], MS-DRG, All Patient Refined DRG [APR-DRG]) are derived using software that groups the diagnosis, procedures, and other claim elements into like resource utilization groups. On these submission mechanisms it is a "situational" data element—required only when the hospital is under contract with the payer to do so.

DRGs are derived with vendor software based on Principal Diagnosis code, Principal Surgical Procedure code, other diagnosis codes, patient age, patient gender, patient discharge status, and complications and comorbidities. There are several different DRG systems in use, the details of which are beyond the scope of this book. Suffice it to say that it is incumbent upon the healthcare analyst working with DRGs to understand which type (e.g., MS-DRG, AP-DRG, or APR-DRG) and version (e.g., 18, 24, 25, or 31) of DRGs are being used.

Medicare, as well as most commercial carriers, is currently using MS-DRG for its Inpatient Prospective Payment System (IPPS). This DRG methodology considers the severity of illness and complications documented upon admission. This consideration is based upon the POA indicator field that accompanies each diagnosis code on the claim submission. Medicare no longer pays for HAC as it did under previous versions of DRG methodology.

Provider IDs

While provider identifiers are not unique to facility claims, the variety and location of the fields are. Specifically, there is no provider ID on the service line detail as we find on the professional claim. At the header level we find Billing Provider, Attending Physician, Operating Physician, and Others (Referring, Ordering, Assisting, etc.). The healthcare data analyst should consider what a particular provider ID field on their data source means to their project. Data sources do not often draw the distinction that the claim source intended.

Others

Other codes of interest found only on facility claim submissions are briefly described in Table 5.9 and should be considered as potentially important for claims analysis and reporting projects.

Table 5.9: Other codes unique to facility claims

Code Set	Description
Condition Code	Code(s) used to identify condition(s) relating to this bill or relating to the patient or the patient's insurance coverage.
Occurrence Code	Code(s) defining a significant event relating to the claim that may affect payer processing. Describes an event and the date it happened.
Occurrence Span Code	Code(s) that identifies an event relating to payment of the claim. This event occurs over a span of days and will have begin and end dates associated with it.
Value Codes and associated amounts	Code(s) that identifies data of a monetary nature that is necessary for processing this claim as required by the payer organization. Describes the type of primary payer and specifies the dollar amount from that payer.
Admission Type	Code indicating the priority of this admission. Defines the nature of the patient's admission, for example, emergency, elective, newborn.
Admission Source	Describes the cause of the admission, for example, transfer from another facility, physician office, birth.

Case Study: Calculating C-Section Rates among Hospitals

Healthy Living, Inc., tracks and monitors providers on key measures. One important measure is C-section deliveries as a percentage of total births. This rate is of interest to two departments within the company: Finance and Patient Management. C-section births are more expensive than vaginal deliveries, so the Finance department tracks C-section rates among hospitals. C-section births have a greater risk of medical complications. Therefore, a larger proportion of C-section deliveries is of concern to the Patient Management department.

This section shows you how to compute C-section rates and payments. These rates are expediently calculated from the FacilityHeader table and require no joins with the FacilityDetail table. Please note that for the sake of simplicity, date parameters are omitted. In a real-world example, rates should be computed with respect to a time period.

To accomplish this computation, you will first filter on FacilityHeader, then summarize data, and, finally, create a bar graph showing different C-section rates. The following is a conceptual roadmap.

1. Extract Hospital-Inpatient claims from FacilityHeader using BillType. For further information on BillType and its nomenclature, refer to Tables 5.1 and 5.2. This is an important step because it reduces the data volume being processed. Strategies to reduce data volume are of paramount importance.
2. Extract claims that correspond to the following DRG codes: 765, 766, 767, 774, and 775. This set of code values is our project's definition of births; other definitions may exist.
3. Sum births and payments by providers. The resultant data set includes all DRG codes specified in step 2 and constitutes the denominator for calculating C-section rate.
4. Limit data to C-section DRG codes (765 and 766) and repeat step 3. This data set is used to calculate the numerator for C-section rate.
5. Join the data sets from steps 3 and 4 to compute C-section rate.
6. Create a bar chart showing differing C-section rates.

> The DRG codes used in this analysis are by no means exhaustive. They were selected because they tend to be the most common codes corresponding to deliveries. It is incumbent upon the analyst to clearly understand what type and version of DRG is being billed to ensure accurate understanding of the claims being pulled. Example: MS-DRG V31 765 Cesarean section w CC/MCC versus CMS-DRG V18 370 Cesarean section w CC.

Create Summary Data Set for All Births

In this section we will create a summary data set containing number of births with total AmtPaid by each hospital.

1. Select **Query Builder** from **Tasks—Data**.
2. Navigate within **SAS Servers** to select FacilityHeader in BOOK library.
3. In **Query Builder**, type AllBirths in **Query Name**. Type AllBirths in **Output Name**. The resultant data set is called AllBirths, and it resides in the temporary WORK library. Select ClaimID, AmtPaid, and ProviderID.
4. Click **Filter Data** tab in **Query Builder** to create two filters.
5. Click  to create a new filter for Hospital-Inpatient claims.
6. In **Select a type** window, select **Advanced Filter**. In **Build an advanced filter** window, either build or type in the following expression.

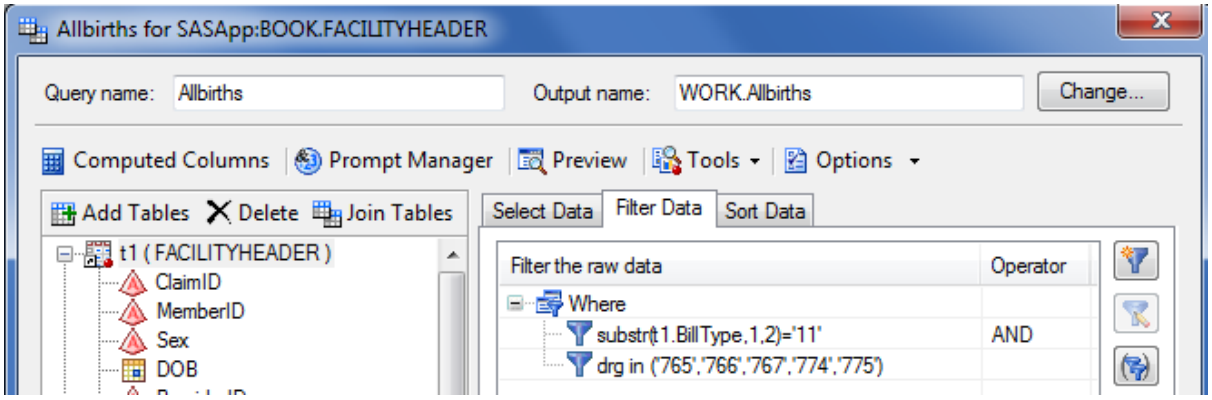
```
substr(t1.BillType,1,2)="11"
```

7. Repeat steps 5 and 6 to create a filter for DRG codes, typing in the following expression:

```
Drg in ("765","766","767","774","775")
```

8. Confirm that the **Filter Data** tab has filters as shown in Figure 5.1.

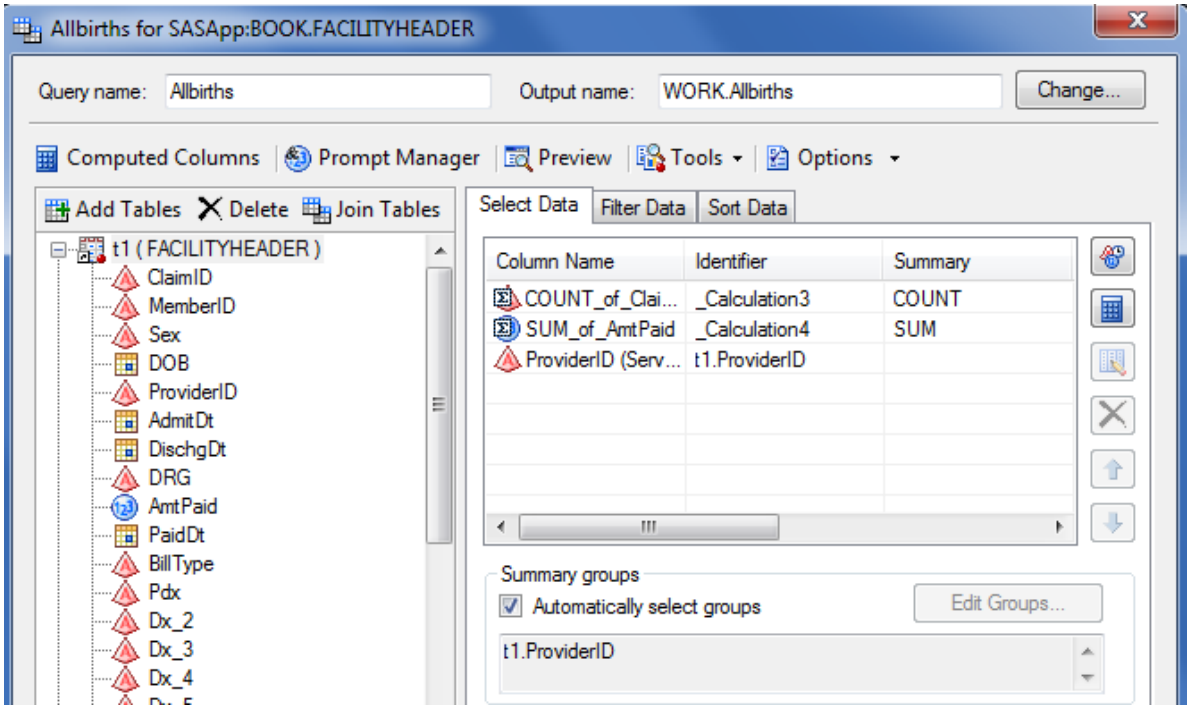
Figure 5.1: Adding filters in Query Builder



9. Summarize the data by ClaimID and AmtPaid by selecting a function from the **Summary** column. Click on the **Summary** column corresponding to ClaimID and select COUNT. Select **SUM** for AmtPaid (see Figure 5.2).

Selecting summary functions triggers an automatic grouping of variables. The **Automatically select groups** checkbox in **Summary Groups** becomes checked and grouped variables are displayed (see Figure 5.2).

Figure 5.2: Summarize data



10. Confirm that results are summarized by ProviderID (see Figure 5.3).

Figure 5.3: All births by ProviderID

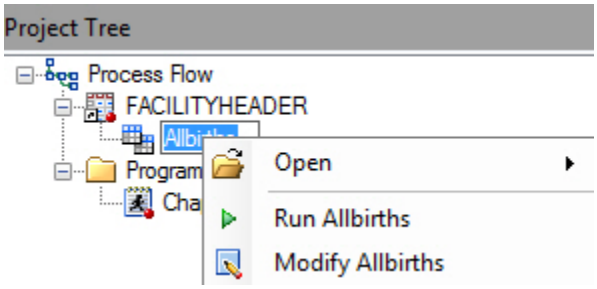
	COUNT_of_ClaimID	SUM_of_AmtPaid	ProviderID
1	80	404776.41	P623457601
2	140	688925.66	P623457604
3	153	667458.60	P623459983
4	20	108608.71	P623486792

Create Summary Data Set for C-Section Births

Enterprise Guide allows you to duplicate your existing tasks. In this section you will create a summary data set containing C-section births using the AllBirths task developed in the previous section.

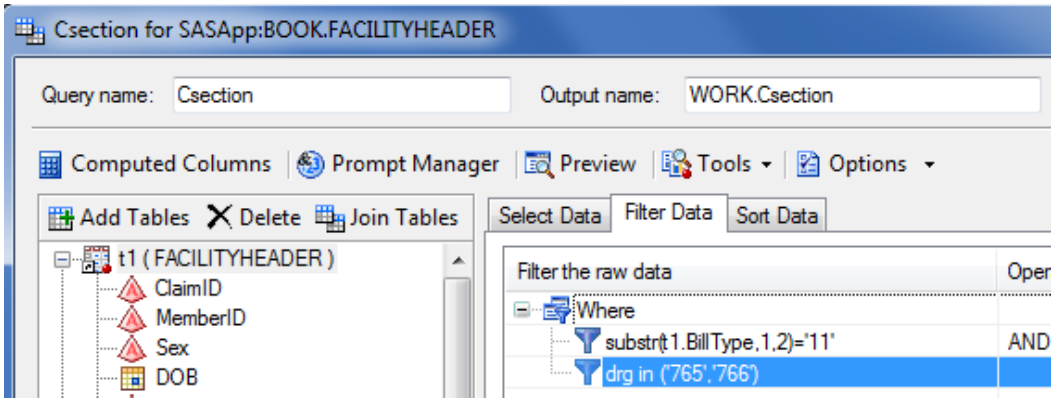
1. Right-click on AllBirths in **Process Flow** and select **Modify AllBirths** as shown in Figure 5.4.

Figure 5.4: Modify an existing task



2. In **Query Builder**, type C-section in **Query Name**. Type C-section in **Output Name**.
3. In the **Filter Data** tab in **Query Builder**, double-click the DRG filter to modify it. Remove the last three DRG codes from the expression, making sure to leave DRG codes relating to C-section births (765,766) (see Figure 5.5).

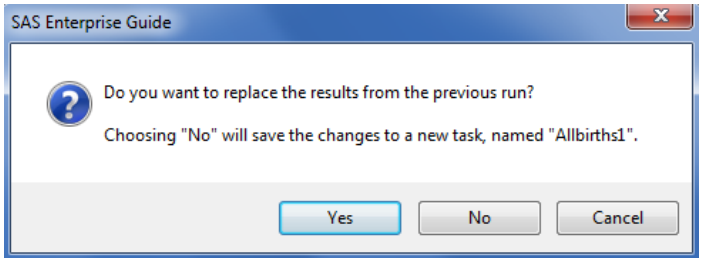
Figure 5.5: Modify an existing filter



4. Click Run to run C-section query.

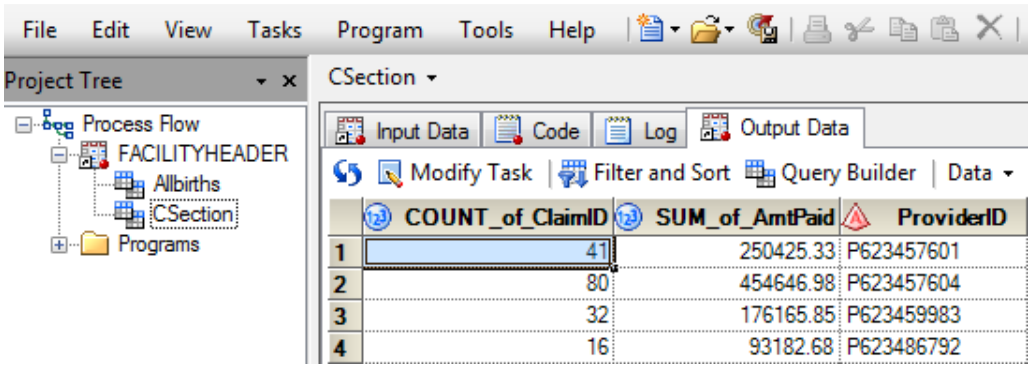
5. Click No at the prompt as shown in Figure 5.6.

Figure 5.6: Modify query



6. View results. C-section task now appears in the **Process Flow**.

Figure 5.7: C-section task and summary data set

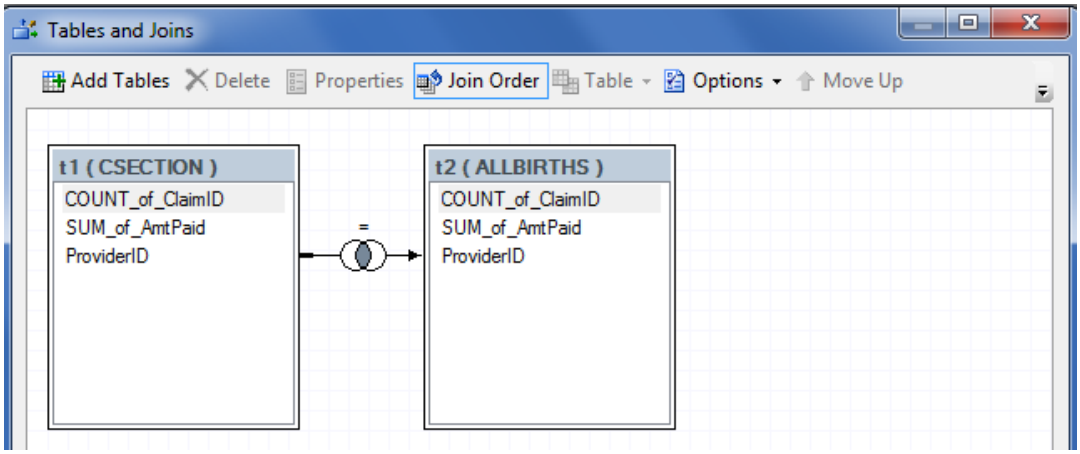


Join Summary Data Sets

This section joins Allbirths and C-section data sets.

1. Select **Query Builder** from **Tasks—Data**.
2. In **Query Builder**, type C-section Rates in **Query Name** and CsectionRates in **Output Name**. Notice that C-section data set is already chosen for this task because that is the last data set created.
3. Click Add Tables. Navigate through the SAS Servers structure to select Allbirths data set.
4. Specify join relationship between the two tables (see Figure 5.8).
   a. Click Join Tables
   b. In **Tables and Joins** window, click Join Order.
   c. Drag ProviderID from C-section and drop it on top of ProviderID in AllBirths. This joins the two tables by ProviderID. Although an inner join is chosen by default, you can control the join by selecting an appropriate join type.
   d. Right-click the default join on Count of ClaimID and choose **Delete Join**.
   e. Confirm that the tables have been joined correctly (see Figure 5.8).

Figure 5.8: Join tables



5. Select ProviderID in **Select Data** tab.
6. Create two computed columns, CsectionRate and CsectionAmtPaidPercent, using the two respective expressions shown below.

```
t1.Count_of_ClaimID / t2.COUNT_of_ClaimID
(t1.SUM_of_AmtPaid) * 100 / t2.SUM_of_AmtPaid
```

7. Confirm results (see Figure 5.9).

Figure 5.9: C-section rates and CSection AmtPaid as a percentage of total AmtPaid

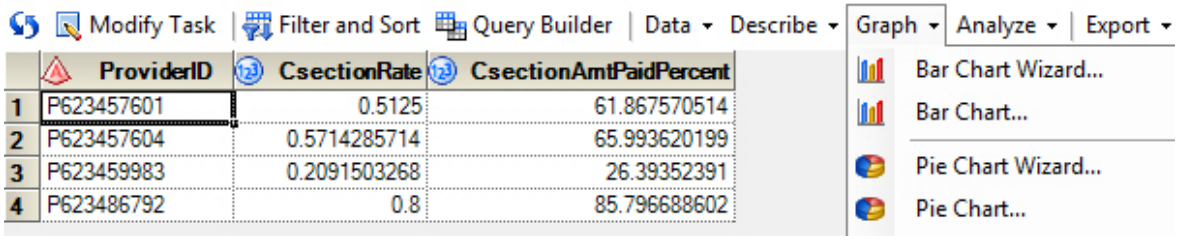
	ProviderID	CsectionRate	CsectionAmtPaidPercent
1	P623457601	0.5125	61.867570514
2	P623457604	0.5714285714	65.993620199
3	P623459983	0.2091503268	26.39352391
4	P623486792	0.8	85.796688602

Create Bar Graphs

This section shows you how to create a bar graph with the data generated in the previous section.

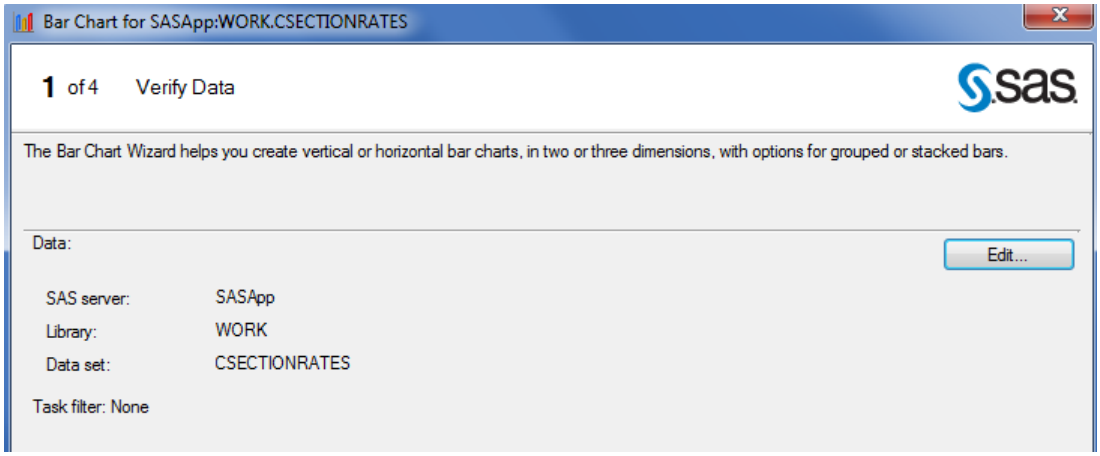
1. While viewing the CsectionRates data set, select **Graph—Bar Chart Wizard** as shown in Figure 5.10.

Figure 5.10: Create a bar chart



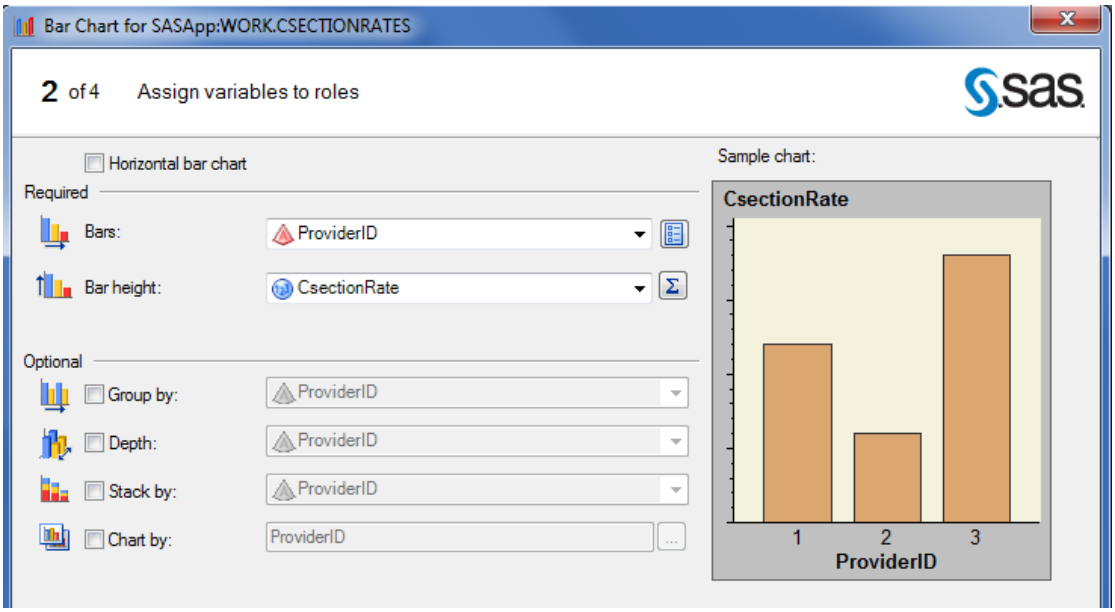
2. The **Verify Data** window allows you to set up additional filters through the **Edit** button. Click Next to continue.

Figure 5.11: Verify data source



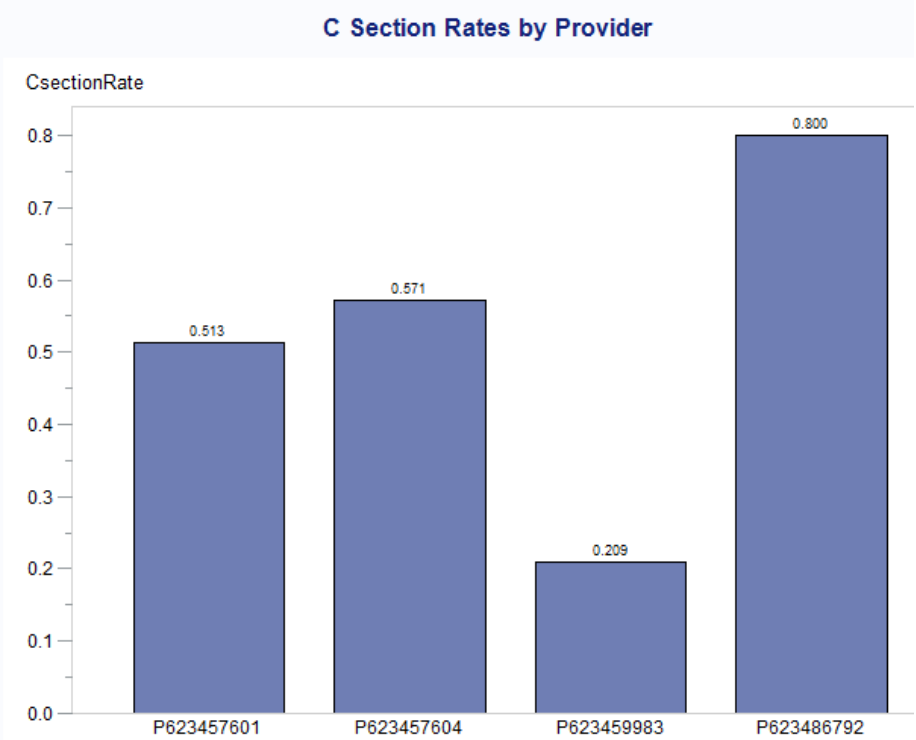
In **Assign variables to roles** window, select ProviderID under **Bars** dropdown and CsectionRate under **Bar height** (Figure 5.12)

Figure 5.12: Assign variables



3. In **Specify appearance** window, check **Data labels** and pick **Average** from the associated drop-down list. This allows you to see C-section rates printed above individual bars.
4. In **Provide a title and footnote** window, type in "C Section Rates by Provider" and delete the default footnote.
5. Click **Finish** and view results.

Figure 5.13: C Section Rates by Provider



The following extensions to the bar graph may be useful and are left to the reader as an exercise.

- Incorporate a time variable into the bar graph shown in Figure 5.13. To explore multiple ways of including a time variable, try placing the variable under Group by: or Stack by:. These are available in the Optional section as shown in Figure 5.12.
- In addition to a bar graph, you can create a bar-line graph. This is useful if you want to display AmtPaid as a line in addition to a bar display of C-section rate.

Case Study: Top Reasons for ER Utilization

Healthy Living, Inc., closely monitors its members' ER utilization because ER visits cost more than their usual alternatives: urgent or primary care. As a first step in reducing ER utilization, the company wants to understand why its members are using the ER.

This section shows you how to list top diagnosis codes associated with ER visits. This process involves three separate conceptual steps:

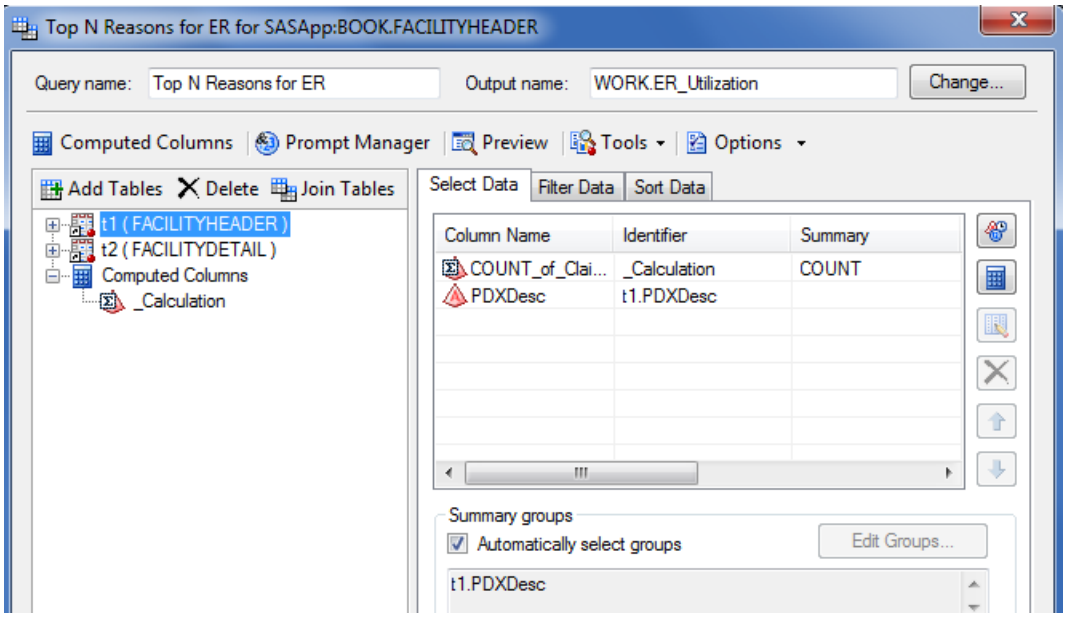
1. Identify ER claims using revenue codes in FacilityDetail table.
2. Join with FacilityHeader on ClaimID.
3. Summarize visits by primary diagnosis code (PDX).

Follow the steps below to generate a list of the top reasons for ER utilization. As with the previous case study, no date restrictions are made for the sake of simplicity.

1. Select **Query Builder** from **Tasks—Data**.
2. Navigate within **SAS Servers** to select FacilityHeader in BOOK library.
3. Enter Top N Reasons for ER in **Query name** and ER_Utilization in **Output Name**.
4. Click Add Tables to add FacilityDetail in BOOK library to the query.

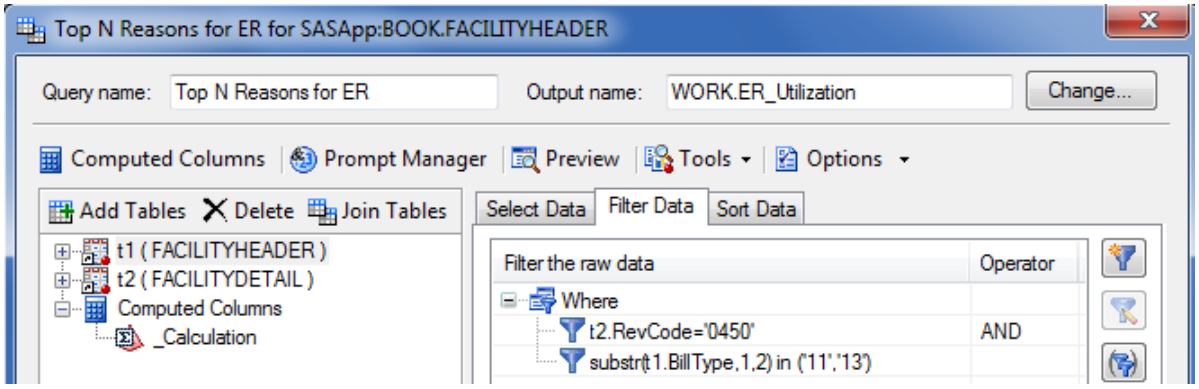
5. To merge FacilityHeader and FacilityDetail, select an inner join based on ClaimID. See previous section on more information about joining tables.
6. Select ClaimID and PDXDesc in **Select Data** tab. Since you are counting claims by primary diagnosis code description, select COUNT in the **Summary** column.

Figure 5.14: Top reasons for ER visits query



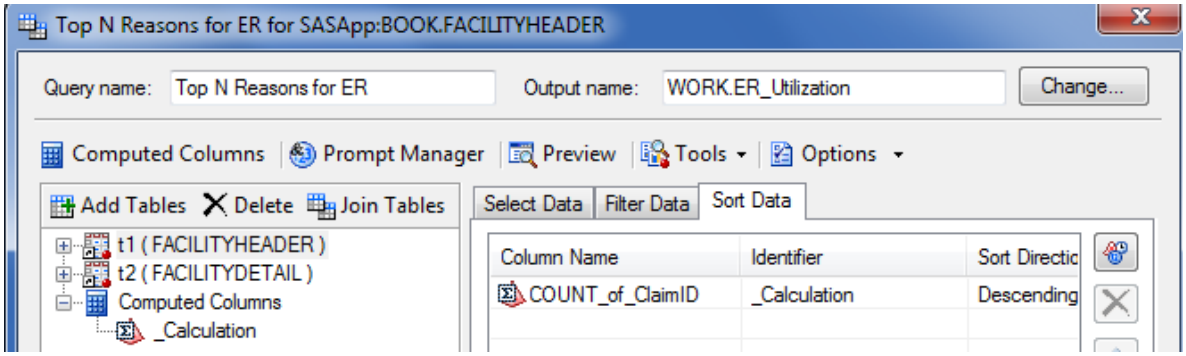
7. Add two filters. The first filter on revenue code ensures that you are extracting only ER claims from FacilityDetail. The second filter uses BillType to extract Hospital Inpatient and Outpatient visits (see Figure 5.15).

Figure 5.15: Filter Top N Reasons for ER query



8. Place _Calculation from **Computed Columns** in **Sort Data** tab and choose Descending under **Sort Direction**. This sorts the data so that PDXDesc with the most visits appear first (see Figure 5.16).

Figure 5.16: Sort data



9. View results in Figure 5.17.

Figure 5.17: Top N Reasons for ER visits

	COUNT_of_ClaimID	PDXDesc
1	131	Pneumonia, Organism Unspecified
2	80	Unspecified Septicemia
3	67	Obst Chronic Bronchitis W/Exacerbat
4	52	Other Chest Pain
5	51	Acute Kidney Failure Unspecified
6	50	Uti Site Not Specified
7	45	Atrial Fibrillation
8	43	Acut Mi Subndocrdl Infarct Init Eoc
9	39	Unspec Cerbrl Art Occl W/Infarct
10	35	Calculus Of Ureter

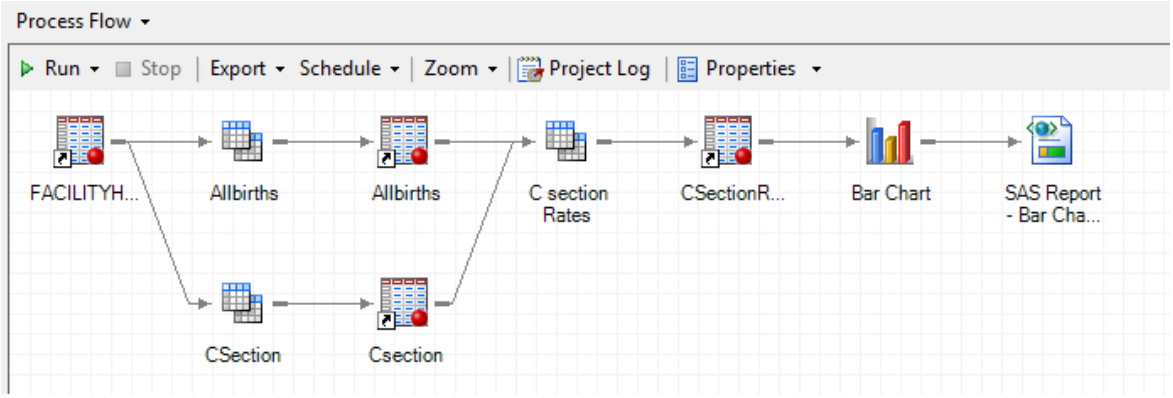
One way to extend the analysis in this section is to classify ER visits into whether they were avoidable or truly emergent. There are several industry algorithms to assist in performing this classification. One publicly available algorithm has been developed by New York University (www.wagner.nyu.edu/faculty/billings/nyued-download).

In an effort to decrease ER utilization, healthcare insurance companies often have programs to educate their members. Some of these involve placing a gatekeeper at the ER to divert avoidable visits to primary or urgent care.

Automating Reports with SAS Enterprise Guide

SAS Enterprise Guide makes it easy for the healthcare analyst to automate reports. Once you have created your task, simply run your entire analysis in process flow. Figure 5.18 shows the process flow created by the C-section case study.

Figure 5.18: C-section case study process flow



The flow diagram is useful in a number of ways. It is a way to produce good documentation effortlessly. A glance at the process flow is sufficient to tell you that there are three **Query Builder** tasks and a **Bar Chart** task. In fact, you can hover on each task or data set to view more information about it. The process flow diagram also allows you to run or re-run the entire analysis or any part of it. To run the entire process flow, select **Run—Run Process Flow**. To run analysis from Csection node, right-click on CSection **Query Builder task** and select **Run Branch** from CSection.

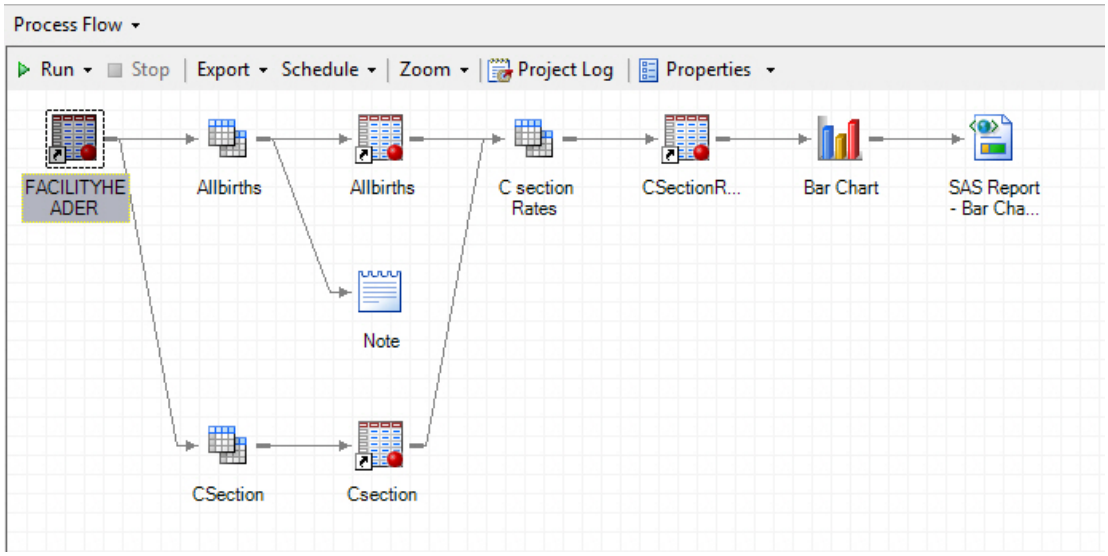
Creating Documentation in SAS Enterprise Guide

You can easily add documentation to the process flow created in Figure 5.18. In fact, you can store detailed instructions on your project and link these instructions to any object in the process flow.

Let us add documentation for the **Allbirths** task.

1. Click on **Allbirths** task shown in Figure 5.18 and select **Note** from **File—New** menu.
2. Key in instructions. Close the window.
3. View the modified process flow in Figure 5.19 to ensure that a **Note** has been connected with **Allbirths** task.

Figure 5.19: Documentation in SAS Enterprise Guide



Conclusion

Institutional providers account for the larger portion of medical claim dollars, but the smaller incidence of member claim volume. This chapter helped us to understand this important player in the healthcare industry and how:

- The claim submission mechanism used by facilities is different from other providers.
- There are data elements specific to the facility's business model.

By looking at some simple yet robust applications in SAS Enterprise Guide we were able to:

- Create bar graphs illustrating different C-section rates among hospitals.
- Identify top reasons for ER utilization.
- Automate and document a process flow in SAS Enterprise Guide.

Chapter 6: Professional and Ancillary Claims

Introduction 59

Medical Claim Submission 60

- CMS-1500 Claim Form 60
- 837P Electronic Claim Format 60

Data Elements Unique to CMS-1500/837P 60

- Demographic Information 60
- Diagnosis Codes 61
- Diagnosis Pointer 61
- Provider Identifiers 61
- Procedure Codes and Modifiers 61
- Place of Service 62

Provider Specialty 62

Payment Methodologies 62

Case Study: Identifying Children Who Miss Their Checkups 63

- Identify Currently Enrolled Children under Six Years of Age 64
- Import Excel Data 66
- Identify Appropriate Professional Claims 68
- Create Outreach Report 70
- Create Internal Report 71

Case Study: Automating Reports with Macro Variables 73

- Extract Code from Enterprise Guide Tasks 73
- Import Data Code 74
- Query Builder Code 75
- Automate Code 76

Conclusion 81

Introduction

While they provide different types of healthcare services, professional and ancillary providers both submit data for reimbursement via the CM-1500 paper form and its electronic analogue, the 837P. It is important to understand this data source, as it is used many times in conjunction with facilities data.

In this chapter you will:

- Understand the claim submission mechanism used by professional and ancillary providers.
- Learn about data elements specific to the CMS-1500 and 837P formats.
- Analyze professional header and detail data in SAS Enterprise Guide to create outreach reports for members lacking appropriate care.
- Extract and parameterize code from SAS Enterprise Guide in order to automate outreach reports.

Medical Claim Submission

Both professional and ancillary providers use the same mechanism for claim submission. Their reimbursement is generally based on procedure codes that describe the healthcare services provided for their patients.

CMS-1500 Claim Form

The CMS-1500 form (Health Insurance Claim Form) is the standard used by a noninstitutional provider or supplier. It is sometimes referred to as the AMA (American Medical Association) form. Despite the HIPAA mandate to move to electronic submission of claims, the paper mechanism is still being used when a provider qualifies for a waiver from the Administrative Simplification Compliance Act (ASCA) requirement for electronic submission. The National Uniform Claim Committee (NUCC—www.nucc.org) is responsible for the design and maintenance of the CMS-1500 form.

Version 02/12 of the CMS-1500 (see Figure A.2 in the Appendix for a facsimile) became fully effective April 1, 2014, at which time payers are to receive and process paper claims submitted only on the revised 1500 Claim Form. The primary intent of this most recently enhanced version was to align with the requirements in the Accredited Standards Committee X12 (ASC X12) Health Care Claim: Professional (837P) Version 5010 Technical Report Type 3.

> The CMS-1500 instructions as provided by NUCC are not specific to a particular public or private payer. We suggest that you refer to specific instructions issued by your payer, clearinghouse, and/or vendor for further clarification of reporting requirements.

837P Electronic Claim Format

The 837P electronic claim format can be used to submit healthcare claim billing information, encounter information, or both, from providers of healthcare services to payers, either directly or via intermediary billing services and claims clearinghouses.

This transaction is intended to originate with the healthcare provider or the provider's designated agent. In some instances, a payer may originate an 837 to report a healthcare encounter to another payer or sponsoring organization. The 837 transaction provides all necessary information to allow the payer to at least begin to adjudicate the claim.

The 837 coordinates with a variety of other transactions, including, but not limited to, the following: Health Care Information Status Notification (277), Health Care Claim Payment/Advice (835), and the Functional Acknowledgment (997).

ASC X12 (www.x12.org) is a standards development organization, accredited by the American National Standards Institute (ANSI), which focuses on developing standards for electronic information exchanges. ASC X12 has subcommittees that focus on different industries, such as finance, government, transportation, and insurance. The AMA is a member of ASC X12 and participates on the Insurance Subcommittee (X12N). X12N develops and maintains standards related to the insurance and healthcare industries, such as the 837 format.

Data Elements Unique to CMS-1500/837P

This section explores some data elements unique to the CMS-1500/837P claim mechanism. There are others to consider, but these bear special mention.

Demographic Information

Unlike the CMS-1450 institutional claim form, the CMS-1500 requires many elements of demographic information: both patient and the insured contract holder. The type of insurance plan as well as secondary insurers is required to be entered and submitted. These data elements are then used by the adjudication systems to validate the beneficiary via the payer's membership database.

Diagnosis Codes

As with institutional claims, diagnosis codes are at the claim level, not the service detail level. Unlike the institutional claim, there aren't multiple categories of diagnosis codes (e.g., admitting, external cause of injury), and there is no explicit ordinality associated with the codes. On the paper form and the electronic format there are up to twelve diagnosis codes. Providers are asked to code to the highest level of specificity and to enter the diagnoses in priority order. Despite this requested "priority order," it is generally taken by the payer to be without ordinality. The 837P documentation states that "the diagnosis listed in this element (the first diagnosis code element) is assumed to be the principal diagnosis."

Diagnosis Pointer

The diagnosis pointer is unique to professional claims. It is a required data field that resides on the detail service line and, on the CMS-1500, contains the diagnosis code reference letter (A thru L) as shown in Item 21, the diagnosis codes at the claim level. This mechanism relates the service line procedure performed to one or more diagnoses. On the 837P the acceptable values are "1" through "12" and correspond to Composite Data Elements 01 through 12 in the Health Care Diagnosis Code HI segment in the Claim Loop ID-2300. Up to four reference letters or numbers per detail line are allowed. The most relevant, or primary, is required to be listed first. The discrepancy in the domains (digits versus letters) is relative to the paper form's need to string single characters together in one field, while the electronic format provides four fields to enter the values.

Payer data warehouses will choose different methods to store the "diagnosis pointer" concept. The most direct method is to simply store the "primary" diagnosis code on the service line detail record. With this method there is no need for the user to do any lookup—the warehouse acquisition team has done it. This direct method was used by our fictitious payer, Healthy Living, Inc. The indirect method would store the value(s) as it was submitted. Storing a single diagnosis code does result in loss of some granularity, whereas the indirect retains detail but adds work for the analyst.

Provider Identifiers

As with the facility claim, care must be taken to understand where provider identifiers originate. On the detail service line the rendering provider's National Provider Identifier (NPI) is required. At the claim header level several possibilities exist: referring provider, ordering provider, or supervising provider. Upon submission there will be a qualifier indicating the role of the provider being reported; we hope this information will be captured by your data source.

A further note about the difference between header providers and detail providers: The billing provider occurs once at the header, but the rendering provider can be on one or more of the detail lines. In fact, there may be as many different providers on the detail as there are lines of detail services. For example, a patient goes to an urgent care center. The patient sees a nurse practitioner first, then a physician, followed by a cardiologist who orders labs. If these visits are all one billing entity, the "header" provider occurs only once, but there will be multiple rendering provider IDs (hopefully NPI) associated with the claim.

Procedure Codes and Modifiers

The source of billing information on the CMS-1500/837P claim format is the procedure code and any associated modifiers found at the detail service line level. The AMA's Common Procedural Terminology (CPT®) or CMS's Healthcare Common Procedure Coding System (HCPCS) code sets are used for billing purposes. A per-unit charge is associated with a procedure and the Units data element serves as a multiplier to arrive at the submitted charge for a detail service line.

Multiple modifiers may be used to clarify or improve the reporting accuracy of the associated procedure code. CMS maintains a list of procedures that must, may, or must not have an associated modifier. Many commercial payers follow similar adjudication rules.

More in-depth discussion of procedure codes and their modifiers is in Chapter 8. The concept at this juncture is the importance and relative uniqueness of procedure codes and their modifiers as found at the service line detail for professional and ancillary providers.

Place of Service

This data element is on the service level detail. It is used to identify the setting, using a Place of Service (POS) code, for each service performed. It is a required field. POS is a standard two-character field that is used nationally by all payers and providers and is maintained by CMS at www.cms.gov/physicianfeesched/downloads/Website_POS_database.pdf. Table A.3 in the Appendix shows the current domain of values as of this book's printing.

The importance of this concept is not to be overlooked. Unlike facilities, professional providers have "feet" and move about providing services for their patients. One day a physician may see Mary Doe in his office, the next day at the emergency department, and then later in an inpatient setting—all very different settings. All of these services may appear on a single medical claim. Many analytic projects will need to categorize the data by the "setting" in which a service is provided; POS allows this capability.

POS is often needed to determine the acceptability of direct billing of Medicare, Medicaid, or commercial insurance services provided by any given provider, that is, the billable procedure code must be consistent with the POS. For example, a tonsillectomy submitted by the surgeon could not have been done in an office setting.

Bill Type in institutional claims comes close to capturing the concept of setting. A value of "11" in its first two characters indicates that the service was provided in a hospital inpatient setting. While analogous, POS and Bill Type are not synonymous, and any attempted relationship between the two must be approached with much caution.

Provider Specialty

Healthcare Provider Taxonomy Codes (see Chapter 4) categorize the type, classification, and/or specialization of healthcare providers. HPTC is currently a required data element on all claim submission formats. This code set is the new, nationally accepted standard for denoting professional provider practice specialty.

Please understand that you may see reference to other code sets that describe provider specialty. CMS has such a set that may be found at www.cms.gov/Medicare/Provider-Enrollment-and-Certification/MedicareProviderSupEnroll/Downloads/TaxonomyCrosswalk.pdf, and, prior to the HIPAA mandate, other payers also maintained a set for their use in the maintenance of provider networks. As with legacy provider identifiers being replaced with NPI, it is fair to say that these "local" provider specialty code sets will disappear in favor of the HPTC. The transition may be hampered by validity issues with taxonomy: It is self-reported, and CMS does not validate for Board Certification.

Payment Methodologies

Of particular note for analysts is the notion that professional providers may be contracted by the payers' network under a capitation arrangement. Capitation is a payment mechanism wherein a fixed payment is made at regular intervals to a medical provider by a managed care organization for an enrolled member. The effect on our administrative healthcare data is that financial data takes on zero values. Analysts need to be able to identify encounter data that was submitted by a capitated professional provider as being different than a denied claim ($0 paid). There may be an indicator in the claim record that suggests it is capitated or a join to provider data may have to be performed in order to interrogate the contract arrangement. In addition to maintaining denied claims, Enterprise Data Warehouses (EDWs) will also keep duplicate claims

that may have $0 paid. Strategies must be developed in any project to identify and net out such ill-behaving claims.

Medicare pays physicians for services based on submission of a claim using one or more specific procedure codes—effectively a fee-for-service arrangement. Each code has a Relative Value Unit (RVU) assigned to it that, when multiplied by a conversion factor (CF) and a geographical adjustment (Geographic Practice Cost Index [GPCI]), creates the compensation level for a particular service. The Medicare Physician Fee Schedule (MPFS) look-up tool is a searchable database that provides payment information for professional claims. This tool is worth a look and may be found at www.cms.gov/Medicare/Medicare-Fee-for-Service-Payment/PFSlookup/index.html.

The various state Medicaid offices publish similarly derived fee schedules for providers of Medicaid services.

Case Study: Identifying Children Who Miss Their Checkups

The Quality Improvement department at Healthy Living, Inc., engages in several member outreach efforts. One focuses on ensuring that members receive routine and regular primary care. Outreach workers at the company contact members who have missed their primary care checkups, reminding them and their physicians to schedule an appointment.

This section shows you how professional claims and membership tables may be used to derive two kinds of reports for children under six years of age.

- The "outreach report" identifies members who have not had any primary care visits within the calendar year. This report is used to contact members and their providers.
- The "internal report" lists members who have had one or more primary care visits. This report is scrutinized further to ensure that members are receiving the care they need.

The following steps lay out a technical roadmap.

1. Identify children under six years of age who are enrolled as of December 1, 2011. This is done by utilizing BOOK.Members, which contains Healthy Living, Inc.'s enrollment data. Figure 6.1 shows a few records from Members.

Figure 6.1: Members table

MEMBERS

Filter and Sort | Query Builder | Data | Describe | Graph | Analyze | Export | Send To

	MemberID	BeginEnroll	EndEnroll	DOB	Sex
1	M000000086000...	2010-01-01	2011-12-31	1930-05-14	F
2	M000000827000...	2009-06-01	2009-12-31	1928-12-17	F
3	M000000827000...	2010-01-01	2010-10-31	1928-12-17	F
4	M000001353000...	2007-01-01	2007-12-31	1937-03-06	M
5	M000001353000...	2008-01-01	2008-12-31	1937-03-06	M
6	M000001353000...	2009-01-01	2009-07-31	1937-03-06	M
7	M000001494000...	2007-01-01	2007-12-31	1928-07-17	F
8	M000001494000...	2008-01-01	2009-12-31	1928-07-17	F

The analysis in this case study is dependent on a hard-coded date of reference, namely December 1, 2011. The next case study shows you how to parameterize date variables so that the analysis may be run periodically.

2. Identify relevant procedure codes to ensure that professional claims selected are appropriate. The selection of procedure codes is dependent upon your project. The procedure codes used in this section are based on a Healthcare Effectiveness Data and Information Set (HEDIS) measure called Children

and Adolescents' Access to Primary Care Practitioners. The codes have been placed in an Excel spreadsheet that is imported into SAS Enterprise Guide. Excel is a good way to maintain and manage small code sets (driver tables). For larger code sets, permanent SAS tables should be used.

HEDIS is a key performance tool used by the vast majority of insurance plans. It lays out specifications for seventy-five performance measures, one of which is the subject of Chapter 11.

3. Identify appropriate professional claims from Professional Claim Header and Detail tables. For simplicity, Professional Claim tables contain visits from 2011 only.
4. Use Provider Taxonomy on the Professional Claim Detail table to limit encounters to internal and family medicine, general practice, and pediatrics.
5. Use POS on the Professional Claim Detail table to filter encounters to those that take place in a clinic or office setting.
6. Limit procedure codes to those identified in (2).
7. Join the result from (3) and (1) to yield the outreach and internal reports.

Identify Currently Enrolled Children under Six Years of Age

This section creates a table containing currently enrolled children under six years of age.

1. Select **Query Builder** from **Tasks—Data**.
2. Navigate within **SAS Servers** to select Members in BOOK library.
3. In **Query Builder**, type "Current Members Kids under 6" in **Query Name**. Type "WORK.CurrentMembersKids" in **Output Name**.
4. Select MemberID, BeginEnroll, EndEnroll, DOB, and Sex.
5. Create a computed column called Age, using the following expression.

```
round(('01DEC2011'd-DOB)/365.25,.01)
```

The numerator in the expression in step 5 is the difference in days between December 1, 2011, and the member's Date of Birth. This is divided by 365.25 to derive Age in years. The ROUND function then rounds off the results to the second decimal place.

The ROUND function has several uses of the second argument.

round(123.63) =124. When there is no second argument, the function rounds to the integer.

round(123.63, 100) = 100. The function rounds to the nearest hundreds place value.

round(123.63,.01) = 123.63. The function rounds to the nearest hundreds place value.

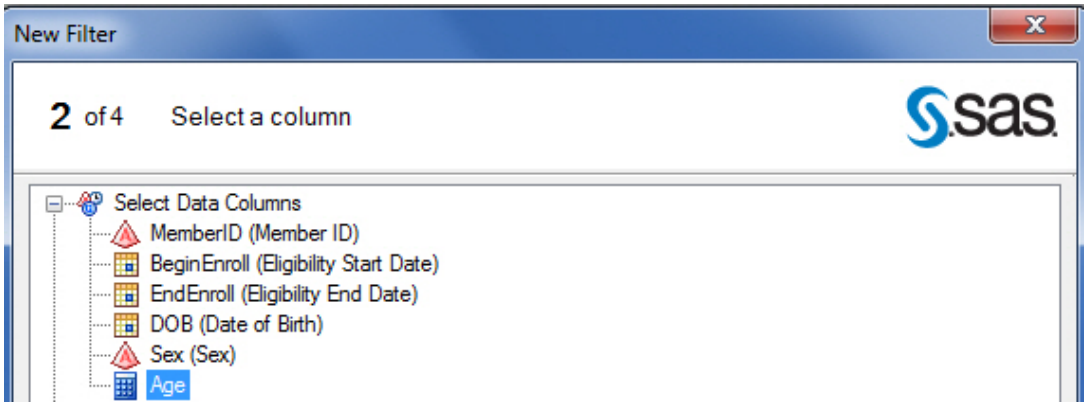
6. Click **Filter Data** tab in **Query Builder** to create an advanced filter, using the expression provided below. For a review on creating advanced filters, see Chapter 5.

```
t1.EndEnroll >= '01DEC2011'd
```

The above filter specifies that the ending enrollment date is greater than or equal to December 1, 2011. SAS is rigid about how the user inputs dates into code. As SAS stores dates as numbers, a user-specified date will be translated into a number only if it is specified in the manner above—a date literal.

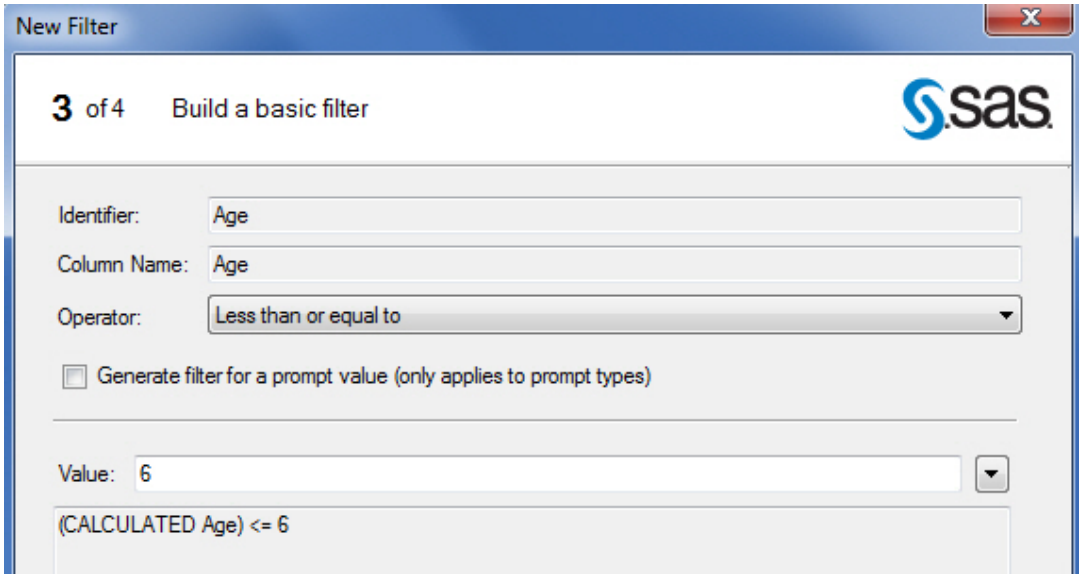
7. Click  to create a new filter for children under six years of age (see Figure 6.2). In **Select a type window**, select **Basic Filter**. Click on Age.

Figure 6.2: Create a basic filter on a computed item



8. In **Step 3 of 4 Build a basic filter**, select **Less than or equal** to under **Operator**. Key in "6" under **Value** (see Figure 6.3).

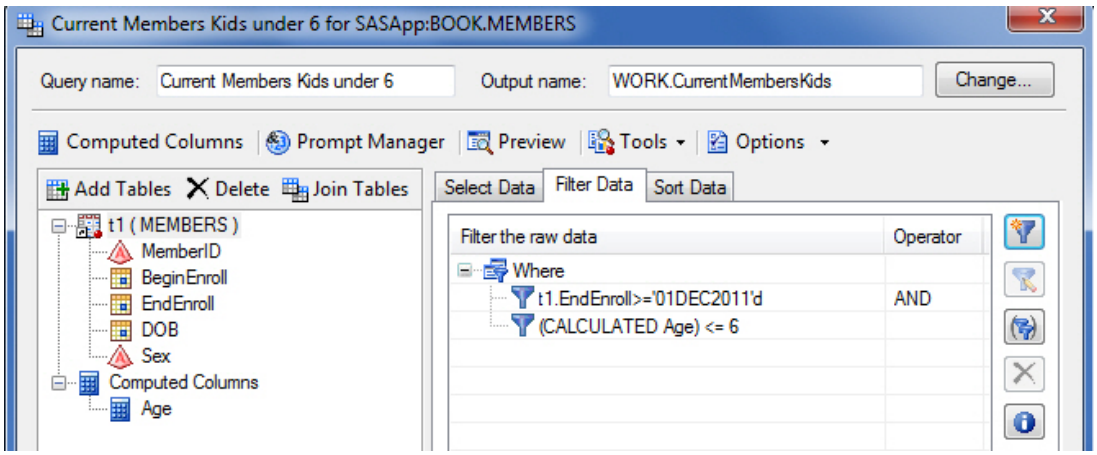
Figure 6.3: Build a basic filter



The CALCULATED keyword references a computed item. A basic filter that includes a computed column uses this keyword. This is different from an advanced filter, which copies the formula of the computed item in the filter. A basic filter is advantageous because if the formula is updated, the reference to the formula in the filter is automatically updated.

9. Confirm that the **Filter Data** tab has filters, as shown in Figure 6.4.

Figure 6.4: Adding filters in Query Builder



10. Confirm results to ensure that only currently enrolled children under six years of age are selected (see Figure 6.5).

Figure 6.5: CurrentMembersKids table

	MemberID	BeginEnroll	EndEnroll	DOB	Sex	Age
1	M003202097102...	2011-04-01	2011-12-31	2005-12-27	F	5.93
2	M003208634103...	2011-01-01	2011-12-31	2006-01-20	M	5.86
3	M003215422103...	2011-01-01	2011-12-31	2006-02-13	F	5.8
4	M003243833104...	2011-01-01	2011-12-31	2006-04-13	F	5.63
5	M003249025104...	2011-01-01	2011-12-31	2006-02-21	M	5.77
6	M003254632104...	2011-01-01	2011-12-31	2006-04-30	F	5.59

Import Excel Data

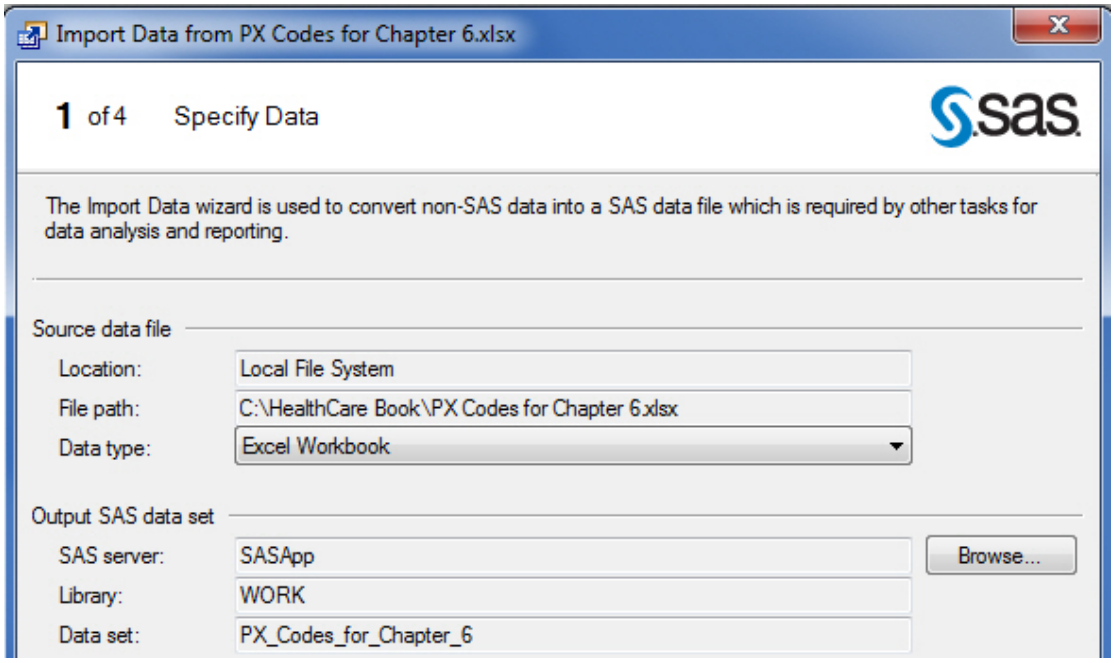
This section shows you how to import data from Excel using Enterprise Guide. Relevant procedure codes are placed in a spreadsheet, as shown in Figure 6.6.

Figure 6.6: Procedure codes used to identity professional claims of interest

	A
1	PX
2	99201
3	99202
4	99203
5	99204
6	99205
7	99211
8	99212

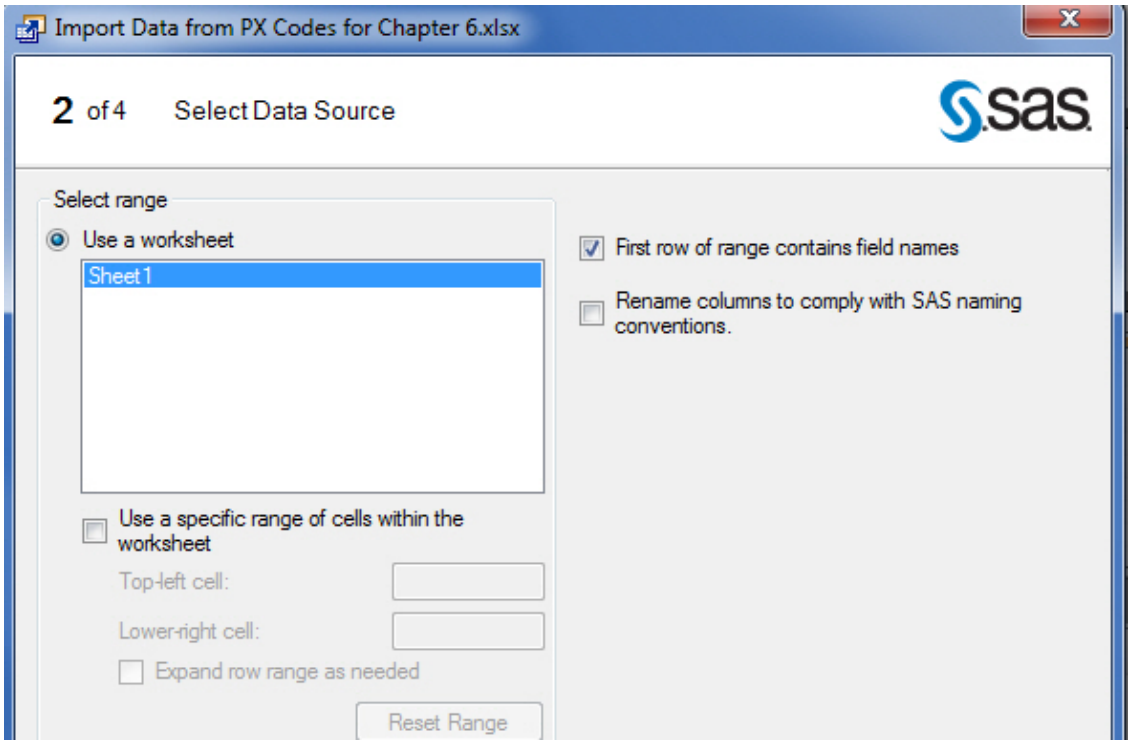
1. Select **Import Data** from **File**.
2. Open PX Codes for Chapter 6.xlsx. This file and other data can be downloaded from the authors' website.
3. Specify WORK as the **Output Library** and PX_Codes_for_Chapter_6 as the name of the data set (Figure 6.7).

Figure 6.7: Import Excel data using Enterprise Guide

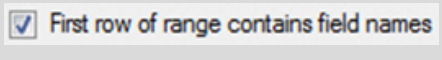


11. In **Select Data Source**, click on **Select range** radio button (see Figure 6.8).

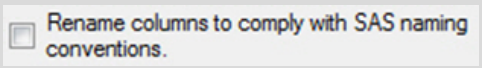
Figure 6.8: Import Excel data using Enterprise Guide



The **Select Data Source** allows you to specify a range of cells in your worksheet. In addition, it contains two useful checkboxes.



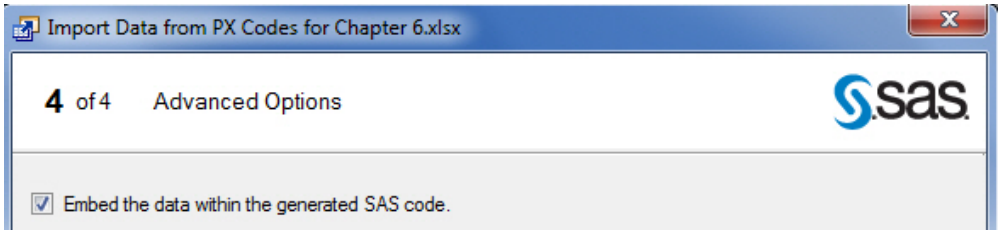
The checkbox above allows Enterprise Guide to create column names from the first row of the spreadsheet.



The column names in the source spreadsheet may violate SAS column name rules. For example, they may have spaces or begin with numbers, both of which are not allowed under SAS naming conventions. The checkbox above renames columns to comply with SAS syntax.

12. In **Advanced Options**, check **Embed the data within the generated SAS code** checkbox as shown in Figure 6.9. This checkbox places data from the source spreadsheet right into SAS code. This is useful if you wish to maintain your data with SAS code, rather than in Excel. You will learn how to harvest this code later on in the chapter.

Figure 6.9: Embed data in SAS code



13. Confirm the results of **Import Data** (see Figure 6.10).

Figure 6.10: PX codes in SAS data set

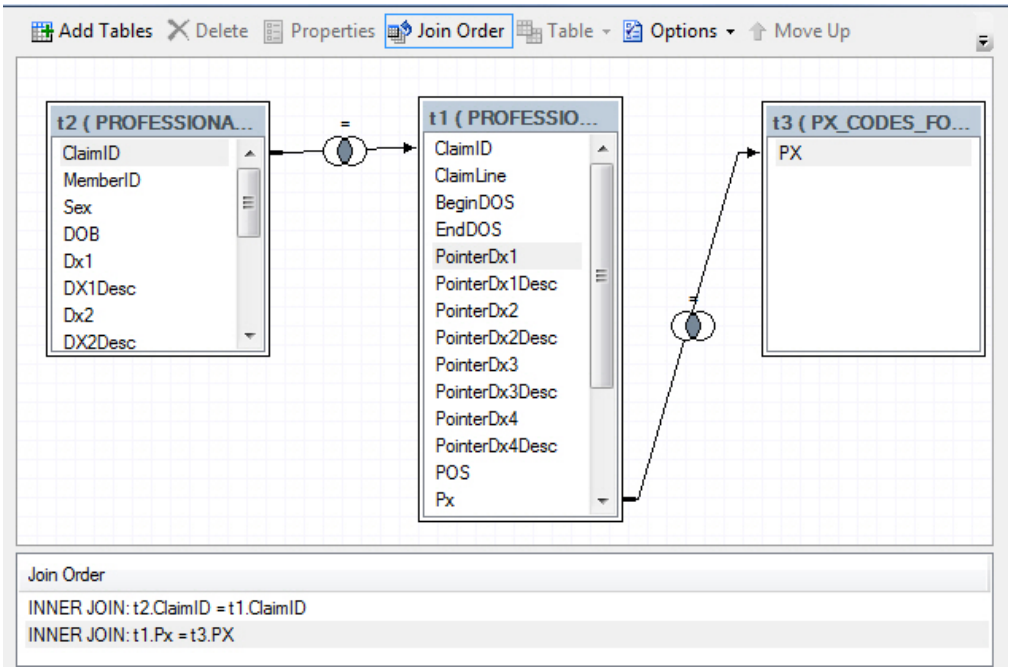
	PX
1	99201
2	99202
3	99203
4	99204
5	99205
6	99211
7	99212
8	99213
9	99214
10	99215

Identify Appropriate Professional Claims

This section shows you how to identify professional claims that take place in the office setting. For the purpose of simplicity, the Professional Claim tables contain visits from 2011 only.

1. Select **Query Builder** from **Tasks—Data**.
2. Click Add Tables. Select BOOK.ProfessionalHeader, BOOK.ProfessionalDetail, and WORK.PX_Codes_for_Chapter_6.
3. Type Professional Visits in **Query Name** and WORK.ProfessionalVisits in **Output Name**.
4. In **Tables and Joins** window, click Join Order. Enterprise Guide has automatically joined the three tables appropriately as shown in Figure 6.11. ClaimID joins BOOK.ProfessionalHeader and BOOK.ProfessionalDetail. PX joins BOOK.ProfessionalDetail and WORK.PX_Codes_for_Chapter_6. As is depicted visually and laid out in code in **Join Order** tab, both joins are inner joins.

Figure 6.11: Join relationships in Enterprise Guide



5. In **Query Builder**, select MemberID from BOOK.ProfessionalHeader and the following columns from BOOK.ProfessionalDetail: ClaimID, ClaimLine, BeginDos, EndDos, PointerDx1-PointerDx4, PointerDx1Desc-PointerDx4Desc, PX, PXDesc, and SrvcProvID.
6. Create a filter on POS using the following expression. A value of 11 indicates that the encounter took place in an "Office" setting.

```
t1.POS = "11"
```

7. Create a filter on SrvcProvTaxonomy in BOOK.ProfessionalDetail table, using the expression shown in step 9. The code values indicate specialties of family and internal medicine, general practice, and pediatrics, respectively. See Chapter 4 for a review of provider taxonomy nomenclature. The filter ensures that professional visits are carried out by primary care physicians.

```
t1.SrvcProvTaxonomy in ("207Q00000X", "207R00000X", "208D00000X",
"208000000X")
```

This section includes Pediatrics in its definition of primary care professional visits because children are the primary focus of the analytics. This will vary with the project. If your project focuses on women, you may consider including Obstetrics & Gynecology. Similarly, a project analyzing older patients should include Geriatrics.

8. Confirm that the resultant data set ProfessionalVisits contains appropriate medical encounters.

Figure 6.12: ProfessionalVisits data set

MemberID	ClaimID	BeginDOS	EndDOS	PointerDx1
M558481762328...	C056076745	2011-02-07	2011-02-07	41401
M498448753307...	C056076879	2011-02-07	2011-02-07	4280
M495308304298...	C056076910	2011-02-06	2011-02-06	5990
M488094836280...	C056076911	2011-02-06	2011-02-06	5959
M431726427169...	C056076914	2011-02-06	2011-02-06	4139
M428563881168...	C056076931	2011-01-29	2011-01-29	25000
M901301153357...	C056076966	2011-02-05	2011-02-05	V812
M816828277337...	C056077080	2011-02-05	2011-02-05	78791

Create Outreach Report

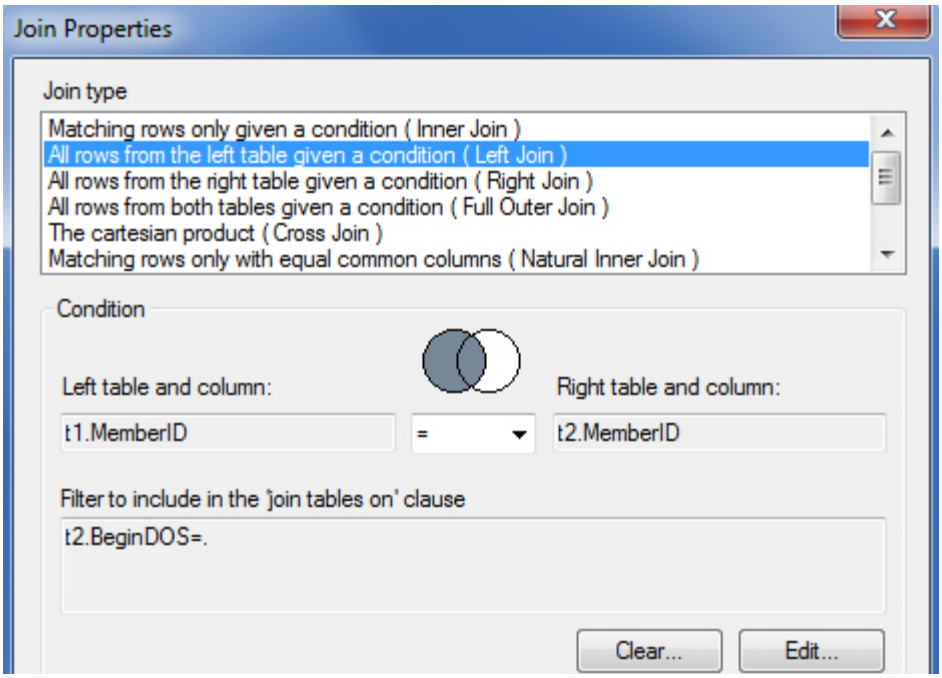
This section joins ProfessionalVisits and CurrentMemberKids tables to create an outreach report.

1. Select **Query Builder** from **Tasks—Data**.
2. Click Add Tables. Select CurrentMemberKids and ProfessionalVisits.
3. Type in Kids Outreach in **Query Name** and WORK.KidsOutreach in **Output Name**.
4. Select MemberID, DOB, and Sex from CurrentMembersKids.
5. In **Tables and Joins** window, click Join Order. A default inner join is chosen by Enterprise Guide. As you need to select children who have not had any professional visits, you will change the join so that you only retrieve records from CurrentMemberKids who do not join with any records from ProfessionalVisits.
6. Double click on Venn diagram to change the default join conditions.
7. Choose **All rows from the left table given a condition (Left Join)** under **Join Type**.
8. Click **Edit** to add a condition on the join. In the **Advanced Filter Builder** window, create the following filter:

```
BeginDos=.
```

9. Confirm the join order (see Figure 6.13).

Figure 6.13: Modify join order



10. Confirm results (see Figure 6.14).

Figure 6.14: Outreach report

MemberID	DOB	Sex
M172525584127...	2005-12-03	M
M003202097102...	2005-12-27	F
M478968632273...	2006-01-06	M
M003208634103...	2006-01-20	M
M471919825211...	2006-01-27	M
M468825494180...	2006-02-01	F

In an industry implementation, the outreach report is combined with demographic data containing contact and address information for both members and their providers so that outreach workers may make immediate contact.

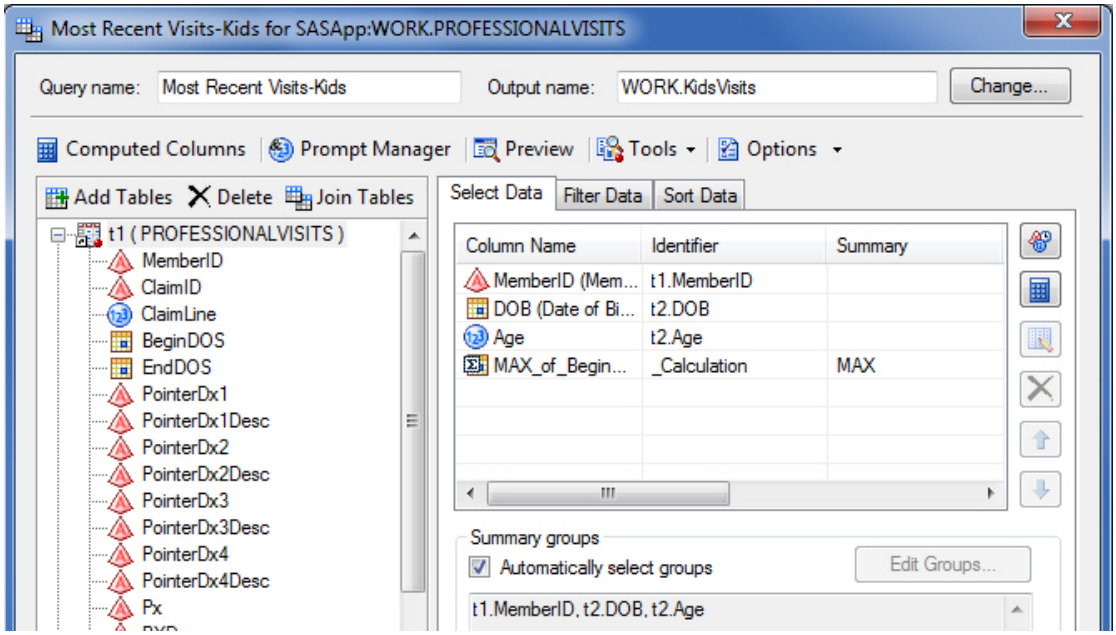
Address data in an outreach report may be analyzed to identify demographic hotspots of noncompliance with standards of preventive medicine. Some insurance companies send health workers out to hotspots regularly in an attempt to promote better health.

Create Internal Report

This section shows you how to build a report of children who have had at least one primary care–related professional visit in the past year. This report is used by healthcare workers in Healthy Living, Inc., to ensure that members are receiving appropriate care.

1. Select **Query Builder** from **Tasks—Data**.
2. Click Add Tables. Select CurrentMemberKids and ProfessionalVisits.
3. Type Most Recent Visits-Kids in **Query Name** and WORK.KidsVisits in **Output Name**.
4. Select columns as shown in Figure 6.15. Select DOB and Age from CurrentMembersKids. Select MemberID and BeginDos from ProfessionalVisits. For BeginDos, choose MAX from dropdown under **Summary**. This ensures that the most recent visit for every child is retrieved by the query.

Figure 6.15: Select columns for internal report



5. In **Tables and Joins** window, click Join Order to confirm that an inner join is used.
6. Confirm results (see Figure 6.16).

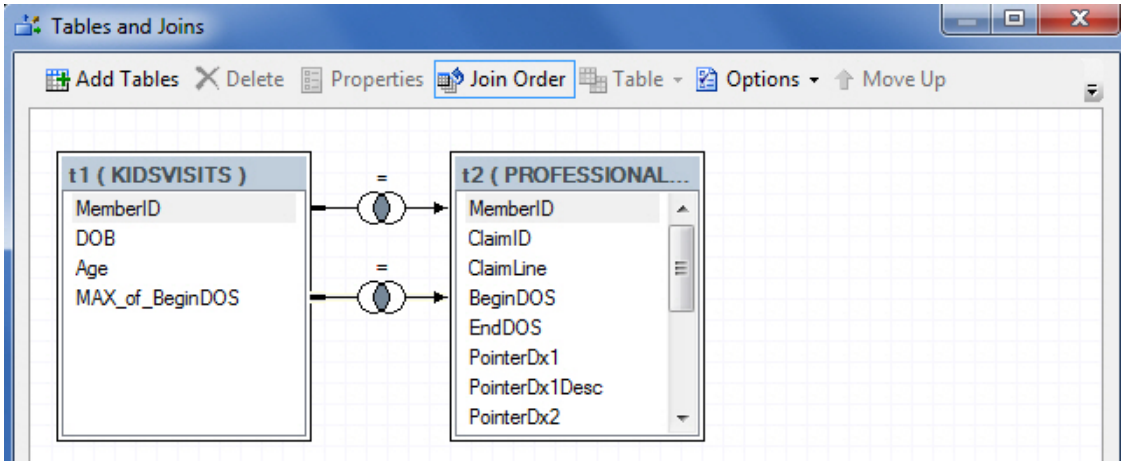
Figure 6.16: KidsVisits table

MemberID	DOB	Age	MAX_of_BeginDOS
M003202097102...	2005-12-27	5.93	2011-02-15
M003249025104...	2006-02-21	5.77	2011-02-04
M003302448105...	2006-08-04	5.33	2011-02-21
M003315628105...	2006-08-14	5.3	2011-02-26
M003363977106...	2006-11-05	5.07	2011-02-19
M003382703107...	2007-03-06	4.74	2011-02-07
M003484035110...	2007-11-10	4.06	2011-02-15
M003497202110...	2007-11-13	4.05	2011-02-20

This report is enhanced so that healthcare workers view the most recent visit of each member along with the visit's diagnosis information.

7. While KidsVisits table is open, click **Query Builder**. This places KidsVisits table in the query.
8. Click Add Tables and select ProfessionalVisits.
9. Type Most Recent Kids Visit with Diagnosis in **Query Name** and WORK.KidsVisitsDiag in **Output Name**.
10. Select MemberID, DOB, and Age from KidsVisits. Select BeginDOS, SrvcProvID, PointerDx1-PointerDx4, and PointerDx1Desc-PointerDx4Desc from ProfessionalVisits.
11. In **Tables and Joins** window, click Join Order. Confirm that the inner join is on both MemberID and BeginDOS, ensuring that the query retrieves results for the most recent visit for every child (see Figure 6.17).

Figure 6.17: Inner join on two columns



12. Confirm results (see Figure 6.18).

Figure 6.18: Internal Report

MemberID	DOB	Age	BeginDOS	SrvcProvID	PointerDx1	PointerDx1Desc
M472921656223...	2007-03-19	4.7	2011-02-15	P143478158	0340	Streptococcal Sore...
M003202097102...	2005-12-27	5.93	2011-02-15	P143456856	78841	Urinary Frequency
M003861754118...	2010-01-27	1.84	2011-01-31	P383457929	2875	Unspecified Throm...

Case Study: Automating Reports with Macro Variables

The Quality Improvement department at Healthy Living, Inc., finds the Outreach and Internal reports to be useful. The department would like to run these reports periodically to check compliance and note areas of improvement.

In Chapter 5, you learned how to run an Enterprise Guide project in its entirety. This section shows you how to automate code. We demonstrate this by first extracting code from Enterprise Guide tasks and then modifying code for automation.

This section uses two new rules.

1. A report generated on any given day looks back at professional encounters occurring in the prior twelve months. This makes sense because children under six should have at least one visit with their doctor in any given twelve-month period.
2. The previous section assumed that professional claim tables contain 2011 encounters only. As companies routinely store several years' worth of encounters in their data warehouse, this is an overly simplistic assumption and is relaxed in this section.

As a result of the above rules, two Query Builder tasks require modification to parameterize hard-coded dates. These are Current Members Kids under 6 and Professional Visits.

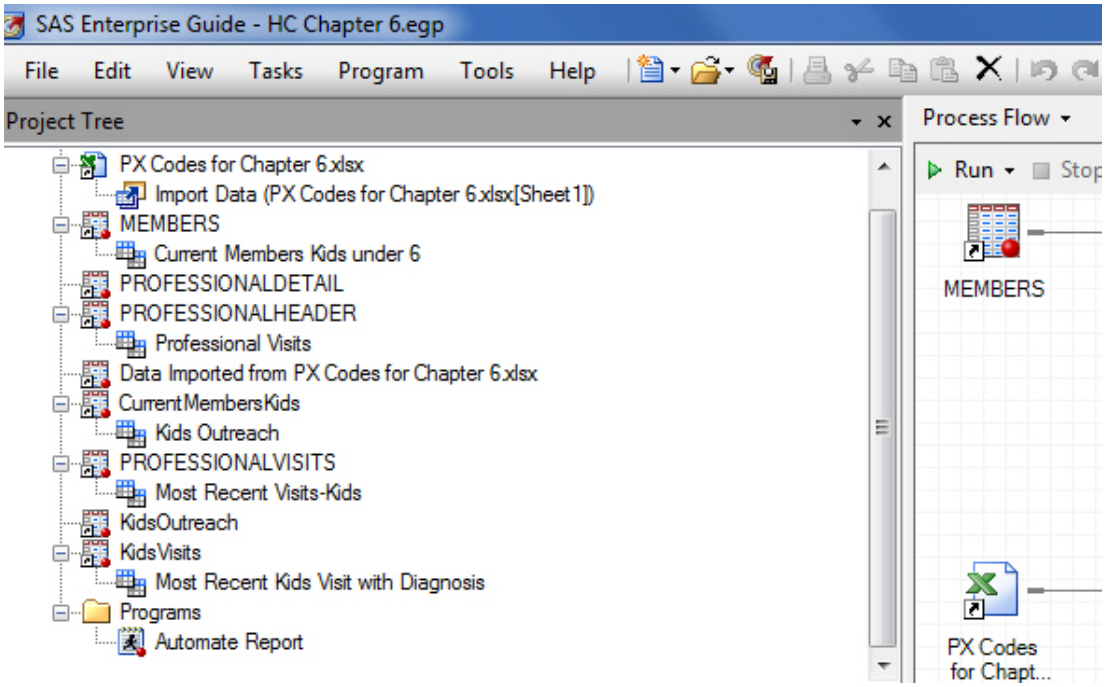
Let's first extract and understand code behind Enterprise Guide tasks created in the previous section.

Extract Code from Enterprise Guide Tasks

Extracted code from Tasks can be placed in a "code node." The following creates a code node.

1. Select **Program** from **File—New**. This creates a directory called **Programs** under **Project Tree**. Your new program sits in this directory.
2. Right click on the program under the **Project Tree** and select **Rename**. Type "Automate Code."
3. Confirm results (see Figure 6.19).

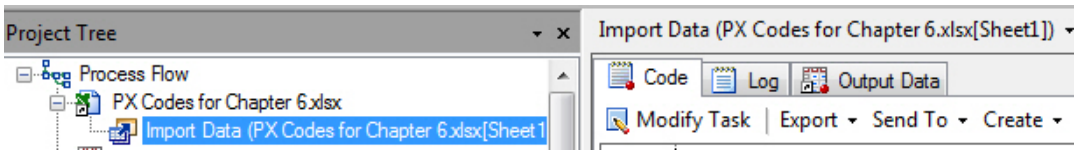
Figure 6.19: The program directory in Project Tree



To extract code, follow the steps below.

1. In **Project Tree**, double click on the task called **Import Data (PX Codes for Chapter 6.xlsx[Sheet1])**.
2. Copy code from Code window, which is shown in Figure 6.20.

Figure 6.20: Code Window



3. Paste code in Automate Code program.
4. Repeat steps 1–3 for all Enterprise Guide tasks created in the previous section.

Import Data Code

The **Import Data** task in Enterprise Guide is based on the data step. Program 6.1 shows this code.

Program 6.1: Import Task Code

```
data work.px_codes_for_chapter_6;                    ❶
    length   px   $ 5 ;                             ❷
    format   px   $char5. ;                          ❸
    informat px   $char5. ;
    infile datalines4 dlm="7f"x missover dsd;        ❹
    input px: $char5. ;                              ❺
datalines4;                                          ❻
99201
99202
… (Data Lines Not Shown)
99429
;;;;                                                 ❻
```

❶ Creates a data set called PX_Codes_for_Chapter_6 and places it in the temporary WORK library. The statement ends with a semicolon. This is a SAS coding convention.

❷ The LENGTH statement specifies the length of PX.

❸ The FORMAT statement specifies how PX should be displayed. The INFORMAT statement tells SAS how PX should be read.

❹ The INFILE statement specifies the source of the data as instream (DATALINES4) and other relevant options such as delimiter (DLM).

❺ The INPUT statement lists the columns to be read.

❻ The DATALINES4 statement marks the beginning of instream data. Four semicolons mark the end of instream data. The DATALINES4 statement is an enhanced version of the DATALINES statement, as it allows semicolons to be placed in the instream data.

The SAS programming language has two main components, the DATA step and the PROC step. You write the DATA step to read, modify, combine, and manage data, incorporating complex programming constructs such as arrays, loops, and hash tables. You use the PROC step for a specific data processing purpose, such as regression analysis. Also known as a procedure, the PROC step is a pre-programmed routine.

Query Builder Code

Query Builder tasks in Enterprise Guide are based on PROC SQL, the SAS implementation of the database query language SQL. PROC SQL code downloaded from remaining Enterprise Guide tasks is shown in Program 6.2. SAS code comments, which begin with * and end with a semicolon, have been added to each query.

Program 6.2: Code Downloaded from Query Builder Tasks

```
* Query 1: Identify currently enrolled children under 6 years of age;
proc sql;
   create table work.currentmemberskids as
   select distinct t1.MemberID, t1.BeginEnroll, t1.EndEnroll,
          t1.DOB, t1.Sex, (round(('01DEC2011'd-DOB)/365.25,.01)) as Age
   from BOOK.MEMBERS t1
   where t1.EndEnroll>='01DEC2011'd and (CALCULATED Age) <= 6;
quit;

* Query 2: Identify appropriate professional visits;
proc sql;
   create table WORK.PROFESSIONALVISITS as
   select t2.MemberID, t1.ClaimID, t1.ClaimLine, t1.BeginDOS,
          t1.EndDOS, t1.PointerDx1, t1.PointerDx1Desc,
          t1.PointerDx2, t1.PointerDx2Desc,
          t1.PointerDx3, t1.PointerDx3Desc,
          t1.PointerDx4, t1.PointerDx4Desc,
          t1.Px, t1.PXDesc, t1.SrvcProvID
   from BOOK.PROFESSIONALHEADER t2,
        BOOK.PROFESSIONALDETAIL t1,  WORK.PX_CODES_FOR_CHAPTER_6 t3
   where (t2.ClaimID = t1.ClaimID and t1.Px = t3.PX) and
   (t1.POS="11" and t1.SrvcProvTaxonomy in ("207Q00000X",
    "207R00000X","208D00000X","208000000X"));
quit;

* Query 3: Create Outreach report;
proc sql;
   create table WORK.KidsOutreach as
   select t1.MemberID, t1.DOB, t1.Sex
   from WORK.CURRENTMEMBERSKIDS t1
       left join WORK.PROFESSIONALVISITS t2
             on (t1.MemberID = t2.MemberID AND (t2.BeginDOS=.));
quit;

* Query 4: Create Internal report;
proc sql;
   create table WORK.KidsVisits as
   select t1.MemberID, t2.DOB, t2.Age,
         (MAX(t1.BeginDOS)) FORMAT=YYMMDD10. as MAX_of_BeginDOS
   from WORK.PROFESSIONALVISITS t1
        inner join WORK.CURRENTMEMBERSKIDS t2
   on (t1.MemberID = t2.MemberID)
   group by t1.MemberID,t2.DOB,t2.Age;
quit;

* Query 5: Enhance Internal report with diagnosis and other information;
proc sql;
   create table WORK.KidsVisitsDiag as
   select t1.MemberID, t1.DOB, t1.Age, t2.BeginDOS, t2.SrvcProvID,
          t2.PointerDx1, t2.PointerDx1Desc,
          t2.PointerDx2, t2.PointerDx2Desc,
          t2.PointerDx3, t2.PointerDx3Desc,
          t2.PointerDx4, t2.PointerDx4Desc
   from WORK.KIDSVISITS t1, WORK.PROFESSIONALVISITS t2
   where (t1.MemberID = t2.MemberID AND
```

```
          t1.MAX_of_BeginDOS = t2.BeginDOS);
quit;
```

Some key notes about Program 6.2.

- PROC SQL invokes the SQL procedure. A QUIT statement terminates this procedure.
- The CREATE table statement creates a SAS data set.
- The SELECT statement selects existing columns and creates new columns. For example, Query 1 creates a unique listing of members, their dates of enrollment, and other demographic data. The query creates a computed column called Age by supplying an algebraic formula.
- The FROM statement specifies the source data for the query. Each table in the statement has an alias that is used to refer to the table in the Select statement. For example, the alias for BOOK.Members in Query 1 is t1.
- The WHERE statement filters records. For example, Query 1 filters out members whose enrollment has ended prior to December 1, 2011. The CALCULATED keyword allows the WHERE statement to reuse the computation for AGE specified in the SELECT statement.
- The GROUP BY statement is used to summarize data with respect to an aggregate function in the SELECT statement. Query 4 aggregates on the basis of the MAX function. It retrieves the maximum BeginDOS for the variables in the group (MemberID, DOB, and Age).

Automate Code

To automate code, you need to replace the hard-coded date of December 1, 2011, in the following places:

- Age computation in the SELECT statement in Query 1, Program 6.2.
- WHERE statement in Query 1, Program 6.2.
- Specify a dynamic twelve-month look-back period in the WHERE statement of Query 2, Program 6.2.

This dynamic replacement is done with the help of two macro variables, EndDOS and StartDOS.

> The SAS macro language allows users to substitute text strings that will be compiled into executable code. Its advanced usage includes iterative and conditional code generation. Its syntax is similar to that of SAS programming language. Many SAS functions have equivalents in the macro language.
>
> The basic building block of SAS macro language is the macro variable, which inserts dynamically changing code into existing SAS code. There are two kinds of macro variables, SAS supplied and user defined. In this case study, you define two macro variables.

The %LET statement initializes or changes the value of a macro variable. Since the macro language is just a text handling facility, the "date value" supplied in the following statement is treated as simply a text string with no inherent meaning until compiled by the macro facility.

```
%let enddos=01DEC2011;
```

Query 1 can be rewritten with &EndDOS, eliminating the need for a hard-coded date. The ampersand is needed for a macro variable to resolve. Also, note the use of double quotes where single quotes were seen in the hard-coded example above. Macro variables will only resolve within double quotes and *not* within single quotes.

```
proc sql;
   create table WORK.CurrentMembersKids as
   select distinct t1.MemberID, t1.BeginEnroll, t1.EndEnroll,
          t1.DOB, t1.Sex, (round(("&EndDOS"d-DOB)/365.25,.01)) as Age
```

```
      from BOOK.MEMBERS t1
      where t1.EndEnroll>="&EndDOS"d and (CALCULATED Age) <= 6;
quit;
```

When the query runs, the value of EndDOS is replaced with the string 01DEC2011. To supply Query 1 with a different value of EndDOS, all you need to do is change its value in the %LET statement.

To make Query 2 dynamic a second macro variable called StartDOS is created by the following %LET statement.

```
%let startdos = %sysfunc(intnx(month,"&EndDOS"d,-12),date9.);
```

The %LET statement uses two functions.

1. The INTNX function is a SAS function that advances a date value by a time interval. The expression intnx(month,"01DEC2011"d,-12) results in a date value of 01DEC2010. This value is stored as a number, not a calendar date.
2. The INTNX function is a SAS data step function, not a SAS macro function. However, the macro function %SYSFUNC enables you to use SAS data step functions within macro code. The function's second argument simply specifies the format. The format translates the number value of date to a calendar date.

4.

Query 2 is rewritten below with StartDOS and EndDOS variables.

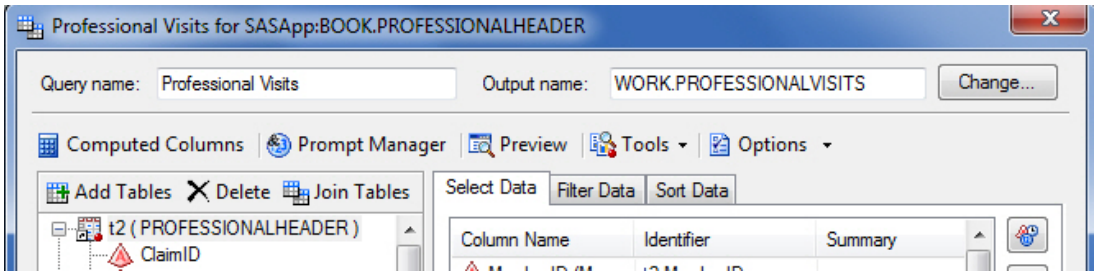
```
proc sql;
   create table WORK.PROFESSIONALVISITS as
   select t2.MemberID, t1.ClaimID, t1.ClaimLine, t1.BeginDOS,
          t1.EndDOS, t1.PointerDx1, t1.PointerDx1Desc,
          t1.PointerDx2, t1.PointerDx2Desc,
          t1.PointerDx3, t1.PointerDx3Desc,
          t1.PointerDx4, t1.PointerDx4Desc,
          t1.Px, t1.PXDesc, t1.SrvcProvID
    from BOOK.PROFESSIONALHEADER      t2,
         BOOK.PROFESSIONALDETAIL      t1,
         WORK.PX_CODES_FOR_CHAPTER_6  t3
   where (t2.ClaimID = t1.ClaimID and t1.Px = t3.PX)
     and t1.POS="11"
     and t1.SrvcProvTaxonomy in
                 ("207Q00000X","207R00000X","208D00000X","208000000X")
     and t1.BeginDOS between "&StartDOS"d and "&EndDOS"d;
quit;
```

To run the code for any given day, you need only make a single change in the %LET statement initializing the value of EndDOS.

SAS Enterprise Guide also allows you to create and use macro variables in its tasks. The advantage of this approach is that you need not convert your project tasks to code before automation. To demonstrate this technique, let us parameterize the **Professional Visits Query** so that the user is prompted for values of EndDOS and StartDOS.

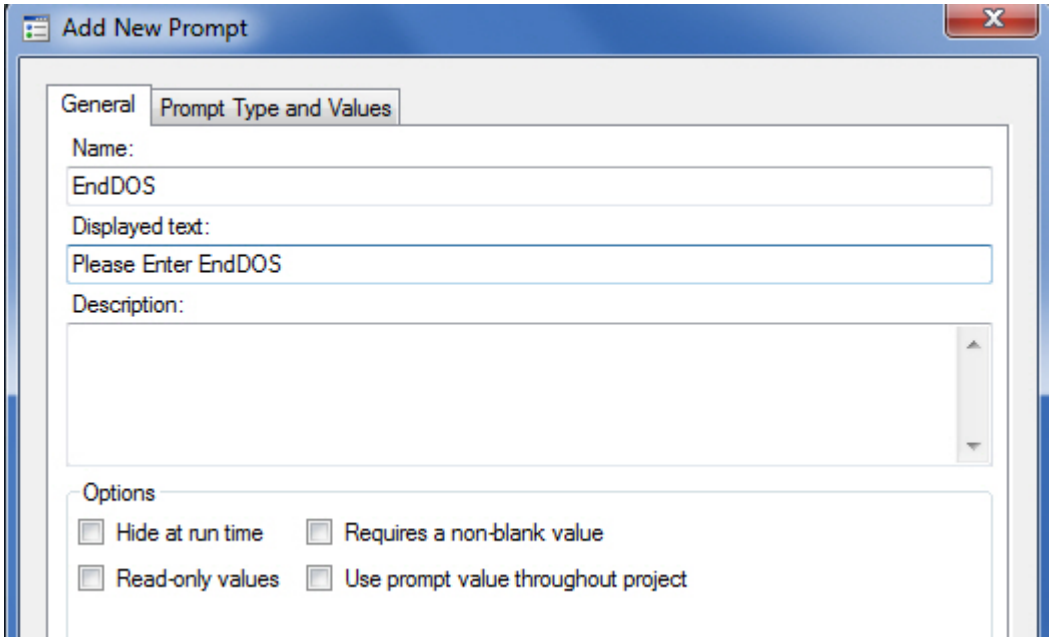
1. Right click on **Professional Visits** under **Process Flow** and select **Modify Professional Visits**.
2. To create macro variables in a Query, click **Prompt Manager** (see Figure 6.21).

Figure 6.21: Prompt Manager in Query Builder



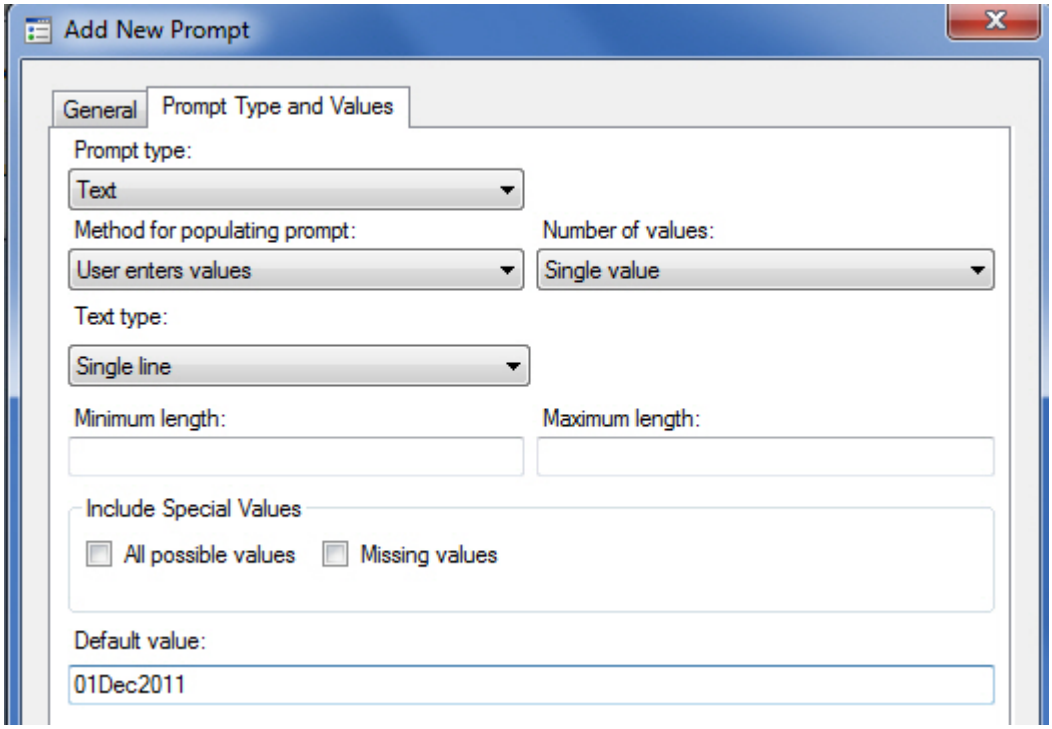
3. In **Prompt Manager** window, click **Add** to create a new prompt. A prompt creates a macro variable whose value can be chosen at run time.
4. In the **General** tab of **Add New Prompt**, enter values for **Name** and **Displayed** text as shown by Figure 6.22.

Figure 6.22: Naming a new prompt



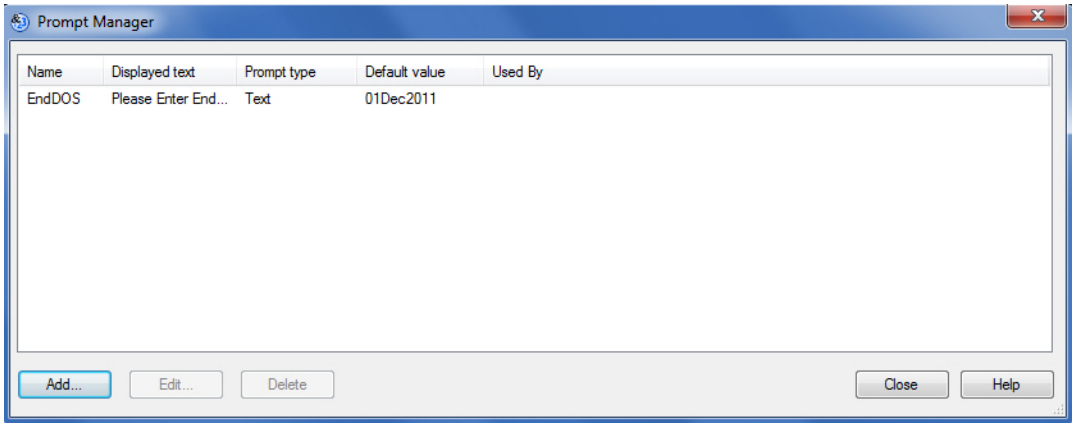
5. In the **Prompt Type and Values** tab of **Add New Prompt**, enter value for **Default value** as shown in Figure 6.23.

Figure 6.23: Creating a new prompt



6. Verify that the prompt EndDOS is created (see Figure 6.24).

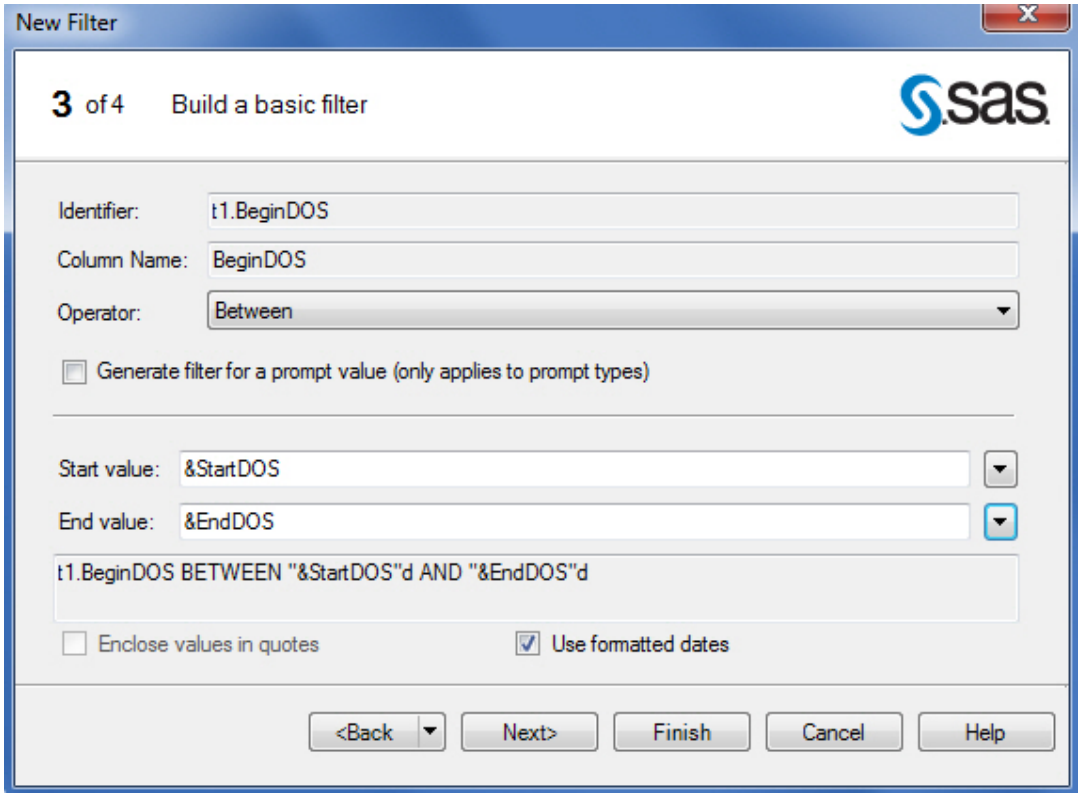
Figure 6.24: Prompt EndDOS



7. Repeat steps 3–5 for **StartDOS** prompt, setting the **Default value** as 01Dec2011.
8. In the **Professional Visits Query**, click the **Filter** tab to create a filter on **BeginDOS**.
9. In Step 1 of 4 of New Filter, select **Basic Filter**.
10. In Step 2 of 4 of New Filter, select **BeginDOS**.

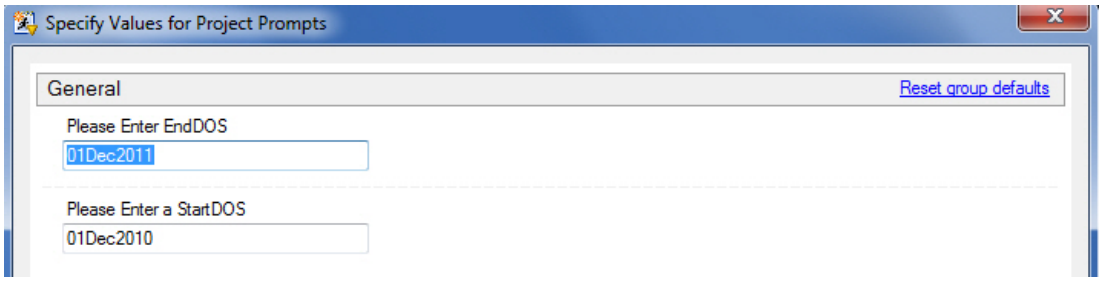
11. As shown in Figure 6.25, select **Between** in **Operator** dropdown. In the **Start** and **End value** dropdowns, select or enter **&StartDOS** and **&EndDOS**, respectively. As the filter is built on a date column, SAS Enterprise Guide automatically encloses user-entered values in double quotes followed by d.

Figure 6.25: Adding prompts in a filter



12. Click **Run** in **Professional Visits Query** to run the query.
13. In **Specify Values for Project Prompts**, click **Run** to accept default prompts (see Figure 6.26).

Figure 6.26: Specifying prompt values



The key advantage to creating prompts in SAS Enterprise Guide is that you do not have to download code and then modify it. However, a drawback to the approach is that each macro variable is independent of others. StartDOS and EndDOS are both independent prompts that the user has to select; the Prompt Manager is unable to create StartDOS as a function of EndDOS and business logic. Two prompts are created as opposed to one macro variable created with SAS code.

Conclusion

This chapter explained the claims submission mechanism used by professional and ancillary providers. It covered some useful data elements such as POS and Diagnosis Pointer, which are unique to the CMS-1500 claim form and its electronic analog 837I. The chapter showed you how to effectively use professional claims data to generate outreach reports. It also demonstrated how SAS code may be downloaded from Enterprise Guide and parameterized for automation.

Chapter 7: Pharmacy Claims

Introduction 83
NCPDP Claim Formats 84
Paper Claim Form 84
Electronic Format 84
Data Elements Unique to Pharmacy Data 86
Provider Identifiers 86
National Drug Code 86
Generic Product Identifier 88
Therapeutic Class Codes 88
Other Fields of Interest 89
Case Study: Computing Medication Adherence 90
PDC Computation 90
Data Expansion Using SAS Data Step 91
Create Study Period Data Set 92
Create Days Covered Data Set 94
Combine and Summarize Data 94
Graphing PDC 96
Automating PDC Graphs Generation 98
Conclusion 101

Introduction

Pharmacists, as a class of service providers, play a significant role in the healthcare industry. They provide their services in two types of settings, stand-alone pharmacies—like Rite-Aid or Walgreens—or institutions such as hospitals. It is important to distinguish between drug claims from these two different settings. Claims from stand-alone pharmacies are the source of our warehoused prescription drug claims. When an institutional patient (e.g., inpatient, outpatient, skilled nursing facility [SNF]) receives a dispensed pharmaceutical it usually comes from the facility's formulary. The source for this claim service line in our warehoused data is a facility claim, that is, CMS-1450 or 837I. The billing mechanism is typically the Revenue Code, as discussed in Chapter 5. Additional information on the service line record about the dispensed drug may be provided as a Healthcare Common Procedure Coding System (HCPCS) code or a National Drug Code (NDC).

This chapter focuses on drug claims from stand-alone pharmacies.

In this chapter you will:

- Understand the claim submission mechanism used by stand-alone pharmacies.
- Understand the difference between pharmacy submission and drugs billed at institutions.
- Learn about the unique content of this prescription drug data source.
- Calculate medication adherence for Healthy Living, Inc.'s members using data expansion routines in SAS Enterprise Guide.
- Analyze, graph, and automate medication noncompliance in SAS Enterprise Guide.

NCPDP Claim Formats

The National Council for Prescription Drug Programs (NCPDP [http://ww.ncpdp.org]) is a not-for-profit, multistakeholder organization that develops industry standards for pharmacy services. Its primary focus is to create standards to facilitate the electronic exchange of information.

As a class of healthcare providers, pharmacists were among the first to adopt electronic transmission methodologies. While a paper claim form is available and maintained for currency, the majority of pharmacists use the electronic standard mandated by the Health Insurance Portability & Accountability Act of 1996 (HIPAA).

Table 7.1 lists documentation that is available to NCPDP members in the "Members" section of their website; nonmembers may purchase the documents.

Table 7.1: NCPDP documentation

Document	Description
Manual Claim Forms Reference Implementation Guide	Information for completing and processing the NCPDP manual claim forms
Telecommunication Standard Implementation Guide	Specifies transmission formats for claim submission and response
Data Dictionary	Full reference to all fields used in NCPDP standards
External Code List	Full reference to all values used in NCPDP standards
Billing Unit Standard	Standardized billing units used for claim submission

Paper Claim Form

Due to changes required for Medicare Part D processing of claims with dates of service after February 27, 2013, new versions of the NCPDP Telecommunication vD.0 Continuous Feed (PUCF-D02PT) and Laser Universal Claim Form (PUCF-D01PT) have been created. While limitations exist, the intent of the recent updated versions is that all fields on the Universal Claim Form align with the field format and definitions of fields defined in the Telecommunication Standard Version D.Ø (the electronic format).

A facsimile of the Universal Claim Form can be found in Figures A.3a and A.3b in the Appendix. These materials are reproduced with the consent of ©National Council for Prescription Drug Programs, Inc., 2013 NCPDP, and are not for distribution or use.

Electronic Format

HIPAA requires that a single standard be used for the electronic submission of retail pharmacy claims—the NCPDP format. Since pharmacy transactions are considered "Point of Sale" data it has been relatively simple for pharmacists to receive a prescription across the counter, check a patient's benefit eligibility, and fill the script. The electronic claim is submitted to the payer before the patient leaves the pharmacy.

All payers will have an internal document known as a "payer sheet" that documents their expectations of pharmacies for complete and accurate claim submissions. It is incumbent upon the providers to follow the instructions of the payer to which they submit claims.

Detailed information about the transactions, segments, fields, rules, and syntax is found in the Telecommunication Standard Implementation Guide. While there is much more information for a programmer/developer contained in the guide, for purposes of this book, note that the Transaction Header Segment in the request and the Response Header Segment in the response are fixed-length segments containing fixed-length fields. The rest of the segments in the request or response (such as Patient Segment, Insurance

Segment, Claim Segment, Response Status Segment, Response Claim Segment, etc.) are variable segments with variable fields (where applicable) and variable field lengths.

Table 7.2 is provided to give the reader a sense of the data being passed from the pharmacist to the payer organization as a Claim Request transaction; other transaction types exist, such as a Claim Reversal. The Key Fields column is not meant to be an exhaustive list of data elements.

Table 7.2: Transaction segments on a pharmacy claim

Segment identifier	Segment name	Description	Key fields
00	Transaction Header	Required—an initiation record that denotes the transaction type	Transaction Code, Provider ID (NPI), Date of Service
AM01	Patient	Required—recipient-level information	Patient Demographics, Place of Service
AM02	Pharmacy Provider	Optional—provider-level information also on the Header segment	Provider ID (NPI)
AM03	Prescriber	Required—physician-level information	Prescriber ID (NPI), Prescriber Demographics
AM04	Insurance	Required—payer-level information	Member ID, Benefit Plan ID
AM05	COB/Other Payments	Required—if other payer coverage exists	Other Payer ID, Other Payer Coverage Type
AM06	Workers' Compensation	Required—if claim is related to a workplace injury	Employer Demographics, Date of Injury, Carrier ID
AM07	Claim	Required—dispensed pharmaceutical information	National Drug Code (NDC), Quantity, Fill Number, Days Supply, Number of Refills, Date Rx Written,
AM08	DUR/PPS	Optional—if Drug Utilization Review is performed the pharmacist may submit findings	
AM09	Coupon	Optional—for use when a coupon is used to reduce the patient out-of-pocket cost	Coupon Type, Number, and Amount
AM10	Compound	Required—when claim is for a compounded drug	NDC, Quantity, Ingredient Cost
AM11	Pricing	Required—for entry of any requested reimbursement	Ingredient Cost, Dispensing Fee, Other Amounts
AM12	Prior Authorization	Required—if member benefit plan so stipulates prior authorization	

Segment identifier	Segment name	Description	Key fields
AM13	Clinical	Optional—for reporting of diagnosis codes if available	Diagnosis Code(s)
AM14	Additional Documentation	Optional—free-form text entry	
AM15	Facility	Required—when the place of service is a nursing facility	Facility ID, Facility Demographics
AM16	Narrative	Optional—free-form text entry	Narrative Message
99	Trailer	Required—a termination record for the transaction	

Data Elements Unique to Pharmacy Data

As with other provider types, pharmacy claims contain data elements that are distinctive to their provided services. This not only includes the submitted data, but also data that is derived by the adjudication and warehouse acquisition processes. In this section we will highlight a few of these data elements.

Provider Identifiers

The NCPDP Provider Identification number (NCPDP Provider ID), formerly known as the NABP number, was developed to provide pharmacies with a nationally unique identifier that would assist them in their interactions with payers. The NCPDP Provider ID is a seven-digit number that is assigned to every licensed pharmacy and qualified Non-Pharmacy Dispensing Site (NPDS) in the United States.

> Founded in 1904, the National Association of Boards of Pharmacy (NABP [http://www.nabp.net]) is an impartial professional organization that supports the state boards of pharmacy in protecting public health.

Even though pharmacies now use the National Provider Identifier (NPI) to bill insurance carriers, it is critical that pharmacies maintain their NCPDP Provider ID. This unique identifier is used by many payers and third-party contracting entities to categorize pharmacies and crosswalk to the NPI.

NCPDP expects Workers' Compensation, and other programs not covered by HIPAA legislation, to continue to use the NCPDP Provider ID for some time into the future. It is expected that many payers will crosswalk the NPI to the NCPDP Provider ID for processing in the near-to-intermediate term. The relationship and demographic information in the NCPDP database is needed more than ever by the industry. NCPDP will continue issuing NCPDP Provider ID numbers—even if the only future use is internal to NCPDP and their database users. There are no known plans to phase out the numbers.

Also of interest to the analyst accessing prescription drug claims data from a post-adjudication data store is the notion that several NPI fields will most likely be available. Hopefully the field names will distinguish between the prescribing provider and dispensing provider.

National Drug Code

The Drug Listing Act of 1972 requires "labelers" (typically pharmaceutical companies) to provide the Food and Drug Administration (FDA) with a list of all drugs manufactured for commercial distribution. Drugs for human use are to be identified using a unique ten-digit, three-segment identifier known as the National Drug Code (NDC).

According to the FDA, the NDC will be in one of the following configurations: 4-4-2, 5-3-2, or 5-4-1, denoting the number of possible digits to be found in each of the three segments; the segments are separated by dashes. Regardless of length or configuration, the NDC is a three-segment number. The built-in intelligence is defined as follows:

- The first segment, the **labeler code**, is assigned by the FDA. A labeler is any firm that manufactures or distributes the drug under its own name.
- The second segment, the **product code**, identifies a specific strength, dosage form, and formulation of a drug. It is assigned by the labeler.
- The third segment, the **package code**, identifies package sizes/types and is assigned by the labeler.

Pharmacies are required to transmit the NDC in the NCPDP standard format for identification of prescription drugs dispensed through a retail pharmacy. The NDC replaces the HCPCS codes for retail pharmacy drugs that were previously used.

You may find NDC codes from facility claims in your post-adjudicated claim source. This may happen because the 837I submission format allows the submission of NDC in situations where payer regulation mandates that prescribed drugs and biologics are reported with NDC numbers or when the provider chooses to report NDC numbers to enhance the claim reporting or adjudication processes. This nuance of the reporting mechanism may cause your data store not to carry the element or to overwrite the associated billable procedure code and carry it in that field if it was used for payment. This is yet another example of why we, as analysts, must be vigilant and understanding of operational distinctions.

The FDA maintains the National Drug Code Directory and updates it frequently. Detailed information on the directory, including definitions, table lookups, and file downloads, can be found at www.fda.gov/Drugs/InformationOnDrugs/ucm142438.htm.

While the NDC is a ten-digit identifier (based on the 1972 legislation), confusion exists because of a proliferation of different notations and variants. Many of these variants have been discarded over time and should not cause today's analysts any problem. The one that does still exist bears mention (and concern). CMS had created an eleven-digit NDC derivative, which pads the labeler, product, or package code segments of the NDC with leading zeroes where needed to result in a fixed-length, 5-4-2 configuration written without dashes. This format was adopted as the data standard pursuant to HIPAA regulation, thus other government and commercial databases may contain the eleven-digit form. The eleven-digit NDC format cannot be transformed back into the ten-digit standard format without the help of a product listing database. See Table 7.3 for a few example NDCs;

Table 7.3: Example NDCs

NDC	Labeler segment		Product segment		Package segment	
00054003621	00054	Roxane Laboratories, Inc.	0036	Clarithromycin—250 mg tablet	21	60-tablet bottle
00025152031	00025	Pfizer US Pharmaceutical	1520	Celebrex—100 mg capsule	31	100-capsule bottle
24090047388	24090	Akrimax Pharmaceutical	0473	Inderal LA—120 mg capsule	88	100-capsule bottle

Of all the industry standard code sets (more will be discussed in Chapter 8), NDCs require the most frequent update. The most widely used commercial services are provided by Medi-Span (www.medispan.com), First DataBank (www.fdbhealth.com), and Red Book (www.redbook.com).

Generic Product Identifier

The Medi-Span proprietary Generic Product Identifier (GPI) is a unique concept that identifies drug products by a hierarchical therapeutic classification structure. The fourteen-character GPI defines pharmaceutically equivalent drug products having the same active ingredients, dosage form, strength, route of administration, and therapeutic use. Drugs that have the same GPI are identical with respect to their active ingredients, dosage form, administration route, and strength.

The code consists of seven two-character "couplets," each providing increasingly more specific information about a drug. The first ten characters of the GPI define the therapeutic class code and the last four digits define route, dosage, or strength. Table 7.4 shows the breakdown for GPI 58200060100105: Nortriptyline, an antidepressant.

Table 7.4: Example GPI

Couplet	Value	Definition
Drug group	58	Antidepressants
Drug class	58-20	Tricyclic agents
Drug subclass	58-20-00	[none]
Drug name	58-20-00-60	Nortiptyline
Drug name extension	58-20-00-60-10	Nortiptyline Hydochloride
Dosage form	58-20-00-60-10-01	Capsule
Strength	58-20-00-60-10-01-05	10mg

This structure allows for the creation of complex drug lists with many levels of granularity The GPI's level of specificity makes it ideal for reviewing and screening data in a clinical setting. For this reason many organizations will provide this value-add in their data warehouses.

While extensively used and with a proven track record, Medi-Span's GPI is not the only available proprietary categorical identifier. First DataBank has the Generic Sequence Number (GSN), Hierarchal Ingredient Code (HIC), and Enhanced Therapeutic Classification System (ETC).

Therapeutic Class Codes

Classifying prescription drugs according similar pharmacologic, therapeutic, and/or chemical characteristics is often another value-add for adjudicated pharmacy claims.

The American Hospital Formulary Service (AHFS [http://www.ahfsdruginformation.com]) Pharmacologic-Therapeutic Classification has been the foundation for organizing drug formularies in institutional, governmental, and other settings since 1959. Booz Allen Hamilton (http://www.boozallen.com/) recently undertook a study of therapeutic classification systems and found that it is a "recognized standard for systemic classification of drugs that uses a coding system to organize drugs" and that it is "a key reference and starting point for formulary development because it is readily available and universally accepted by physicians."

The AHFS classification allows the grouping of drugs with similar pharmacologic, therapeutic, and/or chemical characteristics in a four-tier hierarchy. There are 31 classifications in the first tier, 187 in the second tier, 263 in the third tier, and 97 in the fourth tier. The code consists of four two-digit "couplets," each providing

increasingly more specific information about a drug category. The colon separator may or may not be stored with the data element. Table 7.5 shows the breakdown for a serotonin modulator such as Trazodone.

Table 7.5: Four-tier structure of AHFS therapeutic class codes

Couplet	Value	Definition
First tier	28:00:00:00	Central Nervous System Agents
Second tier	28:16:00:00	Psychotherapeutic Agents
Third tier	28:16:04:00	Antidepressants
Fourth tier	28:16:04:24	Serotonin Modulators

The use of such information is readily apparent to the programmer/analyst wishing to filter data to a specific level of categorization. The use of a substring function in a WHERE statement will yield quick results against large volumes of data.

Other Fields of Interest

Other data fields not found in pharmacy claims that bear mention are described in Table 7.6 and should be of interest to the astute programmer/analyst.

Table 7.6: Other data of interest

Variable	Definition/Comment
Days Supply	Estimated number of days the prescription will last.
Quantity Dispensed	Amount doled out, expressed in metric decimal units. In conjunction with the Days Supply field the quantity per day can be imputed.
Number of Refills Authorized	Number of refills authorized by the prescriber.
Fill Number	Indication of whether prescription is an original (00) or a refill (01–99).
Date of Service	The date the prescription was filled. This is not to be confused with the date of service (e.g., an office visit) on which the prescription was written.
Date Prescription Written	Date prescription was written. This date may not coincide with any other service (e.g., an ER visit) in the case where a prescription may have been phoned in.

Also of interest is that the post-adjudicated claim will have many monetary fields related to the dispensation of the prescription. By contract with the payer, the pharmacy may receive Dispensing and Packaging Fees in addition to the Drug Cost or the sum of the Ingredient Costs in the case of a compounded drug. These are in addition to the standard Total Submitted Charge, Total Paid Amount, and any patient liability (copayment, coinsurance, deductible, etc.). If there is a financial component to your project, be sure to consider all, and more, of these possibilities.

It is common to see negative claim dollar amounts in pharmacy data. This is most often due to the federal "pull-back rule," which states that if a prescription has not been picked up within seven days of dispensation

(prescription abandonment), then the pharmacist must pull the prescription back and file a "negative" claim to back out the first. This problem is generally larger than assumed, and much research has been conducted, from the patient's perspective, into why this happens. The analyst must devise a method to capture the net paid claims and filter out the data noise caused by this rule.

Case Study: Computing Medication Adherence

The Pharmacy Services Department at Healthy Living, Inc., monitors medication usage of the company's insured members. The primary reason they do is to ensure that members adhere to their medication regimes. Proper medication compliance leads to better management of chronic conditions and may even lower ER/inpatient utilization. Other reasons for tracking medication usage include preventing abuse of certain drugs such as opiates and simplifying medication regimes of members with multiple complex conditions.

This case study shows you how to calculate medication adherence using Proportion of Days Covered (PDC) methodology. The PDC technique is an industry standard. CMS uses it to compute its medication adherence–related performance measures. CMS uses these and other measures to rate Medicare insurance plans. The Pharmacy Quality Alliance (PQA [http://www.pqaalliance.org]), a large nonprofit alliance whose goal is to promote appropriate and safe medication use, has also endorsed the PDC methodology.

Let us begin by examining the Pharmacy table.

Figure 7.1: Select columns from the Pharmacy data set

MemberID	NDC	label	AhfsClass Code	gpi14	Pharmacy Name	qty	filldate
M000007425001...	00093550101	BUDEPRION TAB 100MG SR	28160492	58300040107420	Pharmacy C	30	2011-04-04
M000007425001...	00093550101	BUDEPRION TAB 100MG SR	28160492	58300040107420	Pharmacy C	30	2011-05-04
M000007425001...	00093550101	BUDEPRION TAB 100MG SR	28160492	58300040107420	Pharmacy C	30	2011-06-03
M000007425001...	00093550101	BUDEPRION TAB 100MG SR	28160492	58300040107420	Pharmacy C	30	2011-07-03
M000007425001...	00093550101	BUDEPRION TAB 100MG SR	28160492	58300040107420	Pharmacy C	30	2011-08-02
M000007425001...	00093550101	BUDEPRION TAB 100MG SR	28160492	58300040107420	Pharmacy C	30	2011-09-01

The Pharmacy table is transactional. Figure 7.1 shows six transactions or medication fills by a member. Each transaction has information about the medication, including where it was filled (Pharmacy Name), when it was filled (FillDate), and days supply (Qty).

While this case study focuses on PDC, other analyses with pharmacy data are no less important. SAS programming techniques for a number of relevant analyses have already been covered in previous chapters.

To identify members who may be abusing opiates, first use GPI and/or American Hospital Formulary Service (AHFS) codes to identify opiates. Store these codes in Excel and import them as needed (see Chapter 6). Next, summarize the data to compute fills by member. This has been shown in Chapters 5 and 6. Finally, graph this data to isolate members with the highest utilization, as shown in Chapter 5.

To identify members with an overly complex medication regimen, count the number of unique prescriptions for a member. The aggregation may be done using NDC codes or varying levels of GPI/AHFS codes.

PDC Computation

The PDC calculation is a ratio of two numbers. The numerator is the number of days a member had a supply of a medication in a study period. This is based on fill dates and days supply associated with each fill. The denominator is the number of days between the first fill date of the medication and the end of the measurement period. This case study assumes that December 1, 2011, marks the end of the study period.

The following steps lay out a technical roadmap:

1. Create data for the study period. Expand data in the Pharmacy table using the first fill date of a medication and the end date of the study period. The resultant data set will contain a row for each day between a medication's first fill date to December 1, 2011, the end date of the study. This data is used to create the denominator of PDC.
2. Create data for the days covered calculation. Expand the Pharmacy table, this time using FillDate and Qty to create data at a member-medication-day level. Replace FillDate with a date column that tracks each day for which there is supply of the medication. This data is used to create the numerator of PDC.
3. Combine data sets from steps (1) and (2).
4. Summarize and create the PDC ratio for each member-medication combination.

Data Expansion Using SAS Data Step

In this section you learn how to expand data using a SAS DATA step. It is important to learn this technique because data expansion has numerous applications in healthcare data management. You cannot expand data using Query Builder or PROC SQL.

You can use a DATA step to read, modify, combine, and manage data, incorporating complex conditional logic. You can also include a loop within the DATA step to create multiple observations from a single observation.

Before attempting to understand the code that expands data in the Pharmacy table, let us first understand a simple DATA step with a DO loop. Program 7.1 shows you how to create four observations from a single row of data.

Program 7.1: Expanding data with DO loops

```
* Create one row of data;
data test;                          ❶
    x=1;
run;

* Expand Test;
data ExpandTest;
    set test;                       ❷
    do i=1 to 4;                    ❸
      /* additional SAS Code here */
      output;                       ❹
    end;                            ❺
run;
```

❶ A SAS DATA step begins with the keyword DATA and ends with the keyword RUN. The name of the data set being created follows the keyword DATA. The data set "test" contains one observation and one column, x, which is programmatically given the value of 1.

❷ The SET statement assigns source data, test, to the new data set, ExpandTest.

❸ A loop begins. In its first iteration, i is initialized to 1. Any SAS code within the loop is processed.

❹ The OUTPUT statement causes the current instance of the single observation in test to be placed in ExpandTest.

❺ The loop's counter, i, is incremented, and the loop iterates again, repeating steps ❸ and ❹. This occurs until i reaches the value of 5 and the loop stops running.

Figure 7.2 shows that a single observation is expanded to four observations. The loop's counter variable, i, is also shown.

Figure 7.2: ExpandTest Data Set

	x	i
1	1	1
2	1	2
3	1	3
4	1	4

Create Study Period Data Set

Program 7.2 shows code to create the study period data set.

Program 7.2: Create the study period

```
* Compile complete study period;

proc sort data=book.Pharmacy
          out=StudyPeriod(keep=MemberID NDC Label FillDate);    ❶
    by MemberID NDC Label FillDate;                             ❷
run;

proc sort data=StudyPeriod nodupkey;                            ❸
    by MemberID NDC Label;
run;

data StudyPeriod2(drop=FillDate);
    set StudyPeriod;
    Interval="01DEC2011"d - FillDate;                           ❹
    do i=1 to Interval + 1;                                     ❺
      Date=intnx("day",FillDate,i-1);                           ❻
      format Date yymmdd10.;                                    ❼
      output;                                                   ❽
    end;
run;
```

❶ PROC SORT is commonly used in SAS programming to sort data and to remove duplicates. The procedure may also be used to create a sorted version of the source data. The DATA= option specifies the source data set, BOOK.Pharmacy. The OUT= option names the resultant, sorted data set, StudyPeriod. The KEEP= data set option in parentheses ensures that the data set StudyPeriod contains just the four columns of interest.

❷ The BY statement lists the columns with which the data is to be sorted. Unless specified, data is sorted in an ascending manner. Each member-medication set of rows is sorted so that the earliest FillDate appears first.

❸ The NODUPKEY option removes duplicate rows with respect to the columns listed in the BY statement, the "key" variables. Only the first row in a duplicative set of rows is kept. This results in a data set that has the first fill date for every member-medication combination as shown in Figure 7.3. As no OUT= option is specified, the data set StudyPeriod is replaced.

Figure 7.3: StudyPeriod data set

MemberID	NDC	label	filldate
M000007425001...	00093550101	BUDEPRION T...	2011-04-04
M000007425001...	00185041005	BUPROPION T...	2011-05-04
M000010632001...	00093550101	BUDEPRION T...	2011-03-05
M000010632001...	00185041005	BUPROPION T...	2011-04-04
M000010632001...	00378021810	ATENOLOL T...	2011-06-03
M000010632001...	59310057920	PROAIR HFA A...	2011-03-05

❹ SAS processes one observation at a time, holding all manipulations in memory until it is ready to place the observation in the output data set. The column Interval is created to capture the number of times each observation in StudyPeriod must be expanded.

❺ The DO loop is set up to run "one plus the value calculated by interval" times.

❻ The INTNX function increments a date value by an interval specified in its first argument. As the first argument of the INTNX function in this case is days, the function adds days to FillDate and places the value in a new column called Date. For the first iteration of the loop, zero days are added to the FillDate. This makes sense, as the first date in every member-medication combination should be the FillDate.

❼ The FORMAT statement defines the format attribute as yymmdd10. so that you can visually verify results.

❽ The OUTPUT statement causes the observation to be placed in StudyPeriod2.

The loop's counter, i, increments by one, causing the loop to run again. Steps ❻ through ❽ are repeated until i reaches the value of Interval plus one. At this point, SAS reads the next observation from StudyPeriod and steps ❹ through ❽ are repeated. This continues until SAS has read and processed all observations from StudyPeriod.

A few observations from StudyPeriod2 are shown in Figure 7.4.

Figure 7.4: StudyPeriod2 data set

MemberID	NDC	label	interval	i	date
M000007425001...	00093550101	BUDEPRION T...	241	1	2011-04-04
M000007425001...	00093550101	BUDEPRION T...	241	2	2011-04-05
M000007425001...	00093550101	BUDEPRION T...	241	3	2011-04-06
M000007425001...	00093550101	BUDEPRION T...	241	4	2011-04-07
M000007425001...	00093550101	BUDEPRION T...	241	5	2011-04-08
M000007425001...	00093550101	BUDEPRION T...	241	6	2011-04-09
M000007425001...	00093550101	BUDEPRION T...	241	7	2011-04-10
M000007425001...	00093550101	BUDEPRION T...	241	8	2011-04-11
M000007425001...	00093550101	BUDEPRION T...	241	9	2011-04-12
M000007425001...	00093550101	BUDEPRION T...	241	10	2011-04-13
M000007425001...	00093550101	BUDEPRION T...	241	11	2011-04-14

This data set is used to calculate the denominator in PDC.

Create Days Covered Data Set

Program 7.3 shows code to create days' covered for each member-medication combination.

Program 7.3: Data expansion for covered days

```
data pharmacy(keep=MemberID NDC Label Date RxAdherent);
    set book.pharmacy;
    do i=1 to Qty;
      Date=intnx("day",FillDate,i-1);
      RxAdherent = 1;
      output;
      format Date yymmdd10.;
    end;
run;
```

Although similar to the code presented in Program 7.2, the code in Program 7.3 has one notable difference. Each member-medication transaction is expanded by the value in Qty. As a result, Pharmacy contains all days for which a member has supply of a specific medication. A column called RxAdherent is initialized to 1. This column is summed later to create a numerator for PDC.

Figure 7.5: Pharmacy data set

MemberID	NDC	label	date	RxAdherent
M000007425001...	00093550101	BUDEPRION T...	2011-04-04	1
M000007425001...	00093550101	BUDEPRION T...	2011-04-05	1
M000007425001...	00093550101	BUDEPRION T...	2011-04-06	1
M000007425001...	00093550101	BUDEPRION T...	2011-04-07	1
M000007425001...	00093550101	BUDEPRION T...	2011-04-08	1
M000007425001...	00093550101	BUDEPRION T...	2011-04-09	1
M000007425001...	00093550101	BUDEPRION T...	2011-04-10	1
M000007425001...	00093550101	BUDEPRION T...	2011-04-11	1
M000007425001...	00093550101	BUDEPRION T...	2011-04-12	1
M000007425001...	00093550101	BUDEPRION T...	2011-04-13	1
M000007425001...	00093550101	BUDEPRION T...	2011-04-14	1

Combine and Summarize Data

The previous chapters have shown ways in Query Builder and PROC SQL to combine and aggregate data. Program 7.4 shows you how to combine two or more tables using the merge statement in a DATA step. In order for the merge to work correctly, all tables being combined need to be sorted correctly.

Program 7.4: Create PDC ratio by Member and Medication

```
* Data sets need to be sorted prior to being merged;

proc sort data=Pharmacy;                                    ❶
    by MemberID NDC Label Date;
run;

proc sort data=StudyPeriod2;                                ❷
    by MemberID NDC Label Date;
run;

data RxAdherence(drop=Interval i);
    merge Pharmacy StudyPeriod2;                            ❸
    by MemberID NDC Label Date;                             ❹
    count=1;                                                ❺
    if RxAdherent=. then RxAdherent=0;                      ❻
run;
```

```
proc sql;                                                    ❼
    create table Summary as
    select MemberID, NDC, Label,
          (sum(RxAdherent)/sum(Count)) as PDC format comma10.3
      from RxAdherence
     group by MemberID, NDC, Label;
quit;
```

❶ Sorts the Pharmacy table so that the MERGE statement works in step ❸.

❷ Sorts the StudyPeriod2 table.

❸ The MERGE statement is used to combine data horizontally in a DATA step. Unless additional code is specified, the default behavior of merge is to keep all observations from source tables.

> Unless additional code is specified, the MERGE statement in a DATA step keeps all observations from its source data sets, including observations that do not match. You can add code to specify whether you want a left join, a right join, or an inner join.
>
> The MERGE statement works for joining tables with a join relationship that is either one to one or one to many. Tables whose merge relationship is many to many should not be joined with the MERGE statement, as results may be erratic.
>
> If your tables have a many-to-many relationship, examine the data carefully to see if it makes sense. More likely than not, you will want to remove duplicates from at least one table following a set of rules. If, however, you decide that you do not want to remove duplicates, you can use PROC SQL to join this data correctly.

❹ The BY statement specifies the join criterion for the merge. A MERGE with a BY statement performs a "match merge," matching on the key variables in the BY statement.

❺ Count is initialized to one for every observation. When summarized by member and medication, this is the denominator of PDC.

❻ Data in Pharmacy is a subset of data in StudyPeriod2. The column RxAdherent, created in Pharmacy, is set to missing for observations in StudyPeriod2 that do not have a corresponding match in Pharmacy. The conditional IF . . . THEN statement changes the missing value of RxAdherent to 0. When summarized, by member and medication, RxAdherent is the numerator of PDC.

❼ The PROC SQL aggregates RxAdherent and Count to create a PDC ratio for every member-medication combination. The results are shown in Figure 7.6.

Figure 7.6: PDC ratio

MemberID	NDC	label	PDC
M0000074250011100	00093550101	BUDEPRION T…	0.882
M0000074250011100	00185041005	BUPROPION T…	0.744
M0000106320012800	00093550101	BUDEPRION T…	1.000
M0000106320012800	00185041005	BUPROPION T…	0.331

As Figure 7.6 shows, the PDC for member M0000106320012800 and medication 00185041005 is 0.33. That means that the member did not have an adequate supply for that medication 66 percent of the time in the study period.

PDC calculations are actionable. Healthy Living, Inc., may decide to reach out to members for whom at least one PDC falls below 0.6. The data may be joined with the Members table used in Chapter 6 to create outreach reports.

The PDC computation quickly becomes complex when other considerations are taken into account. For example, what if the member re-fills his prescription before his previous supply has run out? This creates an oversupply issue. Programmatically, this means that you will have duplicate records for which there is supply. As a healthcare analyst, you will need to analyze your data to figure out how often this happens. If the issue of oversupply is small and spread out, then removing duplicates might be the best solution. Otherwise, you can add DATA step code that tacks on oversupply of a drug to days where the member did not have enough supply.

What if a member had a hospitalization? The code in this chapter will show a lower PDC, when in reality the member was medication-compliant and had an oversupply because of his days in the hospital. Again estimate how severe the problem is and adjust your code accordingly. The power of the DATA step makes it possible to add complex business logic to existing algorithms.

Graphing PDC

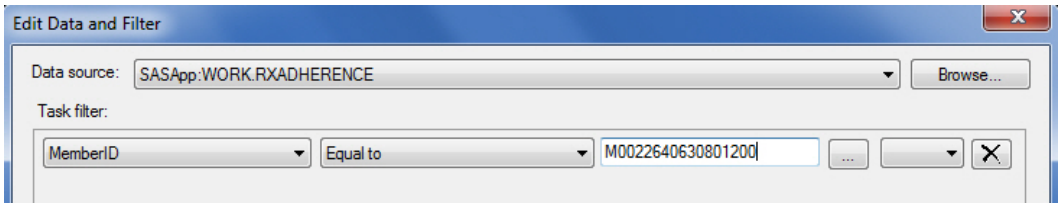
The pharmacists at Healthy Living, Inc., believe that there are specific patterns to medication noncompliance. To investigate this further, they have decided to graph the data that underlies a member's PDC calculation.

This section shows you how to create medication compliance graphs for a specific member.

To create graphs for MemberID M0022640630801200, follow the steps outlined below.

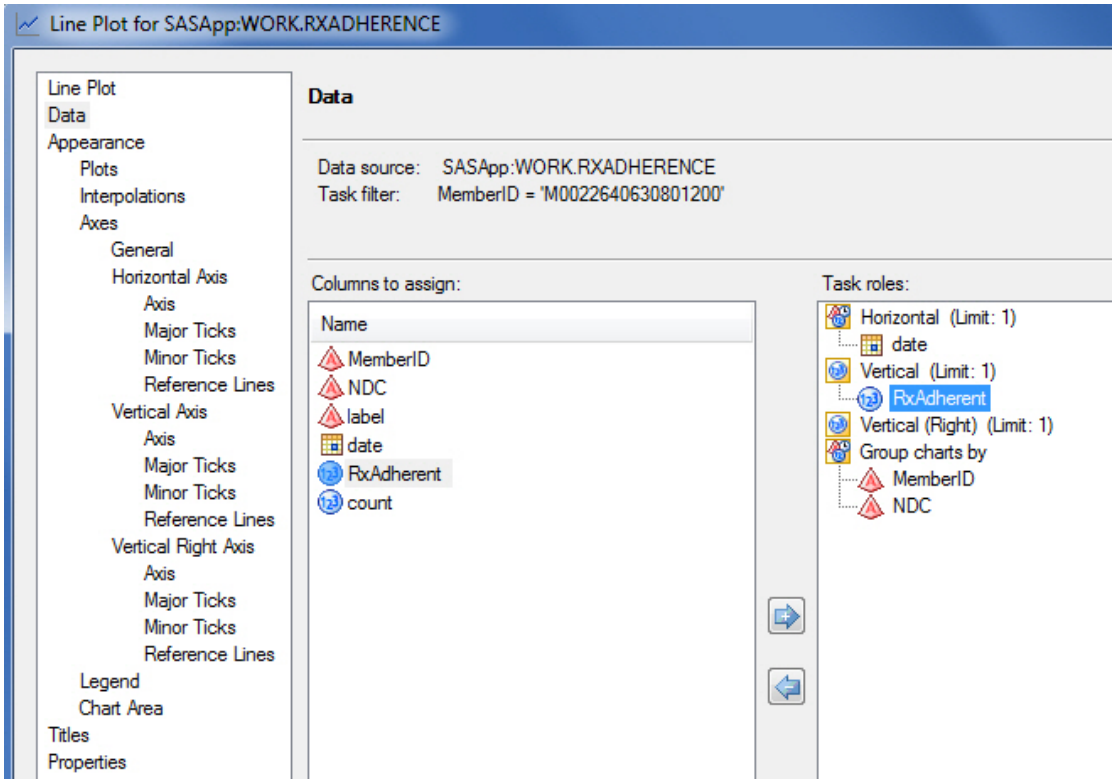
1. Select **Line Plot** from **Tasks—Graph**.
2. Select **Line Plot** in **Line Plot** tab.
3. In **Data** tab, click **Edit** to set up a filter on the data source as shown in Figure 7.7. Note that the **Browse** button in Figure 7.7 allows you to change the default data source. WORK.RxAdherence was chosen by default because it is the last data set to be created.
   a. As the filter is based on MemberID, choose MemberID from the first drop down.
   b. Choose **Equal to** from the second drop down.
   c. Enter M0022640630801200 and click **OK**.

Figure 7.7: Create a task filter



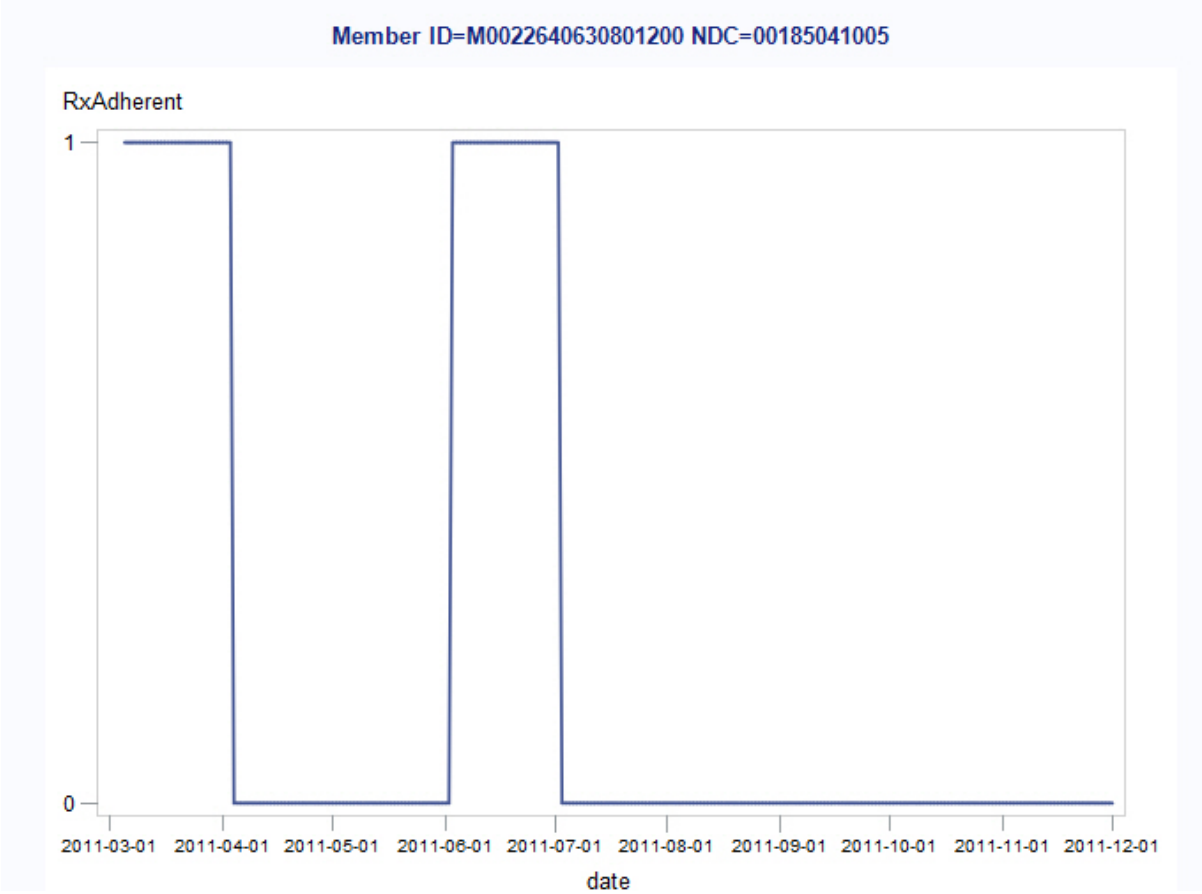
   d. Move columns to **task roles** as shown in Figure 7.8. MemberID and NDC are placed in **Group charts by Task role**. This allows separate charts to be created for every combination of the two columns.

Figure 7.8: Assign columns to task roles



4. In **titles** tab, uncheck **Use Default Text** for both **Graph** and **Footnote**.
5. Note that three graphs are created, one of which is displayed in Figure 7.9.

Figure 7.9: PDC graph



As Figure 7.9 shows, the member had adequate supply of the medication in March and June 2011 only.

Automating PDC Graphs Generation

The pharmacists at Healthy Living, Inc., have decided that PDC graphs give them useful information about the compliance habits of members. They want to peruse PDC graphs for every member for whom at least one medication PDC rate is below 0.6.

This section shows you how to create code to generate graphs for every member for whom the PDC is below 0.6 and to capture these graphs in a PDF document.

Let us begin by examining the code extracted from the **Line Plot** task presented in the previous section. This code has been indented and spaced appropriately for ease of reading. For a review of how to extract code from an Enterprise Guide task and place it in a code node, see Chapter 6.

Program 7.5: Code extracted from Line Plot

```
proc sort
    data=WORK.RXADHERENCE(where=(MemberID = "M0022640630801200")
                          keep=Date RxAdherent MemberID NDC    )
    out=WORK.SORTTempTableSorted;
    by MemberID NDC Date;
run;                                                              ❶

symbol1 interpol=join height=10pt value=none
        line=1 width=2 cv = _style_;                              ❷
axis1 style=1 width=1 minor=none;
axis2 style=1 width=1 minor=none;                                 ❸
```

```
title;                                                                    ❹
footnote;                                                                 ❺

proc gplot data=WORK.SORTTempTableSorted nocache ;                        ❻
    plot RxAdherent * Date/vaxis=axis1 haxis=axis2 frame;                 ❼
    by MemberID NDC;                                                      ❽
run;
quit;
```

❶ PROC SORT sorts the data by MemberID and NDC. This is required for the GPLOT procedure to create separate graphs for each combination of member and drug code. The WHERE= data set option ensures that only MemberID #M0022640630801200 is to be found in WORK.SortedTempTable.

❷ The SYMBOL statement defines characteristics of the symbols to be used by the GPLOT procedure. INTERPOL=JOIN allows points along a line to be joined. VALUE=NONE specifies that no symbols are used to define each point. CV specifies the color of the symbol.

❸ The AXIS statements specify characteristics for the axis, which are associated with the vertical and horizontal axes in GPLOT procedure code.

❹ A TITLE statement without arguments wipes out any existing titles.

❺ A FOOTNOTE statement without argument negates any existing footnotes.

❻ Invokes the GPLOT procedure.

❼ Defines the plot. FRAME draws a frame around the plot.

❽ Plots are generated for each Member-NDC combination.

Program 7.6 is modified in two ways.

- Data for the GPLOT procedure needs to include all members for whom the PDC falls below 0.6. This can be done by joining Summary to RxAdherence and filtering for PDC below 0.6. As this technique has been covered in a previous chapter, a new technique is shown in Program 7.6.
- Output from the GPLOT procedure is channeled to a PDF document via an Output Delivery Systems (ODS) "sandwich."

Program 7.6: Graphs for members with PDC less than .6

```
* Example of Sub-Query;

proc sql;                                                                 ❶
    create table GraphPDC as
    select MemberID, NDC, Date, RxAdherent
      from RxAdherence
     where MemberID in (select distinct MemberID
                          from summary
                         where PDC<.6     )
     order by MemberID, NDC, Date;
quit;

ods pdf file="C:\HealthCare Book\PDC Graphs for Members.pdf";              ❷

symbol1 interpol=join height=10pt value=none
        line=1 width=2 cv = _style_;
axis1 style=1 width=1 minor=none;
axis2 style=1 width=1 minor=none;
title;
footnote;
```

```
proc gplot data = GraphPDC nocache;
    plot RxAdherent * Date/vaxis=axis1 haxis=axis2 frame;
    by MemberID NDC;
run;
quit;

ods pdf close;                                                    ❸
```

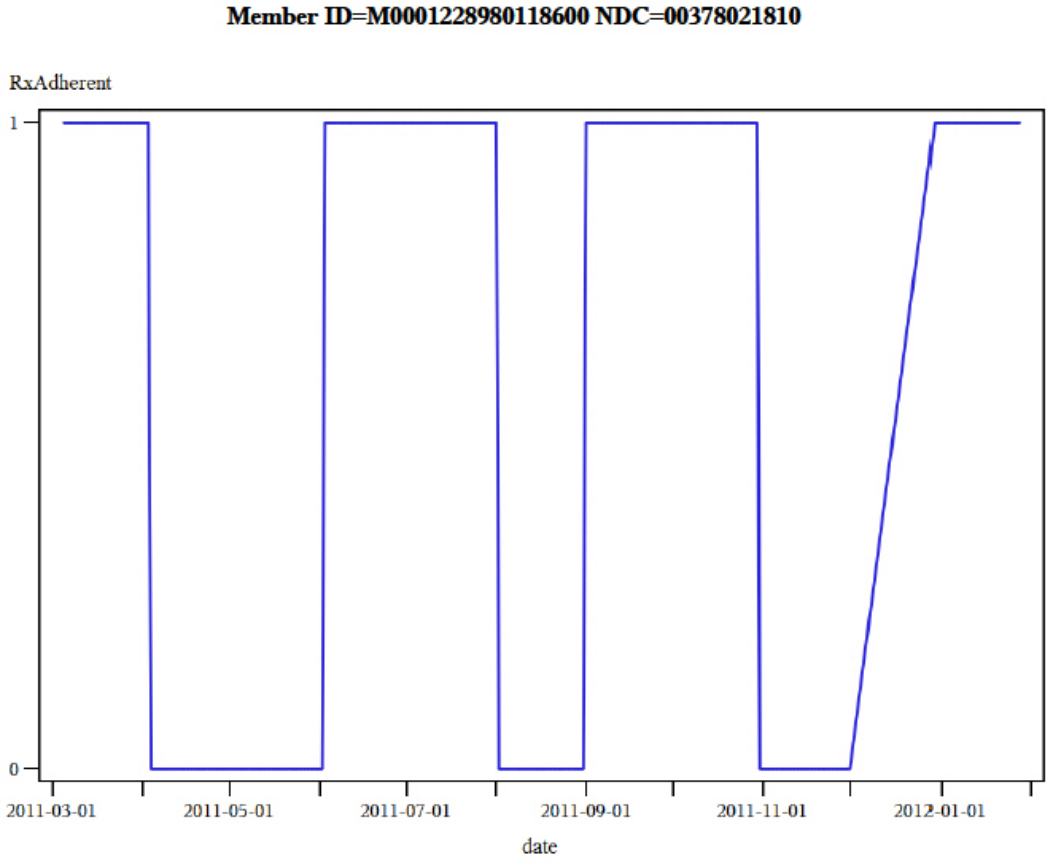
❶ The SQL procedure shows an instance of a subquery, or a query inside a query. It is a powerful technique to quickly join data from one table to another. This subquery is placed in the WHERE statement of the main query and executes first, returning a list of MemberIDs. The main query is executed subsequently with a filter on MemberIDs.

❷ The ODS PDF statement directs any output to a file listed in the FILE= option. This is the opening of the ODS "sandwich"—a block of code that begins with an ODS OPEN and ends with an ODS CLOSE.

❸ The ODS PDF destination is closed.

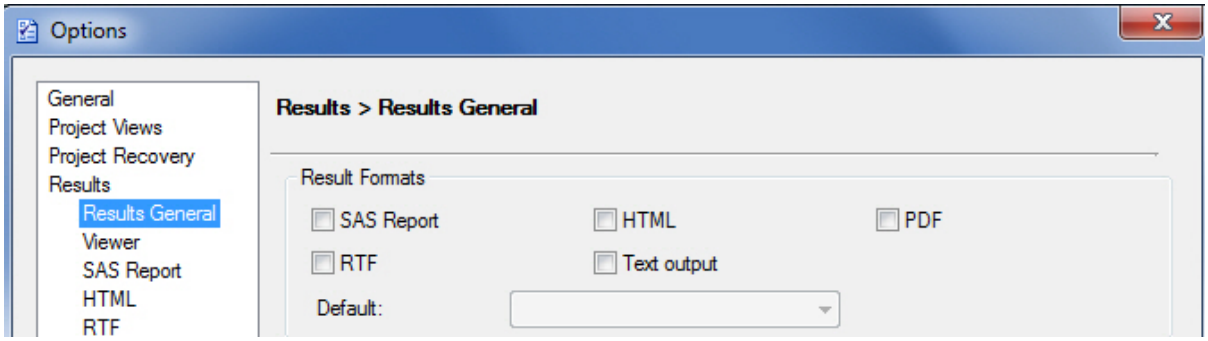
Figure 7.10 shows a PDC graph from PDC Graphs for Members.pdf.

Figure 7.10: PDC Graphs for Members.pdf



By default any output is placed in your SAS Enterprise Guide project. To avoid large project files, you can turn off the default behavior. To accomplish this, select **Options** from the **Tools** menu. In the **Results General** tab, uncheck **Results Formats** as shown in Figure 7.10.

Figure 7.10: Control output placed in SAS Enterprise Guide project



> The ODS sandwich statements can direct SAS output to a number of destinations, including RTF (Word), Excel, and HTML. Although a discussion of these destinations is beyond the scope of this book, it is worth mentioning that ODS is extremely well integrated with Excel through a predefined template called ExcelXP Tagset.

Programs 7.2, 7.3, 7.4, and 7.6 may be concatenated and executed together to create an automated listing of graphs for members whose PDC falls below 0.6. Using macro variables for beginning and ending dates of the study period, this code may also be parameterized. As parameterization of code has been covered in Chapter 6, this is left to the reader as an exercise.

Conclusion

This chapter has shown you that pharmacy data is an integral part of the healthcare database. You learned how claims submission works to create pharmacy data. You also acquired knowledge of data elements unique to pharmacy data. Through the case study presented, you learned how to calculate and analyze medication adherence.

Chapter 8: Healthcare Claim Codes

Introduction 103

International Classification of Diseases 104

Diagnosis Codes 105

ICD-9-CM 105

ICD-10-CM 107

Surgical Procedure Codes 109

ICD-9-PCS 109

ICD-10-PCS 110

Current Procedural Terminology (CPT) 112

Category I 113

Category II 113

Category III 114

HCPCS 114

Level I 115

Level II 115

Level III 115

Modifiers 115

HIPPS 116

Other PPS Code Sets 117

NDC 117

LOINC 118

Case Study: Identifying a Patient with Complex Conditions 118

Code Simplification with SAS Array Processing 119

Identifying Members with Complex Conditions 120

Parameterizing Program 8.3 with Macro Variables 122

Case Study: Using Formats to Create Data Hygiene Routines 123

Conclusion 126

Introduction

Previously we learned of the origin of medical claims data—CMS-1500/837P, CMS-1400/837I, and NCPDP UCF/D.0—the paper and electronic claim submission formats. Having that firm foundation, let us build on it by considering the content of the submissions.

Medical claims data is full of codes. It is important not only to understand the codes' purpose, but also the codes' architecture. For the most part, the designers of the code sets built them with purpose and forethought. One of the many outcomes of HIPAA legislation was the codification of national code sets to describe medical conditions, treatments, and other information that support the accurate adjudication of a claim. Without exception, the codes we will discuss (see Table 8.1) are nominal-level character data. Understanding the embedded meaning within these codes makes them useful

Table 8.1: Medical claim code sets

Code set	Acronym	What	Source
International Classification of Diseases, Version x, Clinical Modifications, Vols. 1 & 2	ICD-x-CM	diagnosis codes	Facility, Professional, and Ancillary
International Classification of Diseases, Version x, Clinical Modifications, Vol. 3	ICD-x-PCS	surgical procedure codes	Facility only
Current Procedural Terminology	CPT	professional provider procedure/service codes	Facility and Professional
Healthcare Common Procedure Coding System	HCPCS	services and products not included in CPT	Facility and Ancillary
Procedure Modifiers	n/a	codes that serve to "modify" the procedure codes (CPT/HCPCS)	Facility, Professional, and Ancillary
Health Insurance Prospective Payment System	HIPPS	rate codes that guide payment determinations under several prospective payment systems, other than DRG	Facility: SNF, Rehabilitation
Diagnostic Related Group	DRG	rate code used for inpatient prospective payment (IPPS) claims	Facility: Inpatient
Ambulatory Payment Classifications	APC	the rate code used for outpatient prospective payment (OPPS)claims	Facility: Outpatient
National Drug Code	NDC	identifier for dispensed human pharmaceuticals	Facility, Professional, and Pharmacy
Logical Observation Identifiers Name and Codes	LOINC	identifier for laboratory results	Automated laboratory data feed

In this chapter you will:

- Learn about the national code sets that describe a medical claim
- Understand the architecture of these codes
- Use code sets and SAS array processing to identify patients with complex conditions
- Create SAS formats that catch invalid code sets

International Classification of Diseases

The International Classification of Diseases (ICD) is designed and maintained by the United Nations's World Health Organization (WHO [www.who.int/classifications/icd/en/]). Its intent is to promote "international comparability in the collection, processing, classification, and presentation of mortality statistics," reporting

causes of death on death certificates. Version 10 (ICD-10) is the currently accepted and widely used revision; WHO is currently developing version 11.

While the U.S. healthcare system uses ICD-10 to code mortality on death certificates, we are still using ICD-9 as a basis for coding diseases in health care settings. The International Classification of Diseases, Clinical Modification (ICD-9-CM), is an adaption created by the U.S. National Center for Health Statistics (NCHS), under the auspices of CMS, and used in assigning diagnostic and surgical procedure codes associated with facility and professional services in the United States. It is updated annually, on October 1, the beginning of the federal fiscal year. Other countries have created similar modifications to the point at which the WHO is working on ICD-10-XM, an international version of clinical modifications.

ICD-9-CM consists of three volumes:

Volume 1—a tabular list of numerical code

Volume 2—an alphabetical list of the disease entries

Volume 3—an alphabetic index and tabular list of surgical and therapeutic procedures

Diagnosis Codes

Diagnostic coding is the conversion of a written description of a medical condition (disease, illness, or injury) into codes from a classification system. The ICD system is most widely used in our rendering of healthcare delivery.

It is important to understand that these codes describe a medical condition, not the treatment, cost, or services rendered for that condition.

ICD-9-CM

According to CMS version 31, ICD-9-CM contains 14,567 distinct codes for use in federal fiscal year 2014. These are the diagnosis codes that are required to be submitted on the Facility, Professional, and Ancillary provider claim forms. They are optionally used by pharmacy providers. Current and historic code sets can be downloaded from the CMS website: www.cms.gov/Medicare/Coding/ICD9ProviderDiagnosticCodes/codes.html.

It is important to understand the architecture of ICD-9-CM diagnosis codes in order to develop effective and efficient programming routines such as data hygiene modules. Table 8.2 describes the salient design features. Note that the generic architecture is "pictured" in the table titles.

Table 8.2: ICD-9 Diagnosis code architecture—xxx.xx

Characteristic	Value
Minimum length	3
Maximum length	5
Character composition	
First character	0–9, V, E
Subsequent characters	0–9
Note 1	Codes should always be stored as character data as they have no arithmetical meaning.
Note 2	Not all codes have a 4th or 5th position; if they do, it is for the purpose of adding specificity to the code description.
Note 3	If present, the decimal point is after the 3rd character.

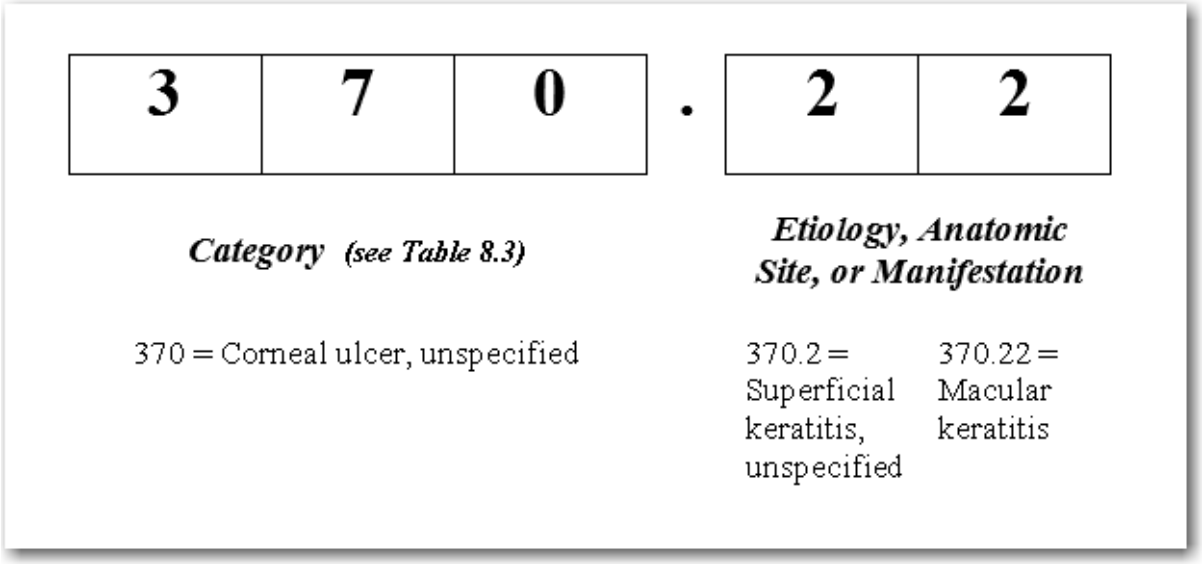
Characteristic	Value
Exception to Note 3	If the code is an "E" code and the decimal is present, it follows the 4th character.

Diagnosis codes beginning with a "V," known as V-codes, are used to denote patient care not related to a particular disease state. They are typically not used as a principal diagnosis code on facility claims. They are commonly used to describe patient status (V01.5—contact with or exposure to rabies), screenings (V75.1—screening examination for malaria), history (V113—personal history of alcoholism), or aftercare (V54.81—aftercare following joint replacement).

Codes beginning with an "E," E-codes, are used to denote an external cause of injury (ECI). They are frequently seen on emergency department claims—facility claim format—and are intended to provide data for injury research and prevention strategies. While E-codes can be used anywhere a diagnosis code is entered on a claim form, they most often originate on the facility claim format in the ECI blocks. Please see the exception to the architecture rule in Table 8.2.

Example 8.1 demonstrates the structure and increased specificity in ICD-9-CM.

Example 8.1: Example ICD-9 diagnosis code



Major Disease Categories (see Table 8.3) is a categorization schema promoted by ICD and simply ranges the codes based on etiology or body system. This information is typically a "value-add" at the time of warehouse or data mart information acquisition.

Table 8.3: Major Disease Categories, 2014 ICD-9 codes

Code Range	Description
001–139	Infectious and Parasitic Diseases
140–239	Neoplasms
240–279	Endocrine, Nutritional, and Metabolic Diseases and Immunity Disorders
280–289	Diseases of the Blood and Blood-Forming Organs
290–319	Mental Disorders
320–389	Diseases of the Nervous System and Sense Organs
390–459	Diseases of the Circulatory System
460–519	Diseases of the Respiratory System
520–579	Diseases of the Digestive System

Code Range	Description
580–629	Diseases of the Genitourinary System
630–679	Complications of Pregnancy, Childbirth, and the Puerperium
680–709	Diseases of the Skin and Subcutaneous Tissue
710–739	Diseases of the Musculoskeletal System and Connective Tissue
740–759	Congenital Anomalies
760–779	Certain Conditions Originating in the Perinatal Period
780–799	Symptoms, Signs, and Ill-Defined Conditions
800–999	Injury and Poisoning
V01–V91	Factors Influencing Health Status and Contact with Health Services
E000–E999	External Causes of Injury and Poisoning

The fifth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5) was released by the American Psychiatric Association in May 2013. The DSM codes are a subindustry criterion for the diagnosis and classification of mental disorders. While these diagnostic codes may be seen in some context, be aware that insurance companies must follow HIPAA, and ICD is the HIPAA-compliant code set.

DSM and ICD have been "harmonized" but do not correlate 100 percent of the time. A crosswalk may be obtained from the American Psychological Association at http://www.apa.org.

ICD-10-CM

The current expectation for adoption of the tenth revision of the U.S. Clinical Modifications to the ICD is October 2015. Given the delays from previously announced implementation dates and pushback from some sectors of the industry, some are skeptical that ICD-10-CM will be adopted in 2015. That being said, ICD-10-CM (and ICD-10-PCS) is going to be adopted eventually, so we need to understand the changes that it will bring to our analytics and reporting.

The change from the ninth revision is dramatic. ICD-10-CM contains 69,823 entries, a fourfold expansion from ICD-9-CM. The change significantly complicates how healthcare software applications capture data and how historical information can be interpreted. Given the widespread expansion in the tenth revision, it is not possible to convert ICD-9 codes directly into ICD-10 codes, although some tools are available to help guide users. ICD-10 has extended its architecture and, therefore, its specificity of definition.

For diagnosis codes, ICD-10-CM is intended to replace volumes 1 and 2 of ICD-9-CM. Annual updates are provided through various commercial vendors and the National Center for Health Statistics (www.cdc.gov/nchs/icd/icd10cm.htm). More information on the transition to the tenth revision can be found at the CMS website: www.cms.gov/Medicare/Coding/ICD10.

Because ICD-10-CM is much more specific, is more clinically accurate, and uses a more logical structure, it should be much easier to use than ICD-9-CM. Table 8.4 details the structure of these diagnosis codes.

Table 8.4: ICD-10 Diagnosis Code Architecture—xxx.xxxx

Characteristic	Value
Minimum length	3
Maximum length	7
Character composition	
First character	A–Z, except U
Second character	0–9
Subsequent characters	A–Z, 0–9
Note 1	If present, the decimal point is after the 3rd character.
Note 2	Alpha characters are not case sensitive.

Characteristic	Value
Note 3—the 7th character, known as the "Extension"	Certain code categories (e.g., injuries and external causes) require an applicable 7th character. This character must always be the 7th character in the data field. If a code that requires a 7th character is not at least 6 characters in length, a placeholder "X" must be used to fill in the empty positions. There is a known domain of values for the 7th position; it is left to the reader to research further as needed.

Example 8.2 demonstrates the architecture and the increased specificity as one reads from left to right. It is incumbent upon the provider to provide as much detail as possible on the claim. The minimum requirement is the three-character category code. In practice, the required level of detail will be enforced by the provider's claim coding software and/or the payer's adjudication system and, in many cases, will be situational.

Example 8.2: Example ICD-10 diagnosis code

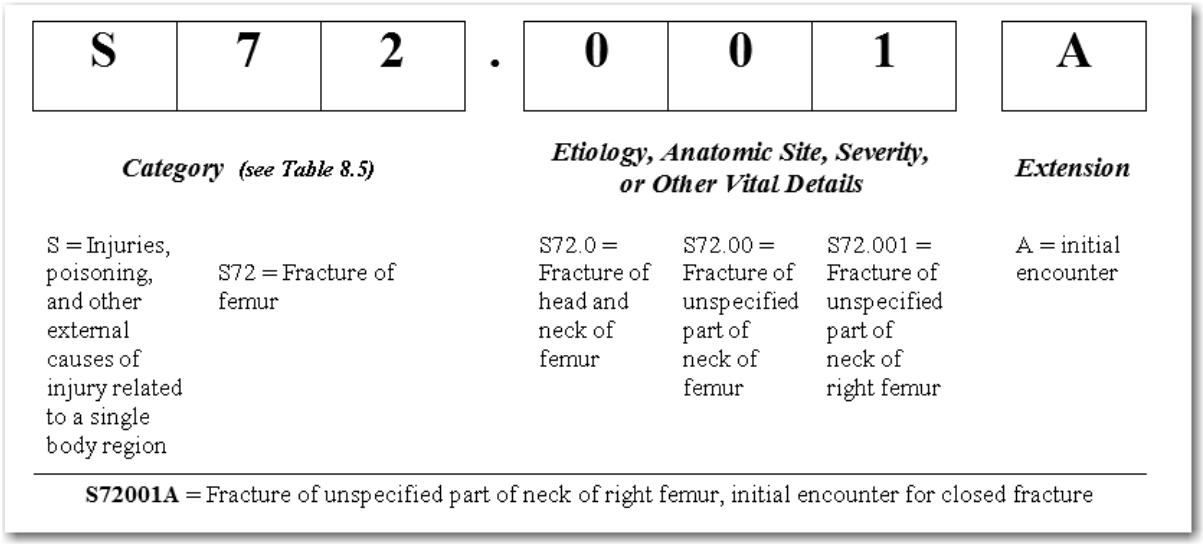


Table 8.5 describes the Major Disease Categories as defined in ICD-10.

Table 8.5: Major Disease Categories, 2014 ICD-10-CM codes

Code range	Description
A00–B99	Certain infectious and parasitic diseases
C00–D49	Neoplasms
D50–D89	Diseases of the blood and blood-forming organs and certain disorders involving the immune mechanism
E00–E89	Endocrine, nutritional, and metabolic diseases
F01–F99	Mental, behavioral, and neurodevelopmental disorders
G00–G99	Diseases of the nervous system
H00–H59	Diseases of the eye and adnexa
H60–H95	Diseases of the ear and mastoid process
I00–I99	Diseases of the circulatory system
J00–J99	Diseases of the respiratory system
K00–K95	Diseases of the digestive system
L00–L99	Diseases of the skin and subcutaneous tissue
M00–M99	Diseases of the musculoskeletal system and connective tissue

Code range	Description
N00–N99	Diseases of the genitourinary system
O00–O9A	Pregnancy, childbirth, and the puerperium
P00–P96	Certain conditions originating in the perinatal period
Q00–Q99	Congenital malformations, deformations, and chromosomal abnormalities
R00–R99	Symptoms, signs, and abnormal clinical and laboratory findings, not elsewhere classified
S00–T88	Injury, poisoning, and certain other consequences of external causes
V00–Y99	External causes of morbidity
Z00–Z99	Factors influencing health status and contact with health services

Surgical Procedure Codes

The surgical procedure codes are an addition to the ICD Clinical Modifications and are known as ICD-x-PCS (Procedural Classification System). The WHO did not introduce this set of codes, given their primary interest in diagnosis codes as an international definition of morbidity data. This is the code set that is used to describe inpatient hospital surgeries only and is to be found at the header level of the institutional claim, that is, they apply to the entire claim and are not service-level detail.

ICD-9-PCS

Revision nine of the ICD surgical procedure codes can be found in volume 3 of the ICD-9-CM documentation. The list published by CMS for use in the 2014 federal fiscal year contains 3,882 entries.

As surgeries move from the inpatient setting to the outpatient setting, it is important to note that ICD-9-CM surgical codes cannot be used and, instead, we see HCPCS or CPT codes used in conjunction with a Revenue Code (the billable code) to describe surgeries in this setting.

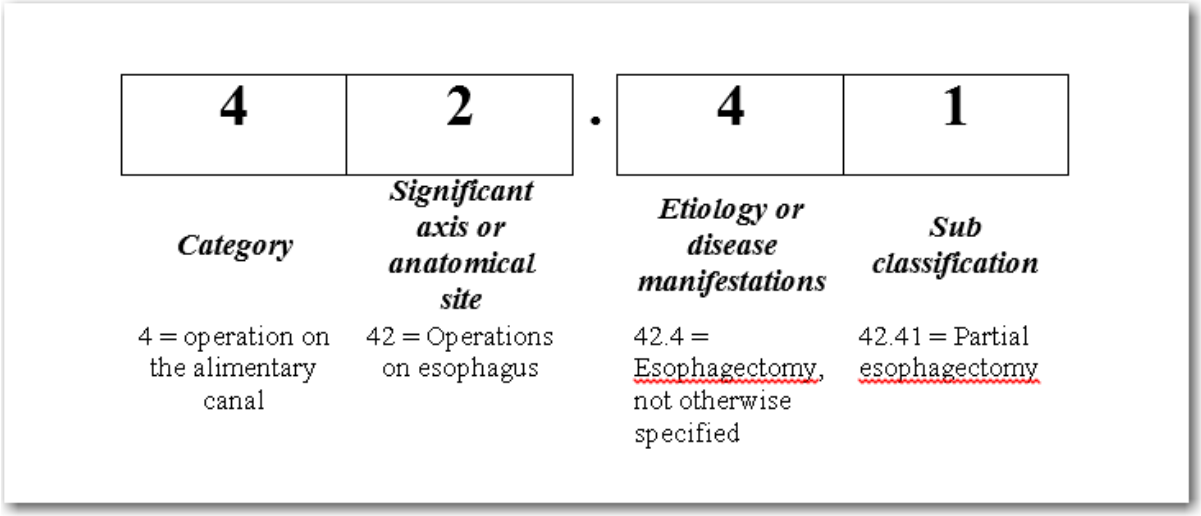
IP:	laparoscopic appendectomy	ICD-9-CM code	47.01
OP:	laparoscopic appendectomy	CPT code	44970

Table 8.6 and Example 8.3 should provide an essential understanding of the structure of the surgical procedure codes.

Table 8.6: ICD-9-CM surgical procedure code architecture—xx.xx

Characteristic	Value
Minimum length	3
Maximum length	4
Character composition	0–9
Note 1	The first two characters define a category and subcategory that is useful for data filtering.
Note 2	Not all codes have a 4th position; if they do, it is for the purpose of adding specificity to the code description
Note 3	If present, the decimal point is after the 2nd character.

Example 8.3: Example ICD-9 surgical procedure code



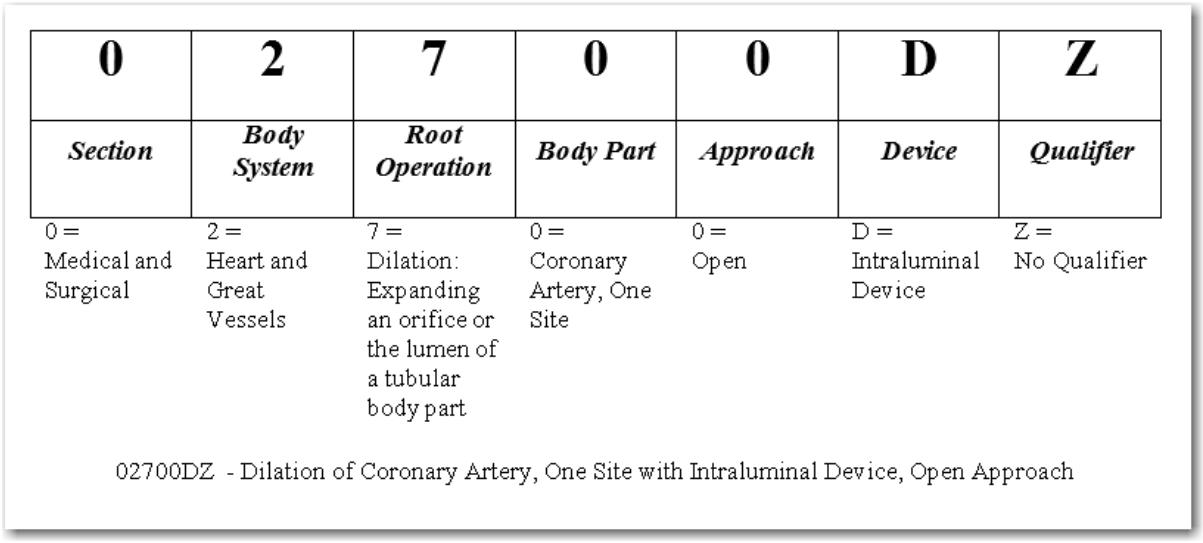
ICD-10-PCS

ICD-10-PCS (Procedural Coding System), for surgical procedure codes, will replace volume 3 of ICD-9-CM. Annual updates are provided through various commercial vendors and CMS. The fiscal year 2014 version, as released by CMS, contains 71,924 entries; an 18-fold increase over the number of ICD-9 surgical procedure codes. It is not difficult to see the improvements (and headaches) that the new systems will bring to the industry. Table 8.7 provides a high-level view of the code architecture. Example 8.4 further explains the component parts.

Table 8.7: ICD-10-PCS surgical procedure code architecture—xxxxxxx

Characteristic	Value
Minimum length	7
Maximum length	7
Character composition	0–9, A–H, J–N, and P–Z
Note 1	The characters O and I are not used in order to avoid confusion with the digits 0 and 1.
Note 2	The term *procedure* refers to the complete specification of the seven characters.

Example 8.4: Example ICD-10 surgical procedure code



0	2	7	0	0	D	Z
Section	***Body System***	***Root Operation***	***Body Part***	***Approach***	***Device***	***Qualifier***
0 = Medical and Surgical	2 = Heart and Great Vessels	7 = Dilation: Expanding an orifice or the lumen of a tubular body part	0 = Coronary Artery, One Site	0 = Open	D = Intraluminal Device	Z = No Qualifier

02700DZ - Dilation of Coronary Artery, One Site with Intraluminal Device, Open Approach

Procedures are divided into "sections" (the first of the seven component characters) that identify the general type of procedure. The sections are shown in Table 8.8.

Table 8.8: ICD-10-PCS sections

Section	Description
0	Medical and Surgical
1	Obstetrics
2	Placement
3	Administration
4	Measurement and Monitoring
5	Extracorporeal Assistance and Performance
6	Extracorporeal Therapies
7	Osteopathic
8	Other Procedures
9	Chiropractic
B	Imaging
C	Nuclear Medicine
D	Radiation Oncology
F	Physical Rehabilitation and Diagnostic Audiology
G	Mental Health
H	Substance Abuse Treatment

When building a claim for submission to a payer, ICD-10-PCS codes are not assigned from a lookup table as with diagnosis codes and the other codes discussed. They are built character-by-character with coding software at the provider's billing organization. "Z" is used when either is not applicable to the procedure. In all sections, the second and third character specifies the general type of procedure performed, while the other characters give additional information such as the body part and approach. To expand upon the above surgical code example, consider the valid combinations of characters (beyond position three) that are possible for "027"—Medical and

Surgical (section: 0), Heart and Great Vessels (body system: 2), and Dilation (root operation: 7). The possibilities can be seen in Table 8.9.

Table 8.9: ICD-10-PCS sections

Body Part	0 = Coronary Artery, One Site
	1 = Coronary Arteries, Two Sites
	2 = Coronary Arteries, Three Sites
	3 = Coronary Arteries, Four or More Sites
Approach	0 = Open
	3 = Percutaneous
	4 = Percutaneous Endoscopic
Device	4 = Drug-eluting Intraluminal Device
	D = Intraluminal Device
	T = Radioactive Intraluminal Device
	Z = No Device
Qualifier	6 = Bifurcation
	Z = No Qualifier

In addition to designing an expandable system, other design goals were to incorporate a method to ensure the following:

- Completeness: All substantially different procedures have a unique code. This is a much more complete system than its predecessor and allows for greater specificity.
- Standardized terminology: Unique definitions for all terminology used are included. This is an ease-of-use issue.
- Multi-axial structure: To the degree possible, each component character has the same meaning within a section and across sections. The second through seventh characters mean the same thing within each section, but may mean different things in other sections.

It is important to clarify that physician practices will not use ICD-10-PCS because physician professional services, delivered in any setting, will continue to be coded using CPT and HCPCS. So, to review, facilities use these surgical procedure codes at the claim header level to describe services provided, but use revenue codes at the claim service line detail level to bill for the surgery. The surgeon, a professional provider, bills for his or her services using the CMS-1500/837P mechanism.

On August 21, 2008, the U.S. Department of Health and Human Services (HHS) proposed new code sets to be used for reporting diagnoses and procedures on healthcare transactions; that proposal said that the ICD-9-CM code sets would be replaced with the ICD-10-CM code sets, effective October 1, 2013. Subsequently, HHS has delayed from October 1, 2013, to October 1, 2014, to October 1, 2015, the compliance date for the ICD-10-CM and ICD-10-PC. Skepticism remains in the industry that the 2015 date will be met.

Current Procedural Terminology (CPT)

The Current Procedural Terminology (CPT) code set is a medical claims code set maintained by the American Medical Association (AMA [http://www.ama-assn.org/]) that describes medical, surgical, and diagnostic services. It is intended to convey uniform information about medical services provided by physicians for administrative, financial, and analytical functions. CPT is a registered trademark of the AMA. Revised editions are released annually.

CPT is currently identified by CMS as "Level I" of the HCPCS. The original HCPCS code set (to be discussed later in this chapter) is known to them as "Level II."

There are three categories of CPT codes, each with a distinct architecture and purpose. These will be discussed in turn.

Category I

Category I CPT codes are the billable codes. Used by professional providers and, in a limited way, by facility outpatient departments, this is the information with which payer adjudication systems determine reimbursement for services provided. Category I is the set that analysts usually identify with when talking about CPT.

To be approved by the AMA and assigned a CPT code, a procedure must:

- Be approved by the Food and Drug Administration (FDA) for a specific purpose
- Be a distinct service performed by many practitioners across the United States
- Have well established and documented clinical efficacy in the U.S. peer review literature
- Be distinct and different from existing procedures and/or codes

Category I codes are updated annually and may be broken down into six segments—Evaluation and Management, Anesthesiology, Surgery, Radiology, Pathology and Laboratory, and Medicine. The architecture is described in Table 8.10.

Table 8.10: CPT Category I code architecture—xxxxx

Characteristic	Value
Minimum length	5
Maximum length	5
Character composition	0–9
Note 1	Codes should always be stored as character data as they have no arithmetic meaning.
Examples	99211—Office or other outpatient visit for the evaluation and management of an established patient, which may not require the presence of a physician. Usually, the presenting problem(s) are minimal. Typically, five minutes are spent performing or supervising these services. 42999—Unlisted procedure, pharynx, adenoids, or tonsils

Category II

Lesser known to analysts are Category II and III CPT codes. Category II are supplemental codes used to track certain services and test results that support nationally established performance measurement. The use of these codes is optional and cannot be a substitute for Category I (billable) codes; they are intended to enable data collection regarding the quality of patient care.

Criteria for the assignment and use of Category II performance codes are that they must be:

- Measurements that have been developed and tested by a national organization
- Measurements that are evidence-based with established links to health outcomes
- Measurements that deal with clinical conditions of high prevalence, risk, or cost
- Measurements that are well-established and currently used by large segments of the healthcare industry

CPT Category II codes are arranged in eight segments—Composite Measures, Patient Management, Patient History, Physical Examination, Diagnostic/Screening, Therapeutic/Preventive, Follow-up or Other Outcomes, and Patient Safety. The architecture is described in Table 8.11.

Table 8.11: CPT Category II code architecture—xxxxF

Characteristic	Value
Minimum length	5
Maximum length	5
Character composition	
First four characters	0–9
Last character	F
Examples	0014F—Comprehensive preoperative assessment performed for cataract surgery with intraocular lens (IOL) placement. 1110F—Patient discharged from an inpatient facility (e.g., hospital, skilled nursing facility, or rehabilitation facility) within the last sixty days

Category III

CPT Category III codes describe new and emerging technologies and are used strictly for data collection and utilization tracking; they are not billable codes. These coded services are not performed by many healthcare providers, do not have FDA approval, and do not have proven clinical efficacy. To be eligible for inclusion as a Category III code, the procedure or service must be involved in ongoing or planned research. The intended use of these codes is to help researchers track emerging technology in order to substantiate widespread usage and clinical efficacy. If not accepted by the FDA as a Category I code within five years, they will be retired. The architecture is described in Table 8.12.

Table 8.12: CPT Category III code architecture—xxxxT

Characteristic	Value
Minimum length	5
Maximum length	5
Character composition	
First four characters	0–9
Last character	T
Examples	0016T—Destruction of localized lesion of choroid (e.g., choroidal neovascularization), transpupillary thermotherapy 0079T—Placement of visceral extension prosthesis for endovascular repair of abdominal aortic aneurysm involving visceral

HCPCS

HCPCS (commonly pronounced "Hick-Picks") originally stood for HCFA Common Procedure Coding System, as CMS was previously known as the Health Care Financing Administration (HCFA) before 2001. The acronym was then changed to stand for Healthcare Common Procedure Coding System.

HCPCS was established in 1978 by CMS to support Medicare claims adjudication and provider reimbursement. It provided a standardized coding system for describing the specific items and services provided in the delivery

of health care. Prior to the implementation of HIPAA, use of the codes was voluntary; it is now mandatory. HCPCS includes three levels of codes: I, II, and III.

Level I

Level I HCPCS codes are the CPT Category I codes discussed above. Nothing more need be said, except to suggest that going forward we may begin to lose the term CPT and simply refer to HCPCS Level I and Level II.

Level II

The Level II HCPCS codes, as established by CMS, primarily represent items, supplies, and non-physician services not covered by the AMA's CPT codes. Examples include ambulance services, prosthetic devices, durable medical equipment (DME), durable medical supplies (DMS), and certain pharmaceuticals. Medicare, Medicaid, and private health insurers use HCPCS procedure and modifier codes (see below) for claims processing.

Table 8.13: HCPCS Level II code architecture—xxxxx

Characteristic	Value
Minimum length	5
Maximum length	5
Character composition	
First character	A–Z
Subsequent characters	0–9
Examples	A0225—Ambulance service, neonatal transport, base rate, emergency transport, one way K0001—Standard wheelchair

Level III

Level III codes, also called local codes, were developed by the states for use in specific programs and areas. Based on a HIPAA mandate, the use of Level III codes was discontinued on December 31, 2003. All local codes were mapped to HCPCS Level I or II; if a mapping did not occur then a new HCPCS code was created to serve the purpose.

Modifiers

Medical procedure codes submitted to support claims adjudication, and therefore reimbursement, may be "modified" under certain circumstances to more accurately represent the rendered service. Modifier codes, when accompanying a HCPCS Level I or II procedure code, add information without changing the definition of the code. The documentation of the service provided must support the use of the modifier. Many payers will have their own rules for the use of modifiers. Payers differ on how many modifiers they will accept, although the CMS-1500 and 837P formats allow for four to be submitted on each detail service line.

Level I (CPT) modifiers are updated annually by the AMA. Level II (CMS) modifiers are updated annually by CMS. A complete list of modifiers may be found in the CPT and HCPCS Coding Manuals. Table 8.14 details the structure of modifier codes.

Table 8.14: Procedure modifier code architecture—xx

Characteristic	Value
Minimum length	2
Maximum length	2
Character composition	
For HCPCS Level I (CPT Category I)	0–9
For HCPCS Level II	AA through VP
Examples	47—Anesthesia by surgeons; no additional reimbursement is allowed for anesthesia by a surgeon, assistant surgeon, nursing staff, or any other non-anesthesiologist professional during a procedure AA—Anesthesia services performed personally by anesthesiologist; 100% anesthesia rate

As part of the National Correct Coding Initiative (NCCI [http://www.cms.gov/Medicare/Coding/NationalCorrectCodInitEd/]), CMS stipulates which modifiers are allowed, not allowed, or not specified for use with a particular procedure. Other payers may also have such adjudication rules in their provider manuals.

HIPPS

Health Insurance Prospective Payment System (HIPPS) rate codes represent specific sets of patient characteristics on which payment is based for the following prospective payment systems (PPS)—Skilled Nursing Facility (SNF), Home Health (HI), and Inpatient Rehabilitation Facility (IRF). These codes are located at the service line level only and are associated with a revenue code on the service line.

HIPPS (pronounced “hips”) codes are not assigned from a coding manual as are other code sets. They are created by CMS grouper software based on a set of information entered by the facility. A standard patient assessment method is used by case-mix grouping software algorithms to assign the case mix group; at least one HIPPS code is defined to represent each case-mix group. These HIPPS codes are reported on claims to payers in conjunction with specific revenue codes (see Chapter 5). One revenue code is defined for each prospective payment system that requires HIPPS codes. HIPPS codes are reported in data element SV202 of the 837I electronic format or in FL 44 on a CMS-1450 paper format.

Each HIPPS code contains built-in intelligence, with certain positions of the code indicating the case-mix group itself and other positions providing additional information (e.g., information about the clinical assessment). The additional information varies among HIPPS codes pertaining to different payment systems. The code position that contains the case-mix group information also varies by payment system.

Table 8.15: HIPPS code architecture—xxxxx

Characteristic	Value
Minimum length	5
Maximum length	5
Character composition	0–9, A–Z
Note 1	The specific composition of HIPPS codes varies by PPS (SNF, HH, or IRF)
SNF example	CE150—Clinically Complex , ADL Index 15-16/90-day
HH example	1AFNX—Early Episode, 0–13 therapies, Clinical Severity Level 1, Functional Severity Level 1,Service Severity Level 4, Supply Severity Level 6, supplies provided

A **case mix group** (CMG) is used to group together patients with similar attributes. This provides a basis for describing the types of patients a provider treats (its case mix).

The reader is encouraged to further investigate the composition of various facility specific HIPPS codes as the need arises.

Other PPS Code Sets

Several additional code sets pertaining to Medicare's Prospective Payment System are listed in Table 8.16. Due to their specific and unique use, it is left to the reader to further understand these codes as needed.

Table 8.16: Additional PPS Codes

Code	Description
DRG	Diagnostic Related Groups—these codes are at the header level of IP hospital claims only, are independent of a revenue code, and are used for Medicare's Inpatient Prospective Payment System (IPPS). Commercial payers are also using them for "DRG-based" payment systems They are discussed further in Chapter 5.
APC	Ambulatory Payment Classifications—these codes are Medicare's basis for paying for facility outpatient services under their Outpatient Prospective Payment System (OPPS). State Medicaid programs and commercial payers are adopting this methodology.
RUG	Resource Utilization Groups—these codes are specific to the Skilled Nursing Facilities (SNFs) PPS and are generated using vendor grouper software. The RUG code comprises the first three positions of the HIPPS. code.
PAI	Patient Assessment Instrument—these codes are specific to Medicare Part A services received from an Inpatient Rehabilitation Facility (IRF).

NDC

The National Drug Code (NDC) was discussed at length in Chapter 7. It is important, however, to remind the reader that this important data element and its associated information are submitted not only on pharmacy claims but also on facility claims. An NDC will appear on the claim wherever pharmaceutical products are being billed to the payer.

LOINC

Logical Observation Identifiers Names and Codes (LOINC) is a universal standard for identifying medical laboratory observations. It was developed in 1994 and is maintained by the Regenstrief Institute, a medical research organization associated with Indiana University. Both the LOINC database and the Regenstrief LOINC Mapping Assistant (RELMA) may be downloaded at no cost from the Regenstrief Institute (www.loinc.org). Since its launch, the database has expanded to include nursing diagnosis, nursing interventions, and outcomes classification.

It is important to note that LOINC (pronounced "L'oink," like the sound a French pig might make) codes are not available through claims adjudication and its normal data sources. They are available via an automated laboratory data feed from a contracted medical laboratory vendor to the payer organization. The data feed will have sufficient information to match to claim records such that the laboratory test and results might be used in an analytic project, for example, outcomes analysis.

Also of interest is that the LOINC code has no built-in intelligence and is useless without its associated nomenclature. This nomenclature consists of six elements as detailed in Table 8.17.

Table 8.17: LOINC code nomenclature structure

Element	Description
Component	What is being measured, evaluated, or observed (e.g., blood flow)
Property	Characteristic of what is being measured (e.g., velocity)
Time aspect	Time interval over which the measurement is taken (e.g., 6 hour)
Body system	Specimen type that is being measured (e.g., serum/plasma)
Measurement scale	Scale of the measurement (e.g., nominal, ordinal, quantitative)
Measurement method	Test technique; this is an optional element (e.g., EKG)

Table 8.18: LOINC code architecture

Characteristic	Value
Minimum length	3
Maximum length	7
Character composition	
2nd to last character	-
Other characters	0–9
Note 1	The last digit is a "check digit" that results as a function of the digits to the left of the "-." It serves to validate those digits.
Example (code—nomenclature)	11674-9—Mean Blood flow : Velocity : Point in time, Random : Cerebral Middle Artery : Quantitative : Doppler Ultrasound

Case Study: Identifying a Patient with Complex Conditions

The primary goal of the Patient Management department at Healthy Living, Inc., is to ensure that patients receive appropriate care. The department is headed by a physician and includes nurses, pharmacists, and social workers. It performs a variety of functions, including coordination of care among multiple providers and social and communal support. If a patient needs a wheelchair, a social worker makes sure that the patient gets it

quickly. If a patient needs assistance sorting multiple medications, a pharmacist reaches out on the phone or in person to help.

The Patient Management department uses several approaches to find patients or patient cohorts who require further intervention.

- They identify patients with multiple complex conditions. Examples include adults with chronic obstructive pulmonary disease (COPD) and congestive heart failure (CHF).
- They identify patients who seek primary care in the ER. Frequent ER users can be directed to doctors' offices and urgent care centers.
- They identify patients who misuse or overuse opiates and other prescription drugs.

As the latter two approaches have been the subject of previous chapters, this case study focuses on identifying patients with complex conditions using diagnosis code sets and SAS array processing.

The following steps lay out a conceptual as well as technical roadmap for the case study.

1. Search the **FacilityHeader**'s diagnosis code columns for codes related to certain conditions. As **FacilityHeader** contains up to eight diagnosis codes for each claim, this search is performed multiple times. A SAS array is used to automate and simplify the search. For a review of the structure of **FacilityHeader**, see Chapter 5.
2. Use a predetermined mapping between diagnosis codes and conditions.
3. Create a member-level data set with condition indicators. This table is used to identify members with complex multiple conditions.

Let us first begin with learning more about the array programming technique.

Code Simplification with SAS Array Processing

In this section you will learn how to repeat the same action on multiple variables using a SAS array. This programming technique has many applications in healthcare analysis. It has the added benefit of simplifying and reducing code. SAS Enterprise Guide's Query Builder or PROC SQL does not contain the capability built into ARRAY statement programming, a DATA step construct.

Program 8.1 shows you how to search for the CHF condition in eight diagnosis code columns without a SAS array. Please note that it is assumed that the diagnosis codes in the data do not contain the decimal point. This assumption should always be checked in any project.

Program 8.1: Searching for CHF without a SAS array

```
if pdx  in: ("4280") then CHF="CHF";
if dx_2 in: ("4280") then CHF="CHF";
if dx_3 in: ("4280") then CHF="CHF";
if dx_4 in: ("4280") then CHF="CHF";
if dx_5 in: ("4280") then CHF="CHF";
if dx_6 in: ("4280") then CHF="CHF";
if dx_7 in: ("4280") then CHF="CHF";
if dx_8 in: ("4280") then CHF="CHF";
```

The IN operator may be combined with the colon modifier to make pattern matching easier, efficient, and powerful. The colon modifier has several different uses in the SAS programming language. In Program 8.1, it is used to search for diagnosis codes beginning with "4280," eliminating the need for a complete list. The colon modifier in this context is read as "beginning with."

The code in Program 8.1 is repetitive, long, and difficult to maintain. Any change in diagnosis code matching requires manually changing eight lines of code, making the program prone to human error.

Using array processing, the code in Program 8.1 can be simplified. This is shown in Program 8.2.

Program 8.2: Using an array to simplify and reduce code

```
array diag(8)$ pdx dx_2 dx_3 dx_4 dx_5 dx_6 dx_7 dx_8;
do i=1 to 8;
 if diag(i) in: ("4280") then CHF="CHF";
end;
```

An ARRAY statement can be thought of as defining a container with "like" variables. Variables in an array can be either character or numeric, but not both. The diagnosis array in Program 8.2 has eight character variables.

The DO loop iterates through each of the variables in the diagnosis array and searches for diagnosis codes beginning with "4280"—in other words, any digit in position 5 or no digit. If a value of "4280" is found in the first four positions during any loop iteration, the value of the new variable CHF is set to "CHF."

Identifying Members with Complex Conditions

Program 8.3 shows you how to use FacilityHeader to derive MemberConditions, which is used to identify members with complex conditions.

Program 8.3: Identifying members with complex conditions

```
* Sort Data by MemberID for BY Group Processing;

proc sort data=book.facilityheader out=facilityheader;                ❶
    by Memberid;
run;

data MemberConditions;
    set facilityheader;
    * BY Group processing;
    by memberid;                                                      ❷
    length Diabetes CHF Depression COPD Asthma $10 Conditions $50;    ❸
    retain Diabetes CHF Depression COPD Asthma;                       ❹
    if first.memberid then do;                                        ❺
      Diabetes="";
      CHF="";
      Depression="";
      COPD="";
      Asthma="";
    end;
    array diag(8) pdx dx_2 dx_3 dx_4 dx_5 dx_6 dx_7 dx_8;             ❻
    do i=1 to 8;
      if diag(i) in: ("250","3572","3620","648","36641")
        then Diabetes="Diabetes";
      if diag(i) in: ("4280")
        then CHF="CHF";
      if diag(i) in: ("29620","29621","29622","29623","29624",
                      "29625","29630","29631","29632","29633",
                      "29634","29635","2980","3004","3091","311")
        then Depression="Depression";
      if diag(i) in: ("4912","4932","496","5064")
        then COPD="COPD";
      if diag(i) in: ("493")
        then Asthma="Asthma";
    end;
    if last.memberid then do;                                         ❼
      TotConditions = sum((Diabetes   ne ""),(CHF ne ""),
                          (Depression ne ""),(COPD ne ""),
                          (Asthma     ne "")
                         );                                           ❽
      Conditions=catx(", ",Diabetes,CHF,Depression,COPD,Asthma);      ❾
      output;                                                         ❿
```

```
        end;
        keep MemberID Sex DOB ToTConditions Conditions Diabetes CHF
             Depression COPD Asthma;
    run;
```

❶ PROC SORT sorts FacilityHeader by MemberID. Recall that FacilityHeader is at a claims-member level. To create a member-level data set, that is, one record per member, you need to collapse records of members with multiple claims. To do this, use BY group processing with respect to MemberID. This allows SAS to treat multiple records per member as a block. A prerequisite for BY group processing is that the source data is sorted appropriately.

❷ The BY statement activates BY group processing.

❸ The LENGTH statement sets up the length of the new variables. Without it, SAS determines the length of a variable through its first initialization, often leading to truncated values. This is known as the "rule of first use"; the LENGTH statement satisfies the rule.

❹ The RETAIN statement allows SAS to remember the value of a variable from one iteration of the DATA step to the next. Without it, the values of new variables are initiated to missing when a new observation is SET into the data step.

❺ The DO loop initializes the condition variables to missing if the observation being processed is the first in a MemberID BY group. This is needed to reset the new condition variables, whose values are being retained, when SAS processes the subsequent MemberID.

❻ An ARRAY and a DO loop search through eight diagnosis code variables to turn the condition variables "on." The search is carried out for every record in FacilityHeader. If a MemberID has multiple records, then the values of the condition variables are passed on (retained) to the next observation.

❼ If the observation being processed is the last in a MemberID BY group, then another DO loop is kicked off. This DO loop accomplishes steps 8 through 10.

❽ Using Boolean logic, TotConditions counts the non-missing occurrences of the condition variables. For a review on Boolean logic, see Chapter 4.

❾ Concatenates the condition variables using the CATX function. For a review of the CATX function, see Chapter 4.

❿ Places the record in the MemberConditions data set.

Figure 8.1 shows some member records from MemberConditions. Specific condition variables are not shown due to space constraints.

Figure 8.1: Select variables from MemberConditions data set

MemberID	Sex	DOB	Conditions	TotConditions
M000003703000...	F	1935-08-12	Diabetes	1
M000005330000...	F	1923-04-27		0
M000006362000...	F	1937-12-23		0
M000007172001...	F	1979-02-12		0
M000013309001...	F	1967-07-31	Diabetes	1
M000020770002...	F	1924-08-11	COPD, Asthma	2
M000021380002...	F	1966-05-22	Depression	1
M000022203002...	M	1946-04-23	Diabetes	1
M000022409002...	M	1961-08-25		0
M000024133002...	F	1919-05-05	Diabetes, Depression, COPD, Asthma	4

The structure of MemberConditions makes it easy to find patients or patient cohorts with certain conditions. To find patients with both Depression and Diabetes, simply look for members whose Depression and Diabetes variables are set to "Depression" and "Diabetes," respectively.

Parameterizing Program 8.3 with Macro Variables

Diagnosis codes contained in the search algorithms presented in Program 8.3 may need to be updated. This makes the entire program susceptible to errors, as the diagnosis codes are located in a central position. It is better to separate portions of code that require updates from other portions that are stable.

In this section, you create macro variables containing diagnosis codes to be used in search algorithms. These macro variables may be placed at the beginning so that any updates may be confined to a small section of the program.

Program 8.4: Diagnosis code searching via macro variables

```
%let Diabetes  = ("250","3572","3620","648","36641");
%let CHF       = ("4280");
%let Depression= ("29620","29621","29622","29623","29624","29625",
                  "29630","29631","29632","29633","29634","29635",
                  "2980","3004","3091","311");
%let COPD      = ("4912","4932","496","5064");
%let Asthma    = ("493");

data MemberConditions;
    set facilityheader;
    by memberid;
    length Diabetes CHF Depression COPD Asthma $10 Conditions $50;
    retain Diabetes CHF Depression COPD Asthma;
    if first.memberid then do;
      Diabetes="";
      CHF="";
      Depression="";
      COPD="";
      Asthma="";
    end;
    array diag(*) pdx dx_2 dx_3 dx_4 dx_5 dx_6 dx_7 dx_8;
    do i=1 to dim(diag);
      if diag(i) in: &Diabetes   then Diabetes="Diabetes";
      if diag(i) in: &CHF        then CHF="CHF";
      if diag(i) in: &Depression then Depression="Depression";
      if diag(i) in: &COPD       then COPD="COPD";
      if diag(i) in: &Asthma     then Asthma="Asthma";
    end;
    if last.memberid then do;
      TotConditions = sum((Diabetes ne ""),(CHF ne ""),
                          (Depression ne ""),(COPD ne ""),
                          (Asthma ne "")
                         );
      Conditions=catx(", ",Diabetes, CHF, Depression, COPD, Asthma);
      output;
    end;
    keep MemberID Sex DOB ToTConditions Conditions Diabetes CHF
         Depression COPD Asthma;
run;
```

By defining macro variables at the beginning, Program 8.4 protects the body of the code from unintentional changes when condition definitions (diagnosis codes) are updated. Also, the search algorithms in the body of the code have a much cleaner look, making the entire code more readable.

Case Study: Using Formats to Create Data Hygiene Routines

A primary goal for the analysts in the Data Warehouse department at Healthy Living, Inc., is to improve data quality. As part of this effort, they constantly scour the databases in order to search and report invalid data.

Chapter 2 showed you how to use SAS Enterprise Guide's **Create Formats** task to create formats based on code sets. One of these, the DX format, provides a translation from diagnosis codes to their descriptors. This case study shows you how the DX format can be enhanced to catch invalid codes.

ICD9 code sets and their descriptors can be downloaded from CMS (http://www.cms.gov/Medicare/Coding/ICD9ProviderDiagnosticCodes/codes.html). For the sake of simplicity, this code set has been converted into a SAS data set, BOOK.ICD9.

Figure 8.2 shows you a few records from BOOK.ICD9, which contains both diagnosis as well as surgical procedure codes.

Figure 8.2: BOOK.ICD9

Code	Short Nomenclature	Medium Nomenclature	Long Nomenclature
00	PROCEDURES AND I...	PROCEDURES AND INT...	Procedures and interve...
00.0	THERAPEUTIC ULTRA...	THERAPEUTIC ULTRAS...	Therapeutic ultrasound
00.01	THERAPEUTIC US VE...	THERAPEUTIC ULTRAS...	Therapeutic ultrasound...
00.02	THERAPEUTIC ULTRA...	THERAPEUTIC ULTRAS...	Therapeutic ultrasound...
00.03	THERAPEUTIC US PE...	THERAPEUTIC ULTRAS...	Therapeutic ultrasound...
00.09	OTHER THERAPEUTI...	OTHER THERAPEUTIC...	Other therapeutic ultra...
00.1	PHARMACEUTICALS	PHARMACEUTICALS	Pharmaceuticals
00.10	IMPLANTATION CHEM...	IMPLANTATION OF CHE...	Implantation of chemot...
00.11	INFUSION OF DROTR...	INFUSION OF DROTREC...	Infusion of drotrecogin...
00.12	ADMINISTRATION INH...	ADMINISTRATION OF IN...	Administration of inhale...
V91.10	TRIPLET GEST UNS #...	TRIPLT GEST UNS # PL...	Triplet gestation, unspe...
V91.11	TRIPLET GEST 2/> M...	TRIPLET GESTATION W/...	Triplet gestation, with t...
V91.12	TRIPLT GEST 2/> MO...	TRIPLET GESTATION W/...	Triplet gestation, with t...
V91.19	TRI GEST UNABL DET...	TRIPLT GEST UNABL DE...	Triplet gestation, unabl...
V91.2	QUADRUPLET GESTA...	QUADRUPLET GESTATI...	Quadruplet gestation pl...

The variables of interest in BOOK.ICD9 are Code and Short Nomenclature.

> The variables in BOOK.ICD9 have embedded spaces. This violates SAS naming conventions. This is acceptable in SAS Enterprise Guide tasks, but not in SAS code. To work with the variable "Short Nomenclature" in SAS code, simply refer to it as 'Short Nomenclature'n. For information related to this topic, see Chapter 6.

In this case study, you first work with the BOOK.ICD9 data set to create DxCntlin data set. Next, you create a new format called DxN using DxCntlin data set as a source. Finally, you use the new format on some test data to verify that it catches bad diagnosis codes.

Program 8.5 shows the code required to create DxCntlin data set.

Program 8.5: Create DxCntlin data set

```
data DxCntlin(keep=start label fmtname hlo);                    ❶
    set BOOK.ICD9(rename=(code=start)) end=last;                ❷
    length fmtname $8;
    label = propcase('Short Nomenclature'n);
    if (   index(start,'.')  = 4                                ❸
        or length(start)     = 3
```

```
          or substr(start,1,1) = 'E'
         )then do;                                                    ❹
                  fmtname = "$dxN";
                  start = compress(start,'.');
                  output dxCntlin;
              end;
      if last then do;                                                ❺
                     hlo='O';
                     fmtname = "$dxN";
                     label='** UNKNOWN DX CODE **';
                     output dxCntlin;
                  end;
run;
```

❶ As the DxCntlin data set is created so that it can serve as an input to a later PROC FORMAT, it requires at least three variables (Start, Label, and FmtName). The fourth variable, Hlo, is created to assist in catching bad data.

❷ The END= option associated with the SET statement keeps track of the end of the file. Last is a user-choice temporary variable (it is not part of the target data set) that is evaluated as "true" when SAS reaches the end of the file.

❸ Evaluates diagnosis codes for valid architecture. Codes are considered valid if one of three conditions is met: a decimal is found in the 4th position, it is 3 characters long, or it begins with the letter "E."

❹ If a diagnosis code is found to be valid by any one of the three criteria listed in (3) then the COMPRESS function squeezes the decimal from it and the record is written to the data set DxCntlin.

❺ When the last record has been processed, then Hlo is set to "O." Label is assigned the value of "** UNKNOWN DX CODE **." The last record is output again. Figure 8.3 shows the last few records in DxCntlin. Notice that the last two records have the same value of Start. The actual value of Start is ignored in the very last record because of the value of "O" (for Other) in Hlo.

Figure 8.3: DxCntlin data set

start	fmtname	label	hlo
V9110	$dxN	Triplet Gest Uns...	
V9111	$dxN	Triplet Gest 2/>...	
V9112	$dxN	Triplt Gest 2/> M...	
V9119	$dxN	Tri Gest Unabl D...	
V912	$dxN	Quadruplet Gest...	
V9120	$dxN	Quad Gest Uns #...	
V9121	$dxN	Quad Gest 2/> M...	
V9122	$dxN	Quad Gest 2/> M...	
V9129	$dxN	Quad Gst Unabl...	
V919	$dxN	Oth Spec Mx Ge...	
V9190	$dxN	Oth Spc Mx Gst...	
V9191	$dxN	Oth Mx Gest 2/>...	
V9192	$dxN	Oth Spec Mx Ge...	
V9199	$dxN	Oth Mx Gst Not...	
V9199	$dxN	** UNKNOWN D...	O

Program 8.6 loads DxCntlin into a format, creates a test data set, and verifies that the new format is catching bad codes.

Program 8.6: Create and verify DxN format

```
proc format library=work cntlin=dxCntlin;                    ❶
run;

data test;                                                   ❷
    infile cards dsd truncover;
    input ClaimID $ Pdx $ Dx2 $ Dx3 $;
cards;
1,0030,0031,0032
2,0031,0032,
3,0032,,
4,00320,,
5,ZZ222,Z555,
;
run;

data InvalidCodes(keep=ClaimID Error);
    set test;
    array dx (*) Pdx Dx:;                                    ❸
    do i = 1 to dim(dx);
      if dx(i) = " " then leave;                             ❹
      if put(dx(i),$dxN.) =: "**";                           ❺
      Error = catx(" ","Invalid diagnosis"
                      ,i,"for claim #",ClaimID
                      ,"-- Code is",Dx(i)
                  );                                         ❻
      output;
    end;
run;
```

❶ PROC FORMAT uses DxCntlin as input and creates DxN format in the WORK library. Although DxCntlin only has one format, there is no reason that it could not have more. In fact, it is efficient to have multiple formats in one source data set.

❷ Test data set is created with instream data using the CARDS technique. See Chapter 6 for a review on creating test data using SAS code. Note that the last record (ClaimID: 5) has two invalid diagnosis codes.

❸ ARRAY DX is created to loop through all diagnosis variables. Note that the use of the colon (begins with), which avoids listing Dx2 and Dx3 columns.

❹ The LEAVE statement forces SAS to exit the loop when it encounters the first missing diagnosis code of a ClaimID. This design presupposes that all diagnosis codes are filled left to right; when one is missing all subsequent ones will be missing.

❺ Keeps only records whose code matches its format beginning with "**." These are invalid codes.

❻ The CATX function strings together an error message, which is stored in the new variable Error. For review on this function, see Chapter 4.

Figure 8.4 shows the InvalidCodes data set, which contains two rows containing invalid codes with their identifying claim.

Figure 8.4: Invalid Codes Data set

ClaimID	Error
5	Invalid diagnosis 1 for claim # 5 -- Code is ZZ222
5	Invalid diagnosis 2 for claim # 5 -- Code is Z555

BOOK.ICD9 contains both diagnosis as well as surgical procedure codes. The case study can be modified to include creation of both diagnosis and surgical procedure codes formats. This extension is left as an exercise for the reader.

Conclusion

In this chapter you explored the definition and architecture of many of the code sets used in medical claims submissions. There is no limit as to how this knowledge might be put to work—from building data hygiene routines to constructing advanced filters. In the first case study, you learned how to use code sets and SAS arrays to select patients with complex conditions. In the second case study, you learned how to create formats to catch invalid data.

Chapter 9: The Members

Introduction 127
Member Demographics 128
Member Enrollment 129
Member Eligibility 130
Membership Issues of Interest 130
- Membership Maintenance 130
- Electronic Eligibility Inquiry 130
- Changing Member ID 131
- "Cross-Client" Projects 131
- Householding 131
- Member Months 131
- Continuous Enrollment 132

Rate Setting and Risk Adjustment 132
- Setting Rates 132
- Adjusting Risk 134

Case Study: Creating Member Months Data 134
- Creating a Callable Macro Program 135
- Member Months Macro Program 137
- Building a Member Months Table 139

Conclusion 141

Introduction

Previously in this book we have discussed the primary players—providers and payers—and the data they generate based on claims submission. But without the insured members there would be no submitted claims.

From the provider's perspective there are "patients" requiring services regardless of the payer involved. When a commercial or government payer is engaged in that provider/patient relationship, we then consider the patient as a "member" of that health plan (payer). In order to correctly adjudicate claims, provide adequate managed care programs, and manage benefit structures a healthcare insurer must maintain large amounts of information on their insured members. This data must be current, historic, accurate, and detailed in order to properly manage the business on behalf of their membership.

This chapter will discuss three components of a quality membership information system and the questions they are intended to answer:

1. Demographic information: Who is the member?
2. Enrollment information: What benefits are being provided?
3. Eligibility information: When are the benefits available?

In this chapter you will:

- Learn about demographic, eligibility, and enrollment data and the relationships among them
- Understand a few of the many issues surrounding membership data
- Use SAS macro programming to generate two useful versions of member months data.

Member Demographics

Member demographics form the core of the data for any health plan, allowing for the identification of an insured individual and his or her categorization for the purpose of analysis and reporting. So the question answered by demographic files is: *Who* is the member?

One does not have to stretch very far in order to understand the concept of membership demographic information—data that describes the characteristics of an individual insured. In most organizations this information is managed via multiple tables of a relational database management system (RDBMS). One table may contain name/sex/birthdate data, another physical address information, and yet another contact information. A unique "member identifier" ties this information together. This intelligent-free code resides on all data tables across the organization that need to be combined based on such a "key" value—member demographics, enrollment, eligibility, claims, and so on.

> HIPAA mandated the creation of four unique national identifiers—Employer, Health Plan, Provider, and Individual. To date, only the National Individual Identifier (NII) has not been implemented, primarily due to privacy concerns.

A data concept analogous to Member ID is that of Subscriber ID. A "subscriber" is the contract-, or policy-, holder who may have one or more dependents on the same contract. For group insurance, the subscriber is typically the employee, with the spouse and children as additional members on that contract. So a subscriber is a member, but a member may not be the subscriber. In many database designs, the Subscriber ID is a foreign key that joins to subscriber specific tables such as benefit enrollment tables—benefits are contract based. In some designs the Member ID may be constructed from the Subscriber ID incrementally suffixed—"00" for the subscriber, "01" for first dependent, "02" for the second dependent, and so on.

Member demographic information is data that is most often thought of in the context of privacy and confidentiality. Without getting into the details of the HIPAA Privacy Rule, suffice it to say that this rule defines two ways to de-identify information: 1) a formal determination by a qualified statistician, and 2) the removal of specified attributes of the individual (the Safe Harbor method). The Safe Harbor method of de-identification removes eighteen direct and indirect data elements from healthcare information. Many of these data elements (shown in Table 9.1) are found in the demographic membership files.

Table 9.1: De-identification elements

Names	Account numbers
Geographic subdivisions smaller than state	Certificate/license numbers
All elements of dates except year	Vehicle identifiers/serial numbers
Telephone numbers	Device identifiers/serial numbers
Fax numbers	URLs
E-mail addresses	IP addresses

Social Security numbers	Biometric identifiers
Medical record numbers	Full-face photographic images
Health plan beneficiary numbers	Any other unique identifying number, characteristic, or code

Claims analysts find it interesting that some minimal and sufficient information from membership demographics are often stored with the claim information due to frequency of use. This prevents us from having to join to membership frequently. Typically these fields include, but are not limited to, sex, birthdate, and zip code. Of course, the Member ID is always available as a join key to find other requisite member information as needed.

Member Enrollment

Member enrollment information answers two primary questions—*what* benefit plan (product) has the member chosen and *which* primary care physician (PCP) has been chosen, if one is required of the product.

It is important to note that typically enrollment precedes eligibility. The reader need only recall his or her own situation when first hired by a new employer. You were asked to pick a package of healthcare benefits (medical, dental, prescription drug, etc.) and were then instructed by the Human Resources Department as to when those benefits would be available to you. Information components of the enrollment process include:

- Employee/applicant demographic information
- Employment details—employer name and location, hire/rehire date, hours per week, and so on
- Reason for enrollment
- Change status (for annual open enrollment)
- Product choice—medical, dental, prescription drug, disability, and so on—usually in the form of a "pick list"
- List of all dependents to be added to the policy, along with their demographic information
- Prior coverage information (if any)

One notable exception to the rule that "enrollment precedes eligibility" is Medicare. For this product, eligibility is established (age, disability, end-stage renal disease [ESRD]) prior to enrollment. Enrollment is then the choice of the beneficiary.

"Annual open enrollment" is the concept that an insurer provides a periodic opportunity for its members to change their health benefits structure—usually upgrading or downgrading a benefit. There are situations that cause a subscriber to amend the policy's attributes outside of the open enrollment period. These "life events" include, but are not limited to, marriage, divorce, birth or adoption, birthday, address change, PCP change, or termination of employment. These latter change reasons may be made outside of an open enrollment period.

Any change in membership data (demographics, enrollment, and eligibility) will cause the date-stamping of the data records. Effective begin and end dates accompany all membership data. When a change is made, an effective end date is populated on the data record and a new record is created with an effective begin date (and open end date). In many organizations the historic record is pushed down in the table; other payers may roll the record off to another table. It is incumbent upon the healthcare data analyst to understand and work with "date-stamped" membership data.

All healthcare data has an associated date—enrollment/eligibility begin and end dates, hospital admit/discharge dates, date of service on claim service detail, claim paid date, prescription fill date, and so on.

Member Eligibility

Member eligibility information answers the primary question: *When* are the various benefit plan components available? It is important to remember that this and other membership information are typically in a highly normalized database structure given that the key variable, Member ID, is always available to join to additional information as required. Foreign keys such as Subscriber ID and Provider ID are defined for additional relationship definitions. For plans that require the choice of a PCP, the Provider ID is a useful link to the Provider database. Also, remember the importance of effective begin and end dates; these are the date fields that allow us to check for currency of a record or to pick a specific time frame for the data of interest.

Eligibility requirements for Medicare and Medicaid were discussed in Chapter 3; these are need-based programs with a somewhat fixed benefit offering. Commercial insurers (employer group and direct-pay), however, provide a variety of benefit structures. There is unlimited variability across the industry in terms of benefit eligibility. Employers and/or the insurer may dictate the rules based on factors such as product design, preexisting conditions, or sex/age combinations, to name a few. For example, as a new employee, your medical benefit may have a thirty-day waiting period, while the prescription drug benefit may not kick in for ninety days. If you are a diabetic, then despite the medical benefit being available after a thirty-day probationary period, diabetic-related services may not pay for sixty days. The Affordable Care Act (ACA) has changed the eligibility landscape, eradicating many if not all limitations to insurance coverage.

Membership Issues of Interest

Given the complexity of membership data, it is important to continually consider what issues may be facing any given healthcare data project. These issues are typically operational within a payer organization and may or may not be known a priori by the project analyst. Below we discuss some of the more common issues to consider, but beware of lingering issues. Always be suspect!

Membership Maintenance

Wherever membership data is stored, it must be carefully maintained for accuracy and currency. State Medicaid offices, CMS (Medicare), commercial insurers, and any intermediary or contractor for the aforementioned all have a critical responsibility. For commercial group insurers, this means working closely with their group's human resource departments to keep records current. Human resource professionals typically have a portal into their insurer for ease of maintenance. Close collaboration between the respective government office and the contractor involved is often required for Medicaid and Medicare recipients.

It is important for the project analyst to understand that membership maintenance departments are not infallible, and many of the data errors are due to operational deficiencies or simple human error. Careful and frequent data validation checks should be an integral part of any project. Data hygiene routines should be built into every effort with functionality to alert those charged with membership data quality. We have the technology to perform such functionality, and we certainly have the mandate to improve data quality. A simple e-mail sent by such a data validation routine to the membership maintenance department will go a long way toward achieving this goal.

Electronic Eligibility Inquiry

We have mentioned the HIPAA-mandated electronic claim formats (837I and 837P) as examples of administrative simplification, but they are also a significant benefit to the quality and quantity of information being collected by payers. In the realm of membership data, an additional set of electronic transaction formats were implemented—Benefits Inquiry and Response: 270 and 271 record formats, respectively. From the provider perspective, this is a significant benefit, as it allows for the determination of eligibility and enrollment

of a patient/member prior to service delivery. A clear example in practice is the pharmacist who determines whether she/he will get reimbursed by a payer long before the prescription is dispensed and delivered.

The key criteria for the 270/271 transaction are Member ID (as found on the patient's insurance card), last name, first name, sex, and date of birth. If any of this key information cannot be validated by the payer upon provider inquiry (the 270 record), then the payer response (the 271 record) will not be positive. So the importance of quality assurance in membership maintenance is readily apparent.

Changing Member ID

The importance of a unique member identifier within a payer organization cannot be emphasized enough. You have only to run into a lack of uniqueness in your project to fully understand the importance of this concept. A corresponding issue that is often overlooked is that a member's identifier may change over time. New Member IDs may be assigned when employees get laid off and brought back, when employees change employers but not insurance carriers, or when a member's product line changes.

The issue for the analyst is to be able to link old IDs to current IDs so that what appears to be a different individual can accurately be determined to be the same person. A good example is the answer to the question, "What is the average cost per member in 2014?" If we find *n* number of Member IDs, do we necessarily have *n* members, or might there be (*n*-*y*) members because some individuals had a reassigned Member ID in 2014? The problem solver is a cross-reference table that links old Member IDs with current Member IDs. It is assumed that all payers maintain a member cross-reference table. With some simple SQL and the use of such a table, we can begin to mitigate the problem of changing member identifiers.

"Cross-Client" Projects

Bringing together multiple payer organizational data is sometimes referred to as a "cross-client" project. Combining Pennsylvania and Ohio Medicaid data for a reporting project or blending all of Maryland's commercial insurer claims for analytical purposes can become problematic on several levels. Of interest to us in this context is that there is no assurance that Member IDs are now unique. Mutual exclusivity can be assumed within a payer organization, but not across multiple payer organizations.

There are several solutions, but recognition of the problem is a good start. The simplest solution might be to prefix the old Member ID with a three-character Payer ID to create a new and unique Member ID for use in the project. More challenging would be to design a business rule that would use member demographics (e.g., name, sex, and birthdate) to arrive at a contrived Member ID that could then be encrypted to de-identify the member. This method is potentially risky due to name changes from marriage and divorce or multiple systems recording spelled names differently.

The final resolution to this issue would be for the HIPAA mandate of a Unique Individual Identifier (UII) to be implemented. Keep an eye to the future for this important solution to be announced.

Householding

Many healthcare reporting and analytic projects need to consider the concept of "householding"—the grouping of individuals by household attribute or relationship patterns. When we report utilization or cost by household, how do we define such a denominator?

A "medical household" would be defined by the Subscriber ID and any linked members. They may not live in the same domicile, but they would share a benefit structure and, by extension, claim utilization patterns. On the other hand, a "physical household" would be defined by physical address (e.g., street, city, and state) whether they share a benefit structure or not. These are not difficult definitions, just project-based business rules that access membership data in specific ways.

Member Months

Rather than counting actual individuals in a health plan, since membership is not static, payers often count "member months" as a basis for comparative statistics. This is simply a date arithmetic programming task

against the enrollment and/or the eligibility files—remembering that each record is date-stamped with an effective begin and end date.

The definition of a member month varies across organizations but typically is defined by a business rule such as being enrolled on the 1st, the 15th, or any day of a month gets the member credit for the entire month. So, considering an annual member month file with Member ID and twelve columns, each monthly column will be a binomial (true or false) indicating credit for the month. Using this information, we can then sum the number of member months in a calendar year, divide by twelve, and arrive at the number of "covered lives" for the year—a covered life being a "virtual member." The number of covered lives is always less than or equal to the number of real lives.

Many utilization measures—Cost per Member per Month, Admits per 1000 Covered Lives, and Inpatient Days per Member per Month—are calculated with member months as the common denominator. The key with this type of calculation is that the numerator (e.g., cost, admits, days) *must* be derived from the same time frame as the denominator. A secondary issue, but nonetheless important, is that it is best to use a twelve-month period so as to erase the seasonality effect seen in the healthcare industry.

Continuous Enrollment

Often projects will have business rules that determine the inclusion of a member's claim information based on some minimum amount of time that a member needs to be enrolled in a product line. This concept is generally known as "continuous enrollment" (CE) criteria and ensures that a claimant's enrollment period is sufficient to meet the requirements of the project. For example, you cannot be expected to find a cholesterol management service in the claims data if the member was not continuously enrolled for the study period. CE rules certainly define the requisite time frame but may also stipulate allowed gaps (number and duration) in enrollment and specified, fixed dates (anchor dates) of enrollment.

The specific criteria for CE may change from project to project but always require that member-specific enrollment records be interrogated for proper alignment of begin and end dates—another date arithmetic exercise with which SAS is particularly proficient. The challenge when spinning through large volumes of membership data is to find and deal with duplicate records, overlapping time frames, and changing product lines within the required time frame. These anomalies may be clear data errors or operational exceptions; enrollment records are never as clean as we expect. The challenge is how to remove anomalous records and aggregate overlapping records into a single coverage span with gap days and number of gaps calculated.

Rate Setting and Risk Adjustment

A health insurance premium is the monthly amount you (individual policy) or your employer (group insurance) pays to an insurance company. The insurer then collects the premiums from all of its subscribers for the purpose of paying member-incurred medical claims, paying operational overhead, and building a reasonable reserve or profit.

Both historic and current rate information is typically stored with membership data. In general, an insurer will calculate a base rate for a particular segment of their membership and store a multiplier at the member level to be applied to that base. These "rate cells" account for family size, relative medical risk, and other factors unique to a benefit structure.

Setting Rates

Prior to the ACA, federal law did not place any limits on the way that insurers may set premium rates or how much of a premium dollar must be spent on claims reimbursement. It has generally been left to the states and their insurance departments to set such rules. Table 9.2 lists some of the more common rating practices.

Table 9.2—Pre-ACA Common Rating Practices

Practice	Definition	Example
Health status rating	Known medical conditions drive premiums up; used specifically for new policies where experience is not available; typically based on a survey methodology	Diabetics are assessed a higher premium.
Gender rating	Gender affects premium rates; used for new and renewing policies	A woman of child-bearing age is rated higher than a male of the same age.
Age rating	Differing age bands affect rates; used for new and renewing policies	Older adults cost more than children.
Geographic rating	Zip code of member influences the rate; used for new and renewing policies	Higher medical cost areas such as inner cities
Industry rating	Standard Industry Code (SIC) is used in rate setting; used for new and renewing policies	Coal miners are at higher risk than sales clerks.
Experience rating	Premiums are relative to claims history; used for renewing policies	Higher cost per-member-per-month generates high renewal premiums.
Community rating	All individuals in the "community" pay the same rate regardless of other factors	Communities may be defined as an association of small employer groups in the same industry.

The ACA stipulated that, beginning January 1, 2014, insurance companies must meet minimum premium rating rules for individual and group health plans. The new rating rules set a floor for compliance by health plans. States can enact a tougher standard than that of the ACA. Health plans will be allowed to adjust premiums based only on the following factors:

- Individual versus family enrollment: Health plans will be allowed to vary rates based on who is enrolled in the plan. In other words, different rates may be charged based on whether the plan covers only an individual or a family.
- Geographic area: Health plans will be allowed to charge more in areas where medical costs are high.
- Age: Health plans will be allowed to vary rates based on age, but by no more than a factor of three.
- Tobacco use: Health plans will be allowed to charge more for people that use tobacco products, but by no more than a factor of 1.5.

Health status, gender rating, and experience rating will no longer be allowed under the ACA. It is felt that certain rating practices encourage healthier people to buy coverage while discouraging the sick from doing so.

To assure that premium dollars are spent on health care services, the ACA requires a minimum amount of all premium dollars collected by insurance companies (85 percent for the large group and 80 percent for the small group market) to be spent on healthcare services. The intent of the law is to ensure that health plans keep their administrative costs and profits within reason.

Adjusting Risk

Prior to the ACA, insurers had broad discretion to deny coverage based on medical condition. Insurers also were allowed to charge higher premiums for riskier individuals and groups, although some states legislated a limit on the premium differentials.

With the ACA, price differences based on health status and medical history are not allowed. There are now strict limits on rate differentials between high- and low-risk individuals. The one exception to this general rule is that lower premium rates can be offered for employee participation in wellness programs. In essence, medical underwriting has been eliminated. Without getting into the details of the law, suffice it to say that the ACA's risk adjustment program is intended to reinforce the prohibition of risk selection by insurers. Under this new federal program, risk adjustment is accomplished by transferring funds from plans with lower-risk enrollees to plans with higher-risk enrollees.

Under the Department of Health and Human Services (HHS) methodology, Medicare's individual risk scores are based on an individual's age, sex, and diagnoses and are assigned to each enrollee. Given that only a medical risk is being assessed, diagnosis information only is used and not treatment (procedural) information. Diagnoses are grouped into a Hierarchical Condition Category (HCC) and assigned a numeric value that represents the relative expenditures a plan is likely to incur for an enrollee with a given category of medical diagnosis. Individual risk scores are then averaged across the health plan to arrive at the plan's average risk score—the plan's predicted expenses. Under this HHS Medicare methodology, adjustments are made for a variety of factors, including variations in actuarial value, allowable rating methodologies, and geographic cost. Under risk adjustment, plans with a relatively low average risk score will make payments into the system, while plans with relatively high average risk scores will receive payments.

The two major goals of the ACA are expansion of coverage and premium regulation—insurance reform. It contains few cost containment measures applicable to the commercial healthcare market. Expanding the marketplace and regulating rates cannot create affordable coverage by themselves. The industry needs effective ways to reduce the prevalence and cost of healthcare services.

Case Study: Creating Member Months Data

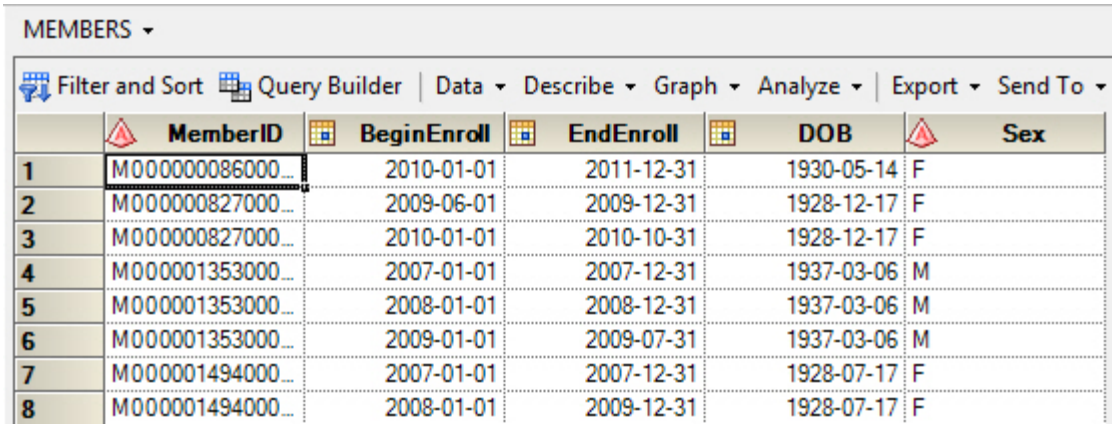
The Finance Department at Healthy Living, Inc., tracks a number of metrics, including costs, visits, and admits. The metrics, indicators of the company's financial health and viability, are shared internally and presented periodically to the company's board of directors as well as to its auditors.

Each metric is represented as the ratio of two numbers. The numerator is usually costs, visits, or admits for a specified period of time, usually twelve months. The denominator is the number of "members" in the *same* time period. For example, Inpatient Costs per Member per Month (PMPM) is the ratio between inpatient costs for members in a twelve-month period and the total number of members eligible in each of the twelve months.

This case study shows you how to compute member months from the BOOK.Members data set. You will learn to calculate key financial metrics in the case study in Chapter 10.

Figure 9.1 shows a few records from BOOK.Members.

Figure 9.1: BOOK.Members table



MEMBERS

Filter and Sort | Query Builder | Data | Describe | Graph | Analyze | Export | Send To

	MemberID	BeginEnroll	EndEnroll	DOB	Sex
1	M000000086000...	2010-01-01	2011-12-31	1930-05-14	F
2	M000000827000...	2009-06-01	2009-12-31	1928-12-17	F
3	M000000827000...	2010-01-01	2010-10-31	1928-12-17	F
4	M000001353000...	2007-01-01	2007-12-31	1937-03-06	M
5	M000001353000...	2008-01-01	2008-12-31	1937-03-06	M
6	M000001353000...	2009-01-01	2009-07-31	1937-03-06	M
7	M000001494000...	2007-01-01	2007-12-31	1928-07-17	F
8	M000001494000...	2008-01-01	2009-12-31	1928-07-17	F

To create member months from BOOK.Members, expand each member's row by the number of relevant months. For example, a member who was enrolled for a twelve-month period will have twelve rows, each indicating the month she was a member. Data may be expanded by using a DO loop within the SAS DATA step. See Chapter 7 for a review on data expansion code.

A member months data file can quickly become very large. Consider an insurance company with 500,000 members. If each of the members has been with the company for an average of five years, the member months data set for this company comprises 30 million records (500,000 x 5 x 12). Joining utilization data with a large file can lead to increased processing times as well as greater memory consumption. To ameliorate both issues, this case study creates a flexible member months macro routine that allows you to achieve the following:

- Generate a member months file for a population cohort of interest for a specified period of time.
- Produce an alternate member months file displaying member months as additional columns instead of rows.

The macro routine approach is efficient for smaller member cohorts. The approach may not be as effective for larger member populations. Analysts working for a large health plan may want to use a combination approach. When working with a smaller member cohort, they may utilize the macro program to generate a member months file. When dealing with larger member populations, they may use stored member month data.

SAS implementation varies by company. For example, a data set taking forty-five seconds to process at one site may take ten seconds at another installation. It is up to the analyst to find the happy medium between using a macro routine for smaller member cohorts and a stored member months file for larger populations.

A large company may choose to store member months data separated by year. A macro program may be used to control multiple file generation, thus avoiding repetitive code.

Creating a Callable Macro Program

Before plunging into the member months macro program, let us take the time to construct a simple callable macro program. In prior chapters, you have been introduced to elements of macro programing. Specifically, you have learned how to use macro variables to create flexible code.

A callable macro program is one that usually resides as an external program and can be accessed from any program.

Program 9.1 shows a macro program that selects members on the basis of specified enrollment date parameters.

Program 9.1: A callable macro program

```
%macro SelectMembers(beg=,end=);
data Member;
    set BOOK.Members;
    if BeginEnroll > "&end"d or EndEnroll < "&beg"d
      then delete;
run;
%mend SelectMembers;
```

In Program 9.1, a DATA step is sandwiched between two macro wrapper statements. This is similar to the ODS wrapper statements covered in Chapter 7. The %MACRO statement, the opening of the macro sandwich, defines a macro program by giving it a name and specifying its parameters. The macro SelectMembers has two parameters or macro variables, BEG and END, respectively. The %MEND statement closes the macro sandwich.

Program 9.1 may be saved in a file and then called from another file. The easiest way to call an external macro program is to use a %INCLUDE statement, which simply retrieves and executes SAS code from the specified location. The result is that the macro is compiled and ready for use. You can then invoke (call) the macro with the desired values of the parameters. Both the %INCLUDE statement as well as macro invocation is shown in Program 9.2.

Program 9.2: Macro inclusion and invocation

```
options symbolgen;
%include "C:\HealthCare Book\Macro programs\Program 9.1.sas"/source2;
%SelectMembers(beg=01JAN2011,end=31DEC2011)
```

The SYMBOLGEN system option shows the resolution of macro variables BEG and END in the log. The SOURCE2 option in the %INCLUDE statement shows in the log the expanded SAS code being "included" and executed. The %SelectMembers statement invokes the macro program with the desired values of BEG and END.

Figure 9.2 shows the log from Program 9.2.

Figure 9.2: SYMBOLGEN and macro invocation in the LOG file

```
17         %include "C:\HealthCare Book\Macro programs\Program 9.1.sas"/source2;
NOTE: %INCLUDE (level 1) file C:\HealthCare Book\Macro programs\Program 9.1.sas is file
      C:\HealthCare Book\Macro programs\Program 9.1.sas.
18         +
19         +%macro SelectMembers(beg=,end=);
20         +data Member;
21         +set BOOK.Members;
22         +if BeginEnroll > "&end"d or EndEnroll < "&beg"d then delete;
23         +run;
24         +%mend SelectMembers;
NOTE: %INCLUDE (level 1) ending.
25         %SelectMembers(beg=01JAN2011,end=31DEC2011)
SYMBOLGEN:  Macro variable END resolves to 31DEC2011
SYMBOLGEN:  Macro variable BEG resolves to 01JAN2011

NOTE: There were 37812 observations read from the data set BOOK.MEMBERS.
NOTE: The data set WORK.MEMBER has 11478 observations and 5 variables.
NOTE: DATA statement used (Total process time):
      real time           0.04 seconds
      cpu time            0.01 seconds
```

If you designate a folder structure in which to save your company's macro programs, then there is an easier option to invoke these myriad macro programs. You can specify the path of the folder(s) where the macro programs reside in the SASAUTOS environmental variable. Then SAS will automatically know where to look for macro programs and you can invoke macro programs without first having to "include" these in your code.

Since SASAUTOS is found in your configuration files, of which there may be many, it is best to leave this change to your SAS administrator.

SYMBOLGEN is a useful macro debugging option. Other options include MPRINT, which prints out SAS statements after macro variables have been resolved, and MLOGIC, which shows the resolution of any iterative macro code logic. It is best to turn these options off once your program is in production in order to avoid large LOG files.

Member Months Macro Program

Program 9.3 shows you the code contained in the member months macro program.

Program 9.3: Member months macro program

```
%macro MM(beg=,end=,inputds=);

data member;
    set &inputds;                                                    ❶
    if BeginEnroll > "&end"d or EndEnroll < "&beg"d then delete;     ❷
    NewBeg = max("&beg"d,BeginEnroll);                               ❸
    NewEnd = min("&end"d,EndEnroll);                                 ❹
    Months = intck('MONTHS',NewBeg,NewEnd) + 1;                      ❺
    format NewBeg NewEnd yymmdd10.;
run;

proc sort data=member;
    by MemberID;
run;

data MMTemp;
    set member;
    keep MemberID date sex DOB;
    format date yymmdd10.;
    do i=1 to months;                                                ❻
      date=intnx('MONTHS',NewBeg,i-1,'B');                           ❼
      output;
    end;
run;

proc sql;                                                            ❽
    create table MM as
    select distinct MemberID, date, sex, DOB
      from MMTemp;
quit;

/* Optional code for deleting intermediate datasets                  ❾

proc datasets library=WORK;
    delete MMTemp Member;
run;

   End of optional code */

%mend MM;
```

```
* Invoke macro;
%MM(beg=01JAN2011,end=31DEC2011,inputds=BOOK.Members)
```

Let us explain macro MM with respect to its invocation, given in the last line of Program 9.3. The last parameter of the macro allows you to specify a member cohort.

❶ Allows you to specify an input data set containing members for whom you want member months data. The input data set can be either permanent or temporary.

❷ Removes members whose enrollment does not span any time segment in 2011. This statement contains two logical conditions. If a member enrolled after December 31, 2011, then the member is removed because she is too new to count. If a member left before January 1, 2011, then the member is also not counted.

❸ Since we are only interested in member months from 2011, NewBeg takes the most recent of either January 1, 2011, or a member's BeginEnroll date.

❹ Following similar logic as defined in ❸, NewEnd takes the most historic of either December 31, 2011, or a member's EndEnroll date.

❺ The INTCK function counts the number of months between NewBeg and NewEnd and adds one to the result. Because of ❷ and ❸, you ensure that the value of Months cannot exceed twelve.

❻ The DO loop iterates from 1 to the value specified in Months.

❼ The INTNX function increments a date value by an interval specified in its first argument. As the first argument is 'MONTHS,' the function adds months to NewBeg and places the value in a new column called Date. The last argument in the INTNX function is optional. 'B' indicates that the beginning value of the month span should be taken. This is done for the purpose of standardization and consistency. If a member enrolled on February 15, 2011, the value for Date in the first iteration of the loop is February 1, 2011. In this manner, the member gets credit if they are eligible on any day of the month.

❽ Removes duplicate records for a member-date combination. Duplicate records occur due to the nature of membership enrollment data, which may contain multiple enrollment records or even overlapping enrollment records. The potentially anomalous nature of the data is handled by the routine without judgment as to data quality.

❾ Deletes unneeded data sets as necessary. The code showing PROC DATASETS, a data management procedure, is commented out because it is good to first check whether your macro is working correctly.

Consider the effect of the MM macro for the first observation shown in Figure 9.1. Member ID M0000000860000100 enrolled on January 1, 2010, and left on December 31, 2011. As Figure 9.3 shows, this MemberID has twelve records in the MM table.

Figure 9.3: MM table

	MemberID	date	Sex	DOB
1	M00000008600000100	2011-01-01	F	1930-05-14
2	M00000008600000100	2011-02-01	F	1930-05-14
3	M00000008600000100	2011-03-01	F	1930-05-14
4	M00000008600000100	2011-04-01	F	1930-05-14
5	M00000008600000100	2011-05-01	F	1930-05-14
6	M00000008600000100	2011-06-01	F	1930-05-14
7	M00000008600000100	2011-07-01	F	1930-05-14
8	M00000008600000100	2011-08-01	F	1930-05-14
9	M00000008600000100	2011-09-01	F	1930-05-14
10	M00000008600000100	2011-10-01	F	1930-05-14
11	M00000008600000100	2011-11-01	F	1930-05-14
12	M00000008600000100	2011-12-01	F	1930-05-14

The MM macro has two key strengths. First, it allows the analyst to feed a data set containing members of her choice. Second, it enables her to specify her own time period. These features avoid costly (in terms of memory and time) joins with a stored member months file.

Building a Member Months Table

As Chapter 10 will show, the MM data set shown in Figure 9.3 is perfect for the analysts in the Finance Department at Healthy Living, Inc., because they can now join this data set with utilization data to compute financial metrics. However, the analysts find member months counts useful in themselves as they help track the company's membership.

This case study shows how the MM macro may be enhanced to generate a table in which member months are represented as columns instead of rows.

Program 9.4 shows the entirety of the code in MM2 macro.

Program 9.4: MM2 macro

```
%macro MM2(yr=,inputds=);

%let beg = 01JAN&yr;                                                    ❶
%let end = 31DEC&yr;

proc sort data=&inputds(where=(not(   BeginEnroll > "&end"d             ❷
                                  or EndEnroll < "&beg"d  ))
                          keep = MemberID Sex DOB BeginEnroll EndEnroll
                         )
           out=temp_mm;
    by MemberID BeginEnroll;
run;

data mmtable(drop=j k BeginEnroll EndEnroll);
    retain MemberID Sex DOB jan&yr feb&yr mar&yr apr&yr may&yr jun&yr   ❸
                            jul&yr aug&yr sep&yr oct&yr nov&yr dec&yr;
    array mbrmon (12) jan&yr feb&yr mar&yr apr&yr may&yr jun&yr         ❹
                      jul&yr aug&yr sep&yr oct&yr nov&yr dec&yr;
    set temp_mm;
    by MemberID;

    if first.MemberID                                                   ❺
      then do j = 1 to dim(mbrmon);
              mbrmon(j) = 0;        *** initialize the array to 0 ***;
           end;
```

```
      do k =    month(max("&beg"d,BeginEnroll))
             to month(min("&end"d,EndEnroll  )) ;                    ❻
         mbrmon(k) = 1;
      end;

      if last.MemberID then do;                                      ❼
                                totmm = sum(of jan&yr. -- dec&yr.);
                                output mmtable;
                             end;
run;

%mend MM2;

%MM2(yr=2011,inputds=BOOK.Members);
```

❶ Because it makes sense to show member months for the entire year as columns with a binomial response, the MM2 macro has one date parameter (YR) instead of two (BEG and END). As BEG and END macro variables are required by the macro, they are recreated with the help of two %LET statements. The reduction in the number of macro parameters makes the macro a little more user friendly. This design assumes a calendar year member month table will be built.

❷ We start the routine with a PROC SORT to bring in the source data, sort by MemberID and BeginEnroll, and store the new version to a temporary target data set. Given that we only need observations that are within our year of interest, the WHERE data set option is used with the appropriate expression defined. Also, we only need a finite set of variables for the task ahead so the KEEP= data set option is also employed. These two techniques assist in removing excess volumes of data.

❸ The RETAIN statement is key to the design. Retaining MemberID, Sex, and DOB seems redundant as they come from an existing data set, but the use here is simply a technique to guarantee position on the target data set. The new variables need to have their values retained (unless a new value is assigned) in memory so that upon output to the target data set we have the desired result.

❹ The ARRAY statement defines our twelve new variables as numeric.

❺ This DATA step has BY MemberID group processing. This means that it treats multiple observations belonging to the same MemberID as a single group. When the first observation in a MemberID BY-group is incurred, the arrayed variables are initialized to zero, or "false"; the member is not enrolled for that month.

❻ For every observation, this DO loop executes for each month in the range of possible eligibility dates in the year. The "Kth" element of the array is then set to one, meaning "true"; the member is enrolled for the month.

❼ When the last observation in the BY-group is incurred, the total number of member months is derived and the observation is output to the target data set.

It is easier to explain the workings of the DATA step if we follow a member's data from source, through interim, to target data sets. Figures 9.4, 9.5, and 9.6 show MemberID M0000014940000700 as she appears in the SAS data sets BOOK.Members, Temp_MM, and MMTable respectively.

Figure 9.4: M0000014940000700 in BOOK.Members

MemberID	BeginEnroll	EndEnroll	DOB	Sex
M000001494000...	2007-01-01	2007-12-31	1928-07-17	F
M000001494000...	2008-01-01	2009-12-31	1928-07-17	F
M000001494000...	2010-01-01	2010-10-31	1928-07-17	F
M000001494000...	2010-11-01	2010-11-30	1928-07-17	F
M000001494000...	2010-12-01	2011-02-28	1928-07-17	F
M000001494000...	2011-03-01	2011-03-31	1928-07-17	F

Figure 9.5: M0000014940000700 in Temp_MM

MemberID	BeginEnroll	EndEnroll	DOB	Sex
M000001494000...	2010-12-01	2011-02-28	1928-07-17	F
M000001494000...	2011-03-01	2011-03-31	1928-07-17	F

Figure 9.6: M0000014940000700 in MMTable (selected columns shown)

MemberID	Sex	DOB	jan2011	feb2011	mar2011	apr2011	may2011
M000001494000...	F	1928-07-17	1	1	1	0	0

The columnar representation of member months shown in Figure 9.6 is useful in several respects. First, it is easy to manipulate this data in Excel, the tool of choice of many business analysts. Second, it is straightforward to sum its columns and derive member months counts by month or year. Finally, the totmm column in the data set provides an easy way to check for continuous eligibility.

Conclusion

This chapter has discussed at length the three components of membership data: demographic, enrollment, and eligibility. In addition, it has shown you how to build SAS macro programs to generate member months files in two different formats.

Chapter 10: Computing and Tracking Financial Metrics

Introduction.....143
Case Study: Bucketing Costs.....143
Case Study: Calculating PMPM Costs.....148
Case Study: Creating Reports.....149
Conclusion 152

Introduction

Healthcare insurance companies are interested in figuring out where and how they are spending their money. With this in mind, the Finance/Actuarial departments of most, if not all, healthcare insurance companies compute, track, and report on key financial metrics. These include but are not limited to inpatient, outpatient, and primary care costs calculated on a Per Member Per Month (PMPM) basis.

Most healthcare insurance companies also track key utilization metrics including, but not limited to, ER visits, primary care visits, specialists visits, and inpatient admits. These visits are usually reported not as PMPM but as Per 1,000 Members Per Year, where "members per year" are based on twelve member months being a member or covered life.

In this chapter you will use SAS Enterprise Guide to:

- Create code to bucket costs in two different ways.
- Integrate member months data with cost data to create PMPM cost metrics.
- Create reports visualizing cost metrics.

Case Study: Bucketing Costs

The day-to-day function of the Finance department at Healthy Living, Inc., is to pay providers for services rendered for the company's members. It is also responsible for analyzing and trending costs. The department generates financial reports for the benefit of the senior management, the company's board of directors, and auditors.

To analyze costs, the Finance department has created the following two levels of cost bucketing.

- Who billed it? Category 1 captures the billing entity, such as "the hospital" or "the primary care physician."
- Where was it billed? Category 2 is based on the location of the services provided. Services provided in an inpatient setting have a value of "inpatient." Other values of Category 2 may include "outpatient," "ER," and "office."

Although cost bucketing is universally applied across the industry, its flavor varies with each payer. A company with a largely Medicare population may wish to break out "nursing home" as a value in Category 1. Another company with a commercial plan population may include pediatricians and obstetricians in its definition of primary care. Many companies also have a third level of cost bucketing. This third way of looking at cost categories is based on the services provided. Depending on setting, such categorizations may be based on DRG, CPT, or Revenue Code; the service bill codes.

This case study shows you how to create service categories.

Program 10.1 shows how service categories are created with Facility data.

Program 10.1: Service categories with Facility data

```
data facility;                                                              ❶
    length Category1 Category2 $30;
    set book.facilityheader;

    BillType_1 = substr(BillType,1,1);
    BillType_2 = substr(BillType,2,1);

        if BillType_1 = '1' then Category1 = 'Hospital';
   else if BillType_1 = '2' then Category1 = 'SNF';
                            else Category1 = 'Other Facility';

  if Category1 = 'Hospital'
    then do;
              if BillType_2 in ('1','2') then Category2 = 'Inpatient';
         else if BillType_2 = '3'        then Category2 = 'Outpatient';
                                         else Category2 = 'Hospital - Other';
        end;
    else Category2 = Category1;
run;

proc sort data=facility;
    by claimid;
run;

* Identify ER;

proc sort data=book.facilitydetail
          out=ERClaims(keep=claimid) nodupkey ;                             ❷
    by claimid;
    where substr(RevCode,1,3) = '045';
run;

data facility2;
     merge facility(in=in1) ERClaims(in=in2);                               ❸
     by claimid;
     if in1 or in2;
     if in2 then do;                                                        ❹
               if Category2 = 'Inpatient'  then Category2 = 'ED Inpatient';
          else if Category2 = 'Outpatient' then Category2 = 'ED Outpatient';
     end;
run;
```

❶ DATA step parses BillType to create Category1 and Category2 at the Claim header level.

❷ To differentiate ER claims, you need to examine Claim detail data and isolate those claim lines having a RevCode beginning with "045." A claim is considered to be an ER claim if one or more of its lines has an ER related RevCode. The SORT procedure isolates these claims in ERClaims data set.

❸ Uses the IN= option to create temporary SAS variables to track the source of the record while combining Facility and ERClaims data sets into Facility2.

❹ Recodes the values of Category2 if the record is from ERClaims.

Program 10.2 creates service categories from Professional data.

Program 10.2: Service categories with Professional data

```
* Professional Claims;                                                    ❶
data Professional;
    set book.Professionaldetail;
    length Category1 Category2 $30;
    select;
           when (    SrvcProvTaxonomy =: '20'
                 and substr(SrvcProvTaxonomy,3,2) in('7Q', '7V', '8D', '80'))
           Category1 = 'Primary Care';
           when (    SrvcProvTaxonomy =: '20'
                 and Category1 ne 'Primary Care' )
           Category1 = 'Other Physician';
           when (SrvcProvTaxonomy in:('29', '33', '34') )
           Category1 = 'Ancillary';
           otherwise Category1 = 'Other Professional';
       end;
    select;
           when (POS = '21')          Category2 = 'Inpatient';
           when (POS = '22')          Category2 = 'Outpatient';
           when (POS in('11','12'))   Category2 = 'Office/Home';
           otherwise                  Category2 = 'Other Setting';
       end;
run;

proc sql;                                                                  ❷
    create table professional2 as
    select a.*
          ,b.memberid
      from professional               a
          ,book.professionalheader    b
     where a.claimid=b.claimid;
quit;

proc sort data=professional2;                                              ❸
    by memberid;
run;

proc sort data=facility2;                                                  ❹
    by memberid;
run;

data AllClaims(keep=MemberID ClaimID Category1
                    Category2 AmtPaid Month   );                           ❺
    set facility2 professional2(rename=(LineAmtPaid=AmtPaid));
    by memberid;
    Month = put(coalesce(BeginDOS,AdmitDt),yymmn6.);                       ❻
run;
```

❶ Professional claims are placed in Category1 and Category2 using detail rather than header data. SELECT/WHEN/OTHERWISE statements, using provider taxonomy (SrvProvTaxonomy) and place of service (POS), are an alternative to the IF/THEN/DO coding construct. For a review on provider taxonomy architecture, see Chapter 4.

❷ Joins data set created in step ❶ to BOOK.Professionalheader in order to bring MemberID onto the result set.

❸ Sorts Professional2 data set so that it may be merged.

❹ Sorts Facility2 data set created in Program 10.1 for merging.

❺ AllClaims data set merges Professional2 and Facility2 data sets.

❻ Creates variable Month by using a combination of COALESCE and PUT functions. This is useful when joining the data to the member month data set in the next section.

Figure 10.1 shows you some records from AllClaims data set.

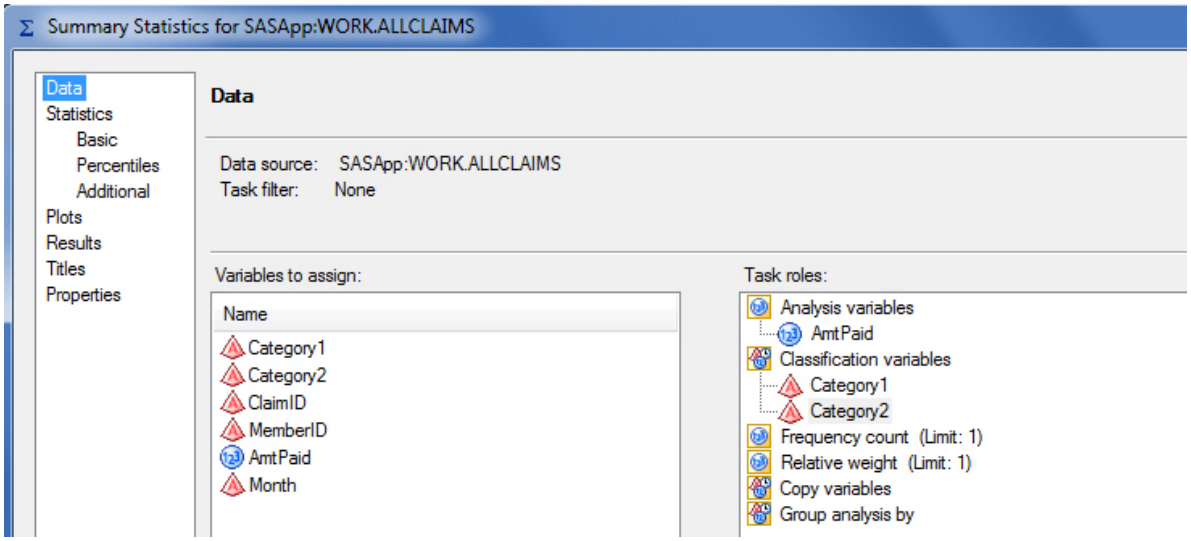
Figure 10.1: AllClaims data set

Category1	Category2	ClaimID	MemberID	AmtPaid	Month
SNF	SNF	C091557872	M000006362000...	11780.95	201101
Primary Care	Other Setting	C077376596	M000006362000...	84.76	201102
Primary Care	Other Setting	C077364138	M000006367000...	63.00	201101
Ancillary	Office/Home	C077390044	M000006367000...	10.94	201103
Ancillary	Office/Home	C077390044	M000006367000...	6.36	201103
Primary Care	Office/Home	C077390044	M000006367000...	7.28	201103
Primary Care	Office/Home	C077390044	M000006367000...	7.44	201103
Ancillary	Office/Home	C077390044	M000006367000...	11.91	201103
Ancillary	Office/Home	C077390044	M000006367000...	13.66	201103
Other Physician	Inpatient	C056070089	M000007168001...	210.26	201101
Other Physician	Inpatient	C056070089	M000007168001...	210.26	201101
Other Physician	Inpatient	C056070089	M000007168001...	96.75	201101
Other Physician	Inpatient	C056166070	M000007168001...	8.60	201101
Hospital	Outpatient	C096026471	M000007172001...	4420.69	201103

Let us verify that all claims have been bucketed successfully. This can be done with a variety of SAS Enterprise Guide tasks.

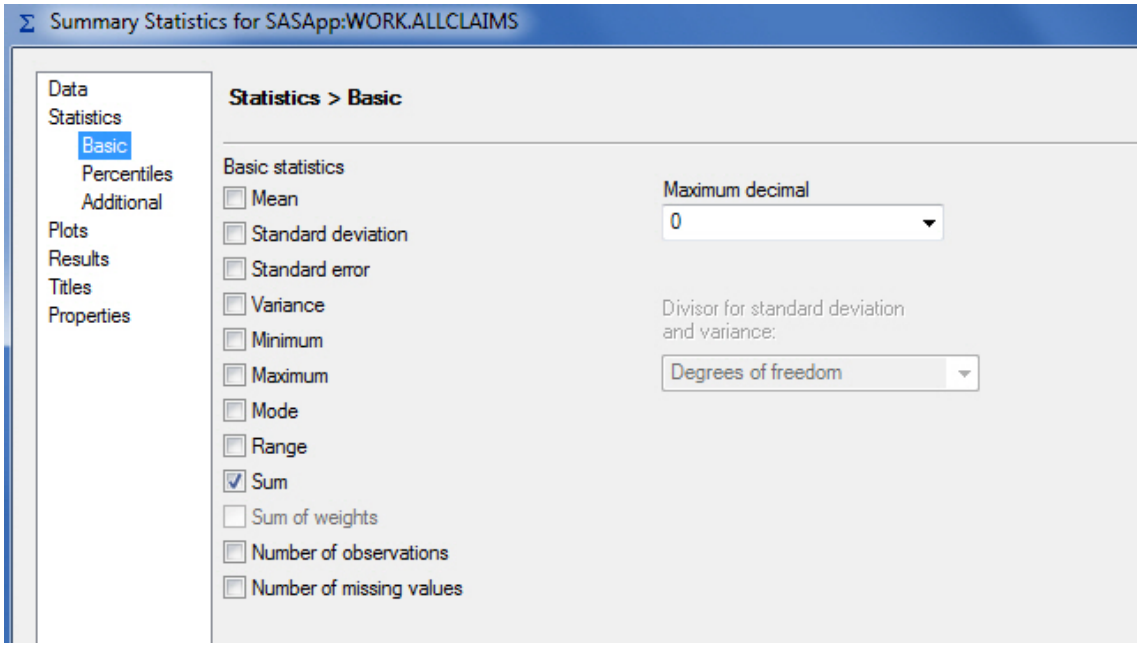
1. While viewing AllClaims data set, select **Summary Statistics** from **Tasks—Describe**.
2. Move AmtPaid to **Analysis variables** and Category1 and Category2 to **Classification variables** as shown in Figure 10.2.

Figure 10.2: Assigning data in summary statistics task



3. In **Statistics-Basic** tab, uncheck all but the **Sum** statistic. Choose 0 under the **Maximum decimal** dropdown. This is shown in Figure 10.3.

Figure 10.3: Statistics-basic tab



4. Verify that claims have been bucketed successfully in Figure 10.4.

Figure 10.4: Sum AmtPaid by Category1 and Category2

The MEANS Procedure

Analysis Variable : AmtPaid Claim Paid Amount		
Category1	Category2	Sum
Ancillary	Inpatient	62863
	Office/Home	523386
	Other Setting	671565
	Outpatient	34795
Hospital	ED Inpatient	18069481
	ED Outpatient	1016140
	Inpatient	22634103
	Outpatient	9403031
Other Facility	Other Facility	1731441
Other Physician	Inpatient	3866720
	Office/Home	694362
	Other Setting	383810
	Outpatient	1004555
Other Professional	Inpatient	47074
	Office/Home	826930
	Other Setting	144341
	Outpatient	48783
Primary Care	Inpatient	1127102
	Office/Home	574959
	Other Setting	489510
	Outpatient	125047
SNF	SNF	3043246

The tasks found in **Tasks—Describe** are not only useful for data checking, but also for reporting. As an extension of this case study, the reader is encouraged to test **Summary Tables** and **Table Analysis** tasks.

Case Study: Calculating PMPM Costs

Though informative, the costs shown in Figure 10.4 do not tell the whole story. A healthcare insurance company may experience increasing costs over time for two reasons. One may be that membership is growing (good), and another might be that cost per each member is increasing (bad). This section shows you how to standardize costs with respect to members.

Program 10.3 shows you how to marry costs and membership data.

Program 10.3: PMPM costs

```
* Invoke Member Months Macro;
%MM(beg=01JAN2011,end=31DEC2011,inputds=BOOK.Members);                    ❶

proc sql;
    create table mm2 as                                                   ❷
    select put(date,yymmn6.) as Month
          ,count(*) as members
      from mm
     group by Month;

    create table allclaims2 as                                            ❸
    select Month, Category1, Category2
          ,sum(AmtPaid) as AmtPaid
      from AllClaims
     group by Month, Category1, Category2;

    create table metrics as                                               ❹
    select a.Month, a.Category1, a.Category2
          ,(a.AmtPaid/b.members) as PaidPMPM
      from allclaims2 a, mm2 b
     where a.month=b.month;
quit;
```

❶ Invoke the member months macro created in Chapter 9.

❷ Create a data set containing the number of members in each month. The variable Month is given a format YYMMN6.

❸ Summarize costs by Month, Category1, and Category2.

❹ Join the data sets created in steps ❷ and ❸ to create the Metrics data set. Create PaidPMPM, a ratio of AmtPaid for members for a certain month divided by the total number of members in the same month.

Figure 10.5 shows a few records from Metrics data set.

Figure 10.5: The Metrics data set

Month	Category1	Category2	PaidPMPM
201101	Ancillary	Inpatient	1.6511244503
201101	Ancillary	Office/Home	11.580501605
201101	Ancillary	Other Setting	20.299239273
201101	Ancillary	Outpatient	1.2886580292
201101	Hospital	ED Inpatient	830.00789289
201101	Hospital	ED Outpatient	26.199073533
201101	Hospital	Inpatient	985.76126767
201101	Hospital	Outpatient	345.35024112
201101	Other Facility	Other Facility	100.66722603
201101	Other Physician	Inpatient	102.98478189
201101	Other Physician	Office/Home	17.890682277
201101	Other Physician	Other Setting	8.0553773921
201101	Other Physician	Outpatient	17.771025793
201101	Other Profession...	Inpatient	1.5881148223
201101	Other Profession...	Office/Home	29.68682634
201101	Other Profession...	Other Setting	10.611602282
201101	Other Profession...	Outpatient	1.7166159515
201101	Primary Care	Inpatient	25.600868893
201101	Primary Care	Office/Home	8.1520468323
201101	Primary Care	Other Setting	15.917597765
201101	Primary Care	Outpatient	3.3321264709
201101	SNF	SNF	289.8721279

Case Study: Creating Reports

There are many ways to visually represent the PaidPMPM metric shown in the Metrics data set. The following three are the most obvious choices.

- Line plots for key combinations of Category1 and Category2. PaidPMPM is placed on the Y axis and Months constitutes the X axis.
- Bar charts for important combinations of Category1 and Category2, with Months as a cluster variable.
- An overall table composed of several tables, each representing a different view of the metrics.

As both line plots and bar charts have been covered in previous chapters, this case study shows you how create a tabular representation with PROC TABULATE.

SAS offers two primary reporting procedures, PROC REPORT and PROC TABULATE. PROC REPORT is best at creating highly customized reports and is widely used in the healthcare industry, especially by pharmaceutical companies. PROC TABULATE excels at combining several mini-tables into one super table, as the case study shows.

Summary Tables in SAS Enterprise Guide's **Tasks—Describe** allows limited access to PROC TABULATE's functionality. Similarly, **List Report Wizard**, also available from **Tasks—Describe**, gives you a flavor of PROC REPORT. If the tasks do not accomplish what you need, do not despair. Teach yourself to program in these procedures, and you will soon be developing highly sophisticated reports.

PROC TABULATE is best learned in small bits rather than in large increments. Program 10.4 gets you started with PROC TABULATE.

Program 10.4: Creating Basic Reports with PROC TABULATE

```
proc tabulate data=Metrics;
    class Category1 Category2;
    var PaidPMPM;
    table category1 category2 category1*category2, PaidPMPM;
run;
```

The CLASS statement is used to list classification or categorical variables. Similarly, the VAR statement's purpose is to list numeric or analysis variables. The key statement in PROC TABULATE is the TABLE statement, which creates the table. A comma separating variables creates a new dimension. A blank space between variables creates stacks of tables in the appropriate dimension.

The TABLE statement in Program 10.4 has one comma. This means that the table is two-dimensional. Variables to the left of the comma appear as the rows of the table whereas variables to the right of the comma comprise the table's columns.

The row dimension of the TABLE statement in Program 10.4 consists of three variables that are separated by a blank space. This creates a stack of three tables. The first table shows Category 1. The second table displays Category2. The third table shows a combination of the two; the asterisk between Category1 and Category2 creates a nesting.

PaidPMPM is listed to the right of the comma. Therefore, it constitutes the sole column of each of the three stacked tables. As a default, numeric variables are summed. This is inappropriate as an average is required and is fixed in Program 10.5.

Figure 10.6 shows the table created by Program 10.4.

Figure 10.6: Creating basic tables with PROC TABULATE (partial output)

		PaidPMPM
		Sum
Category1		
Ancillary		150.88
Hospital		5990.45
Other Facility		203.41
Other Physician		694.64
Other Professional		124.94
Primary Care		270.25
SNF		360.19
Category2		
ED Inpatient		2119.15
ED Outpatient		118.60
Inpatient		3248.86
Office/Home		305.68
Other Facility		203.41
Other Setting		197.47
Outpatient		1241.40
SNF		360.19
Category1	Category2	
Ancillary	Inpatient	7.34
	Office/Home	61.00
	Other Setting	78.46
	Outpatient	4.07
Hospital	ED Inpatient	2119.15
	ED Outpatient	118.60
	Inpatient	2652.80
	Outpatient	1099.90

Program 10.5 enhances the table in Figure 10.6.

Program 10.5: Creating tables with PROC TABULATE

```
proc tabulate data=Metrics;
    class Category1 Category2;
    var PaidPMPM;
    table category1 all='Category1 Total'
          category2 all='Category2 Total'
          category1*category2  ,  PaidPMPM*MEAN=''*format=comma12.0
                                  PaidPMPM*pctsum='Percent'*format=Comma12.2
         /row=float;
run;
```

The ALL= option adds totals to Category1 and Category2 mini-tables and labels it "Total." The FORMAT= option specifies a format for PaidPMPM. The PaidPMPM variable is used twice in the column dimension. In its first use, PaidPMPM is averaged, as denoted to the MEAN statistic, and given a blank label. In its second use, a percent of total is computed with the PCTSUM statistic, and a label of "Percent' is given.

Figure 10.7 shows the table created by Program 10.5.

Figure 10.7: Creating tables with PROC TABULATE (partial output)

		PaidPMPM	PaidPMPM Percent
Category1			
Ancillary		13	1.94
Hospital		499	76.85
Other Facility		68	2.61
Other Physician		58	8.91
Other Professional		10	1.60
Primary Care		23	3.47
SNF		180	4.62
Category1 Total		120	100.00
Category2			
ED Inpatient		706	27.19
ED Outpatient		40	1.52
Inpatient		217	41.68
Office/Home		25	3.92
Other Facility		68	2.61
Other Setting		16	2.53
Outpatient		83	15.93
SNF		180	4.62
Category2 Total		120	100.00
Category1	Category2		
Ancillary	Inpatient	2	0.09
	Office/Home	20	0.78
	Other Setting	26	1.01
	Outpatient	1	0.05
Hospital	ED Inpatient	706	27.19
	ED Outpatient	40	1.52
	Inpatient	884	34.03
	Outpatient	367	14.11

Conclusion

In this chapter you used SAS Enterprise Guide to group costs into service categories, join cost data with membership information, and then report financial cost metrics.

Chapter 11: HEDIS

Introduction 153
The Business Case 154
The Technical Challenges 154
Reporting System Components 155
Colorectal Cancer Screening 156
- Definition 156
- Eligible Population 156
- Exclusions to Eligibility 157
- Compliant Population 157
- Hybrid Specification 158

Case Study: Developing a Rate for Colorectal Cancer Screening 158
- Create a Driver Table 159
- Clean Up of Membership Data 159
- Check Continuous Enrollment 162
- Identify the Denominator 165
- Determine Compliant Population 172
- Compute Rate 173

Conclusion 174

Introduction

In today's ever-changing healthcare landscape, organizations are being required to demonstrate quality across the health care delivery continuum. One early result of this mandate is that 90 percent of America's health plans use HEDIS (Healthcare Effectiveness Data and Information Set) as a tool to measure and reliably compare performance on many important dimensions of care. HEDIS is also one component upon which Medicare bases its Stars ratings—another tool for gauging performance across health plans.

HEDIS is a registered trademark of the National Committee for Quality Assurance (NCQA), a not-for-profit organization dedicated to measuring and improving health care quality since 1990. Through a consensus-building approach with employers, policymakers, providers, patients, and health plans, NCQA has developed a nationally accepted methodology to determine the importance of, measurement of, and promotion of healthcare system improvements.

The intent of this chapter is not to teach HEDIS (that is another book), but rather to introduce the reader to the notion of performance reporting. In addition, it is a great opportunity to look at a tangible industry application that ties together much of what this book is all about.

In this chapter you will:

- Learn what HEDIS is and why it exists.
- Understand various components of a HEDIS performance measure.

- Learn about the data-driven approach to storing and using diagnoses and procedural codes associated with a HEDIS measure.
- Gain in-depth knowledge of the business rules and logic for the Colorectal Cancer Measure.
- Learn SAS code needed to compute the Colorectal Cancer Measure.
- Understand the Continuous Enrollment Macro, which is useful in computing many HEDIS measures.

The Business Case

The origin and evolution of HEDIS, the acronym, tells the story of the importance of the business concept. In the early 1990s, employers were looking for a way to best compare HMOs (the early managed care model) toward the goal of purchasing the most cost-effective health insurance—hence, HEDIS stood for "HMO Employer Data and Information Set." The next iteration (1993) addressed the variety of managed care plans becoming available by changing the "H" to "Health Plan." Version 3.0 of HEDIS (1997) further changed the acronym to replace "Employer" with "Effectiveness," expressing the concept that HEDIS has gone beyond just the needs of the employers.

In essence, HEDIS is an integrated system for reporting performance and quality in the managed care industry; it is a report card for health care delivery and management. As we all know, a good "report card" provides benefits for the "student." In the case of health plans, a good performance report provides accreditation by NCQA, positioning a Plan well in the market place and, potentially, selling more business. HEDIS is voluntary. Health plans do not have to take part, but the benefits outweigh the significant annual cost of participating.

> Monitoring and reporting on performance quality provide benefit. This is an application of the Heisenberg Principle—the very observation of something changes its nature.

The Technical Challenges

As with most large computer-based reporting systems, HEDIS is not without technical challenges. Given that administrative healthcare data can be characterized by its volume, velocity, variety, variability, and complexity, we can begin to see that building and maintaining such a system is not for the faint of heart.

- **Volume**: Depending on the size of the health plan, the volume of data can be astronomical. A seemingly simple business rule like "as far back in the member's claim history as possible" can present processing problems. Typically health plans keep a rolling three years of claims data. This can preclude the use of possibly valuable data. A good programmer/analyst will always be cognizant of the data volume issue and find the most efficient and effective methodologies.
- **Velocity**: Data is online faster. With electronic claims submission formats and improved processing systems, claims complete faster and become available to reporting processes sooner.
- **Variety**: Improved data streams are increasingly able to provide a larger assortment of data. Improved electronic formats and external data sources (e.g., automated laboratory vendor data feeds) are adding to our data warehouses at an unprecedented rate.
- **Variability**: Combining data sources, as we often must do, causes programmer angst when mapping variable sources to a common architecture. Data scientists and data stewards are in increasing demand as health plans address the variety and variability of administrative healthcare data.
- **Complexity**: If nothing else, administrative healthcare data is complex. Originating as financial data (a billed service), the systems analyst must transform it into meaningful clinical data. Today's business rules ask more and more of an already multifaceted system of data.

In addition to the data challenges, reporting system design issues abound. Because the HEDIS business rules are so voluminous and provided by NCQA in tabular form, a data-driven design is recommended. Keeping the rules outside of the reporting system software in a "driver" table provides several benefits. Annual system

maintenance is lessened when minimizing hard-coded rules. "What if" analysis is easily done by using a different driver table.

> A driver table is a data table that contains information that "drives" the contents of the resulting set when joining to (or merging with) the data of interest. It generally contains the key variables necessary to affect the join and other variables (or rules) that provide the data-filtering criteria.

Accessibility and actionability should be guiding principles in any system design. Toward this goal, interim results should be captured. It is one thing to produce the reportable performance score; it is another to be able to show how you got there. Quality control and disease management teams want to see how operational improvements can be made. If they cannot access the information, they cannot take meaningful action.

Reporting System Components

So, what are the principal components of a HEDIS reporting system as defined by NCQA? Considering the grade school report card analogy alluded to above, a "measure" would be history, geography, or mathematics, and the "rate" would be an A, B, or C. Health plans report rates for many different measures.

As with any robust reporting system, HEDIS is a detailed road map toward a uniquely defined measure of health and wellness performance by the managed care organizations. Stepping through the business rules efficiently and effectively provides much information and knowledge to a payer organization. Table 11.1 further describes the elements of HEDIS reporting that comprise this "road map."

Table 11.1: Major components of a HEDIS measure

Component	Description
Product Line	Health plans are required to report commercial insurance, Medicaid, and Medicare separately. This is partially due to the distinct nature of these populations and their use of the healthcare system. Not all health insurers manage all product lines.
Eligible Population	Every measure uniquely defines an eligible population that then serves as the denominator in the rate calculation(s). "Population-based" measures define eligibility solely through membership attributes. "Event-driven" measures start with a medical event (from claims data) and then further refines the member's eligibility through membership attributes.
Medical Event	Event-driven measures start with a defined medical event that initially defines the eligible members in the denominator. An example is a cardiac event (heart attack, open heart surgery, etc.) that initializes the eligible population for the measure Cholesterol Management for Patients with Cardiovascular Conditions (CMC). These members are then further defined as eligible or not.
Rendered Services	Once the denominator (the eligible population) is determined, the numerator is then established by looking in the claims data for qualifying services that were rendered for the eligible members.
Optional Exclusions	In some measures, it is possible that an otherwise eligible member (denominator) could never have acquired the requisite service to qualify for the numerator. In these recognized cases NCQA defines exclusion criteria that can only improve a plan's score.
Value Set Data (VSD)	These are the "business rules" that define the excludable events, rendered services, and other measure specific guidelines. NCQA defines the rules and distributes them

Component	Description
	to users as an Excel workbook to use as a driver table.

HEDIS rates are reported by the plans to NCQA in June of each year. So, the "reporting year" is the year of that June report. The "measure year" is the calendar year previous to the reporting year. For example, "HEDIS 2014" reports on data from the 2013 measurement year. Many of the measures consider years prior to the measure year as the time frame for the member to have acquired a compliant service.

Colorectal Cancer Screening

To round out our understanding of HEDIS performance measurement, let's consider one of the more straightforward measures, Colorectal Cancer Screening (COL). The importance of this measure is not to be understated. Colorectal cancer is the third most commonly diagnosed cancer (excluding skin cancer) in the United States. It is the third leading cause of cancer-related deaths when men and women are considered separately and the second leading cause when both sexes are combined. Purchasers of health insurance would no doubt prefer a plan with high scores on this measure. The death rate for this cancer has been steadily dropping in recent years, due in no small part to the increased emphasis on screening by the provider community and savvy health plans looking to trim the cost of chronic disease.

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Definition

The rate for COL is simply the percentage of members 50 to 75 years of age who have had appropriate screening for colorectal cancer. A simple, prosaic goal is a great place to get started on any healthcare analytic or reporting project. It then becomes a "white board" exercise to break down and understand what it will take to reach the goal.

Eligible Population

Constructing the denominator for the rate calculation is the first step. For this measure, it is a matter of processing membership data to find all those members that satisfy the following criteria:

- **Product line**: Commercial or Medicare; each is reported with a separate rate. The Medicaid population is not reported for this measure.
- **Age**: Between fifty-one and seventy-five years of age as of December 31 of the measurement year. Note the apparent discrepancy in this age band and the prosaic definition above. Clinical guidelines state that adults greater than or equal to fifty years of age should be screened for colorectal cancer. The description portion in the specifications document describes the general age groups included in the measure. However, to take into account the look-back period for the measure, the age criterion has been adjusted. For example, members who were fifty as of December 31 were only forty-nine during the measurement year and did not require a screening. The age criteria specified in the eligible population requirements should be used to report HEDIS measures.
- **Continuous enrollment**: The measurement year and the year prior to the measurement year.
- **Allowable gap**: No more than one gap in continuous enrollment of up to forty-five days during each year of continuous enrollment.
- **Anchor date**: December 31 of the measurement year. The member must be enrolled on this date.
- **Benefit**: Medical.

- **Event/diagnosis**: None.

COL is an example of a population-driven measure, as there is no requirement that a medical event be incurred by the member to be included in the denominator. Once these members are found, they become the basis for further work with claims data.

Exclusions to Eligibility

Some HEDIS measures allow health plans the option to exclude members from the denominator, and hence the numerator, based on a medical condition or situation that suggests that they would never be found in the measure year with a numerator compliant event.

For the COL measure, a member may be excluded based on a claims history of either having a diagnosed colorectal cancer or a total colectomy. In fact, looking as far back in the member's claims history as possible for these two events is allowed. NCQA defines these two excludable events in their Value Set Data (VSD) as detailed in Table 11.2. The reporting program will select from claims data, any records containing codes found in the VSD.

Table 11.2: Exclusion criteria

Value Set	CPT	HCPCS	ICD9-PCS	ICD9-CM	Claims year
Colorectal Cancer		G0213–G0215 G0231		153 153.0–153.9 154 154.1 197.5 V10.05	all
Total Colectomy	44150–44153 44155–44158 44210–44212		45.8 45.81–45.83		all

Compliant Population

Having constructed the denominator (with exclusions considered) as the eligible population, the numerator may then be determined by finding evidence of one or more screenings for colorectal cancer for this population in the administrative claims data. The three screening tests of interest are fecal occult blood test (FOBT), flexible sigmoidoscopy, or colonoscopy.

NCQA defines these three screening tests in their Value Set Data as detailed in Table 11.3. The claims code sets can be recognized for their data origin, providing the reader with the notion that detailed business rules should always suggest the data source and content. Looking within the specified time frame of claims data for any one occurrence of the code values would place that patient/member in compliance with the measure and increase the plan's rate for this measure.

Table 11.3: Numerator compliance criteria

Value Set	CPT	HCPCS	ICD9-PCS	LOINC	Claims Year
FOBT	82270 82274	G0328		12503-9 12504-7 14563-1 14564-9 14565-6 2335-8 27396-1 27401-9 27925-7 27926-5 29771-3 56490-6 56491-4 57905-2 58453-2	Measure Year
Flexible Sigmoidoscopy	45330–45335 45337–45342 45345	G0104	45.24		MsrYr + 4 prior yrs.
Colonoscopy	44388–44394 44397 45355 45378–45387 45391 45392	G0105 G0121	45.22 45.23 45.25 45.42 45.43		MsrYr + 9 prior yrs.

Notice that the look-back period for compliant events (or exclusionary events) can be quite lengthy and seemingly at odds with the typical "three years of online claims data." The savvy health plan will devise a strategy to maintain this interim information each year that it spins though the limited claims data. Maintaining a relatively small table of excludable or compliant members that can be interrogated each reporting season will mitigate the issue of not being able to look back more than three years.

Hybrid Specification

Some of the HEDIS measures will go beyond the "administrative" methodology discussed above and allow the health plan to check the member's medical record to see if a compliant event occurred without creating an administrative claim record. This is known as the "hybrid" methodology.

Without getting too deep in the weeds of this method, suffice it to say that a systematic sample is drawn from the eligible population (the denominator from the administrative method) for each product line. This sample of members is then used as a basis for medical record review. This resource-intensive process of physically looking in the provider's records is apparently worth the effort, as rates are most often improved by finding compliant or exclusionary events.

Case Study: Developing a Rate for Colorectal Cancer Screening

Let's now convert the business knowledge, rules, and logic presented thus far in this chapter to SAS code required to compute the COL rate for Healthy Living, Inc. Because the SAS code is lengthy and at times complex, it is presented in a series of steps, outlined below.

1. Create a Driver table through the IMPORT procedure or the DATA step.
2. Clean up and manage membership data (Program 11.1).
3. Run the Continuous Enrollment macro to define the denominator for rate computation (Program 11.2).
4. Apply exclusions to the denominator (Program 11.3).
5. Determine the compliant population (Program 11.4).
6. Compute the rate (Program 11.5).

Steps 2, 3, and 4 compute the denominator. Step 5 creates the numerator, and step 6 calculates the rate for COL.

Create a Driver Table

Let's begin by creating a permanent SAS data set in the BOOK library that captures the information from Tables 11.2 and 11.3. This is a good practice, as the HEDIS VSD may change; any changes are easier to manage in a table than in SAS code.

The information from Tables 11.2 and 11.3 is placed in an Excel spreadsheet after simple rearrangement and then imported as a data set using SAS Enterprise Guide. For a review of this technique, see Chapter 6. Figure 11.1 shows a limited view of the driver table for the COL measure. The records are chosen at random to show how the table is organized.

Figure 11.1: BOOK.ColDriver

ValueSet	Value Set Version	CodeType	CodeValue
Colonoscopy	01OCT2013	CPT	45391
Flexible Sigmoidoscopy	01OCT2013	CPT	45333
Colorectal Cancer	01OCT2013	ICD9CM	153.5
FOBT	01OCT2013	LOINC	27401-9
Colorectal Cancer	01OCT2013	ICD9CM	153.6
Colorectal Cancer	01OCT2013	HCPCS	G0214

BOOK.ColDriver is an instrumental table that plays a key role in creating exclusions to the denominator and in determining the numerator. It is important to note that the CodeValue column contains embedded decimals for pertinent values of CodeType. As decimals have been removed from our claims data, we will need to strip decimals from CodeValue when joining with claims data.

Clean Up of Membership Data

BOOK.Members contains enrollment data for members of Healthy Living, Inc. However, some records have data quality issues; two are shown in Figure 11.2. This is not an unusual occurrence for many payers. Always consider whether you will need to manipulate data in preparation for its use.

Figure 11.2: Data quality issues in BOOK.Members

	MemberID	BeginEnroll	EndEnroll	Overlap
1	M0002576630194200	2006-04-01	2008-12-31	
2	M0002576630194200	2009-01-01	2010-12-31	
3	M0002576630194200	2010-12-01	2011-01-31	P
4	M0002576630194200	2011-02-01	2011-12-31	
5	M0004102910263700	2007-01-01	2007-12-31	
6	M0004102910263700	2008-01-01	2008-06-30	
7	M0004102910263700	2008-05-01	2008-12-31	P
8	M0004102910263700	2008-07-01	2008-08-31	C
9	M0004102910263700	2008-09-01	2008-10-31	
10	M0004102910263700	2008-11-01	2009-12-31	
11	M0004102910263700	2010-01-01	2010-09-30	
12	M0004102910263700	2010-10-01	2010-10-31	
13	M0004102910263700	2010-11-01	2011-12-31	
14	M0004465340272800	2008-01-01	2008-07-31	
15	M0004465340272800	2008-05-01	2008-12-31	P
16	M0004465340272800	2008-08-01	2008-08-31	C
17	M0004465340272800	2008-09-01	2009-02-28	
18	M0004465340272800	2009-03-01	2009-12-31	
19	M0004465340272800	2010-01-01	2011-01-31	
20	M0004465340272800	2011-02-01	2011-02-28	
21	M0004465340272800	2011-03-01	2011-12-31	
22	M0020762040749600	2009-06-01	2009-06-30	
23	M0020762040749600	2009-07-01	2009-07-31	
24	M0020762040749600	2009-08-01	2009-12-31	
25	M0020762040749600	2010-01-01	2010-12-31	
26	M0020762040749600	2011-01-01	2011-12-31	
27	M0020762040749600	2011-03-01	2011-05-31	C

Members' enrollment records may overlap either partially or completely, as shown by column **Overlap** in Figure 11.2. For example, Member M0002576630194200's enrollment record spanning December 01, 2010, to January 31, 2011, overlaps partially with his enrollment record covering January 1, 2009, to December 31, 2010. Similarly, Member M0020762040749600's enrollment record shown in observation 27 in Figure 11.2 is completely spanned by his enrollment record shown in observation 26. Also notice a potential for collapsing the data in BOOK.Members. Observations 1 and 2 in Figure 11.2 can be summarized by one enrollment record spanning April 1, 2006, to December 31, 2010.

It is extremely difficult to build an algorithm to check for continuous enrollment from data with overlapping enrollment periods. This is why it is important to take care of these data issues before checking for continuous enrollment. Reducing the membership data to unique enrollment segments is also beneficial for future processing. Program 11.1 shows SAS code that "fixes" partially overlapping enrollment records, deletes those that overlap completely, and collapses contiguous enrollment records.

Program 11.1: Pre-process BOOK.Members

```
proc sort data=book.Members out=Members nodupkey;                    ❶
    by MemberID BeginEnroll EndEnroll;
run;

data Members_Clean(drop=prior_: temp_: orig_:);
    retain MemberID temp_beg temp_end;                               ❷
    format Prior_End Orig_BeginEnroll Orig_EndEnroll
           temp_beg temp_end BeginEnroll EndEnroll yymmdd10.;
```

```
      set members(rename=(BeginEnroll=Orig_BeginEnroll
                          EndEnroll  =Orig_EndEnroll  ));          ❸
      by memberid Orig_BeginEnroll Orig_EndEnroll;

      if Orig_BeginEnroll > Orig_EndEnroll then delete;            ❹
      Prior_End = lag(Orig_EndEnroll);                             ❺

           if first.memberid
             then do;                                              ❻
                     temp_beg  = Orig_BeginEnroll;
                     temp_end  = Orig_EndEnroll;
                     Prior_End = .;
                  end;
      else if 0 <= (Orig_BeginEnroll - Prior_End) <= 1
             then do;                                              ❼
                     temp_end = Orig_EndEnroll;
                  end;
      else if (Orig_BeginEnroll - temp_end) >  1
             then do;                                              ❽
                     BeginEnroll = temp_beg;
                     EndEnroll   = temp_end;
                     output;                                       ❾
                     temp_beg = Orig_BeginEnroll;                  ❿
                     temp_end = Orig_EndEnroll;
                  end;
      else if temp_beg <= Orig_BeginEnroll <= temp_end
             then do;                                              ❶*
                     if Orig_EndEnroll > temp_end
                       then temp_end = Orig_EndEnroll;             ❷*
                  end;
     if last.memberid then do;                                     ❸*
                             BeginEnroll = temp_beg;
                             if Orig_EndEnroll > temp_end
                               then EndEnroll = Orig_EndEnroll;
                               else EndEnroll = temp_end;
                             output;
                          end;
run;
```

The keystone of Program 11.1 is a practical use of the LAG function in SAS. This function retrieves the value of a variable from the previous observation. Also critical to the program is the use of the RETAIN statement, which allows SAS to hold values of a variable from a previous observation in an area of memory.

❶ Orders the data by MemberID, BeginEnroll, and EndEnroll. This is critical for the LAG function to work appropriately.

❷ Sets up variables whose values need to be "retained." Initiates two new variables to be used as temporary holders of "assumed" dates and retains their values from previous iterations of the DATA step until reassigned by logic. RETAINing MemberID simply serves the purpose of guaranteeing physical position on the data set. Variables are placed in physical order as they are recognized at compile time

❸ Renames variables. The renamed variables will be manipulated in memory to fix enrollment issues. This also allows use of a meaningful name for assignment of the final value.

❹ Removes unreasonable enrollment records.

❺ Creates a column, Prior_End, that stores the value of EndEnroll from the previous observation. This is accomplished with the LAG function with the "original" EndEnroll data as the argument.

❻ Initializes values of some key variables to missing for the first observation of each MemberID BY group.

❼ Sets Temp_End to equal Orig_EndEnroll when the time frames are contiguous, expanding the length of the enrollment segment.

❽ Identifies gaps in enrollment and closes the current segment by setting the endpoints.

❾ Outputs the enrollment record.

❿ Resets the value of Temp variables to begin a new enrollment segment with assumed endpoints.

❶* Identifies overlapping records that need to be included in the open segment.

❷* Sets Temp_End to equal Orig_EndEnroll for records that overlap partially.

❸* At the last record in a MemberID BY group the final member enrollment segment is output to the data set with appropriate endpoints.

Figure 11.3 shows the Members_Clean data set. Confirm that the issues highlighted in Figure 11.2 have been fixed.

Figure 11.3: Members_Clean data set

	MemberID	BeginEnroll	EndEnroll
1	M0002576630194200	2006-04-01	2011-12-31
2	M0004102910263700	2007-01-01	2011-12-31
3	M0004465340272800	2008-01-01	2011-12-31
4	M0020762040749600	2009-06-01	2011-12-31

Check Continuous Enrollment

The clean enrollment records are now ready for a continuous enrollment check, an algorithm so useful in HEDIS reporting that it merits parameterization into a SAS macro. Program 11.2 shows one such macro solution and its application.

Program 11.2: Continuous enrollment macro application and code

```
%macro CE( sasds=       /* SAS dataset to start with     */
          ,msryr=       /* measure year of interest      */
          ,prioryrs=    /* # of prior years of interest  */
          ,anchordt=    /* anchor date for the measure   */
         );

%let beginyr = %eval(&msryr - &prioryrs);                          ❶

%do yr = &beginyr %to &msryr;                                      ❷

data calyr&yr;
    set &sasds(where=(    BeginEnroll <= "31DEC&yr"d
                      and EndEnroll   >= "01JAN&yr"d   ));         ❸
    Start = max(BeginEnroll,"01JAN&yr"d);                          ❹
    End   = min(EndEnroll  ,"31DEC&yr"d);                          ❺
    format Start End yymmdd10.;
run;

proc sort data=calyr&yr;
    by MemberID start end;
run;

data calyr&yr;
    set calyr&yr;
    by MemberID Start End;
```

```
    LEnd = lag(end);                                                    ❻
    format LEnd yymmdd10.;

* Initiate gap counters;
    if first.MemberID
      then do;                                                          ❼
              LEnd       = .;
              GapsNbr&yr = 0;
              GapDays&yr = 0;
          end;

* Gap at the beginning;                                                ❽
    if     first.MemberID
       and (start - "01JAN&yr"d) > 0
      then do;
              GapDays&yr + (start - "01JAN&yr"d);
              GapsNbr&yr + 1;
          end;

* Gap(s) in the middle;                                                ❾
    if not first.MemberID
      then do;
              GapDays&yr +  (Start - LEnd - 1);
              GapsNbr&yr + ((Start - LEnd - 1) > 1);
          end;

* Gap at the end;                                                      ❿
    If last.MemberID and ("31DEC&yr"d-end) >= 1
     then do;
             GapDays&yr + ("31DEC&yr"d-end);
             GapsNbr&yr + 1;
         end;

* Output the MemberID record;
    if last.MemberID
      then do;                                                          ❶*
              if    GapsNbr&yr > 1
                 or (GapsNbr&yr = 1 and GapDays&yr > 45)
                then CE&yr = 'FAIL';
                else CE&yr = 'PASS';
              output;
          end;
run;

%end;

data ce_final(drop=start end );                                         ❷*
%if &beginyr < &msryr %then
    merge;
                     %else
    set;
%do yr = &beginyr %to &msryr;
          calyr&yr(drop=lend in=x&yr)
%end;  ;
    by MemberID ;
%do yr = &beginyr %to &msryr;
    if not x&yr then ce&yr= 'FAIL';
 %if %sysfunc(year(&anchordt)) = &yr %then
    anchored = (start <= &anchordt <= end);
                                    %else
    anchored = 0;;                                                      ❸*
 %end;
   if    anchored = 0
 %do yr = &beginyr %to &msryr;
      or CE&yr = 'FAIL'
```

```
 %end;
    then CE = 'FAIL';
    else CE = 'PASS';
run;
%mend CE;
```

As macro inclusion, invocation, and options have been discussed in Chapter 9, let us focus on the CE macro code.

There are three essential elements to the CE macro. First, it relies on iterative processing for every calendar year under consideration. This is done via a macro DO loop. Second, it uses the LAG function to identify three possible kinds of gaps in coverage. Third, it keeps track of both the number of gaps plus the total days of gap in each calendar year, the two parameters for judging continuous enrollment in a specified time frame.

❶ Uses %EVAL macro function to carry out arithmetic (integer only) computations on macro variables. The macro variable BeginYr resolves to 2010 from (BeginYr = 2011 - 1). The two user parameters allow the necessary offset from measure year to get begin year.

❷ Sets up a DO loop that processes each calendar year iteratively. As the value of BeginYr is computed to 2010 and the value of MsrYr is specified as 2011, the DO loop runs twice.

❸ Limits member enrollment records to those that span, either completely or partially, the calendar year. To see the effect of this statement clearly, let us consider it during the first iteration of the DO loop. With &yr resolving to 2010, the statement is as follows:

```
where=( BeginEnroll <= "31DEC2010"d and EndEnroll >= "01JAN2010"d);
```

❹ Changes the value of BeginEnroll to the start of the calendar year if the enrollment begins prior to the start of the calendar year.

❺ Changes the value of EndEnroll to the end of the calendar year if the enrollment ends after the end of the calendar year. Both steps ❹ and❺ are important in determining gaps in coverage.

❻ Uses the LAG function so that a current enrollment record may query the previous record's value of EndEnroll.

❼ For the beginning of a MemberID BY group Lend is initialized to missing so as not to retain the lagged value from another member. The gap counter variables are set to 0.

❽ Identifies gap in coverage at the beginning of the enrollment period. For example, if a member's enrollment began on Feb 1, 2010, she has a gap of 31 days for the calendar year 2010. In this example, column GapDays2010, initialized to 0 in Step ❼ is incremented by 31. Column GapsNbr2010 is incremented by 1. To see the effect of this DO group, let us view it when &yr resolves to 2010.

```
* Gap at the beginning;
if first.MemberID and (start - "01JAN2010"d) > 0
  then do;
          GapDays2010 + (start - "01JAN2010"d);
          GapsNbr2010 + 1;
       end;
```

❾ Identifies gap(s) in coverage during the middle of the member's enrollment and increments the gap variables. Notice the Sum statements and that they are accumulating the results of expressions.

❿ Identifies gap in coverage at the end of the member's enrollment and increments the gap variables.

❶* Outputs a member's last enrollment record/segment. When &yr resolves to 2010, a value of "PASS" or "FAIL" is assigned to the variable CE2010. A value of "FAIL" is given when the member has more than 1 gap or if the member's gap is more than 45 days in duration.

❷* Merges the two calendar data sets, calyr2010 and calyr2011, checking to ensure that anchor date is included in the member's enrollment period. As the DATA step uses macro coding to generate DATA step statements, it is best to understand this code by viewing it after all macro code has been resolved.

```
data ce_final(drop=start end );
    merge calyr2010(drop=lend in=x2010)
          calyr2011(drop=lend in=x2011) ;
    by MemberID ;
    if not x2010 then ce2010= 'FAIL';
    anchored = 0;
    if not x2011 then ce2011= 'FAIL';
    anchored = (start <= "31DEC2011"d <= end);
    if anchored = 0 or CE2010 = 'FAIL' or CE2011 = 'FAIL'
      then CE = 'FAIL';
      else CE = 'PASS';
run;
```

Notice that had there been only one year (the measure year), then a SET statement would have been generated by the macro code.

The IN= data set option creates temporary variables that track the source of each observation. These temporary variables have either a True (1) or False (0) value. A value of "False" is used to set CE2010/CE2011 to "Fail." For each calendar year, the DATA step code also checks whether the anchor date is included. This is done by comparing the value of &yr to the year extracted from anchor date. If the two values are not equal to one another, Anchored is set to 0. If they are equal and the anchor date is included in the enrollment record, then Anchored is set to 1. Finally, an overall CE column is created that sets to "Fail" for any one of three reasons.

❸* Two semicolons are needed for the macro code to generate SAS DATA step statements appropriately. Each macro statement ends with a semicolon, and each generated DATA step statement also ends with a semicolon. Depending on the result of the comparison between &yr and YEAR extracted from Anchordt, SAS macro statements write out either "anchored=0" or "anchored = (start<= "31DEC2011"d <= end)." The first semicolon is needed for the macro statement. The second is required to punctuate the generated DATA step statement.

Identify the Denominator

Now that we have a way to check for continuous enrollment, we can determine the denominator. Then we can apply exclusions to the denominator. This is because a member may be excluded from the COL measure based on a claims history of either having a diagnosed colorectal cancer or a total colectomy. Program 11.3 shows you how to compute the denominator and then apply exclusions to it.

Program 11.3: Determine the denominator

```
%let Msryr=2011;

*** ------------------------------------------------------------ ***
*** Part 1 - include and execute the CE macro                   ***
***        - create the initial denominator                    ***
*** ------------------------------------------------------------ **;

options mprint;
%include "C:\HealthCare Book\Macro programs\Program 11.2.sas"/source2;
%CE(sasds=Members_Clean,msryr=2011,prioryrs=1,anchordt="31DEC2011"d);

proc sql;
    create table Col_Denominator as
    select distinct memberid
          ,floor( (intck('month',dob,"31DEC&MsrYr"d)
```

```
                -(day("31DEC&MsrYr"d)<day(dob)))/12) as age
      from book.members
     where     calculated age between 51 and 75
           and memberid in (select distinct memberid
                              from CE_final
                             where CE="PASS");
quit;

*** ------------------------------------------------------------ ***
*** Part 2 - create column-wise versions of the claims data     ***
*** ------------------------------------------------------------ **;

data fac_header_codes (keep=ClaimID MemberID DischgDt
                            CodeValue CodeType CodeRole);
    set book.FacilityHeader;
    array dx $ pDx Dx_: ;
    array px $ pPx Px_: ;
    length CodeValue $5 CodeType $7 CodeRole $1;
    do i = 1 to dim(dx);
                        if dx(i)  = " " then leave;
                        CodeValue = dx{i};
                        CodeType  = "ICD9CM";
                        if i = 1 then CodeRole = "1";
                                 else CodeRole = "2";
                        output work.fac_header_codes;
    end;
    do j = 1 to dim(px);
                        if px(j)  = " " then leave;
                        CodeValue = px{j};
                        CodeType  = "ICD9PCS";
                        if j = 1 then CodeRole = "1";
                                 else CodeRole = "2";
                        output work.fac_header_codes;
    end;
run;

proc sort data=fac_header_codes;
    by MemberID ClaimID;
run;

data pro_detail_codes(keep=ClaimID CodeValue CodeType BeginDOS);
    set book.ProfessionalDetail;
    array dx $ PointerDx1 - PointerDx4 ;
    length CodeValue $5 CodeType $7 ;
    do i = 1 to dim(dx);
                        if dx(i)  = " " then leave;
                        CodeValue = dx{i};
                        CodeType  = "ICD9CM";
                        output work.pro_detail_codes;
    end;
    CodeValue = Px;
    if anyalpha(Px) = 1 then CodeType  = "HCPCS";
    if anyalpha(Px) = 0 then CodeType  = "CPT";
    output work.pro_detail_codes;
run;

proc sort data=pro_detail_codes nodupkey;
    by ClaimID CodeValue CodeType BeginDOS;
run;

*** ------------------------------------------------------------ ***
*** Part 3 - create a list of the excludable members          ***
***        - create the final denominator                    ***
*** ------------------------------------------------------------ **;
```

```
proc sql;
    create table Excludable_Mbrs as
    select distinct
           a.MemberID
          ,a.CodeType     length=7
          ,a.CodeValue
          ,a.CodeRole
          ,"Facility"              as Source
          ,b.ValueSet
      from fac_header_codes   a
          ,BOOK.ColDriver     b
     where a.CodeValue = compress(b.CodeValue,'.')
       and upcase(b.CodeType)  in("ICD9CM","ICD9PCS")
       and upcase(b.ValueSet)  in("COLORECTAL CANCER","TOTAL COLECTOMY")
       and a.DischgDt <= "31DEC&msryr."D
     UNION
    select distinct
           a.MemberID
          ,b.CodeType     length=7
          ,b.CodeValue
          ," "                     as CodeRole
          ,"Professional"          as Source
          ,c.ValueSet
      from book.Professionalheader  a
      join pro_detail_codes         b
        on a.ClaimID = b.ClaimID
      join BOOK.ColDriver           c
        on b.CodeValue = compress(c.CodeValue,'.')
       and upcase(b.CodeType)  in("ICD9CM","CPT","HCPCS")
       and upcase(c.ValueSet)  in("COLORECTAL CANCER","TOTAL COLECTOMY")
       and b.BeginDOS <= "31DEC&msryr"D
     UNION
    select distinct
           a.MemberID
          ,c.CodeType     length=7
          ,b.Px                    as CodeValue
          ," "                     as CodeRole
          ,"Facility"              as Source
          ,c.ValueSet
      from book.Facilityheader  a
      join book.Facilitydetail  b
        on a.ClaimID = b.ClaimID
      join BOOK.ColDriver       c
        on b.Px = c.CodeValue
     where upcase(c.CodeType)  in("CPT","HCPCS")
       and upcase(c.ValueSet)  in("COLORECTAL CANCER","TOTAL COLECTOMY")
       and a.DischgDt <= "31DEC&msryr"D
    ;

    create table Col_Denominator_Final as
    select *
      from Col_Denominator
     where memberid not in(select distinct memberid
                           from Excludable_Mbrs)   ;
quit;
```

Program 11.3 is both complex and lengthy. To understand it, let us break it down into three separate chunks. Program 11.3 (Part 1) shows you the code in the first part—creating the initial denominator.

Program 11.3 (Part 1)

```
%let Msryr=2011;

options mprint;
%include "C:\HealthCare Book\Macro programs\Program 11.2.sas"/source2; ❶
```

```
%CE(sasds=Members_Clean,msryr=2011,prioryrs=1,anchordt="31DEC2011"d);
proc sql;
    create table Col_Denominator as
    select distinct memberid
          ,floor( (intck('month',dob,"31DEC&MsrYr"d)
                -(day("31DEC&MsrYr"d)<day(dob)))/12) as age          ❷
      from book.members
     where calculated age between 51 and 75
       and memberid in (select distinct memberid
                          from CE_final
                         where CE="PASS");                           ❸
quit;
```

Program 11.3 (Part 1) retrieves a subset of members who meet the continuous enrollment and the age criteria. This is the "eligible population" prior to exercising any exclusions.

❶ Bring in the macro code from an external location and call for its execution.

❷ Uses the INTCK function to determine the interval in months between the member's Date of Birth and December 31, 2011. This gives us the age in months. If a member is born on the 31st of a month, then age in months is decremented by 1, as the member has not yet passed the full month period. The resultant number of months is divided by 12 to give the age in years. The FLOOR function is used to round down the age in years to an integer.

❸ A subquery is used to retrieve a list of members who meet the continuous enrollment criteria from CE_Final data set. This list is fed into the WHERE statement of the main query to round out the selection filter.

Before we apply exclusions based on claims history to the denominator, we need to pre-process two claims data sets, BOOK.FacilityHeader and BOOK.ProfessionalDetail. This is important in order to examine all diagnoses and procedure codes attributable to a member regardless of setting. Remember that the NCQA business rules do not necessarily distinguish claim source—only the claim fields and their values. So we must compile both Facility and Professional data as the complete claims history to interrogate later.

Our pre-processing strategy involves transforming data so that each of the diagnosis or procedure codes is on a separate row (row-wise architecture). Figures 11.4 and 11.5 show records from BOOK.FacilityHeader and BOOK.ProfessionalDetail.

Figure 11.4: BOOK.FacilityHeader (only diagnoses and surgical procedure codes columns shown)

Pdx	Dx_2	Dx_3	Dx_4	Dx_5	Dx_6	Dx_7	Dx_8	Ppx	Px_2	Px_3	Px_4	Px_5	Px_6
V5481	V4365	70713	45981	7197	7812	2859	40290						
0388	3911	262	514	44489	99591	2761	78060	3722	8856	8853			
0389	78552	99592	5845	51881	2762	2767	42789	4620	5122				
99666	2851	5859	40390	53081	5939	60000	41401	8006	0391	9904	0481	8456	
4821	51881	5849	5120	V8541	2762	2841	2760	9672	9604	3404	3324	9960	3893
41401	4019	4139	42731	2724	V8801	V103	4240	3612	3615	3722	8856	3961	8853

Figure 11.5: BOOK.ProfessionalDetail (only diagnoses and procedure code columns shown)

PointerDx1	PointerDx2	PointerDx3	PointerDx4	Px
7392	7840	7391		98941
7394				98941
12345	45340			37620
45340				36011
12345				75940
12345				99144
45340				99145
5715	2724			36415
5715				80051
7962	5715			82150
2724				82565
7962	5715	2724		84520
5715	2724			85025

Program 11.3 (Part 2) converts BOOK.FacilityHeader and BOOK.ProfessionalDetail from a wide (column-wise) to a long structure (row-wise). This enables us to find exclusions and compliant events more efficiently by querying one column where we would have had to query multiple columns.

Program 11.3 (Part 2)

```
data fac_header_codes (keep=ClaimID MemberID DischgDt
                           CodeValue CodeType CodeRole);
    set book.FacilityHeader;
    array dx $ pDx Dx_: ;                                          ❶
    array px $ pPx Px_: ;
    length CodeValue $5 CodeType $7 CodeRole $1;
    do i = 1 to dim(dx);
                        if dx(i)  = " " then leave;                ❷
                        CodeValue = dx{i};                         ❸
                        CodeType  = "ICD9CM";                      ❹
                        if i = 1 then CodeRole = "1";              ❺
                                 else CodeRole = "2";
                        output work.fac_header_codes;              ❻
    end;
    do j = 1 to dim(px);                                           ❼
                        if px(j) = " " then leave;
                        CodeValue = px{j};
                        CodeType  = "ICD9PCS";
                        if j = 1 then CodeRole = "1";
                                 else CodeRole = "2";
                        output work.fac_header_codes;
    end;
run;

proc sort data=fac_header_codes;
    by MemberID ClaimID;
run;

data pro_detail_codes(keep=ClaimID CodeValue
                           CodeType BeginDOS);
    set book.ProfessionalDetail;
    array dx $ PointerDx1 - PointerDx4 ;                                ❽
    length CodeValue $5 CodeType $7 ;
    do i = 1 to dim(dx);
                        if dx(i)  = " " then leave;
                        CodeValue = dx{i};
                        CodeType  = "ICD9CM";
                        output work.pro_detail_codes;
    end;
```

```
    CodeValue = Px;
    if anyalpha(Px) = 1 then CodeType  = "HCPCS";          ❾
    if anyalpha(Px) = 0 then CodeType  = "CPT";
    output work.pro_detail_codes;
run;

proc sort data=pro_detail_codes nodupkey;                  ❿
    by ClaimID CodeValue CodeType BeginDOS;
run;
```

As ARRAY statements have been discussed in Chapter 9 and DO loops have been covered in several chapters, we explain only the new elements in Program 11.3 (Part 2).

❶ Place all diagnoses columns in the ARRAY statement. "Dx:" is a handy shortcut for grabbing all columns beginning with "Dx."

❷ Exit the loop when the first missing value in a diagnosis column is encountered. The LEAVE statement is efficient, as it avoids needlessly processing missing diagnoses columns.

❸ Sets CodeValue to be equal to the diagnosis column being processed by the array.

❹ Creates CodeType to preserve the source of CodeValue.

❺ Creates CodeRole to preserve whether it is a primary or other diagnosis.

❻ Places each nonmissing value of diagnosis as a row in the output data set.

❼ Repeats steps ❶ through ❻ for surgical procedure codes.

❽ Uses ARRAY and DO loop to iterate through diagnosis columns in BOOK.ProfessionalDetail to create CodeValue and CodeType columns.

❾ As Px is a sole procedure column, no ARRAY and DO loop statements are needed. The ANYALPHA function searches Px for the first occurrence of a lowercase or uppercase alphabetic character (a letter) and the returned value is the physical position in which the function finds such a character. As we know from our knowledge of code architecture, if there is an alpha in the first position (1), the procedure code is a HCPCS. If no character is found (0), it is a CPT Level 1 code.

❿ Removes duplicates as one claim on any given date of service may have repeat values of procedure codes.

Program 11.3 (Part 3) creates a list of members who should be excluded from the denominator and then applies this list to derive the final denominator.

Program 11:3 (Part 3)

```
proc sql;
    create table Excludable_Mbrs as                        ❶
    select distinct
           a.MemberID
          ,a.CodeType     length=7
          ,a.CodeValue
          ,a.CodeRole
          ,"Facility"              as Source
          ,b.ValueSet
      from fac_header_codes         a
          ,BOOK.ColDriver           b                      ❷
     where a.CodeValue = compress(b.CodeValue,'.')         ❸
       and upcase(b.CodeType)  in("ICD9CM","ICD9PCS")
       and upcase(b.ValueSet)  in("COLORECTAL CANCER","TOTAL COLECTOMY")
       and a.DischgDt <= "31DEC&msryr."D                   ❹
```

```
 UNION
select distinct                                          ❺
       a.MemberID
      ,b.CodeType     length=7
      ,b.CodeValue
      ," "                    as CodeRole
      ,"Professional"         as Source
      ,c.ValueSet
  from book.Professionalheader  a
  join pro_detail_codes         b
    on a.ClaimID = b.ClaimID
  join BOOK.ColDriver           c
    on b.CodeValue = compress(c.CodeValue,'.')
   and upcase(b.CodeType)  in("ICD9CM","CPT","HCPCS")
   and upcase(c.ValueSet)  in("COLORECTAL CANCER","TOTAL COLECTOMY")
   and b.BeginDOS <= "31DEC&msryr"D
 UNION
select distinct                                          ❻
       a.MemberID
      ,c.CodeType     length=7
      ,b.Px                   as CodeValue
      ," "                    as CodeRole
      ,"Facility"             as Source
      ,c.ValueSet
  from book.Facilityheader      a
  join book.Facilitydetail      b
    on a.ClaimID = b.ClaimID
  join BOOK.ColDriver           c
    on b.Px = c.CodeValue
 where upcase(c.CodeType)  in("CPT","HCPCS")
   and upcase(c.ValueSet)  in("COLORECTAL CANCER","TOTAL COLECTOMY")
   and a.DischgDt <= "31DEC&msryr"D
;

create table Col_Denominator_Final as                    ❼
select *
  from Col_Denominator
 where memberid not in(select distinct memberid
                         from Excludable_Mbrs)   ;
quit;
```

❶ Creates Excludable_Mbrs data set. This data set is populated by three SQL queries, which are concatenated by the UNION operator in PROC SQL. The data set creates metadata about why the member is excluded.

❷ The first query to populate the Excludable_Mbrs data set joins Fac_Header_Codes and BOOK.ColDriver data sets to find diagnoses and surgical procedure codes that cause the member to be excluded. The "driver" file is named as such because it contains the business rules that drive the process.

❸ The COMPRESS function removes decimals from CodeValue in BOOK.ColDriver in order to compare its values correctly to that of CodeValue in Fac_Header_Codes, which contains no decimals.

❹ Searches all prior claim history.

❺ The second query, to populate the Excludable_Mbrs data set, uses Pro_Detail_Codes to find diagnoses and procedure codes that cause the member to be excluded. BOOK.Professional_Header is used to retrieve MemberID column.

❻ The third query, to populate the Excludable_Mbrs data set, uses BOOK.Facility_Detail to find facility codes that cause the member to be excluded. BOOK.Facility_Header is used to retrieve the MemberID column.

❼ Create the Col_Denominator_Final data set by removing the members in Excludable_Mbrs data set from Col_Denominator.

Determine Compliant Population

Program 11.4 shows you how to identify the numerator.

Program 11.4: Identify the compliant population

```
proc sql;
    create table Col_Numerator_Final as
    select hdr.MemberID
          ,dtl.BeginDos
          ,dtl.Px              as CodeValue
          ,rul.CodeType
          ,"Professional"      as Source
          ,rul.ValueSet
      from book.professionalheader   hdr                          ❶
      join book.professionaldetail   dtl
        on hdr.ClaimID  = dtl.ClaimID
       and hdr.MemberID in                                        ❷
                          (select distinct MemberID
                             from Col_Denominator_Final)
      join BOOK.ColDriver            rul                          ❸
        on dtl.Px  = rul.CodeValue
     where (    rul.CodeType          = "CPT"
            and upcase(rul.ValueSet)  = "FOBT"
            and year(dtl.BeginDOS)    = &MsrYr   )                ❹
        or (    rul.CodeType= "CPT"
            and upcase(rul.ValueSet) = "FLEXIBLE SIGMOIDOSCOPY"
            and year(dtl.BeginDOS)   between (&MsrYr - 4)
                                         and &MsrYr
           )                                                      ❺
        or (    rul.CodeType= "CPT"
            and upcase(rul.ValueSet) = "COLONOSCOPY"
            and year(dtl.BeginDOS)   between (&MsrYr - 9)
                                         and  &MsrYr
           )                                                      ❻
UNION
    select hdr.MemberID                                         ❼
          ,dtl.BeginDos
          ,dtl.Px              as CodeValue
          ,rul.CodeType
          ,"Facility"          as Source
          ,rul.ValueSet
      from book.facilityheader     hdr
      join book.facilitydetail     dtl
        on hdr.ClaimID  = dtl.ClaimID
       and hdr.MemberID in
                           (select distinct MemberID
                              from Col_Denominator_Final)
      join BOOK.ColDriver          rul
        on dtl.Px  = rul.CodeValue
     where (    rul.CodeType in("CPT", "HCPCS")
            and upcase(rul.ValueSet) = "FOBT"
            and year(dtl.BeginDOS)   = &MsrYr     )
        or (    rul.CodeType in("CPT", "HCPCS")
            and upcase(rul.ValueSet) = "FLEXIBLE SIGMOIDOSCOPY"
            and year(dtl.BeginDOS) between (&MsrYr - 4)
                                       and  &MsrYr     )
        or (    rul.CodeType in("CPT", "HCPCS")
            and upcase(rul.ValueSet) = "COLONOSCOPY"
            and year(dtl.BeginDOS)   between (&MsrYr - 9)
                                         and  &MsrYr     )
;
quit;
```

The Col_Numerator_Final data set is populated by two queries whose results are concatenated by the UNION operator in PROC SQL. The two queries each use BOOK.ColDriver to drive the identification of a compliant event for the eligible population from BOOK.ProfessionalDetail and BOOK.FacilityDetail, respectively.

❶ Joins BOOK.ProfessionalHeader and BOOK.ProfessionalDetail to retrieve MemberID.

❷ The subquery here ensures that only members who are also in the Col_Denominator_Final data set are considered.

❸ Uses BOOK.ColDriver to drive the identification of procedure codes.

❹ Specifies the time period for FOBT procedure to be the measurement year.

❺ Specifies the time period for Flexible Sigmoidoscopy to be the measurement year and the four prior years.

❻ Specifies the time period for Colonoscopy to be the measurement year and the nine prior years.

❼ Modifies logic from 1–6 to identify compliant population from facility claims.

Compute Rate

Program 11.5 combines results from Col_Denominator_Final and Col_Numerator_Final data sets to calculate a COL measure rate.

Program 11.5: Combine results

```
proc sql;
    create table dummy(dummy character(1));                  ❶
    insert into  dummy values(' ');

    title 'Rate Calculation for Colorectal Cancer Screening';
    select (select count(distinct MemberID)
               from COL_Numerator_Final     )       as numer   ❷
          ,(select count(distinct MemberID)
               from COL_Denominator_Final)          as denom   ❸
          ,calculated numer / calculated denom
                              format=percent7.3     as rate    ❹
     from dummy;                                                ❺
quit;
```

❶ Creates a one-row, one-column "dummy" data set that can be used in the subsequent step. This is necessary, as SAS SQL does not allow a SELECT from nothing.

❷ Computes the numerator as a subquery of the Numerator data set in the SELECT clause.

❸ Computes the denominator as a subquery of the Denominator data set in the SELECT clause.

❹ Uses the CALCULATED keyword to compute the rate from recently created (calculated) fields.

❺ Uses the placeholder data set in the FROM statement.

Figure 11.6: Rate calculation results

Rate Calculation for Colorectal Cancer Screening

numer	denom	rate
31	1464	2.12%

Note: The COL rate of 2.12% shown in Figure 11.6 is dismally low. In the real world of HEDIS reporting, this rate would suggest that a programming error had been made. While accurately calculated for this set of bogus data, it is not a good representation of the real world.

Conclusion

In this chapter, you learned the following:

- Purpose of HEDIS
- Business rules and logic of a HEDIS measure
- Data-driven approach to computing a HEDIS measure
- Continuous Enrollment Macro useful for a number of HEDIS measures
- SAS Code needed to compute the COL measure

Chapter 12: Future Healthcare Data Issues

Introduction 175
Impact of the Affordable Care Act 175
Transparency in Pricing 176
ICD-10 176
Patient Centered Medical Home 176
Accountable Care Organization 177
Pharmacy Benefits Manager 177
Evolving Patient Medical Records 178
- Electronic Medical Record 178
- Electronic Health Record 178
- Personal Health Record 178
- Meaningful Use 179

Global Billing 179
All Payer Claims Database 179
Conclusion 180

Introduction

By now you must be convinced that the U.S. healthcare system and the data it generates are incredibly complex. While the future is uncertain, we can expect the data to grow in complexity and volume. Mandated new approaches to care delivery, transparency in prices, and improved payment mechanisms make it imperative that the healthcare industry continue to find new insights and opportunities with reporting and analytics. In this penultimate chapter, we would like to alert the reader to currently evolving issues that will affect the generation and management of administrate healthcare data.

Impact of the Affordable Care Act

In 2014, the Affordable Care Act (ACA) was fully implemented. As a result of the law's mandate to provide affordable insurance coverage, 8 million Americans enrolled in private health insurance plans and an additional 3 million enrolled in individual states' Medicaid plans. As of this book's writing, many health insurance companies are struggling with the influx of new members. The working capacity of many departments within health insurance companies is being taxed. The departments that are most affected include but are not limited to Member Enrollment, Customer Service, and Claims Processing. Anecdotes report that members have been experiencing long waiting periods to see a primary care doctor. In addition, many companies are concerned that the healthcare utilization of new members may be higher than average. This is supported by early findings that new members tend to be both older and sicker. Only time will tell how this law will affect the industry in the long term. One thing remains certain: The industry needs to be flexible and ready to change in order to remain healthy.

Transparency in Pricing

The market for healthcare services does not lend itself to transparency in prices. The forces of supply and demand do not determine prices of healthcare services. Why not? In other markets, buyers and sellers determine prices. In healthcare, buyers are split into two distinct entities: consumers and payers. Consumers or patients utilize healthcare services but generally do not pay for them. Payers (health insurance companies, the government) pay for healthcare services. Prices are negotiated between payers and providers and are well guarded by both.

Two events in 2014 attest to our belief that the healthcare market is moving toward greater transparency. First, Medicare disclosed and publicized millions of its payment records to physicians. For the first time, analysts can examine this gold mine of data and understand variation in prices between physicians and the specific services they render. Second, three leading health insurance companies—Aetna, Humana, and UnitedHealth Group—have agreed to develop an online tool that will give consumers information about prices and quality of healthcare services. Although it has not yet been determined if these two events will constitute a trend, the savvy healthcare analyst will recognize the renewed importance of analytics in the face of a transparent and more competitive industry.

ICD-10

On July 31, 2014, the Department of Health and Human Services announced an additional delay of the implementation of ICD-10, until October 2015. The ICD-10 transition was originally scheduled for October 1, 2013, but the date has been pushed out a number of times, most recently by the Protecting Access to Medicare Act, which the president signed into law on April 1, 2014. The shift from ICD-9 to ICD10 will enrich the information around code sets and exponentially improve insight and understanding.

Many organizations are ready for the change. The data capture by adjudication and warehousing systems should not be a problem. Data management routines will need to change for the new architecture—a tedious but not difficult task. The more difficult transition will be the application of new and revised business rules. Mapping ICD-9 to and from ICD-10 will be problematic. The changeover will be abrupt, as Medicare will no longer accept ICD-9. Other payers may allow a grace period and accept both code sets. Are you ready for the change?

Patient Centered Medical Home

The Patient-Centered Medical Home (PCMH; also known as *primary care* medical home) model was proposed in 2007 by the American Academy of Family Physicians, American Academy of Pediatrics, American College of Physicians, and American Osteopathic Association. This new model gained traction with the passage of the Patient Protection and Affordable Care Act (PPACA). The legislation discusses PCMH as a new model for the organization of primary care delivery. The perceived benefits are decreased cost with increased quality, improved access to care, and enhanced preventive care services.

A medical home is not only defined as a place, but also as an organizational model that delivers the core functions of primary health care. Payers may incentivize providers with a "bonus" for improving primary care services for each patient in the medical home. As a business model, PCMH is intended to address the following:

- Patient focus: A complete "care team" incorporates not only a member's primary care physician, but also the member's family, clergy, and specialty providers.
- Comprehensive care: All providers (specialist, hospital, labs, and so on) are coordinated by a member's primary care physician.
- Coordinated care: From prevention and wellness care to acute and chronic episode management, all care is coordinated by the member's PCMH.
- Improved access: PCMHs provide longer hours of operation, telephone support, and web-enabled facilities for their patients.
- Systems-based approach: Health information technology (HIT) plays a central role.

The healthcare industry continues to recognize the foundational role of the PCMH in delivery models such as Accountable Care Organizations (ACO) and the emerging medical neighborhood model. PCMHs are well-positioned to lead and drive change across the industry. Our challenge is to support that change with meaningful data reporting and analytics.

> The Health Home concept is analogous to, but not the same as, Medical Home. The primary difference is that Health Homes are specific to the Medicaid population with chronic conditions. It is a population-based integrated care model targeting consumers with chronic conditions that coordinates medical and behavioral health care and community and social supports. The ACA established the Health Home as a new state Medicaid option for service delivery. The term is used to suggest that the comprehensive coordinated care model builds on the concept of medical homes but must be unambiguous about the critical role of mental health, substance abuse, and community support.

Accountable Care Organization

Accountable Care Organizations (ACOs) are defined by CMS as groups of hospitals, clinicians, and other healthcare organizations bound by contractual agreements. They share mutual responsibility for improving the quality of care, improving patient outcomes, and reducing costs and inefficiencies. CMS requires ACOs to have a strong network of primary care providers, making the PCMH concept a critical foundation for high-performing ACOs.

ACO is another concept emerging from the PPACA. The intent of the legislation is to reduce Medicare expenditures by enabling ACOs to contract with Medicare through the new Medicare Shared Savings Program. This program provides financial incentives to collaborate and save money by avoiding unnecessary tests and procedures. ACO participants understand the need to have the following in place:

- Formal Engagement Strategies: Enhancement of the responsibility and transparency in the patient-provider relationship
- HIT: Tools provide the effective and efficient sharing of a patient's health information among all stakeholders.
- Population Health Management: Strategies are in place to effectively use technology in support of informed health care decisions, to identify and manage health problems in real-time, and to provide timely access to care.
- Performance Measurement: Programs must be in place to report on service quality, patient outcomes, cost/benefit, and stakeholder satisfaction in order to steer the development and refinement of effective care and reimbursement models.
- Pioneering Payment Paradigms: Incentives for clinicians and patients alike need to be developed that provide for the improvement of health outcomes.

By one definition, ACOs are comprised of many "medical homes." Some have even called the ACO a "medical neighborhood" accountable for the improvement of health outcomes across the entire patient care continuum. With so many stakeholders and their good intent involved in the future of ACOs, it is not hard to see the importance of data and its management, reporting, and analytics for the success of this health care delivery model.

Pharmacy Benefits Manager

A Pharmacy Benefits Manager (PBM) is, generally, a third-party administrator of prescription drug programs; it may, in some cases, be a service inside of an integrated health plan. PBMs are primarily accountable for processing and paying prescription drug claims. They also develop and maintain the formulary, contract with pharmacies, and negotiate discounts and rebates with drug manufacturers. It is this latter feature that provides the benefit for health plan enrollees: lower prices for prescription drugs through discounts from pharmacy vendors.

While not new, PBMs will play an expanded role under healthcare reform as well as enhanced government oversight and compliancy requirements. Part of this expanded role is an increased focus on preventive services and wellness. The improved transparency and value-add is new to the industry, and PBMs will need to step up their game.

What does this mean to the healthcare data analyst?

- New reporting requirements are now in place for PBMs working with Medicare Part D health plans. More, and potentially different, data will become available.
- Medicaid drug benefits will dramatically change. New drug classes are mandated to be covered, and new pricing strategies will have to be implemented.
- The Physician Payment Sunshine Act is part of the PPACA. This act mandates increased transparency regarding financial relationships between providers (pharmacies in this context) and payers. This change will affect treatment choices and costs and we hope will effect positive changes.

Evolving Patient Medical Records

A combination of improving technology and government mandates through legislation has brought the industry several flavors of patient medical records that are gaining traction over time, but the industry has yet to realize their full benefits. It is essential to follow changes and understand the usefulness of patient medical records.

Electronic Medical Record

The electronic *medical* record (EMR) is the health-related information about an individual that is created, gathered, managed, and used by clinicians and staff from a *single* organization involved in the individual's health care. It is a patient's medical history from one provider and is mostly used for diagnosis and treatment.

The benefits of electronic versus the paper records are not to be underestimated. Providers are now able to track a patient's data over time, proactively schedule preventive visits, monitor a patient's progress, and gain an overall improvement in quality of care and outcomes. The blend of electronic information and the technology to use it is expected to provide improvements in process and outcome.

As of January 1, 2014, federal legislation requires that all healthcare providers must have adopted and demonstrated "meaningful use" of EMR in order to maintain their existing Medicaid and Medicare reimbursement levels.

Electronic Health Record

The electronic *health* record (EHR) is the aggregate record of health-related information about an individual that is created and gathered cumulatively across *more than one* healthcare organization and is managed and consulted by clinicians involved in the individual's health care. EHRs go beyond the data collected in the provider's office and include a more comprehensive patient history.

Unlike EMRs, EHRs allow a patient's health record to move with them to other healthcare providers, specialists, hospitals, and nursing homes. The word "health" covers a lot more territory than the word "medical," and EHRs go a lot farther than EMRs with regard to robustness of information.

By many estimates gleaned from the Internet, it is going to be many years before the EHR is robust enough to share information across multiple healthcare organizations. It is not well understood where these records are to be housed.

Personal Health Record

A *personal* health record (PHR) is an electronic format used by patients to maintain and manage their health information in a private, secure, and confidential environment.

PHRs go beyond the common provider portals available today. In addition to allowing patients to view provider information or communicate with providers, portals will allow the patient to update their PHR with diet plans, data from home monitoring systems, contact information, diagnosis lists, medication lists, allergy lists, and immunization histories.

When properly implemented, PHRs enable the patient to manage their health information and become active partners in the pursuit of good health.

Meaningful Use

"Meaningful use," as defined by HealthIT.gov, consists of using EMRs and EHRs to realize:

- Improved quality, safety, and efficiency
- Reduced healthcare disparities
- Engaged patients and family
- Improved care coordination
- Privacy and security of patient health information

One of the many benefits of the American Recovery and Reinvestment Act of 2009 (the Stimulus or Recovery Act) is the inclusion of financial incentives for providers who prove meaningful use of electronic health records. Providers who have not implemented EMR/EHR systems and demonstrated their meaningful use by 2015 will be penalized with a 1 percent reduction in Medicare reimbursements, with reduction rates rising annually thereafter.

The EMR/EHR meaningful use mandate is effecting significant growth in healthcare informatics. Professionals with the skill and knowledge necessary to implement and manage healthcare IT software and applications are in great demand. For more on this important initiative, see the CMS website (www.cms.gov) or the government health IT site (www.healthit.gov).

Global Billing

Under the current, traditional system of billing for services incurred during an episode of care, the bill is split between the "professional" and "technical" components. For a cardiac surgery, for example, the Split Billing model generates two bills for the episode. The payer (or patient) receives one from the surgeon (professional services) and one from the hospital (facility or technical services), each billing for their cost incurred during the episode.

Under the Global Billing model, a global rate is submitted and includes the professional and technical components. Reimbursement goes to the professional provider, who then pays the facility. In our example above, the cardiac surgeon is responsible for all costs associated with the patient's episode of care, from the presurgical consult and work-up, to the surgery, and through any hospital or outpatient follow-up.

The Global Billing model is another cost-cutting measure with incentives to provide timely, high quality services with appropriate outcomes. We will see more of these CPT global billing codes in use as additional healthcare reform legislation is rolled out and fine-tuned. The concern of interest for data managers and analysts is the effect that the loss of granularity will have on our analytics. Where there used to be multiple sources and content of service billings, we will now only see the one service line with a single billable code.

All Payer Claims Database

All Payer Claims Databases (APCDs) are comprehensive databases that methodically collect paid medical claims, pharmacy claims, and dental claims along with supporting member eligibility and provider data from commercial and government payers. They are built and managed at the state level, the first being established in Maine in 2003. Currently more than thirty states have implemented, or have a strong interest in implementing, a state-sponsored APCD.

The content of an APCDs is, as you may suspect, all of the data elements discussed in this book, and more. The challenge in building such a robust database is the inclusion of all requisite data (present and future) and the provision of appropriate access for stakeholders. Stakeholders run the gamut from policy wonks to consumers—anyone having a need to understand information about the current utilization and costs of health care in their state.

APCDs are being used to evaluate the cost and quality of PCMH and ACO pilots in support of healthcare reform. They are providing states with information about regional health care utilization patterns and the needs of specific populations. In general, states with APCDs have a better reference point from which to evaluate the impact of reform efforts.

Going forward, state APCDs will provide the data necessary to the surfacing of information about the cost, effectiveness, and performance of health care delivery at the local, state, and national levels. For more on this important topic, see the APCD Council's website (www.apcdcouncil.org).

Conclusion

This chapter has highlighted some changes coming down the pipeline for the healthcare industry. These include, but are not limited to, rise in health insurance enrollment due to ACA, upgrade to ICD-10, focus on primary care coordination, and electronic medical records. The data generated by the industry continues to increase in complexity and volume. Traditional or uninspired methods of data analytics and reporting will not be sufficient. The industry has already begun to evaluate and adopt new visualization and analytics technologies so that it can keep up with the volume and complexity of big data.

Chapter 13: Extended Coding Examples

Introduction 181
Utility Macros 181
Age-As-Of Calculation 181
Identifying Sparse Variables 182
Arrays of Detail Record Elements on the Header Record 187
Linking to the Diagnosis Pointer 190
Conclusion 193

Introduction

This book is primarily about the business knowledge of administrative healthcare data. The more we know about the data, the more we will think of programmatic solutions to manage, analyze, and report this data.

We have shown example solutions in the context of the various chapters, but we were not able to address many useful—in our experience—tools and techniques. So we take this opportunity to share some extended coding examples. Some are presented as macros, while others are open-code solutions, with many possibilities for transformation into a utility macro.

In SAS programming, we realize that there are many ways to solve a problem. Efficiency, programmer preference, and clarity of purpose generally guide code development. The code examples in this book are simply one way toward a solution. Improvements to code are always welcome by way of constructive criticism sent to the authors.

Utility Macros

Many organizations, or even project teams, are served well with the development and vetting of a set of utility macros. These are generally developed and shared because they provide an "organizational truth," that is, every programmer uses the accepted method to perform a commonly defined task. Once vetted by the organization, they should be stored in a macro library available to all.

In this section, we demonstrate a few such macros. The intent here is to prompt the reader to think about the utility of this organizational approach and the use of a common set of tools.

Age-as-of Calculation

Administrative healthcare data, unlike other industry data, is interminably tied to dates; every data element you will see has a date associated with it. For many business rule implementations we need the age of the patient/member "as of" some index date—age as of hospital discharge, age as of the end of a year, and so on. The macro shown in Program 13.1 provides such a useful tool. Please note that the executable part of this macro is a code snippet (most of it is comments), meaning that it cannot stand alone; it must be used as an expression resolution in an assignment statement or wherever the resolution makes sense.

Program 13.1: Age-as-of macro

```
%macro ageasof(asofdt=, dob=);
  %*--------------------------------------------------------------------*
   * Computes the age in years (as a person does it,                    *
   *                             marking birthdays exactly.)            *
   *                                                                    *
   * Note: This macro was copied from SAS-L and is the work of          *
   *       William Kreuter (algorithm) and Ian Whitlock (macrotization) *
   *        It has been widely used since 1995.                         *
   *                                                                    *
   * Usage: age = %ageasof(asofdt=refdate,dob=brthdate)                 *
   *                                                                    *
   * Parameters named because reversed arguments do not give negative   *
   *            of correct age.                                         *
   *--------------------------------------------------------------------*
  ;
  floor( (  intck('month',&dob,&asofdt)
          - (day(&asofdt)<day(&dob))                          ❶
         )/12
       );
%mend ageasof;

/**** USE EXAMPLE ****/

proc sort data=book.members
           out=members(keep=memberid dob) nodupkey;
    by memberid;
run;

data age;
    set members;
    age = %ageasof(asofdt='07JUL2014'd,dob=DOB);
run;
```

❶ Uses the INTCK function to determine the interval in months between the member's Date of Birth and July 7, 2014, the value that macro variable &asofdt resolves to. This gives us the age in months. If the 7th of the month is prior to the member's day of birth, then 1 month is subtracted from the age in months. The result is divided by 12 to give the age in years. The FLOOR function is used to round down the age in years to an integer number.

Figure 13.1 shows the Age data set.

Figure 13.1: Usage of the age macro

MemberID	DOB	age
M000000086000...	1930-05-14	84
M000000827000...	1928-12-17	85
M000001353000...	1937-03-06	77
M000001494000...	1928-07-17	85
M000001695000...	1941-09-01	72
M000002171000...	1924-08-12	89

Identifying Sparse Variables

Healthcare data sources may contain fields that are so sparsely populated as to be useless in a particular project. As a first step in a project, it may be beneficial to check on the missing value structure of the data. Based on the findings, or as an a priori business rule, you may wish to drop the offending variables from the data.

This macro does many things. It is driven by the following user-provided parameters:

An input data set
An output data set so that the input data set can remain unchanged
The level of sparseness of interest
Whether or not to alter the data, that is, drop the sparse variables
Whether or not to create a data set that documents the sparse variables

Program 13.2 creates the Sparsetest data set, the SparsVars macro, and shows its usage.

Program 13.2: Testing the sparseness of a data set

```
data sparsetest;
infile datalines firstobs=3;
input n100       n90       n80       n10      constant
     c100 $ 18  c90 $ 22  c80 $ 26  c10 $ 30         ;
datalines;
         1         2         3
12345678901234567890123456789 0
.   .   .   1   9    a       a
.   2   .   2   9            c
.   .   .   3   9        c
.   .   3   2   9            d
.   .   2   3   9        b   e
.   .   .   1   9            a
.   .   .   2   9            d
.   .   .   3   9            c
.   .   .   4   9            a
.   .   .   .   9            d
;
run;

OPTIONS MPRINT SYMBOLGEN NOQUOTELENMAX;

*** ------------------------------------------------------------------ ***
*** Goal - demonstrate a technique for determining the sparsity of    ***
***           dataset variables.                                       ***
*** A macro is constructed so as to be able to check any dataset at   ***
***   any level (percentage) of sparseness.  The macro also allows    ***
***   for dropping the variables, or not, as part of  the macro       ***
***   execution.  The macro variable containing the list of vars to   ***
***   be dropped can be used at a later time if dropping them now is  ***
***   not required.                                                    ***
*** ------------------------------------------------------------------ **;

%MACRO SPARSEVARS( dsin=             /* input dataset                  */
                  ,dsout=            /* output ds                      */
                  ,pct_sparse=100    /* threshold to address           */
                  ,alter=N           /* drop the sparse variables? Y/N
                                        &dsout is used as target       */
                  ,print=N           /* create a ds of the dropped
                                        sparse variables?          Y/N */
                 );

%if %symexist(sparsevars) %then %symdel sparsevars;          ❶

proc format;                                                 ❷
     value nmiss
                . = " "
           other = "1"
     ;
     value $cmiss
               " " = " "
```

```
              other = "1"
     ;
run;

ods select none;                                                    ❸
ods output onewayfreqs = oneways;                                   ❹
ods output nlevels = nvalues ;

proc freq data = &dsin nlevels order= formatted ;                   ❺
    format _character_ $cmiss. _numeric_ nmiss.;
    table _all_ / missing;
run;

%let opendsid=%sysfunc(open(nvalues));                              ❻
%let num=%sysfunc(varnum(&opendsid,nmisslevels));
%let closedsid=%sysfunc(close(&opendsid));

%global sparsevars;                                                 ❼

%if &num > 0 %then %do;                                             ❽
proc sql noprint ;
    select scan(table,2," ") as var
      into :sparsevars separated by " "
      from oneways
     where percent  >= &pct_sparse
       and frequency = cumfrequency
       and (select nmisslevels
              from nvalues
             where calculated var = tablevar) = 1
    ;

%if &dsout ne &dsin %then %do;
    create table &dsout as
    select *
      from &dsin;
                         %end;
%if (%upcase(&alter) = Y and &sparsevars ne ) %then %do;
 %let sparsevars = %sysfunc(translate(&sparsevars,","," "));
     alter table &dsout.
      drop &sparsevars;
                                                        %end;
quit;

%if (%upcase(&print) = Y and &sparsevars ne ) %then %do;
data droppedvars(drop=i varcnt string strlen);
    varcnt = count("&sparsevars",
%if %upcase(&alter) = Y %then    ",") + 1;
                        %else    " ") + 1;;
    length string $20000;
    string = "&sparsevars";
    strlen = length(string);
    do i = 1 to varcnt;
                      droppedvar = scan(string,i);
                      output;
    end;
run;
                                                      %end;
%if &sparsevars ne %then %put
NOTE: THE FOLLOWING VARIABLES MET THE &pct_sparse.% SPARSENESS CRITERIA:
      &sparsevars;
                    %else %put
NOTE: NO VARIABLES MET THE &pct_sparse.% SPARSENESS CRITERIA. ;
                    %end;
%else %put NOTE: There are no missing levels in &dsin      ❾
-- This is a good thing;
```

```
options quotelenmax;
ods select all;

%MEND SPARSEVARS;

%SPARSEVARS( dsin=sparsetest
            ,dsout=testout
            ,pct_sparse=100
            ,alter=Y
            ,print=Y
           );

%SPARSEVARS( dsin=sparsetest
            ,dsout=sparsetest
            ,pct_sparse=50
            ,alter=Y
            ,print=N
           );
```

Before we explain the macro code and its usage, let us view the SparseTest data set (Figure 13.2) and the two data sets created by the macro (Figure 13.3).

Figure 13.2: SparseTest data set

	n100	n90	n80	n10	constant	c100	c90	c80	c10
1		.	.	1	9		a		a
2	.	2	.	2	9				c
3	.	.	.	3	9			c	
4	.	.	3	2	9				d
5	.	.	2	3	9			b	e
6	.	.	.	1	9				a
7	.	.	.	2	9				d
8	.	.	.	3	9				c
9	.	.	.	4	9				a
10	.	.	.	.	9				d

Notice from Figure 13.2 that the data set has both numeric and character variables that are named after their sparseness percentage (n100 and c100 are completely empty).

Figure 13.3: The NValues and OneWays data sets

NVALUES

Filter and Sort | Query Builder | Data | Describe | Graph | Analyz

	TableVar	NLevels	NMissLevels	NNonMissLevels
1	n100	1	1	0
2	n90	2	1	1
3	n80	2	1	1
4	n10	2	1	1
5	constant	1	0	1
6	c100	1	1	0
7	c90	2	1	1
8	c80	2	1	1
9	c10	2	1	1

ONEWAYS

Filter and Sort | Query Builder | Data | Describe | Graph | Analyze | Export | Send To

	Table	F_n100	n100	Frequency	Percent	CumFrequency	CumPercent
1	Table n100			10	100.00	10	100.00
2	Table n90			9	90.00	9	90.00
3	Table n90			1	10.00	10	100.00
4	Table n80			8	80.00	8	80.00
5	Table n80			2	20.00	10	100.00
6	Table n10			1	10.00	1	10.00
7	Table n10			9	90.00	10	100.00
8	Table constant			10	100.00	10	100.00
9	Table c100			10	100.00	10	100.00
10	Table c90			9	90.00	9	90.00
11	Table c90			1	10.00	10	100.00
12	Table c80			8	80.00	8	80.00
13	Table c80			2	20.00	10	100.00
14	Table c10			1	10.00	1	10.00
15	Table c10			9	90.00	10	100.00

The end of Program 13.2 shows that the macro SparseVars is invoked twice, at differing levels of the sparseness threshold. The below annotations are considered with the first invocation of the macro.

❶ Checks whether the macro variable SparseVars exists and if so deletes it. It is nice to know that we are starting fresh.

❷ The FORMAT procedure creates two formats; nmiss is for numeric data, and cmiss is for character data. As we are only interested in missing versus nonmissing values, OTHER is used to identify nonmissing values.

❸ Suppresses output as there is no need to see it.

❹ The two ODS statements set up data sets, Oneways and NValue. These receive the data from two ODS objects, Onewayfreqs and Nlevels, respectively, created by the next step.

❺ The FREQ procedure is the workhorse of the macro. The option NLEVEL in the PROC FREQ statement generates a summary of the level counts and causes the Nlevels output object to be created. The FORMAT statement applies the Nmiss format to the numeric variables and the Cmiss format to the character variables. The _ALL_ keyword in the TABLES statement generates a separate table for each variable in the Sparsetest data set. The FREQ procedure generates two ODS objects, Nlevels and Onewayfreqs, for each variable in the Sparsetest data set. These are captured by the two data sets, NValue and Oneways, set up in step ❹.

❻ Looks for the location of Nmissvalues variable in Nlevels data set. The OPEN function used in conjunction with the %SYSFUNC function opens the data set. The VARNUM function returns the position of Nmisslevels in Nlevels data set. The macro variable Num has a positive value if Nmisslevels exists. If Nmisslevels column is not found, that is, no variables in Nlevels have missing values, Num resolves to 0. The data set is then closed with the CLOSE function. &num is used later with conditional logic. Had we not set up this capability and the source data set had no missing value structure, the compiler would generate an error.

❼ Creates a macro variable called SparsVars. The GLOBAL statement allows the macro variable to be referenced outside the macro.

❽ If &Num is a positive number, the code within the DO loop is processed. The macro code within the DO loop simply writes out SQL statements. In order to understand it, let us view the code once it resolves for the first invocation of the macro. Following is the log view with the MPRINT system option turned on.

```
MPRINT(SPARSEVARS):   proc sql noprint ;
MPRINT(SPARSEVARS):   select scan(table,2," ") as var into :sparsevars
separated by " " from
oneways where percent >= 100 and frequency = cumfrequency and (select
nmisslevels from nvalues
where calculated var = tablevar) = 1;
MPRINT(SPARSEVARS):   create table testout as select * from sparsetest;
NOTE: Table WORK.TESTOUT created, with 10 rows and 9 columns.
SYMBOLGEN:  Macro variable SPARSEVARS resolves to n100,c100
MPRINT(SPARSEVARS):   alter table testout drop n100,c100; *
NOTE: Table WORK.TESTOUT has been modified, with 7 columns.
MPRINT(SPARSEVARS):   data droppedvars(drop=i varcnt string strlen);
MPRINT(SPARSEVARS):   varcnt = count("n100,c100", ",") + 1;
MPRINT(SPARSEVARS):   length string $20000;
MPRINT(SPARSEVARS):   string = "n100,c100";
MPRINT(SPARSEVARS):   strlen = length(string);
MPRINT(SPARSEVARS):   do i = 1 to varcnt;
MPRINT(SPARSEVARS):   droppedvar = scan(string,i);
MPRINT(SPARSEVARS):   output;
```

```
MPRINT(SPARSEVARS):   end;
MPRINT(SPARSEVARS):   run;

NOTE: The data set WORK.DROPPEDVARS has 2 observations and 1 variables.
```

The SELECT statement places variables with 100 percent missing values into the macro variable Sparsevars, which resolves to "n100 c100." The WHERE clause ensures that only variables with 100 percent missing values are selected. The CREATE statement creates the TestOut data set as a simple copy of the source data.

Since the value of ALTER is "Y," we modify the TestOut data set, dropping the two variables, n100 and c100. The TRANSLATE function places commas between the variable names in macro variable SparseVars. This makes SparseVar PROC SQL syntax compliant.

The value of PRINT is "Y," so we create DroppedVars data set with one column containing the names of the dropped variables.

❾ If the DO loop in Step ❽ is not processed, then we know that there are no variables that are 100 percent sparse. A message is written to the LOG file.

Please note that this macro only handles the missing value structure. What about a variable having a constant value? For example, the variable Product is always valued at "MEDICAID." This is analogous to the variable being 100 percent missing, equally useless for analytics. How might the macro be enhanced to account for a constant variable value? These questions will be left for the reader to consider.

Arrays of Detail Record Elements on the Header Record

Many times a work product will require that we create an array of code elements that originate as row-wise data on the claim detail file (837I or 837P) and are targeted to be an array of column-wise data with one record per claim. This, in effect, is a loss of detail as we transform the fields of interest onto the header record. While this does not sound like the right thing to do, many appropriate designs call for this type of data manipulation.

Examples might include design rules such as the following:

All possible Revenue Codes from Facility detail to be arrayed on the header record.

Ten highest paying procedure codes from professional detail to be arrayed on the header record.

Any number of reasons may suggest such a data transformation, but the underlying goal is that a target "solution" that receives the data needs this architecture. Several such target solutions may be:

Sending outpatient claims to grouping or scrubbing software such as APC (Ambulatory Payment Classifications) grouping and NCCI/MUE (National Correct Coding Initiative/Medically Unlikely Edit) editing. Several commercial vendor solutions exist that need, as a data feed, one line per claim with the CPT codes from the detail lines arrayed as individual columns on the claim header record.

Clinical subject matter experts (SMEs) many times want to see all the room and board revenue codes for an inpatient stay on one record, looking for specific high-cost beds.

> Ambulatory Payment Classifications (APC) is the basis for reimbursing some acute care facilities for outpatient services (e.g., Outpatient Prospective Payment System or OPPS). This methodology was developed because of the success of DRGs as a basis for the Inpatient Prospective Payment System (IPPS).
>
> The National Correct Coding Initiative (NCCI) is a set of billing edits developed by CMS to identify coding patterns resulting in overpayment to the providers.

While this routine could be hardcoded for a particular project, let's consider the solution as a macro that entertains several user-provided parameters, allowing for a more robust and reusable solution.

Program 13.3 shows the macro code as well as its usage.

Program 13.3: Macro HdrArrayCodes

```
%macro hdrarraycodes( libref       =              /* library reference */
                     ,hdrdsn       =              /* header table name */
                     ,dtldsn       =              /* detail table name */
                     ,patientidvar = MemberID     /* hdr member id var */
                     ,clmidvar     = ClaimID      /* claim id var      */
                     ,clmlinevar   = ClaimLine    /* claim line id var */
                     ,dtlvar2array =              /* var of interest
                                                        to array       */
                     ,maxnumoflines=              /* # of lines to use,
                                                     blank for all     */
                     ,hdrkeepvars  =              /* hdr vars to keep  */
                    );

*** ------------------------------------------------------------------ ***
*** Goal - create column-wise structure from a row-wise structure     ***
***        specifically, put each field value from the detail file    ***
***          into an array of fields on an image of the header record ***
***        example - bring all Revenue Codes, or the first N,         ***
***                  from the Facility Detail Table (rowwise) to      ***
***                  a copy of the Facility Header Table (columnwise) ***
***        this macro                                                 ***
***            handles only one arrayed variable although it could    ***
***                    be expanded for more                           ***
***            assumes a claim line number sort order, a further      ***
***                    enhancement is to make this user choice        ***
*** ------------------------------------------------------------------ ***;
proc sql noprint;
    select max(maxlines)
      into :linecnt  separated by ","
      from (select &clmidvar ,count(*)   as maxlines
              from &libref..&dtldsn
              group by &clmidvar)
    ;
    create table claimdtl as
    select *
      from &libref..&dtldsn(keep=&clmidvar &clmlinevar &dtlvar2array)
     order by &clmidvar, &clmlinevar
    ;
    create table claimhdr as
    select *
      from &libref..&hdrdsn(keep= &patientidvar &clmidvar &hdrkeepvars)
     order by &clmidvar
    ;
quit;

%if &maxnumoflines ne
   %then %let linecnt = &maxnumoflines.;

%put NOTE: Need &linecnt new columns to store the arrayed information;

data &hdrdsn._colwise;
    retain &patientidvar. &clmidvar. &hdrkeepvars.;
    drop i &dtlvar2array.;
    array temp $ &dtlvar2array.1 - &dtlvar2array.&linecnt.;
    do i=1 to &linecnt.;
                        merge claimhdr claimdtl;
                        by &clmidvar.;
                        temp(i) = &dtlvar2array.;
                        if last.&clmidvar.
                          then do;
                                  output;
```

```
                                leave;
                              end;
    end;
run;

%mend hdrarraycodes;


%hdrarraycodes( libref       = book
               ,hdrdsn       = Facilityheader
               ,dtldsn       = Facilitydetail
               ,patientidvar = MemberID
               ,clmidvar     = ClaimID
               ,clmlinevar   = ClaimLine
               ,dtlvar2array = RevCode
               ,maxnumoflines= 10
               ,hdrkeepvars  = BillType DRG
              );
```

The easiest way to explain this macro is to go through the log file when it is invoked (Figure 13.4). Note that the keyword MPRINT has been removed for ease of reading.

Figure 13.4: HrdArrayCodes macro resolution

```
proc sql noprint;                                                    ❶
select max(maxlines) into :linecnt separated by "," from (select ClaimID
,count(*) as maxlines from book.Facilitydetail group by ClaimID);

create table claimdtl as select * from
book.Facilitydetail(keep=ClaimID ClaimLine RevCode) order by ClaimID,
ClaimLine;
NOTE: Table WORK.CLAIMDTL created, with 98818 rows and 3 columns.

create table claimhdr as select * from book.Facilityheader(keep=
MemberID ClaimID BillType DRG) order by ClaimID;
NOTE: Table WORK.CLAIMHDR created, with 6654 rows and 4 columns.

quit;
NOTE: Need 10 new columns to store the arrayed information
data Facilityheader_colwise;
retain MemberID ClaimID BillType DRG;                                ❷
drop i RevCode;
array temp $ RevCode1 - RevCode10;                                   ❸
   do i=1 to 10;
     merge claimhdr claimdtl;                                        ❹
     by ClaimID;
     temp(i) = RevCode;
     if last.ClaimID then do;
       output;
       leave;
     end;
   end;
run;

NOTE: There were 6654 observations read from the data set WORK.CLAIMHDR.
NOTE: There were 98818 observations read from the data set WORK.CLAIMDTL.
NOTE: The data set WORK.FACILITYHEADER_COLWISE has 6654 observations and 15
variables.
```

❶ The SELECT statement queries for the maximum number of claim detail lines and stores the value in macro variable LineCnt. If the user does not supply the value of MaxNumofLines at macro invocation, then LineCnt from the SELECT query is used instead. The two CREATE TABLE statements create claimdtl and claimhrd tables, respectively, ensuring that only variables that are needed are kept.

❷ The RETAIN statement ensures the order of the variables in the output data set FacilityHeader_Colwise.

❸ The ARRAY statement sets up RevCode1 through RevCode10 variables; 10 is the maximum number of lines of detail for this data set as determined by the SQL query creating the macro variable. These variables serve to capture RevCode from ClaimDtl as its rows are converted into columns.

❹ The MERGE statement is placed inside the DO loop. This is key to the design of the DATA step as the MERGE statement iterates up to 10 times for each ClaimID. When i=1 for a ClaimID, then the first claim line is read from ClaimDtl and is merged with its appropriate record from ClaimHdr; the RevCode is extracted and placed into RevCode1 column. The loop iterates up to 10 times (the maximum number of lines per claim), and the 10th observation is placed in the output data set along with RevCode1–RevCode10. If the current record is the last in ClaimID BY group and the loop has not reached its 10th iteration, then it is placed in the output data set and the loop terminates due to the LEAVE statement.

Figure 13.5 shows the FacilityHeader_Colwise data set.

Figure 13.4: FacilityHeader_Colwise data set (not all RevCode columns shown)

MemberID	ClaimID	BillType	DRG	RevCode1	RevCode2	RevCode3	RevCode4	RevCode5	R
M944001983369...	C042929171	332	000	0023					
M274319753132...	C042929554	111	871	0483	0622	0730	0921		
M432744509170...	C042931278	111	853	0801					
M858220924345...	C042931758	111	464	0410	0420	0424	0710	0730	
M836592016341...	C042931936	111	207	0450	0480	0730	0921	0940	0942
M921679084363...	C042931937	111	234	0420	0430	0434	0460	0480	0481
M811072290335...	C042932129	831	000	0490	0490	0278			
M947862448371...	C042932435	111	743	0370	0636	0637	0710	0730	
M878982926350...	C072871521	111	191	0460	0732				
M847439047342...	C072871530	111	641	0450	0730	0732			
M827928886339...	C072871605	111	391	0350	0402	0450	0730	0732	0740

Linking to the Diagnosis Pointer

As we learned in Chapter 6, the detail service lines of a professional/ancillary claim contain a "pointer" that associates one or more diagnosis codes from the header table with the service. Up to four reference letters or numbers per detail line are allowed; in other words, up to four diagnoses may be associated with a service line. In our sample data, we have chosen, as many payers do, to put the actual diagnosis code on the detail table rather than the pointer values (A–L for the CMS-1500 and 1–12 for the 839P). This strategy inflates the storage requirement to store more than just the pointer but, on the upside, prevents the problem program from having to do the merge/join work each time.

Should you find just the pointers on your detail table, the code in Program 13.5 can be used/modified to look up the associated diagnosis codes. Note that for this exercise we will assume our data originates from the CMS-1500 form with a character field (taken from Item 24E on the form) of maximum length 4. If the data source is the 837P, then you will find four fields of two bytes each with which to work.

Since we do not have existing data in the form we need it, we will generate some test data to play with. This, of itself, is a useful exercise, as a good programmer will always test code against known data. The code to generate test data is shown in Program 13.4

Program 13.4: Generating test data

```
*** ------------------------------------------------------------------ ***
***          Build test claim files from scratch (header and detail)  ***
***          this is a useful exercise as it not only provides data   ***
***          to play with, but we exercise some tools for randomly    ***
***          generating data                                          ***
***        - the random seed is a constant to get reproducible data   ***
*** ------------------------------------------------------------------ **;

data test_hdr(keep=MemberID ClaimID dx01 - dx12)
     test_dtl(keep=MemberID ClaimID ClaimLine dxpointer Px);
    length MemberID $3 ClaimID $9 ClaimLine 3 dxpointer $12;
```

```
    array dx (*) $ dx01 - dx12;
    do mbr = 1 to 3;                                              *loop #1;
      MemberID = cats('00',mbr);
      do clm = 1 to (int(8 * ranuni(2)) + 1);                     *loop #2;
        ClaimID = cats(MemberID,'ccc00',clm);
        do x = 1 to dim(Dx);
          Dx(x) = ' ' ;
        end;
        do k = 1 to (int(12 * ranuni(4)) + 1);
          dx(k) = put((int(99999 * ranuni(12)) + 1),5.);              ❶
        end;
        output test_hdr;
        do claimline = 1 to (int(10 * ranuni(8)) + 1);            *loop #3;
          dxpointer = ' ';
          do z = 1 to 4;
          substr(dxpointer,z) = byte(round(65+((k-1)*ranuni(0)),1)); ❷
          end;
          len=length(dxpointer);
          do x = 1 to len-1;
            if substr(dxpointer,x,1) ^= ' '
              then do y = x+1 to len;
                     if substr(dxpointer,x,1)=substr(dxpointer,y,1)
                       then substr(dxpointer,y,1)=' ';               ❸
                   end;
          end;
          dxpointer=compress(dxpointer,' ');
          Px =  put((int(99507 * ranuni(12)) + 100),z5.);
          output test_dtl;

            end;                                                *ends loop #3;
        end;                                                    *ends loop #2;
    end;                                                        *ends loop #1;
run;
```

The single DATA step shown in Program 13.4 generates Test_Hdr and Test_Dtl data sets (Figure 13.6).

Figure 13.4: The Test_Hdr and Test_Dtl data sets

TEST_HDR

Filter and Sort | Query Builder | Data | Describe | Graph | Analyze | Export | Send To

	MemberID	ClaimID	dx01	dx02	dx03	dx04	dx05	dx06
1	001	001ccc001	79965	51880	84320	93855	8596	6339
2	001	001ccc002	89387	42580	20655	35035	53781	22972
3	001	001ccc003	54128	30643	84274	7596	54161	
4	002	002ccc001	37287	98393	65998	8879	40693	15800
5	002	002ccc002	11384	31650	36930	2870	53538	21657
6	002	002ccc003	30548	33150	12467	16381	41247	84953
7	003	003ccc001	55681	60799	81187	61445	24041	50911
8	003	003ccc002	95021	89741	27993	96618	30115	3846
9	003	003ccc003	45620	89097	88573	30527	72106	30727
10	003	003ccc004	45507	52506	72363	2949	84173	1951
11	003	003ccc005	80761	90595	35994	1017	1927	35447
12	003	003ccc006	12438	46931	94634	25158		

TEST_DTL

Filter and Sort | Query Builder | Data | Describe | Graph | Analyze | Expor

	MemberID	ClaimID	ClaimLine	dxpointer	Px
1	001	001ccc001	1	HFL	45178
2	001	001ccc001	2	FKBH	94788
3	001	001ccc001	3	IDC	75130
4	001	001ccc001	4	FAK	17943
5	001	001ccc001	5	HJLE	33551
6	001	001ccc001	6	EJHK	79799
7	001	001ccc001	7	CBDA	86313
8	001	001ccc001	8	EKD	90350
9	001	001ccc001	9	FCGL	09924
10	001	001ccc001	10	BIF	87081
11	001	001ccc002	1	BGHC	08576
12	001	001ccc002	2	KJBF	80745
13	001	001ccc002	3	CEH	24292

The key to the DATA step shown in Program 13.4 is the ingenious use of three DO loops. The first loop creates data for three Members. The second loop generates up to eight claims per member. The third loop creates up to ten claim lines for each ClaimID.

❶ Creates up to twelve diagnoses for each ClaimID.

❷ Creates Dxpointer, which has up to four alpha values (A through L). The BYTE function changes random numbers, constrained within the ranges 65 to 76, to values A through L.

❸ Removes duplicate values within the four strings of DxPointer. If a previous string value is the same as the current, then the current value is changed to missing.

Program 13.5 shows the code to replace DxPointer with up to four Diagnosis codes.

Program 13.5: Linking to diagnosis pointer

```
*** ------------------------------------------------------------------ ***
***          Put the correct Dxs from the header on to a copy of the  ***
***          detail table using the DxPointer string on each line    ***
*** ------------------------------------------------------------------ **;

data detail(drop=dx01-dx12 dxpointer i);
    merge test_hdr
          test_dtl;
    by MemberID ClaimID;
    array dx    (*) $ dx01  - dx12 ;
    array dxpt  (*) $ dxpt1 - dxpt4;
    do i = 1 to length(dxpointer);
        dxpt(i) = ' ';
        dxpt(i) = dx(rank(substr(dxpointer,i,1))-64);
```

```
    end;
run;
```

The key to the DATA step in Program 13.5 is the use of the RANK function, which can be thought of as an inverse of the BYTE function. Each character of DxPointer is converted to a number using the RANK function; the number 64 is subtracted from the resultant to constrain the final number to values ranging from 1 to 12. This enables DxPt(i) to be assigned to the right diagnosis from ARRAY Dx. Figure 13.7 shows the Detail data set.

Figure 13.7: Detail data set

MemberID	ClaimID	ClaimLine	Px	dxpt1	dxpt2	dxpt3	dxpt4
001	001ccc001	1	45178	13314	6339	13437	
001	001ccc001	2	94788	6339	70679	51880	13314
001	001ccc001	3	75130	63864	93855	84320	
001	001ccc001	4	17943	6339	79965	70679	
001	001ccc001	5	33551	13314	4775	13437	8596
001	001ccc001	6	79799	8596	4775	13314	70679
001	001ccc001	7	86313	84320	51880	93855	79965
001	001ccc001	8	90350	8596	70679	93855	
001	001ccc001	9	09924	6339	84320	9959	13437
001	001ccc001	10	87081	51880	63864	6339	

Conclusion

This chapter has covered a number of programmatic solutions that are necessary in the manipulation of administrative healthcare data. These include:

Age-as-of macro
Data sparseness macro
Attaching detail rows as columns to header data
Linking diagnoses via diagnosis pointer

Appendix

Glossary of Terms **195**
CMS-1450 Claim Form **201**
CMS-1500 Claim Form **202**
Universal Claim Form for Prescription Drugs **203**
Facility Type **205**
Bill Sequence **206**
Place of Service **207**
Patient Status Code **210**
Revenue Code **211**

Glossary of Terms

Access to Care
: A person's ability to obtain affordable medical care on a timely basis.

Ancillary Services
: Auxiliary or supplemental services, such as diagnostic services, home health services, physical therapy, and occupational therapy, used to support diagnosis and treatment of a patient's condition.

Capitation
: A method of paying for health care services based on the number of patients who are covered for specific services over a specified period of time rather than the cost or number of services that are actually provided.

Case Management
: A process of identifying plan members with special healthcare needs, developing a healthcare strategy that meets those needs, and coordinating and monitoring the care, with the ultimate goal of achieving the optimum health care outcome in an efficient and cost-effective manner.

Case-Mix Adjustment
: See Risk Adjustment.

Centers for Medicare & Medicaid Services (CMS)
: See Health Care Financing Administration (HCFA).

Claim
: An itemized statement of healthcare services and their costs provided by a hospital, physician's office, or other provider facility. Claims are submitted to the insurer or managed care plan by either the plan member or the provider for payment of the costs incurred.

Claim Form
: An application for payment of benefits under a health plan.

Claims Adjudication
: The process by which providers are paid for services rendered. The source for the process is claim forms and the result is a feed to the payment process.

Claims Administration
: The process of receiving, reviewing, adjudicating, and processing claims.

Coinsurance
: A percentage of the costs of covered health-related services paid after the annual deductible is met.

Commercial Insurance
: Healthcare coverage paid for by employers or individual consumers.

Copayment
: A flat fee, or set amount, that is paid for a specific health-related service. Copayments are very common in managed care plans such as HMOs and PPOs.

Current Dental Terminology (CDT)
: A five-character alphanumeric code that begins with the letter D and is followed by four numerals. Each code identifies a specific dental procedure. This code set is maintained by the American Dental Association.

Current Procedural Terminology (CPT)
: A five-character alphanumeric code. Each code identifies a specific medical procedure. This code set is maintained by the American Medical Association.

Deductible
: The amount paid out of pocket each year for health-related expenses before the insurance policy begins to pay.

Diagnostic Related Group (DRG)
: A method of classifying inpatient hospital services. It is used as a method of determining financing to reimburse various providers for services performed.

Disease Management
: A coordinated system of preventive, diagnostic, and therapeutic measures intended to provide cost-effective, quality health care for a patient population who have or are at risk for a specific chronic illness or medical condition.

Disposable Medical Supplies (DMS)
: Non-reusable supplies ordered by a provider for patient use in the home. Examples include diabetic, colostomy, and tracheostomy disposable material.

Drug Utilization Review (DUR)
: A review program that evaluates whether drugs are being used safely, effectively, and appropriately.

Durable Medical Equipment (DME)
: Medical equipment that is ordered by a doctor for use in the home. These items must be reusable, such as walkers, wheelchairs, or hospital beds.

Early and Periodic Screening, Diagnosis, and Treatment (EPSDT)
: Services, including screening, vision, hearing, and dental services, provided under Medicaid to children under age twenty-one at intervals that meet recognized standards of medical and dental practices and at other intervals as necessary in order to determine the existence of physical or mental illnesses or conditions. Plans offering Medicaid coverage to EPSDT participants must provide any service that is necessary to treat an illness or condition that is identified by screening.

Electronic Data Interchange (EDI)
: A protocol for the exchange of standardized document forms between computer systems for business use.

Encounter
: Each time that a person meets with a healthcare provider to receive services.

Episode
: A unit of patient assessment is the episode of care, defined as a health problem from its first encounter with a healthcare provider through the completion of the last encounter. An episode of care is distinct from an episode of disease or illness.

Evaluation and Management (E/M)
: A set of CPT codes that are used specifically for preventive care and well-child visits.

Fee-for-Service
: A traditional method of paying for medical services. A doctor charges a fee for each service provided, and the insurer or patient pays all or part of that fee.

Formulary
: A listing of drugs, classified by therapeutic category or disease class, that are considered preferred therapy for a given managed population and that are to be used by an MCO's providers in prescribing medications.

Grouper
: A software system that results in each code being assigned to a group that makes sense. For example, DRGs are assigned by proprietary software based on a set of diagnostic and procedural codes that was generated by an inpatient episode of care.

HCFA Common Procedure Coding System (HCPCS)
: A set of medical procedures for coding and payment determinations for clinical laboratory tests and for durable medical equipment

Health Care Financing Administration (HCFA)
: Now Centers for Medicare & Medicaid Services (CMS), this is a federal agency within the U.S. Department of Health and Human Services. Programs for which CMS is responsible include Medicare, Medicaid, State Children's Health Insurance Program, and HIPAA.

Health Insurance Portability and Accountability Act (HIPAA)
: A federal act that protects people who change jobs, are self-employed, or who have pre-existing medical conditions. HIPAA standardizes an approach to the continuation of healthcare benefits for individuals and members of small group health plans and establishes parity between the benefits extended to these individuals and those benefits offered to employees in large group plans. The act also contains provisions designed to ensure that prospective or current enrollees in a group health plan are not discriminated against based on health status.

Health Maintenance Organization (HMO)
: The oldest form of managed care plan. HMOs offer members a range of health benefits for a set monthly fee. HMOs give you a list of doctors from which to choose a primary care doctor. This doctor coordinates your care, which means that generally you must contact him or her to be referred to a specialist.

HEDIS
: Health Plan Effectiveness Data and Information Set—NCQA's tool used by health plans to collect data about the quality of care and service they provide.

ICD-9-CM
: The International Classification of Diseases, 9th Revision, Clinical Modification. This code set contains both diagnosis and surgical procedures.

ICD-10-CM
: The International Classification of Diseases, 10th Revision, Clinical Modification. This code set contains both diagnosis and surgical procedures.

Individually Identifiable Health Information (IIHI)
: If one could figure out to whom a particular piece of healthcare information refers by "looking" at it, then it is IIHI. Common identifiers include name, address, and social security number, date of birth, zip code, and county location. IIHI only becomes PHI when a covered entity creates, receives, or maintains the information.

Logical Observation Identifiers Names and Codes (LOINC)
: A database and universal standard for identifying laboratory observations. Developed and maintained by the Regenstrief Institute, Inc., in response to the demand for an electronic database for clinical care and management. It is publicly available at no cost.

Major Diagnostic Categories (MDC)
: A categorization of ICD-9-CM that is formed by dividing all possible principal diagnoses into twenty-five mutually exclusive diagnosis areas. Synonymously known as Major Disease Categories.

Managed Behavioral Healthcare Organization (MBHO)
: A system of behavioral health care delivery that manages quality, utilization, and cost of services and that measures performance in the area of mental and substance abuse disorders.

Managed Care Organization (MCO)
: A general term for health plans that provide healthcare in return for preset monthly payments and coordinate care through a defined network of primary care physicians and hospitals.

Medicaid
: A jointly funded federal and state program that provides hospital expense and medical expense coverage to the low-income population and certain aged and disabled individuals.

Medicare
: A federal government hospital and medical expense insurance plan primarily for elderly and disabled persons.

Medicare Part A
: The part of Medicare that provides basic hospital insurance coverage automatically for most eligible persons. See also Medicare.

Medicare Part B
: A voluntary program that is part of Medicare and provides benefits to cover the costs of physicians' services. See also Medicare.

Medicare Part C
: The part of Medicare that expands the list of different types of entities allowed to offer health plans to Medicare beneficiaries. See also Medicare.

Medicare Part D
: A voluntary program that is part of Medicare and provides benefits to cover the costs of prescription drugs. See also Medicare.

National Drug Code (NDC)
: The National Drug Code (NDC) System was originally established as an essential part of an out-of-hospital drug reimbursement program under Medicare. The NDC serves as a universal product identifier for human drugs. The U.S. Federal Drug Administration maintains the system.

Outcomes Measures
: Healthcare quality indicators that gauge the extent to which healthcare services succeed in improving patient health.

Outpatient Care
: Treatment that is provided to a patient who is able to return home after care without an overnight stay in a hospital or other inpatient facility.

Physician Profiling
: In the context of a pharmacy benefit plan, the process of compiling data about physician prescribing patterns and comparing physicians' actual prescribing patterns to expected patterns within select drug categories. Other practice patterns might also be profiled. Also known as Provider Profiling.

Place of Service (POS)
: This designates where the actual health services are performed, whether it be home, hospital, office, clinic, and so on.

Point of Service (POS)
: A healthcare option that allows members to choose, at the time medical services are needed, whether they will go to a provider within the plan's network or seek medical care outside the network.

Preferred Provider Organization (PPO)
: A healthcare benefit arrangement designed to supply services at a discounted cost by providing incentives for members to use designated healthcare providers (who contract with the PPO at a discount), but which also provides coverage for services rendered by healthcare providers who are not part of the PPO network.

Primary Care Provider (PCP)
: A physician or other medical professional who serves as a group member's first contact with a plan's healthcare system. This is the medical provider chosen by the enrollee in a managed care organization. Also known as a primary care physician, personal care physician, or personal care provider.

Product Line
: Refers to the population that the health plan covers. Examples are Commercial, Medicare, and Medicaid. See their respective definitions.

Prospective Payment System (PPS)
: A system of Medicare reimbursement for Part A benefits that bases most hospital payments on the patient's diagnosis at the time of hospital admission.

Protected Health Information (PHI)
: Protected health information (PHI) under HIPAA means individually identifiable health information. Identifiable refers not only to data that is explicitly linked to a particular individual (that is identified information). It also includes health information with data items that reasonably could be expected to allow individual identification.

Quality Management (QM)
: An organization-wide process of measuring and improving the quality of the health care provided by an MCO.

Rating
: The process of calculating the appropriate premium to charge purchasers, given the degree of risk represented by the individual or group, the expected costs to deliver medical services, and the expected marketability and competitiveness of the MCO's plan. Also known as Rate Setting.

Revenue Code
: Payment codes for services or items found on the CMS-1450 claim form.

Risk Adjustment
: The statistical adjustment of outcome measures to account for risk factors that are independent of the quality of care provided and beyond the control of the plan or provider, such as the patient's gender and age, the seriousness of the patient's illness, and any other illnesses the patient might have. Also known as case-mix adjustment.

Risk Stratification
: A statistical process by which quality of care can be assessed independently of patient case-mix. Evaluation of risk-adjusted patient outcome has become an important part of managed care contracting in some markets, and risk-adjusted outcome rates for hospitals are reported more frequently in the popular press and on the Internet.

Type of Bill
: This three-digit code is found on the CMS-1450 claim form. It has three pieces of built-in intelligence: facility type, type of care, and bill frequency (or bill sequence).

Type of Service (TOS)
: A categorization of HCPCS codes to describe the type of medical service provided.

Unbundling
: A coding inconsistency that involves separating a procedure into parts and charging for each part rather than using a single code for the entire procedure.

Underwriting
: The process of estimating what the expected covered health costs will be for a particular person or group. Medical underwriting includes assessment of the health or chronic illness of a person or group.

Upcoding
: A coding inconsistency that involves using a code for a procedure or diagnosis that is more complex than the actual procedure or diagnosis and that results in higher reimbursement to the provider.

Utilization Management (UM)
: Managing the use of medical services to ensure that a patient receives necessary, appropriate, high-quality care in a cost-effective manner.

Utilization Review (UR)
: The evaluation of the medical necessity, efficiency, and/or appropriateness of health care services and treatment plan.

Many thanks to the following Internet resources for providing some of the above definitions:

"The Free On-Line Dictionary of Computing." http://www.foldoc.org/. Editor: Denis Howe

www.ncqa.org

http://www.cms.gov/apps/glossary/

http://www.agencyinfo.net/iv/medical/health-glossary.htm

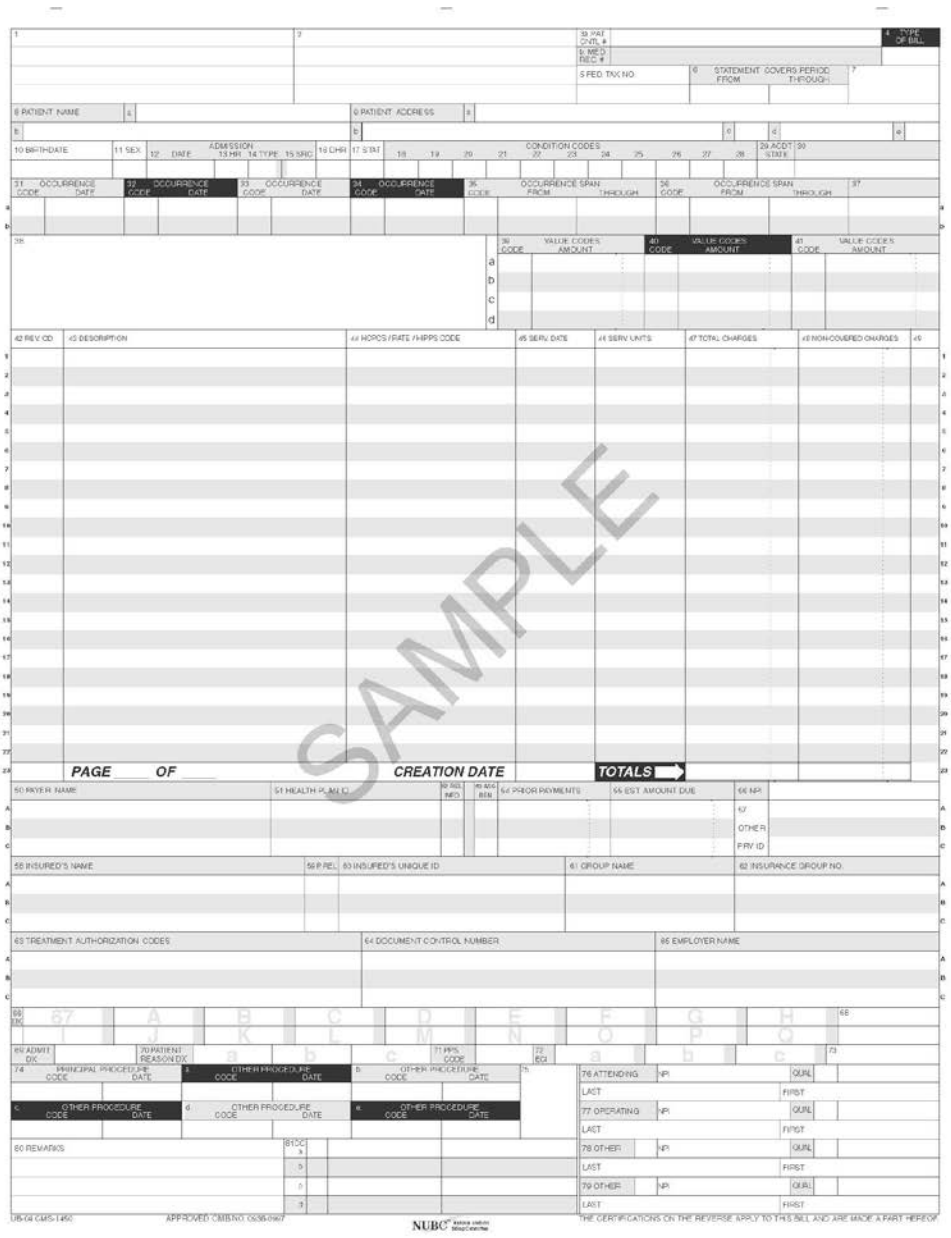
http://www.bcbs.com/glossary/

http://www.wikipedia.org/

CMS-1450 Claim Form

Figure A.1 is a facsimile CMS-1450 claim form supported by the National Uniform Billing Committee (www.nubc.org).

Figure A.1: CMS-1450 claim form



CMS-1500 Claim Form

Figure A.2 is a facsimile CMS-1500 claim form supported by the National Uniform Claim Committee (www.nucc.org).

Figure A.2: CMS-1500 claim form



HEALTH INSURANCE CLAIM FORM

APPROVED BY NATIONAL UNIFORM CLAIM COMMITTEE (NUCC) 02/12

PICA

1. MEDICARE (Medicare#) MEDICAID (Medicaid#) TRICARE (ID#/DoD#) CHAMPVA (Member ID#) GROUP HEALTH PLAN (ID#) FECA BLK LUNG (ID#) OTHER (ID#)

1a. INSURED'S I.D. NUMBER (For Program in Item 1)

2. PATIENT'S NAME (Last Name, First Name, Middle Initial)

3. PATIENT'S BIRTH DATE MM DD YY SEX M F

4. INSURED'S NAME (Last Name, First Name, Middle Initial)

5. PATIENT'S ADDRESS (No., Street)

6. PATIENT RELATIONSHIP TO INSURED Self Spouse Child Other

7. INSURED'S ADDRESS (No., Street)

CITY STATE

8. RESERVED FOR NUCC USE

CITY STATE

ZIP CODE TELEPHONE (Include Area Code) ( )

ZIP CODE TELEPHONE (Include Area Code) ( )

9. OTHER INSURED'S NAME (Last Name, First Name, Middle Initial)

10. IS PATIENT'S CONDITION RELATED TO:

11. INSURED'S POLICY GROUP OR FECA NUMBER

a. OTHER INSURED'S POLICY OR GROUP NUMBER

a. EMPLOYMENT? (Current or Previous) YES NO

a. INSURED'S DATE OF BIRTH MM DD YY SEX M F

b. RESERVED FOR NUCC USE

b. AUTO ACCIDENT? YES NO PLACE (State)

b. OTHER CLAIM ID (Designated by NUCC)

c. RESERVED FOR NUCC USE

c. OTHER ACCIDENT? YES NO

c. INSURANCE PLAN NAME OR PROGRAM NAME

d. INSURANCE PLAN NAME OR PROGRAM NAME

10d. CLAIM CODES (Designated by NUCC)

d. IS THERE ANOTHER HEALTH BENEFIT PLAN? YES NO If yes, complete items 9, 9a, and 9d.

READ BACK OF FORM BEFORE COMPLETING & SIGNING THIS FORM.

12. PATIENT'S OR AUTHORIZED PERSON'S SIGNATURE. I authorize the release of any medical or other information necessary to process this claim. I also request payment of government benefits either to myself or to the party who accepts assignment below.

SIGNED DATE

13. INSURED'S OR AUTHORIZED PERSON'S SIGNATURE I authorize payment of medical benefits to the undersigned physician or supplier for services described below.

SIGNED

14. DATE OF CURRENT ILLNESS, INJURY, or PREGNANCY (LMP) MM DD YY QUAL.

15. OTHER DATE QUAL. MM DD YY

16. DATES PATIENT UNABLE TO WORK IN CURRENT OCCUPATION MM DD YY FROM TO MM DD YY

17. NAME OF REFERRING PROVIDER OR OTHER SOURCE

17a.

17b. NPI

18. HOSPITALIZATION DATES RELATED TO CURRENT SERVICES MM DD YY FROM TO MM DD YY

19. ADDITIONAL CLAIM INFORMATION (Designated by NUCC)

20. OUTSIDE LAB? YES NO $ CHARGES

21. DIAGNOSIS OR NATURE OF ILLNESS OR INJURY Relate A-L to service line below (24E) ICD Ind.

A. B. C. D. E. F. G. H. I. J. K. L.

22. RESUBMISSION CODE ORIGINAL REF. NO.

23. PRIOR AUTHORIZATION NUMBER

24. A. DATE(S) OF SERVICE From MM DD YY	To MM DD YY	B. PLACE OF SERVICE	C. EMG	D. PROCEDURES, SERVICES, OR SUPPLIES (Explain Unusual Circumstances) CPT/HCPCS	MODIFIER	E. DIAGNOSIS POINTER	F. $ CHARGES	G. DAYS OR UNITS	H. EPSDT Family Plan	I. ID. QUAL.	J. RENDERING PROVIDER ID. #
1										NPI	
2										NPI	
3										NPI	
4										NPI	
5										NPI	
6										NPI	

25. FEDERAL TAX I.D. NUMBER SSN EIN

26. PATIENT'S ACCOUNT NO.

27. ACCEPT ASSIGNMENT? (For govt. claims, see back) YES NO

28. TOTAL CHARGE $

29. AMOUNT PAID $

30. Rsvd for NUCC Use

31. SIGNATURE OF PHYSICIAN OR SUPPLIER INCLUDING DEGREES OR CREDENTIALS (I certify that the statements on the reverse apply to this bill and are made a part thereof.)

SIGNED DATE

32. SERVICE FACILITY LOCATION INFORMATION

a. NPI b.

33. BILLING PROVIDER INFO & PH # ( )

a. NPI b.

CARRIER

PATIENT AND INSURED INFORMATION

PHYSICIAN OR SUPPLIER INFORMATION

NUCC Instruction Manual available at: www.nucc.org

PLEASE PRINT OR TYPE

APPROVED OMB-0938-1197 FORM 1500 (02-12)

SAMPLE

Universal Claim Form for Prescription Drugs

Figures A.3a and A.3b represent a facsimile Universal Claim Form supported by the National Council for Prescription Drug Programs (http://www.ncpdp.org). Materials reproduced here and in Chapter 7 are done so with the consent of ©National Council for Prescription Drug Programs, Inc., 2013 NCPDP; not for distribution or use.

Figure A.3a: Universal Claim Form for prescription drugs

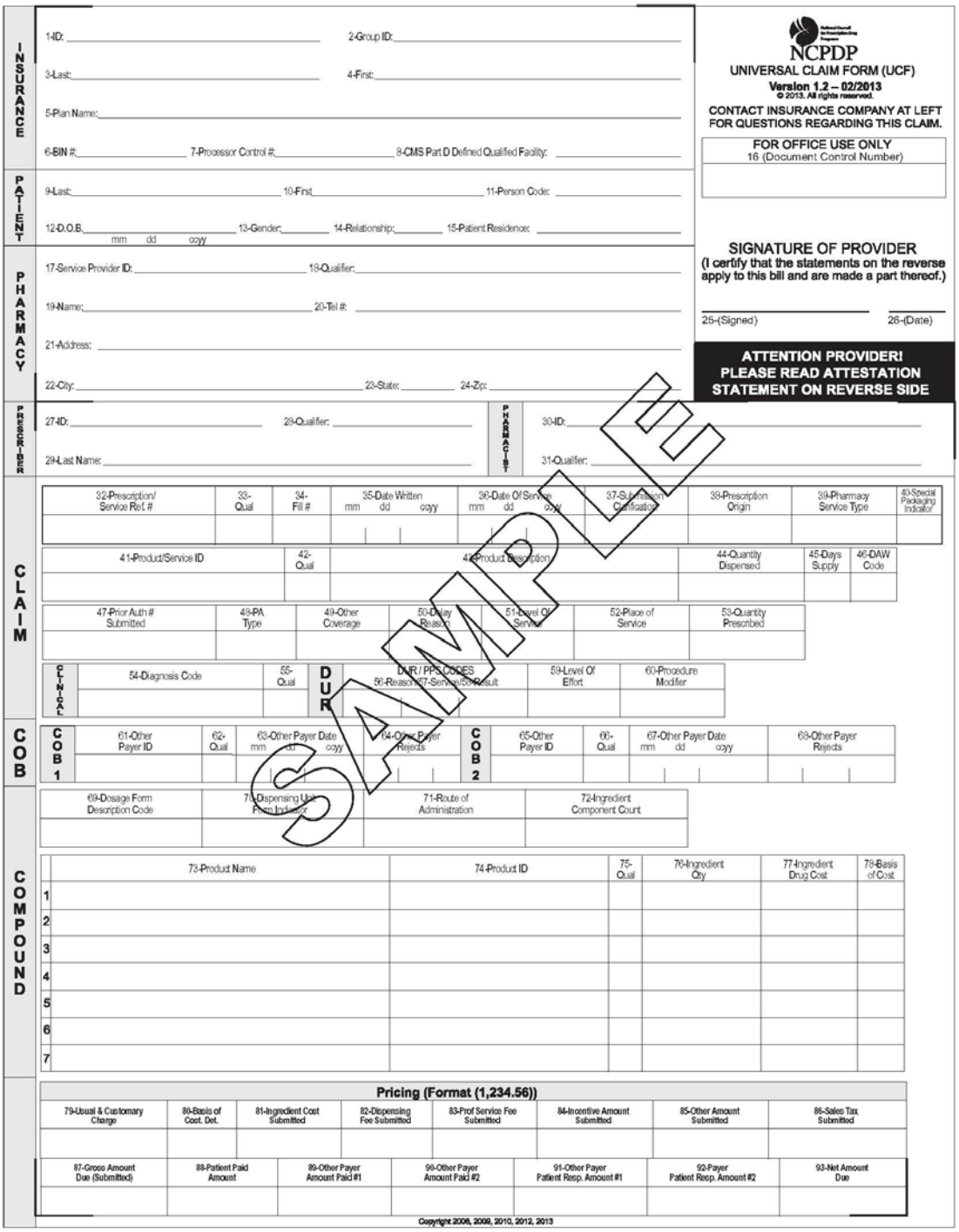


Figure A.3b: Universal Claim Form for prescription drugs (reverse)



Universal Claim Form (Reverse)

The provider agrees to the following:

- Certifies that required beneficiary signatures, or legally authorized signatures of beneficiaries, are on file;
- That the submitted claim is accurate, complete, and truthful; and
- That it will research and correct claim discrepancies.

For more instructions on this form, see the NCPDP *Manual Claim Forms Reference Implementation Guide* available where forms are ordered or with NCPDP membership at www.ncpdp.org.

Code List

For fields not listed below, or more values which may be available, see the *NCPDP Manual Claim Forms Reference Implementation Guide* or the NCPDP *External Code List*.

8 - CMS Part D Defined Qualified Facility
"Y" - Yes
"N" - No

13 - Patient Gender Code
"0" - Not Specified
"1" - Male
"2" - Female

14 - Patient Relationship Code
"0" - Not Specified
"1" - Cardholder
"2" - Spouse
"3" - Child
"4" - Other

15 - Patient Residence
"1" - Home
"3" - Nursing Facility
"4" - Assisted Living Facility
"9" - Intermediate Care Facility/Mentally Retarded
"11" - Hospice
"15" - Correctional Institution

18 - Service Provider ID Qualifier
"blank" - Not Specified
"01" - NPI
"05" - Medicaid
"07" - NCPDP
"99" - Other

28 - Prescriber ID Qualifier
"01" - NPI
"08" - State License
"12" - DEA
"99" - Other

31 - Provider ID Qualifier
"01" - DEA
"02" - State License
"03" - Social Security Number
"04" - Name
"05" - NPI
"06" - HIN
"07" - State issued
"99" - Other

33 - Prescription/Service Reference # Qualifier
"1" - Rx Billing
"2" - Service Billing

37 - Submission Clarification Code
"1" - No Override
"2" - Other Override
"3" - Vacation Supply
"4" - Lost Prescription
"5" - Therapy Change
"6" - Starter Dose
"7" - Medically Necessary
"8" - Process Compound for Approved Ingredients
"9" - Encounters
"10" - Meets Plan Limitations
"11" - Certification on File
"12" - DME Replacement Indicator
"13" - Payer-Recognized Emergency/Disaster Assistance Request
"14" - Long Term Care Leave of Absence
"15" - Long Term Care Replacement Medication
"16" - Long Term Care Emergency Box or Automated Dispensing Machine
"17" - Long Term Care Emergency Supply Remainder
"18" - Long Term Care Patient Admit / Readmit Indicator
"19" - Split Billing
"20" - 340B

37 - Submission Clarification Code (Continued)
"21" - LTC dispensing: 14 days or less not applicable
"22" - LTC dispensing: 7 days
"23" - LTC dispensing: 4 days
"24" - LTC dispensing: 3 days
"25" - LTC dispensing: 2 days
"26" - LTC dispensing: 1 day
"27" - LTC dispensing: 4-3 days
"28" - LTC dispensing: 2-2-3 days
"29" - LTC dispensing: daily and 3-day weekend
"30" - LTC dispensing: Per shift dispensing
"31" - LTC dispensing: Per med pass dispensing
"32" - LTC dispensing: PRN on demand
"33" - LTC dispensing: 7 day or less cycle not otherwise represented
"34" - LTC dispensing: 14 days dispensing
"35" - LTC dispensing: 8-14 day dispensing method not listed above
"36" - LTC dispensing: dispensed outside short cycle
"47" - Shortened Days Supply Fill
"48" - Fill Subsequent to a Shortened Days Supply Fill
"99" - Other

38 - Prescription Origin Code
"0" - Not Known
"1" - Written
"2" - Telephone
"3" - Electronic
"4" - Facsimile
"5" - Pharmacy

39 - Pharmacy Service Type
"1" - Community/Retail Pharmacy Services
"2" - Compounding Pharmacy Services
"3" - Home Infusion Therapy Provider Services
"4" - Institutional Pharmacy Services
"5" - Long Term Care Pharmacy Services
"6" - Mail Order Pharmacy Services
"7" - Managed Care Organization Pharmacy Services
"8" - Specialty Care Pharmacy Services
"99" - Other

40 - Special Packaging Indicator
"1" - Not unit dose
"2" - Manufacturer Unit Dose
"3" - Pharmacy Unit Dose
"4" - Pharmacy Unit Dose Patient Compliance Packaging
"5" - Pharmacy Multi-drug Patient Compliance Packaging
"6" - Remote Device Unit Dose
"7" - Remote Device Multi-drug Compliance
"8" - Manufacturer Unit of Use Package (not unit dose)

42 & 75 - Product/Service ID Qualifier
"00" - Not Specified
"01" - UPC
"02" - HRI
"03" - NDC
"04" - HIBCC
"06" - DUR/PPS
"07" - CPT4
"08" - CPT5
"09" - HCPCS

42 & 75 - Product/Service ID Qualifier (Continued)
"10" - PPAC
"11" - NAPPI
"12" - GTIN
"15" - GCN
"28" - FDB Med Name ID
"29" - FDB Routed Med ID
"30" - FDB Routed Dosage Form Med ID

46 - Dispense as Written (DAW) / Product Selection
"0" - No Product Selection Indicated
"1" - Substitution Not Allowed by Prescriber
"2" - Substitution Allowed - Patient Requested Product Dispensed
"3" - Substitution Allowed - Pharmacist Selected Product Dispensed
"4" - Substitution Allowed - Generic Drug Not in Stock
"5" - Substitution Allowed - Brand Drug Dispensed as a Generic
"6" - Override
"7" - Substitution Not Allowed - Brand Drug Mandated by Law
"8" - Substitution Allowed - Generic Drug not Available in Marketplace
"9" - Substitution Allowed by Prescriber but Plan Requests Brand - Patient's Plan Requested Brand Product to be Dispensed

48 - Prior Authorization Type Code
"0" - Not Specified
"1" - Prior Authorization
"2" - Medical Certification
"3" - EPSDT
"4" - Exemption from Copay and/or Coinsurance
"5" - Exemption from Rx
"6" - Family Planning Indicator
"7" - TANF (Temporary Assistance for Needy Families)
"8" - Payer Defined Exemption
"9" - Emergency Preparedness

49 - Other Coverage Code
"0" - Not Specified by patient
"1" - No Other Coverage
"2" - Other Coverage Exists - Payment Collected
"3" - Other Coverage Billed - Claim Not Covered
"4" - Other Coverage Exists - Payment Not Collected
"8" - Claim is billing for patient financial responsibility only

50 - Delay Reason Code
"1" - Proof of eligibility unknown or unavailable
"2" - Litigation
"3" - Authorization delays
"4" - Delay in certifying provider
"5" - Delay in supplying billing forms
"5" - Delay in delivery of custom-made appliances
"7" - Third party processing delay
"8" - Delay in eligibility determination
"9" - Original claims rejected or denied due to a reason unrelated to the billing limitation rules
"10" - Administration delay in the prior approval process
"11" - Other
"12" - Received late with no exceptions
"13" - Substantial damage by fire, etc to provider records
"14" - Theft, sabotage/other willful acts by employee

51 - Level of Service
"0" - Not Specified
"1" - Patient Consultation
"2" - Home Delivery
"3" - Emergency
"4" - 24 Hour Service
"5" - Patient consultation regarding generic product selection
"6" - In-Home service

52 - Place of Service (For values see https://www.cms.gov/Medicare/Coding/place-of-service-codes/index.html)

55 - Diagnosis Code Qualifier
"00" - Not Specified
"01" - ICD9
"02" - ICD10
"03" - NCCI
"04" - SNOMED
"05" - CDT
"06" - Medi-Span Product Line Diagnosis Code
"07" - DSM IV
"08" - First DataBank Disease Code (FDBDX)
"09" - First DataBank FML Disease Identifier (FDB DxID)
"99" - Other

56 - Reason for Service &
57 - Professional Service Code &
58 - Result of Service Code (For values refer to NCPDP Reference Guide or current External Code List)

59 - DUR/PPS Level of Effort
"0" - Not Specified
"11" - Level 1 (Lowest)
"12" - Level 2
"13" - Level 3
"14" - Level 4
"15" - Level 5 (Highest)

60 - Procedure Modifier Code (See http://www.cms.hhs.gov/hcpcsreleasecodesets/anhcpcs/list.asp)

62 - Other Payer ID Qualifier
& "01" - National Payer ID
66 "02" - HIN
"03" - BIN
"04" - NAIC
"05" - Medicare Carrier Number
"99" - Other

64 - Other Payer Reject Codes
& (For values refer to current
68 NCPDP External Code List)

71 - Route of Administration (See SNOMED CT® http://www.snomed.org)

78 - Compound Basis of Cost Determination
&
80 - Basis of Cost Determination (For values refer to NCPDP Reference Guide or current External Code List)

SAMPLE

Version 1.2

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February 2013

Facility Type

Table A.1 (effective October 1, 2013) shows the first two characters of Type of Bill; commonly referred to as Facility Type.

Table A.1: Facility Type code values

Value	Description
11	Hospital Inpatient (including Medicare Part A)
12	Hospital Inpatient (Medicare Part B Only)
13	Hospital Outpatient
14	Hospital Laboratory Services Provided to Non-Patients
18	Hospital Swing Bed
21	Skilled Nursing—Inpatient (including Medicare Part A)
22	Skilled Nursing—Inpatient (Medicare Part B Only)
23	Skilled Nursing—Outpatient
28	Skilled Nursing—Swing Beds
32	Home Health Services under a Plan of Treatment
34	Home Health Services not under a Plan of Treatment
41	Religious Non-Medical Healthcare Institutions—Hospital Inpatient
43	Religious Non-Medical Healthcare Institutions—Outpatient Services
65	Intermediate Care—Level I
66	Intermediate Care—Level II
71	Clinic—Rural Health
72	Clinic—Hospital-Based or Independent Renal Dialysis Center
73	Clinic—Freestanding
74	Clinic—Outpatient Rehabilitation Facility (ORF)
75	Clinic—Comprehensive Outpatient Rehabilitation Facility (CORF)
76	Clinic—Community Mental Health Center
77	Clinic—Federally Qualified Health Center (FQHC)
78	Clinic—Licensed Freestanding Emergency Medical Facility
79	Clinic—Other
81	Special Facility—Hospice (Non-Hospital Based)
82	Special Facility—Hospice (Hospital Based)
83	Special Facility—Ambulatory Surgery Center
84	Special Facility—Freestanding Birthing Center
85	Special Facility—Critical Access Hospital
86	Special Facility—Residential Facility
89	Special Facility—Other

Bill Sequence

Table A.2 (effective October 1, 2013) shows the third and final character of Type of Bill. This byte of data contains information about the sequence of the facility bill or its disposition.

Table A.2: Bill sequence values

Value	Description
0	Non-Payment/Zero Claim
1	Admit Thru Discharge Claim
2	Interim—First Claim
3	Interim—Continuing Claim
4	Interim—Last Claim
5	Late Charge(s) Only
7	Replacement of Prior Claim
8	Void/Cancel of Prior Claim
9	Final Claim for a Home Health PPS Episode
A	Admission/Election Notice
B	Hospice/Medicare Coordinated Care Demonstration/Religious Non-Medical Healthcare Institution/Centers of Excellence Demonstration/Provider Partnerships Demonstration—Termination/Revocation Notice
C	Hospice Change of Provider Notice
D	Hospice/Medicare Coordinated Care Demonstration/Religious Non-Medical Healthcare Institution/ Centers of Excellence Demonstration/ Provider Partnerships Demonstration—Void/Cancel
E	Hospice Change of Ownership
F	Beneficiary Initiated Adjustment Claim
G	CWF Initiated Adjustment Claim
H	CMS Initiated Adjustment Claim
I	Intermediary Adjustment Claim
J	Initiated Adjustment Claim—Other
K	OIG Initiated Adjustment Claim
M	MSP Initiated Adjustment Claim
O	Non-Payment/Zero Claim
P	QIO Adjustment Claim
Q	Claim Submitted For Reconsideration Outside of Timely Limits (Payer Use Only, Providers Do Not Report This Code)
X	Void/Cancel a Prior Abbreviated Encounter Submission (Medicare Advantage Only)
Y	Replacement of Prior Abbreviated Encounter Submission (Medicare Advantage Only)
Z	New Abbreviated Encounter Submission (Medicare Advantage Only)

Place of Service

Table A.3 shows the CMS Place of Service codes found on the CMS-1500 and 837P claim formats.

Table A.3: CMS Place of Service codes

POS	Name	Description
01	Pharmacy	A facility or location where drugs and other medically related items and services are sold, dispensed, or otherwise provided directly to patients (effective October 1, 2005).
03	School	A facility whose primary purpose is education.
04	Homeless Shelter	A facility or location whose primary purpose is to provide temporary housing to homeless individuals (for example, emergency shelters, individual or family shelters).
05	Indian Health Service Freestanding Facility	A facility or location, owned and operated by the Indian Health Service, which provides diagnostic, therapeutic (surgical and non-surgical), and rehabilitation services to Native American and Alaskan Native peoples who do not require hospitalization.
06	Indian Health Service Provider-based Facility	A facility or location, owned and operated by the Indian Health Service, which provides diagnostic, therapeutic (surgical and non-surgical), and rehabilitation services rendered by, or under the supervision of, physicians to Native American and Alaskan Native peoples admitted as inpatients or outpatients.
07	Tribal 638 Freestanding Facility	A facility or location owned and operated by a federally recognized Native American or Alaskan Native tribe or tribal organization under a 638 agreement, which provides diagnostic, therapeutic (surgical and non-surgical), and rehabilitation services to tribal members who do not require hospitalization.
08	Tribal 638 Provider-based Facility	A facility or location owned and operated by a federally recognized Native American or Alaskan Native tribe or tribal organization under a 638 agreement, which provides diagnostic, therapeutic (surgical and non-surgical), and rehabilitation services to tribal members admitted as inpatients or outpatients.
09	Prison/Correctional Facility	A prison, jail, reformatory, work farm, detention center, or any other similar facility maintained by either federal, state, or local authorities for the purpose of confinement or rehabilitation of adult or juvenile criminal offenders.
11	Office	Location, other than a hospital, skilled nursing facility (SNF), military treatment facility, community health center, state or local public health clinic, or intermediate care facility (ICF), where the health professional routinely provides health examinations, diagnosis, and treatment of illness or injury on an ambulatory basis.
12	Home	Location, other than a hospital or other facility, where the patient receives care in a private residence.
13	Assisted Living Facility	Congregate residential facility with self-contained living units providing assessment of each resident's needs and onsite support twenty-four hours a day, seven days a week, with the capacity to deliver or arrange for services, including some health care and other services (effective October 1, 2003).
14	Group Home	A residence, with shared living areas, where clients receive supervision and other services such as social and/or behavioral services, custodial service, and minimal services (for example, medication administration).

POS	Name	Description
15	Mobile Unit	A facility/unit that moves from place to place and is equipped to provide preventive, screening, diagnostic, and/or treatment services.
16	Temporary Lodging	A short-term accommodation such as a hotel, campground, hostel, cruise ship, or resort where the patient receives care, and which is not identified by any other POS code.
17	Walk-in Retail Health Clinic	A walk-in health clinic, other than an office, urgent care facility, pharmacy, or independent clinic and not described by any other POS code, that is located within a retail operation and provides, on an ambulatory basis, preventive and primary care services.
20	Urgent Care Facility	Location, distinct from a hospital emergency room, an office, or a clinic, whose purpose is to diagnose and treat illness or injury for unscheduled, ambulatory patients seeking immediate medical attention.
21	Inpatient Hospital	A facility, other than psychiatric, which primarily provides diagnostic, therapeutic (both surgical and non-surgical), and rehabilitation services by, or under, the supervision of physicians to patients admitted for a variety of medical conditions.
22	Outpatient Hospital	A portion of a hospital that provides diagnostic, therapeutic (both surgical and nonsurgical), and rehabilitation services to sick or injured persons who do not require hospitalization or institutionalization.
23	Emergency Room—Hospital	A portion of a hospital where emergency diagnosis and treatment of illness or injury is provided.
24	Ambulatory Surgical Center	A freestanding facility, other than a physician's office, where surgical and diagnostic services are provided on an ambulatory basis.
25	Birthing Center	A facility, other than a hospital's maternity facilities or a physician's office, which provides a setting for labor, delivery, and immediate postpartum care as well as immediate care of newborn infants.
26	Military Treatment Facility	A medical facility operated by one or more of the Uniformed Services. Military Treatment Facility (MTF) also refers to certain former U.S. Public Health Service (USPHS) facilities now designated as Uniformed Service Treatment Facilities (USTF).
31	Skilled Nursing Facility	A facility that primarily provides inpatient skilled nursing care and related services to patients who require medical, nursing, or rehabilitative services but does not provide the level of care or treatment available in a hospital.
32	Nursing Facility	A facility that primarily provides to residents skilled nursing care and related services for the rehabilitation of injured, disabled, or sick persons, or, on a regular basis, health-related care services above the level of custodial care to other than mentally retarded individuals.
33	Custodial Care Facility	A facility that provides room, board, and other personal assistance services, generally on a long-term basis, and that does not include a medical component.
34	Hospice	A facility, other than a patient's home, in which palliative and supportive care for terminally ill patients and their families is provided.
41	Ambulance—Land	A land vehicle specifically designed, equipped, and staffed for lifesaving and transporting the sick or injured.

POS	Name	Description
42	Ambulance—Air or Water	An air or water vehicle specifically designed, equipped, and staffed for lifesaving and transporting the sick or injured.
49	Independent Clinic	A location, not part of a hospital and not described by any other POS code, that is organized and operated to provide preventive, diagnostic, therapeutic, rehabilitative, or palliative services to outpatients only (effective October 1, 2003).
50	Federally Qualified Health Center	A facility located in a medically underserved area that provides Medicare beneficiaries preventive primary medical care under the general direction of a physician.
51	Inpatient Psychiatric Facility	A facility that provides inpatient psychiatric services for the diagnosis and treatment of mental illness on a twenty-four-hour basis, by or under the supervision of a physician.
52	Psychiatric Facility-Partial Hospitalization	A facility for the diagnosis and treatment of mental illness that provides a planned therapeutic program for patients who do not require full-time hospitalization, but who need broader programs than are possible from outpatient visits to a hospital-based or hospital-affiliated facility.
53	Community Mental Health Center	A facility that provides the following services: outpatient services, including specialized outpatient services for children, the elderly, individuals who are chronically ill, and residents of the CMHC's mental health services area who have been discharged from inpatient treatment at a mental health facility; twenty-four-hour-a-day emergency care services; day treatment, other partial hospitalization services, or psychosocial rehabilitation services; screening for patients being considered for admission to state mental health facilities to determine the appropriateness of such admission; and consultation and education services.
54	Intermediate Care Facility/Mentally Retarded	A facility that primarily provides health-related care and services above the level of custodial care to mentally retarded individuals but does not provide the level of care or treatment available in a hospital or SNF.
55	Residential Substance Abuse Treatment Facility	A facility that provides treatment for substance (alcohol and drug) abuse to live-in residents who do not require acute medical care. Services include individual and group therapy and counseling, family counseling, laboratory tests, drugs and supplies, psychological testing, and room and board.
56	Psychiatric Residential Treatment Center	A facility or distinct part of a facility for psychiatric care that provides a total twenty-four-hour therapeutically planned and professionally staffed group living and learning environment.
57	Non-Residential Substance Abuse Treatment Facility	A location that provides treatment for substance (alcohol and drug) abuse on an ambulatory basis. Services include individual and group therapy and counseling, family counseling, laboratory tests, drugs and supplies, and psychological testing (effective October 1, 2003).
60	Mass Immunization Center	A location where providers administer pneumococcal pneumonia and influenza virus vaccinations and submit these services as electronic media claims, paper claims, or using the roster billing method. Although these generally are housed in a mass immunization setting, such as a public health center, pharmacy, or mall, they might occur in a physician office setting.

POS	Name	Description
61	Comprehensive Inpatient Rehabilitation Facility	A facility that provides comprehensive rehabilitation services under the supervision of a physician to inpatients with physical disabilities. Services include physical therapy, occupational therapy, speech pathology, social or psychological services, and orthotics and prosthetics services.
62	Comprehensive Outpatient Rehabilitation Facility	A facility that provides comprehensive rehabilitation services under the supervision of a physician to outpatients with physical disabilities. Services include physical therapy, occupational therapy, and speech pathology services.
65	End-Stage Renal Disease Treatment Facility	A facility other than a hospital that provides dialysis treatment, maintenance, and/or training to patients or caregivers on an ambulatory or home-care basis.
71	Public Health Clinic	A facility maintained by either state or local health departments that provides ambulatory primary medical care under the general direction of a physician (effective October 1, 2003).
72	Rural Health Clinic	A certified facility located in a rural, medically underserved area that provides ambulatory primary medical care under the general direction of a physician.
81	Independent Laboratory	A laboratory certified to perform diagnostic and/or clinical tests independent of an institution or a physician's office.
99	Other Place of Service	Other place of service not previously identified.

Patient Status Code

Table A.4 shows the NUBC Patient Status codes (effective 2014) found on the CMS-1450 and 837I claim formats.

Table A.4: NUBC Patient Status codes

Status Code	Description
01	Discharged to home or self-care (routine discharge)
02	Discharged/transferred to a short-term general hospital for inpatient care
03	Discharged/transferred to SNF with Medicare certification in anticipation of covered skilled care
04	Discharged/transferred to a facility that provides custodial or supportive care
05	Discharge/transfer to a designated cancer center or children's hospital
06	Discharged/transferred to home under care of organized home health service organization in anticipation of covered skilled care
07	Left against medical advice or discontinued care
09	Admitted as an inpatient to this hospital
20	Expired (or did not recover—Religious Non-Medical Health Care Patient)
21	Discharged/transferred to Court/Law Enforcement
30	Still patient
40	Expired at home (Hospice claims only)
41	Expired in a medical facility, such as a hospital, SNF, ICF, or freestanding hospice (Hospice claims only)
42	Expired—place unknown (Hospice claims only)

Status Code	Description
43	Discharged/transferred to a Federal hospital
50	Discharged/transferred to Hospice—home
51	Discharged/transferred to Hospice—medical facility
61	Discharged/transferred to a hospital-based Medicare approved swing bed
62	Discharged/transferred to an inpatient rehabilitation facility including distinct part units of a hospital
63	Discharged/transferred to long-term care hospital
64	Discharged/transferred to a nursing facility certified under Medicaid but not Medicare
65	Discharged/transferred to a psychiatric hospital or psychiatric distinct part unit of a hospital
66	Discharged/transferred to a Critical Access Hospital
69	Discharged/transferred to a designated disaster alternative care site
70	Discharge/transfer to another type of health care institution not defined elsewhere in the code list
81	Discharged to home or self-care with a planned acute care hospital readmission
82	Discharged/transferred to a short-term general hospital for inpatient care with a planned acute care hospital inpatient readmission
83	Discharged/transferred to a skilled nursing facility (SNF) with Medicare certification with a planned acute care hospital inpatient readmission
84	Discharged/transferred to a facility that provides custodial or supportive care
85	Discharged/transferred to a designated cancer center or children's hospital with a planned acute care hospital inpatient readmission
86	Discharged/transferred to home under care of organized home health service
87	Discharged/transferred to court/law enforcement with a planned acute care hospital inpatient readmission
88	Discharged/transferred to a federal health care facility with a planned acute care hospital inpatient readmission
89	Discharged/transferred to a hospital-based Medicare approved swing bed with a planned acute care hospital inpatient readmission
90	Discharged/transferred to an inpatient rehabilitation facility (IRF) including rehabilitation distinct part units of a hospital with a planned acute care hospital inpatient readmission
91	Discharged/transferred to a Medicare certified long-term care hospital (LTCH) with a planned acute care hospital inpatient readmission

Revenue Code

Table A.5 shows the NUBC Revenue codes (effective 2014) found on the CMS-1450 and 837I claim formats.

Table A.5: NUBC Revenue codes

Revenue Code	Description
0022	Skilled nursing facility prospective payment system
0023	Home health prospective payment system
0024	Inpatient rehabilitation facility prospective payment system
0100	All-Inclusive Rate—All-Inclusive Room and Board Plus Ancillary
0101	All-Inclusive Rate—All-Inclusive Room and Board
0110	Room & Board—Private (one bed)—General

Revenue Code	Description
0111	Room & Board—Private (one bed)—Medical/Surgical/GYN
0112	Room & Board—Private (one bed)—OB
0113	Room & Board—Private (one bed)—Pediatric
0114	Room & Board—Private (one bed)—Psychiatric
0115	Room & Board—Private (one bed)—Hospice
0116	Room & Board—Private (one bed)—Detoxification
0117	Room & Board—Private (one bed)—Oncology
0118	Room & Board—Private (one bed)—Rehabilitation
0119	Room & Board—Private (one bed)—Other
0120	Room & Board—Semiprivate (two beds)—General
0121	Room & Board—Semiprivate (two beds)—Medical/Surgical/GYN
0122	Room & Board—Semiprivate (two beds)—OB
0123	Room & Board—Semiprivate (two beds)—Pediatric
0124	Room & Board—Semiprivate (two beds)—Psychiatric
0125	Room & Board—Semiprivate (two beds)—Hospice
0126	Room & Board—Semiprivate (two beds)—Detoxification
0127	Room & Board—Semiprivate (two beds)—Oncology
0128	Room & Board—Semiprivate (two beds)—Rehabilitation
0129	Room & Board—Semiprivate (two beds)—Other
0130	Room & Board—Three and Four Beds—General
0131	Room & Board—Three and Four Beds—Medical/Surgical/GYN
0132	Room & Board—Three and Four Beds—OB
0133	Room & Board—Three and Four Beds—Pediatric
0134	Room & Board—Three and Four Beds—Psychiatric
0135	Room & Board—Three and Four Beds—Hospice
0136	Room & Board—Three and Four Beds—Detoxification
0137	Room & Board—Three and Four Beds—Oncology
0138	Room & Board—Three and Four Beds—Rehabilitation
0139	Room & Board—Three and Four Beds—Other
0140	Room & Board—Deluxe Private—General
0141	Room & Board—Deluxe Private—Medical/Surgical/GYN
0142	Room & Board—Deluxe Private—OB
0143	Room & Board—Deluxe Private—Pediatric
0144	Room & Board—Deluxe Private—Psychiatric
0145	Room & Board—Deluxe Private—Hospice
0146	Room & Board—Deluxe Private—Detoxification
0147	Room & Board—Deluxe Private—Oncology
0148	Room & Board—Deluxe Private—Rehabilitation
0149	Room & Board—Deluxe Private—Other
0150	Room & Board—Ward—General
0151	Room & Board—Ward—Medical/Surgical/GYN
0152	Room & Board—Ward—OB
0153	Room & Board—Ward—Pediatric
0154	Room & Board—Ward—Psychiatric

Revenue Code	Description
0155	Room & Board—Ward—Hospice
0156	Room & Board—Ward—Detoxification
0157	Room & Board—Ward—Oncology
0158	Room & Board—Ward—Rehabilitation
0159	Room & Board—Ward—Other
0160	Other Room & Board—General
0164	Other Room & Board—Sterile Environment
0167	Other Room & Board—Self-Care
0169	Other Room & Board—Other
0170	Nursery—General
0171	Nursery—Newborn—Level I
0172	Nursery—Newborn—Level II
0173	Nursery—Newborn—Level III
0174	Nursery—Newborn—Level IV
0179	Nursery—Other
0180	Leave of Absence—General
0182	Leave of Absence—Patient Convenience—Charges Billable
0183	Leave of Absence—Therapeutic Leave
0185	Leave of Absence—Nursing Home (for Hospitalization)
0189	Leave of Absence—Other Leave of Absence
0190	Subacute Care—General
0191	Subacute Care—Level I
0192	Subacute Care—Level II
0193	Subacute Care—Level III
0194	Subacute Care—Level IV
0199	Subacute Care—Other Subacute Care
0200	Intensive Care Unit—General
0201	Intensive Care Unit—Surgical
0202	Intensive Care Unit—Medical
0203	Intensive Care Unit—Pediatric
0204	Intensive Care Unit—Psychiatric
0206	Intensive Care Unit—Intermediate ICU
0207	Intensive Care Unit—Burn Care
0208	Intensive Care Unit—Trauma
0209	Intensive Care Unit—Other Intensive Care
0210	Coronary Care Unit—General
0211	Coronary Care Unit—Myocardial Infarction
0212	Coronary Care Unit—Pulmonary Care
0213	Coronary Care Unit—Heart Transplant
0214	Coronary Care Unit—Intermediate CCU
0219	Coronary Care Unit—Other Coronary Care
0220	Special Charges—General
0221	Special Charges—Admission Charge
0222	Special Charges—Technical Support Charge

Revenue Code	Description
0223	Special Charges—UR Service Charge
0224	Special Charges—Late Discharge, Medically Necessary
0229	Special Charges—Other Special Charges
0230	Incremental Nursing Charge—General
0231	Incremental Nursing Charge—Nursery
0232	Incremental Nursing Charge—OB
0233	Incremental Nursing Charge—ICU
0234	Incremental Nursing Charge—CCU
0235	Incremental Nursing Charge—Hospice
0239	Incremental Nursing Charge—Other
0240	All-Inclusive Ancillary—General
0241	All-Inclusive Ancillary—Basic
0242	All-Inclusive Ancillary—Comprehensive
0243	All-Inclusive Ancillary—Specialty
0249	All-Inclusive Ancillary—Other Inclusive Ancillary
0250	Pharmacy—General
0251	Pharmacy—Generic Drugs
0252	Pharmacy—Nongeneric Drugs
0253	Pharmacy—Take-Home Drugs
0254	Pharmacy—Drugs Incident to Other Diagnostic Services
0255	Pharmacy—Drugs Incident to Radiology
0256	Pharmacy—Experimental Drugs
0257	Pharmacy—Nonprescription
0258	Pharmacy—IV Solutions
0259	Pharmacy—Other Pharmacy
0260	IV Therapy—General
0261	IV Therapy—Infusion Pump
0262	IV Therapy—IV Therapy/Pharmacy Services
0263	IV Therapy—IV Therapy/Drug/Supply Delivery
0264	IV Therapy—IV Therapy/Supplies
0269	IV Therapy—Other IV Therapy
0270	Medical/Surgical Supplies and Devices—General
0271	Medical/Surgical Supplies and Devices—Nonsterile Supply
0272	Medical/Surgical Supplies and Devices—Sterile Supply
0273	Medical/Surgical Supplies and Devices—Take-Home Supplies
0274	Medical/Surgical Supplies and Devices—Prosthetic/Orthotic Devices
0275	Medical/Surgical Supplies and Devices—Pacemaker
0276	Medical/Surgical Supplies and Devices—Intraocular Lens
0277	Medical/Surgical Supplies and Devices—Oxygen—Take-Home
0278	Medical/Surgical Supplies and Devices—Other Implants
0279	Medical/Surgical Supplies and Devices—Other Supplies/Devices
0280	Oncology—General
0289	Oncology—Other
0290	DME (Other than Renal)—General

Revenue Code	Description
0291	DME (Other than Renal)—Rental
0292	DME (Other than Renal)—Purchase of New DME
0293	DME (Other than Renal)—Purchase of Used DME
0294	DME (Other than Renal)—Supplies/Drugs for DME
0299	DME (Other than Renal)—Other Equipment
0300	Laboratory—General
0301	Laboratory—Chemistry
0302	Laboratory—Immunology
0303	Laboratory—Renal Patient (Home)
0304	Laboratory—Nonroutine Dialysis
0305	Laboratory—Hematology
0306	Laboratory—Bacteriology and Microbiology
0307	Laboratory—Urology
0309	Laboratory—Other Laboratory
0310	Laboratory Pathology—General
0311	Laboratory Pathology—Cytology
0312	Laboratory Pathology—Histology
0314	Laboratory Pathology—Biopsy
0319	Laboratory Pathology—Other
0320	Radiology—Diagnostic—General
0321	Radiology—Diagnostic—Angiocardiography
0322	Radiology—Diagnostic—Arthrography
0323	Radiology—Diagnostic—Arteriography
0324	Radiology—Diagnostic—Chest X-ray
0329	Radiology—Diagnostic—Other
0330	Radiology—Therapeutic and/or Chemotherapy Administration—General
0331	Radiology—Therapeutic and/or Chemotherapy Administration—Chemotherapy
0332	Radiology—Therapeutic and/or Chemotherapy Administration—Chemotherapy
0333	Radiology—Therapeutic and/or Chemotherapy Administration—Radiation
0335	Radiology—Therapeutic and/or Chemotherapy Administration—Chemotherapy
0339	Radiology—Therapeutic and/or Chemotherapy Administration—Other
0340	Nuclear Medicine—General
0341	Nuclear Medicine—Diagnostic
0342	Nuclear Medicine—Therapeutic
0343	Nuclear Medicine—Diagnostic Radiopharmaceuticals
0344	Nuclear Medicine—Therapeutic Radiopharmaceuticals
0349	Nuclear Medicine—Other
0350	CT Scan—General
0351	CT Scan—Head Scan
0352	CT Scan—Body Scan
0359	CT Scan—Other CT Scans
0360	Operating Room Services—General
0361	Operating Room Services—Minor Surgery
0362	Operating Room Services—Organ Transplant—Other than Kidney

Revenue Code	Description
0367	Operating Room Services—Kidney Transplant
0369	Operating Room Services—Other Operating Room Services
0370	Anesthesia—General
0371	Anesthesia—Anesthesia Incident to Radiology
0372	Anesthesia—Anesthesia Incident to Other Diagnostic Services
0374	Anesthesia—Acupuncture
0379	Anesthesia—Other Anesthesia
0380	Blood and Blood Products—General
0381	Blood and Blood Products—Packed Red Cells
0382	Blood and Blood Products—Whole Blood and Blood Products
0383	Blood and Blood Products—Plasma
0384	Blood and Blood Products—Platelets
0385	Blood and Blood Products—Leukocytes
0386	Blood and Blood Products—Other Components
0387	Blood and Blood Products—Other Derivatives (Cryoprecipitates)
0389	Blood and Blood Products—Other Blood
0390	Administration, Processing, and Storage for Blood and Blood Components—General
0391	Administration (e.g., Transfusion) of Blood Products
0392	Processing and Storage of Blood Products
0399	Other Blood Handling
0400	Other Imaging Services—General
0401	Other Imaging Services—Diagnostic Mammography
0402	Other Imaging Services—Ultrasound
0403	Other Imaging Services—Screening Mammography
0404	Other Imaging Services—Positron Emission Tomography
0409	Other Imaging Services—Other Imaging Services
0410	Respiratory Services—General
0412	Respiratory Services—Inhalation Services
0413	Respiratory Services—Hyperbaric Oxygen Therapy
0419	Respiratory Services—Other Respiratory Services
0420	Physical Therapy—General
0421	Physical Therapy—Visit Charge
0422	Physical Therapy—Hourly Charge
0423	Physical Therapy—Group Rate
0424	Physical Therapy—Evaluation or Reevaluation
0429	Physical Therapy—Other Physical Therapy
0430	Occupational Therapy—General
0431	Occupational Therapy—Visit Charge
0432	Occupational Therapy—Hourly Charge
0433	Occupational Therapy—Group Rate
0434	Occupational Therapy—Evaluation or Reevaluation
0439	Occupational Therapy—Other Occupational Therapy
0440	Speech Therapy-Language Pathology—General
0441	Speech Therapy-Language Pathology—Visit Charge

Revenue Code	Description
0442	Speech Therapy-Language Pathology—Hourly Charge
0443	Speech Therapy-Language Pathology—Group Rate
0444	Speech Therapy-Language Pathology—Evaluation or Reevaluation
0449	Speech Therapy-Language Pathology—Other Speech-Therapy
0450	Emergency Room—General
0451	Emergency Room—EMTALA Emergency Medical Screening Services
0452	Emergency Room—ER Beyond EMTALA Screening
0456	Emergency Room—Urgent Care
0459	Emergency Room—Other Emergency Room
0460	Pulmonary Function—General
0469	Pulmonary Function—Other Pulmonary Function
0470	Audiology—General
0471	Audiology—Diagnostic
0472	Audiology—Treatment
0479	Audiology—Other Audiology
0480	Cardiology—General
0481	Cardiology—Cardiac Cath Lab
0482	Cardiology—Stress Test
0483	Cardiology—Echocardiology
0489	Cardiology—Other Cardiology
0490	Ambulatory Surgical Care—General
0499	Ambulatory Surgical Care—Other Ambulatory Surgical Care
0500	Outpatient Services—General
0509	Outpatient Services—Other Outpatient Services
0510	Clinic—General
0511	Clinic—Chronic Pain Center
0512	Clinic—Dental Clinic
0513	Clinic—Psychiatric Clinic
0514	Clinic—OB/GYN Clinic
0515	Clinic—Pediatric Clinic
0516	Clinic—Urgent Care Clinic
0517	Clinic—Family Practice Clinic
0519	Clinic—Other Clinic
0520	Freestanding Clinic—General
0521	Freestanding Clinic—Clinic visit by member to RHC/FQHC
0522	Freestanding Clinic—Home visit by RHC/FQHC practitioner
0523	Freestanding Clinic—Family Practice Clinic
0524	Freestanding Clinic—Visit by RHC/FQHC Practitioner in a Part A covered SNF
0525	Freestanding Clinic—Visit by RHC/FQHC Practitioner to a non-covered SNF
0526	Freestanding Clinic—Urgent Care Clinic
0527	Freestanding Clinic—Visiting Nurse Service to a Member's Home
0528	Freestanding Clinic—Visit by RHC/FQHC Practitioner to other non-RHC/FQHC
0529	Freestanding Clinic—Other Freestanding Clinic
0530	Osteopathic Services—General

Revenue Code	Description
0531	Osteopathic Services—Osteopathic Therapy
0539	Osteopathic Services—Other Osteopathic Services
0540	Ambulance—General
0541	Ambulance—Supplies
0542	Ambulance—Medical Transport
0543	Ambulance—Heart Mobile
0544	Ambulance—Oxygen
0545	Ambulance—Air Ambulance
0546	Ambulance—Neonatal Ambulance Services
0547	Ambulance—Pharmacy
0548	Ambulance—EKG Transmission
0549	Ambulance—Other Ambulance
0550	Skilled Nursing—General
0551	Skilled Nursing—Visit Charge
0552	Skilled Nursing—Hourly Charge
0559	Skilled Nursing—Other Skilled Nursing
0560	Home Health (HH)—Medical Social Services—General
0561	Home Health (HH)—Medical Social Services—Visit Charge
0562	Home Health (HH)—Medical Social Services—Hourly Charge
0569	Home Health (HH)—Medical Social Services—Other Medical Social Services
0570	Home Health (HH) Aide—General
0571	Home Health (HH) Aide—Visit Charge
0572	Home Health (HH) Aide—Hourly Charge
0579	Home Health (HH) Aide—Other Home Health Aide
0580	Home Health (HH)—Other Visits—General
0581	Home Health (HH)—Other Visits—Visit Charge
0582	Home Health (HH)—Other Visits—Hourly Charge
0583	Home Health (HH)—Other Visits—Assessment
0589	Home Health (HH)—Other Visits—Other Home Health Visits
0590	Home Health (HH)—Units of Service—General
0600	Home Health (HH) Oxygen—General
0601	Home Health (HH) Oxygen—Oxygen—Stat/Equip/Supply or Contents
0602	Home Health (HH) Oxygen—Oxygen—Stat/Equip/Supply/Under 1 LPM
0603	Home Health (HH) Oxygen—Oxygen—Stat/Equip/Over 4 LPM
0604	Home Health (HH) Oxygen—Oxygen—Portable Add-On
0609	Home Health (HH) Oxygen—Other
0610	Magnetic Resonance Technology—General
0611	Magnetic Resonance Imaging—Brain/Brain Stem
0612	Magnetic Resonance Imaging—Spinal Cord/Spine
0614	Magnetic Resonance Imaging—Other MRI
0615	Magnetic Resonance Angiography—Head and Neck
0616	Magnetic Resonance Angiography—Lower Extremities
0618	Magnetic Resonance Angiography—Other MRA
0619	Magnetic Resonance Technology—Other MRT

Revenue Code	Description
0621	Medical/Surgical Supplies (Extension of 027X)—Supplies Incident to Radiology
0622	Medical/Surgical Supplies (Extension of 027X)—Supplies Incident to Other
0623	Medical/Surgical Supplies (Extension of 027X)—Surgical Dressings
0624	Medical/Surgical Supplies (Extension of 027X)—FDA Investigational Devices
0631	Pharmacy—Extension of 025X—Single Source Drug
0632	Pharmacy—Extension of 025X—Multiple Source Drug
0633	Pharmacy—Extension of 025X—Restrictive Prescription
0634	Pharmacy—Extension of 025X—Erythropoietin (EPO) Less than 10,000 Units
0635	Pharmacy—Extension of 025X—Erythropoietin (EPO) 10,000 or More Units
0636	Pharmacy—Extension of 025X—Drugs Requiring Detailed Coding
0637	Pharmacy—Extension of 025X—Self-Administrable Drugs
0640	Home IV Therapy Services—General
0641	Home IV Therapy Services—Nonroutine Nursing, Central Line
0642	Home IV Therapy Services—IV Site Care, Central Line
0643	Home IV Therapy Services—IV Start/Care, Peripheral Line
0644	Home IV Therapy Services—Nonroutine Nursing, Peripheral Line
0645	Home IV Therapy Services—Training Patient/Caregiver, Central Line
0646	Home IV Therapy Services—Training, Disabled Patient, Central Line
0647	Home IV Therapy Services—Training, Patient/Caregiver, Peripheral Line
0648	Home IV Therapy Services—Training, Disabled Patient, Peripheral Line
0649	Home IV Therapy Services—Other IV Therapy Services
0650	Hospice Service—General
0651	Hospice Service—Routine Home Care
0652	Hospice Service—Continuous Home Care
0655	Hospice Service—Inpatient Respite Care
0656	Hospice Service—General Inpatient Care (Nonrespite)
0657	Hospice Service—Physician Services
0658	Hospice Service—Hospice Room & Board—Nursing Facility
0659	Hospice Service—Other Hospice
0660	Respite Care—General
0661	Respite Care—Hourly Charge/Nursing
0662	Respite Care—Hourly Charge/Aide/Homemaker/Companion
0663	Respite Care—Daily Respite Charge
0669	Respite Care—Other Respite Care
0670	Outpatient Special Residence Charges—General
0671	Outpatient Special Residence Charges—Hospital-owned
0672	Outpatient Special Residence Charges—Contracted
0679	Outpatient Special Residence Charges—Other Special Residence Charges
0681	Trauma Response—Level I
0682	Trauma Response—Level II
0683	Trauma Response—Level III
0684	Trauma Response—Level IV
0689	Trauma Response—Other trauma response
0690	Pre-hospital Palliative Care Services—General Classification

Revenue Code	Description
0691	Pre-hospital Palliative Care Services—Visit Charge
0692	Pre-hospital Palliative Care Services—Hourly Charge
0693	Pre-hospital Palliative Care Services—Evaluation
0694	Pre-hospital Palliative Care Services—Consultation and Education
0695	Pre-hospital Palliative Care Services—Inpatient Care
0696	Pre-hospital Palliative Care Services—Physician Services
0699	Pre-hospital Palliative Care Services—Other Care Services
0700	Cast Room—General
0710	Recovery Room—General
0720	Labor Room/Delivery—General
0721	Labor Room/Delivery—Labor
0722	Labor Room/Delivery—Delivery
0723	Labor Room/Delivery—Circumcision
0724	Labor Room/Delivery—Birthing Center
0729	Labor Room/Delivery—Other Labor Room/Delivery
0730	EKG/ECG (Electrocardiogram)—General
0731	EKG/ECG (Electrocardiogram)—Holter Monitor
0732	EKG/ECG (Electrocardiogram)—Telemetry
0739	EKG/ECG (Electrocardiogram)—Other EKG/ECG
0740	EEG (Electroencephalogram)—General
0750	Gastrointestinal Services—General
0760	Specialty Services—General
0761	Specialty Services—Treatment Room
0762	Specialty Services—Observation Hours
0769	Specialty Services—Other Specialty Services
0770	Preventive Care Services—General
0771	Preventive Care Services—Vaccine Administration
0780	Telemedicine—General
0790	Extra-Corporeal Shock Wave Therapy (formerly Lithotripsy)—General
0800	Inpatient Renal Dialysis—General
0801	Inpatient Renal Dialysis—Inpatient Hemodialysis
0802	Inpatient Renal Dialysis—Inpatient Peritoneal (Non-CAPD)
0803	Inpatient Renal Dialysis—Inpatient Continuous Ambulatory Peritoneal Dialysis
0804	Inpatient Renal Dialysis—Inpatient Continuous Cycling Peritoneal
0809	Inpatient Renal Dialysis—Other Inpatient Dialysis
0810	Acquisition of Body Components—General
0811	Acquisition of Body Components—Living Donor
0812	Acquisition of Body Components—Cadaver Donor
0813	Acquisition of Body Components—Unknown Donor
0814	Acquisition of Body Components—Unsuccessful organ search—donor bank charges
0819	Acquisition of Body Components—Other Donor
0820	Hemodialysis—Outpatient or Home Dialysis—General
0821	Hemodialysis—Outpatient or Home Dialysis—Hemodialysis/Composite or
0822	Hemodialysis—Outpatient or Home Dialysis—Home Supplies

Revenue Code	Description
0823	Hemodialysis—Outpatient or Home Dialysis—Home Equipment
0824	Hemodialysis—Outpatient or Home Dialysis—Maintenance/100%
0825	Hemodialysis—Outpatient or Home Dialysis—Support Services
0829	Hemodialysis—Outpatient or Home Dialysis—Other Outpatient Hemodialysis
0830	Peritoneal Dialysis—Outpatient or Home—General
0831	Peritoneal Dialysis—Outpatient or Home—Composite or Other Rate
0832	Peritoneal Dialysis—Outpatient or Home—Home Supplies
0833	Peritoneal Dialysis—Outpatient or Home—Home Equipment
0834	Peritoneal Dialysis—Outpatient or Home—Maintenance/100%
0835	Peritoneal Dialysis—Outpatient or Home—Support Services
0839	Peritoneal Dialysis—Outpatient or Home—Other Outpatient Peritoneal Dialysis
0840	Continuous Ambulatory Peritoneal Dialysis (CAPD)—OP or Home—General
0841	Continuous Ambulatory Peritoneal Dialysis (CAPD)—OP or Home—Composite Rate
0842	Continuous Ambulatory Peritoneal Dialysis (CAPD)—OP or Home—Supplies
0843	Continuous Ambulatory Peritoneal Dialysis (CAPD)—OP or Home—Equipment
0844	Continuous Ambulatory Peritoneal Dialysis (CAPD)—OP or Home—Maintenance
0845	Continuous Ambulatory Peritoneal Dialysis (CAPD)—OP or Home—Support Services
0849	Continuous Ambulatory Peritoneal Dialysis (CAPD)—OP or Home—Other
0850	Continuous Cycling Peritoneal Dialysis (CCPD)—OP or Home—General
0851	Continuous Cycling Peritoneal Dialysis (CCPD)—OP or Home—Composite Rate
0852	Continuous Cycling Peritoneal Dialysis (CCPD)—OP or Home—Supplies
0853	Continuous Cycling Peritoneal Dialysis (CCPD)—OP or Home—Equipment
0854	Continuous Cycling Peritoneal Dialysis (CCPD)—OP or Home—Maintenance
0855	Continuous Cycling Peritoneal Dialysis (CCPD)—OP or Home—Support Services
0859	Continuous Cycling Peritoneal Dialysis (CCPD)—OP or Home—Other
0860	General classification
0861	MEG
0880	Miscellaneous Dialysis—General
0881	Miscellaneous Dialysis—Ultrafiltration
0882	Miscellaneous Dialysis—Home Dialysis Aid Visit
0889	Miscellaneous Dialysis—Other Miscellaneous Dialysis
0900	General
0901	Electroshock Treatment
0902	Milieu Therapy
0903	Play Therapy
0904	Activity Therapy
0905	Intensive Outpatient Services—Psychiatric
0906	Intensive Outpatient Services—Chemical Dependency
0907	Community Behavioral Health Program (Day Treatment)
0911	Behavioral Health Treatments/Services—Extension of 090X—Rehabilitation
0912	Partial Hospitalization—Less Intensive
0913	Partial Hospitalization—Intensive
0914	Behavioral Health Treatments/Services—Extension of 090X—Individual Therapy
0915	Behavioral Health Treatments/Services—Extension of 090X—Group Therapy

Revenue Code	Description
0916	Behavioral Health Treatments/Services—Extension of 090X—Family Therapy
0917	Behavioral Health Treatments/Services—Extension of 090X—Bio Feedback
0918	Behavioral Health Treatments/Services—Extension of 090X—Testing
0919	Behavioral Health Treatments/Services—Extension of 090X—Other
0920	Other Diagnostic Services—General
0921	Other Diagnostic Services—Peripheral Vascular Lab
0922	Other Diagnostic Services—Electromyelogram
0923	Other Diagnostic Services—Pap Smear
0924	Other Diagnostic Services—Allergy Test
0925	Other Diagnostic Services—Pregnancy Test
0929	Other Diagnostic Services—Other Diagnostic Service
0931	Medical Rehabilitation Day Program—Half Day
0932	Medical Rehabilitation Day Program—Full Day
0940	Other Therapeutic Services—General
0941	Other Therapeutic Services—Recreational Therapy
0942	Other Therapeutic Services—Education/Training
0943	Other Therapeutic Services—Cardiac Rehabilitation
0944	Other Therapeutic Services—Drug Rehabilitation
0945	Other Therapeutic Services—Alcohol Rehabilitation
0946	Other Therapeutic Services—Complex Medical Equipment—Routine
0947	Other Therapeutic Services—Complex Medical Equipment—Ancillary
0948	Other Therapeutic Services—Pulmonary Rehabilitation
0949	Other Therapeutic Services—Other Therapeutic Services
0951	Other Therapeutic Services—Athletic Training
0952	Other Therapeutic Services—Kinesiotherapy
0953	Other Therapeutic Services—Chemical Dependency
0960	Professional Fees—General
0961	Professional Fees—Psychiatric
0962	Professional Fees—Ophthalmology
0963	Professional Fees—Anesthesiologist (MD)
0964	Professional Fees—Anesthetist (CRNA)
0969	Professional Fees—Other Professional Fees
0971	Professional Fees—Laboratory
0972	Professional Fees—Radiology—Diagnostic
0973	Professional Fees—Radiology—Therapeutic
0974	Professional Fees—Radiology—Nuclear Medicine
0975	Professional Fees—Operating Room
0976	Professional Fees—Respiratory Therapy
0977	Professional Fees—Physical Therapy
0978	Professional Fees—Occupational Therapy
0979	Professional Fees—Speech Pathology
0981	Professional Fees—Emergency Room
0982	Professional Fees—Outpatient Services
0983	Professional Fees—Clinic

Revenue Code	Description
0984	Professional Fees—Medical Social Services
0985	Professional Fees—EKG
0986	Professional Fees—EEG
0987	Professional Fees—Hospital Visit
0988	Professional Fees—Consultation
0989	Professional Fees—Private-Duty Nurse
0990	Patient Convenience Items—General
0991	Patient Convenience Items—Cafeteria/Guest Tray
0992	Patient Convenience Items—Private Linen Service
0993	Patient Convenience Items—Telephone/Telegraph
0994	Patient Convenience Items—TV/Radio
0995	Patient Convenience Items—Nonpatient Room Rentals
0996	Patient Convenience Items—Late Discharge Charge
0997	Patient Convenience Items—Admission Kits
0998	Patient Convenience Items—Beauty Shop/Barber
0999	Patient Convenience Items—Other Patient Convenience Items
1000	Behavioral Health Accommodations—General
1001	Behavioral Health Accommodations—Residential—Psychiatric
1002	Behavioral Health Accommodations—Residential—Chemical Dependency
1003	Behavioral Health Accommodations—Supervised Living
1004	Behavioral Health Accommodations—Halfway House
1005	Behavioral Health Accommodations—Group Home
2100	Alternative Therapy Services—General
2101	Alternative Therapy Services—Acupuncture
2102	Alternative Therapy Services—Acupressure
2103	Alternative Therapy Services—Massage
2104	Alternative Therapy Services—Reflexology
2105	Alternative Therapy Services—Biofeedback
2106	Alternative Therapy Services—Hypnosis
2109	Alternative Therapy Services—Other Alternative Therapy Services
3101	Adult care—Adult daycare, medical and social—hourly
3102	Adult care—Adult daycare, social—hourly
3103	Adult care—Adult daycare, medical and social—daily
3104	Adult care—Adult daycare, social—daily
3105	Adult care—Adult foster care—daily
3109	Adult care—Other adult care

Index

A

access to care, defined 195
Accountable Care Organization (ACO) 177
Additional Documentation segment, on pharmacy claims 86t
addresses, on claims 32t
administrative healthcare data, flow of 3–4
Administrative Simplification Compliance Act (ASCA) (2001) 60
admission codes 45, 45t
admission dates 42–43
admission diagnosis code, on claims 44t
admission source, on claims 46t
admission type, on claims 46t
age
  in HEDIS 156
  rating 133*t*
Age-as-of-Calculation utility macro 181–182
ALL keyword 186
ALL= option 151
All Payer Claims Database (APCD) 6–7, 179–180
allowable gap, in HEDIS 156
ambulances, Place of Service codes for 208–209t
Ambulatory Payment Classifications (APCs) 25, 33, 104t, 117t, 187
ambulatory surgical centers, Place of Service codes for 208t
American Hospital Formulary Service (AHFS) 88–89, 90
American Medical Association (AMA) 61, 112
American National Standards Institute (ANSI) 60
anchor date, in HEDIS 156
ancillary claims
  about 10*t,* 59
  case studies 63–72, 73–80
  data elements unique to CMS-1500/837P 60–62
  payment methodologies 62–63
  provider specialty 62
  submitting medical claims 60
ancillary providers/services 4, 31, 195
ANYALPHA function 170
ARRAY DX 125
ARRAY statement 119, 120, 121, 140, 170, 190
ASC X12 60
Assign Project Library task 12
assisted living facilities, Place of Service codes for 207t
automating
  code 76–80
  PDC graphs generation 98–101
  reports with macro variables 73–80
AXIS statement 99

B

bar graphs, creating 51–53
BEG macro variable 140
beneficiary of medical services, synonyms for 3t
benefit, in HEDIS 157
bill sequence 206t
Billing Unit Standard 84t
birthing centers, Place of Service codes for 208t
Black Lung Benefits Act (BLBA) (2000) 26
BOOK keyword 12
Boolean logic 38
Booz Allen Hamilton 88
business case 154
BY statement 92, 95, 121
BYTE function 192, 193

C

CALCULATED keyword 65, 76, 173
callable macro programs, creating 135–136
capitation 23, 24, 195
CARDS technique 125
case management, defined 195
Case Mix Groups (CMGs) 25, 117
case studies
  ancillary claims 63–72, 73–80
  financial metrics 143–148, 148–149, 149–151
  healthcare claim codes 118–122, 123–126

Healthcare Effectiveness Data and Information Set (HEDIS) 158–170
members 134–141
pharmacy claims 90–101
professional claims 63–72, 73–80
providers 34–36, 36–38, 47–53, 53–55
case-mix adjustment, defined 195
CATS function 36
CATX function 35, 121, 125
Centers for Medicare & Medicaid Services (CMS)
See also specific claim forms
defined 195
Healthcare Common Procedure Coding System (HCPCS) 43, 61, 83, 104*t,* 114–115
ICD9 code sets 123
Medicare administered by 24
National Plan and Provider Enumeration System (NPPES) 32
website 105, 107, 179
CHAMPVA 26
character variables 13–14
children, identifying missed checkups by 63–72
Civilian Health and Medical Program of the Uniformed Services (CHAMPUS) 26
claim form, defined 195
See also specific forms
Claim segment, on pharmacy claims 85t
ClaimID 10
ClaimLine 10
claims
See also ancillary claims
See also facility claims
See also pharmacy claims
data types 10*t*
defined 195
processing 5
submitting 4–5, 60
synonyms for 3*t*
claims adjudication, defined 196
claims administration, defined 196
CLASS statement 150
Clinical segment, on pharmacy claims 86t
CLOSE function 186
CMS
See Centers for Medicare & Medicaid Services (CMS)
CMS-1450 paper claim form
about 5, 40
for facilities 30
patient status codes for 210–211*t*
revenue codes for 211–223*t*
sample 201
CMS-1500 paper claim form
about 5, 60
for ancillary providers 31
data elements unique to 60–62
Place of Service codes for 207–210*t*
for professional providers 30
sample 202*f*
COALESCE function 35, 146
COB/Other Payments segment, on pharmacy claims 85t
code
automating 76–80
simplifying with SAS Array processing 119–120
coinsurance, defined 196
colon modifier 38
colorectal cancer screening 156–170
commercial insurance
about 26
defined 196
as provider ID 33*t*
Common Procedural Terminology (CPT) 61
community mental health centers, Place of Service codes for 209t
community rating 133t
complexity, HEDIS and 154
compliant population 157–158
Compound segment, on pharmacy claims 85t
comprehensive inpatient rehabilitation facilities, Place of Service codes for 210t
COMPRESS function 19, 124, 171
conceptual data model 11f
Condition Code 46t
continuous enrollment
about 132
in HEDIS 156
conversion factor (CF) 63
copayment, defined 196
correctional facilities, Place of Service codes for 207t
Coupon segment, on pharmacy claims 85t
CREATE statement 76, 187
CREATE TABLE statement 190
"cross-client" projects 131
C-section rates, calculating among hospitals 47
Current Dental Terminology (CDT), defined 196
Current Procedural Terminology (CPT)
about 112–114
code set 104*t*
defined 196
custodial care facilities, Place of Service codes for 208t

D

data
expanding using SAS data step 91–92
used in this book 2
Data Dictionary 84t

data elements
unique to CMS-1500/837P 60–62
unique to facilities 40–46
unique to pharmacies 86–90
data hygiene routines, creating with formats 123–126
data sets 12–14
DATA step 74, 165, 191, 192, 193
DATALINES4 statement 74
DATASETS procedure 138
dates, on claim forms 32t
DEA, as provider ID 33t
deductible, defined 196
Defense Health Agency (DHA) 26
demographics
on claim forms 60–61
of members 128–129
detail files, distinguishing between header files and 10
diagnosis codes
about 44–45
on claim forms 61
format of 16–19
in healthcare claims 105–109
diagnosis pointer
on claim forms 61
linking to 190–193
Diagnosis Related Group (DRG)
about 15–16
code 117*t*
code set 104*t*
defined 196
formats 14
Medicare and 25, 33, 45–46
Diagnostic and Statistical Manual of Mental Disorders (DSM-5) 107
discharge dates, on claim forms 42–43
DischgDtChar character variable 20
disease management, defined 196
Disposable Medical Supplies (DMS) 31, 196
DO loop 91, 93, 120, 121, 138, 164, 170, 186, 187, 192
documentation, creating in SAS Enterprise Guide 56–57
$ sign 20
DRG
See Diagnosis Related Group (DRG)
driver table 155
Drug Listing Act (1972) 86–87
Drug Utilization Review (DUR), defined 196
"dual eligibles" 26
Durable Medical Equipment (DME) 31, 196
DUR/PPS segment, on pharmacy claims 85t

E

Early and Periodic Screening, Diagnosis, and Treatment (EPSDT), defined 196
E-codes 106
837I electronic claim format
about 5, 40
for facilities 30
patient status codes for 210–211*t*
revenue codes for 211–223*t*
837P electronic claim format
about 5, 60
for ancillary providers 31
data elements unique to 60–62
Place of Service codes for 207–210*t*
for professional providers 30
EIN, as provider ID 33t
Electronic Data Interchange (EDI), defined 196
electronic eligibility inquiry 130–131
electronic format 84–86
Electronic Health Record (EHR) 178
Electronic Medical Record (EMR) 178
eligibility
exclusions to 157
of members 130
eligible population, in HEDIS 155t, 156–157
emergency rooms, Place of Service codes for 208t
encounters 197
See also claims
See also visit
END macro variable 140
END= option 124
EndDOS variable 76–80
end-stage renal disease treatment facilities, Place of Service codes for 210t
Enhanced Therapeutic Classification System (ETC) 88
enrollment, of members 129–130
Enterprise Data Warehouse (EDW) 5
Entity Type Code 32t
episode, defined 197
episode of care
See visit
ER utilization, reasons for 53–55
%EVAL function 164
Evaluation and Management (E/M), defined 197
event/diagnosis, in HEDIS 157
"evidence presented" 5
experience rating 133t
extended coding
about 181
arrays of detail record elements on header record 187–190
linking to diagnosis pointer 190–193
utility macros 181–187

External Cause of Injury (ECI) 44t, 106
External Code List 84t

F

facilities
  data elements unique to 40–46
  as providers 4, 30
  types of 205*t*
facility claims
  837I electronic claim format 40
  about 10*t,* 30, 30*t,* 39
  CMS-1450 paper claim form 40
  data elements unique to facilities 40–46
Facility segment, on pharmacy claims 86t
FECA Black Lung 26–27
Federal Employees' Compensation Act (FECA) (1958) 27
federal poverty level (FPL) 25
federally qualified health centers, Place of Service codes for 209t
Fee-for-Service (FFS)
  about 24
  defined 197
  model 3–4, 4*f*
financial metrics
  about 143
  case studies 143–148, 148–149, 149–151
First DataBank 87, 88
FLOOR function 168, 182
FMTSEARCH option 16
FOBT procedure 173
Food and Drug Administration (FDA) 86
FOOTNOTE statement 99
format catalogs 20
FORMAT= option 151
FORMAT procedure 124, 125, 186
FORMAT statement 74, 93, 186
formats
  about 14
  applying to SAS variables 19–21
  diagnosis code 16–19
  Diagnosis Related group (DRG) 15–16
  using to create data hygiene routines 123–126
formatting existing variables 19–20
formulary, defined 197
FRAME 99
FREQ procedure 186
FROM statement 76, 173
functions
  See specific functions
future healthcare data issues
  about 175
  Accountable Care Organization (ACO) 177
  All Payer Claims Database (APCD) 179–180
  evolving patient medical records 178–179
  global billing 179
  ICD-10 176
  impact of Affordable Care Act 175
  Patient Centered Medical Home (PCMH) 176–177
  Pharmacy Benefits Manager (PBM) 177–178
  transparency in pricing 176

G

gender rating 133t
Generic Product Identifier (GPI) 88
Generic Sequence Number (GSN) 88
geographic rating 133t
geographical adjustment (Geographic Practice Cost Index [GPCI]) 63
global billing 179
GLOBAL statement 186
government health IT 179
GPLOT procedure 99
graphing PDC 96–98
GROUP BY statement 76
group homes, Place of Service codes for 207t
grouper, defined 197

H

HCFA Common Procedure Coding System, defined 197
header files, distinguishing between detail files and 10
Health and Human Services (HHS) 24, 112
"health care," compared with "healthcare" 3
Health Care Financing Administration (HCFA), defined 197
Health Home concept 177
health insurance 23–24
health insurance plan
  See reimburser of service cost
Health Insurance Portability and Accountability Act (HIPAA) (1996) 4–5, 6, 32, 84, 128, 197
Health Insurance Prospective Payment System (HIPPS) 43, 104t, 116–117
Health Maintenance Organization (HMO) 23–24, 197
health status rating 133t
"healthcare," compared with "health care" 3
healthcare claim codes
  about 103–104
  case studies 118–122, 123–126
  Current Procedural Terminology (CPT) 112–114
  diagnosis codes 105–109
  Health Insurance Prospective Payment System (HIPPS) 116–117
  Healthcare Common Procedure Coding System (HCPCS) 114–115

International Classification of Diseases 104–105
Logical Observation Identifiers Names and Codes (LOINC) 118
modifiers 115–116
National Drug Code (NDC) 117
surgical procedure codes 109–112
Healthcare Common Procedure Coding System (HCPCS) 43, 61, 83, 104t, 114–115
Healthcare Effectiveness Data and Information Set (HEDIS)
about 64, 153–154
business case 154
case studies 158–170
colorectal cancer screening 156–158
defined 197
reporting system components 155–156
technical challenges 154–155
Healthcare Provider Taxonomy Code (HPTC) 32–33
HEDIS
See Healthcare Effectiveness Data and Information Set (HEDIS)
Hierarchical Condition Category (HCC) 134
Hierarchical Ingredient Code (HIC) 88
HIPAA
See Health Insurance Portability and Accountability Act (HIPAA) (1996)
Home Health Agency (HHA) 30
homeless shelters, Place of Service codes for 207t
homes, Place of Service codes for 207t
hospice, Place of Service codes for 208t
Hospital Acquired Conditions (HAC) cases 45
householding 131
hybrid specification 158

I

ICD-9-CM 105–107, 197
ICD-9-PCS 109–110
ICD-10 176
ICD-10-CM 107–109, 197
ICD-10-PCS 110–112
identification numbers, in National Provider Registry 32t
IF...THEN statement 95, 145
Import Data task 12, 74
IN operator 37, 38, 119
IN= option 145, 165
%INCLUDE statement 136
independent clinics, Place of Service codes for 209t
independent laboratories, Place of Service codes for 210t
Indian Health Service, Place of Service codes for 207t
Individually Identifiable Health Information (IIHI), defined 198
industry algorithms 55
industry rating 133t
INFILE statement 74
INFORMAT statement 74
Inpatient Costs per Member per Month 134
inpatient hospitals, Place of Service codes for 208t
Inpatient Prospective Payment System (IPPS) 25, 46, 187
inpatient psychiatric facilities, Place of Service codes for 209t
INPUT statement 74
Insurance segment, on pharmacy claims 85t
insurer
See reimburser of service cost
INTCK function 138, 168, 182
Intermediate Care Facility (ICF) 30, 209t
International Classification of Diseases (ICD) 104–105, 104t
International Classification of Diseases, Clinical Modification (ICD-9-CM) 105
International Classification of Diseases, Nth Revision, Clinical Modification (ICD CM) codes 44
International Classification of Diseases, Nth Revision, Procedure Coding System (ICD PCS) codes 45
INTNX function 77, 93, 138

J

"judgment pronounced" 5

K

Kaiser Family Foundation 25
KEEP= data set option 140
key players 4
keywords
See specific keywords

L

labeler code, on pharmacy claims 87
LAG function 164
Laser Universal Claim Form (PUCF-D01PT) 84
LEAVE statement 125, 170, 190
legislation 6–7
LENGTH statement 74, 121
%LET statement 76–77
LIBNAME statement 12
libraries, creating permanent 12
LIBRARY library 16
Logical Observation Identifiers Names and Codes (LOINC) 104t, 118, 198

M

%MACRO statement 136
macro variables, automating reports with 73–80
maintenance, of members 130
Major Diagnostic Categories (MDC), defined 198

Managed Behavioral Healthcare Organization (MBHO), defined 198
Managed Care Organization (MCO) 24, 198
See also reimburser of service cost
Manual Claim Forms Reference Implementation Guide 84t
mass immunization centers, Place of Service codes for 209t
MAX function 76
meaningful use 179
Medicaid
about 25–26
defined 198
as provider ID 33*t*
medical claims
See claims
medical event, in HEDIS 155t
Medicare
about 24–25, 24–25*t*
defined 198
as provider ID 33*t*
Medicare Part A, defined 198
Medicare Part B, defined 198
Medicare Part C, defined 198
Medicare Part D, defined 84, 198
Medicare Physician Fee Schedule (MPFS) 63
medication, computing adherence to 90–101
Medigap Supplemental Insurance 25
Medi-Span 87, 88
member
See beneficiary of medical services
member ID, changing 131
"member identifier" 128
"member months" 131–132, 134–141
members
about 127–128
case studies 134–141
demographics of 128–129
eligibility of 130
enrollment of 129–130
issues of interest to 130–132
maintenance of 130
rate setting for 132–134
risk adjustment 132–134
%MEND statement 136
MERGE statement 95, 190
military treatment facilities, Place of Service codes for 208t
MLOGIC option 137
mobile units, Place of Service codes for 208t
modifiers 115–116
MPRINT option 137, 186–187, 189

N

names, in National Provider Registry 32t
Narrative segment, on pharmacy claims 86t
National Association of Boards of Pharmacy (NABP) 86
National Center for Health Statistics 107
National Committee for Quality Assurance (NCQA) 153, 156
National Correct Coding Initiative (NCCI) 116
National Council for Prescription Drug Programs (NCPDP) 5, 31, 84–86
National Drug Code (NDC)
about 83, 117
code set 104*t*
defined 198
Directory 87
in pharmacy claims 86–87
National Individual Identifier (NII) 128
National Plan and Provider Enumeration System (NPPES) 32
National Provider Identifier (NPI) 32, 32t
National Provider Registry 32–34
standardizing provider names from 34–36
National Uniform Billing Committee (NUBC) 40, 42, 43, 201f
National Uniform Claim Committee (NUCC) 60, 202
NCCI/MUE (National Correct Coding Initiative/Medically Unlikely Edit) 187
NCPDP Telecommunication vD.0 Continuous Feed (PUCF-D02PT) 84
NLEVEL option 186
NODUPKEY option 92
non-residential substance abuse treatment centers, Place of Service codes for 209t
numeric variables 13–14
nursing facilities, Place of Service codes for 208t

O

Occurrence Code 46t
Occurrence Span Code 46t
ODS statement 100, 186
Office of Workers' Compensation Programs (OWCP) 26–27
offices, Place of Service codes for 207t
OPEN function 186
optional exclusions, in HEDIS 155t
options
See specific options
Other diagnosis code 44t
other fields of interest, in pharmacy claims 89–90
OTHERWISE statement 145
OUT= option 92
outcomes measures, defined 198
outpatient care, defined 198
outpatient hospitals, Place of Service codes for 208t
outpatient prospective payment system (OPPS) 25

OUTPUT statement 91, 93

P

package code, on pharmacy claims 87
PAI code 117t
paper claims, forms for 84
    See also specific forms
parenthesis 37
Patient Centered Medical Home (PCMH) 176–177
patient discharge status 43
Patient Protection and Affordable Care Act (2010) 6, 26, 130, 175
Patient Reason diagnosis code 44t
Patient segment, on pharmacy claims 85t
patient status code 210–211t
patients
    See also beneficiary of medical services
    evolving medical records of 178–179
    identifying with complex conditions 118–122
"payer sheet" 84
payers
    See also reimburser of service cost
    about 23
    CHAMPVA 26
    commercial insurance 26
    FECA Black Lung 26–27
    health insurance 23–24
    Medicaid 25–26
    Medicare 24–25
    TRICARE 26
payment methodologies 62–63
PDC
    automating generation of graphs 98–101
    computing 90–91
    graphing 96–98
Personal Health Record (PHR) 178–179
pharmacies
    Place of Service codes for 207*t*
    as providers 4, 31
Pharmacy Benefits Manager (PBM) 31, 177–178
pharmacy claims
    about 10*t,* 83
    case studies 90–101
    data elements unique to pharmacy data 86–90
    data expansion using SAS data step 91–92
    NCPDP claim formats 84–86
    transaction segments on 85–86*t*
Pharmacy Provider segment, on pharmacy claims 85t
Pharmacy Quality Alliance (PQA) 90
physician profiling, defined 199
Place of Service (POS) 31, 62, 199, 207–210t
PMPM costs, calculating 148–149
Point of Service (POS) 24, 31, 199
practitioner
    See supplier of medical services
Preferred Provider Organization (PPO) 24, 199
Prescriber segment, on pharmacy claims 85t
prescription drugs, universal claim form for 203–204f
present on admission, code values for 45
pricing, transparency in 176
Pricing segment, on pharmacy claims 85t
primary care medical home 176–177
Primary Care Physician (PCP)
    about 23
    defined 199
    identifying with Taxonomy Codes 36–38
Principal diagnosis code 44t
principle diagnosis code (PDx) 14
Prior Authorization segment, on pharmacy claims 85t
prisons, Place of Service codes for 207t
PROC step 74
procedure codes, on claim forms 61–62
procedure modifiers 61–62, 104t
process flow 12
processing claims 5
product code, on pharmacy claims 87
Product Line
    defined 199
    in HEDIS 155*t,* 156
professional claims
    about 10*t,* 59
    case studies 63–72, 73–80
    data elements unique to CMS-1500/837P 60–62
    payment methodologies 62–63
    provider specialty 62
    submitting medical claims 60
professional providers 4, 30–31
programming, used in this book 2
PROPCASE function 35
Prospective Payment System (PPS), defined 199
Protected Health Information (PHI), defined 199
provider identifiers
    on claim forms 61
    in pharmacy claims 61, 86
Provider IDs 46
provider names, standardizing from National Provider Registry 34–36
provider specialty 62
ProviderID 11, 46
providers
    See also supplier of medical services
    about 29
    ancillary 4, 31, 195
    automating reports with SAS Enterprise Builder 56
    case studies 34–36, 36–38, 47–53, 53–55

creating documentation in SAS Enterprise Guide 56–57
facilities 4, 30
National Provider Registry 32–34
pharmacies 4, 31
professional 4, 30–31
types of 4, 4*f,* 29–31, 30*f*
psychiatric facilities, Place of Service codes for 209t
psychiatric residential treatment centers, Place of Service codes for 209t
public health clinics, Place of Service codes for 210t
PUT function 20, 146

Q

Quality Management (QM), defined 199
Query Builder 34

R

RANK function 193
rate setting, for members 132–134
rating, defined 199
recipient
See beneficiary of medical services
Red Book 87
Regenstrief Institute 118
reimburser of service cost, synonyms for 3t
Relative Value Unit (RVU) 25, 63
rendered services, in HEDIS 155t
REPORT procedure 149–150
reports
automating with macro variables 73–80
automating with SAS Enterprise Builder 56
residential substance abuse treatment facilities, Place of Service codes for 209t
Resource Utilization Groups (RUGs) 25, 33, 117t
resource-based relative value scale (RBRVS) 25
RETAIN statement 121, 140, 190
revenue code 30t, 43–44, 199, 211–223t
risk adjustment
defined 199
for members 132–134
risk stratification, defined 199–200
ROUND function 64
rural health clinics, Place of Service codes for 210t

S

sample data 9–11
SAS Array Processing, code simplification with 119–120
SAS data sets, viewing 13
SAS data step, data expansion using 91–92
SAS Enterprise Builder, automating reports with 56
SAS Enterprise Guide
about 11–12
creating documentation in 56–57
SAS libraries 12–14
SASAUTOS environmental variable 137
schools, Place of Service codes for 207t
SELECT statement 76, 145, 173, 187, 189–190
SET statement 91
Skilled Nursing Facilities (SNF) 25, 30, 43, 208t
SORT procedure 92, 99, 121, 140, 144
SQL procedure 75, 76, 91, 95, 100, 119, 171, 187
SQL statement 186
SSN, as provider ID 33t
StartDOS variable 77–80
statements
See specific statements
subject matter experts (SMEs) 187
summary data
creating 47–50
joining sets 50–51
summary functions 48
Summary Tables task 12
supplier of medical services, synonyms for 3t
surgical procedure codes 45, 109–112
SYMBOL statement 99
SYMBOLGEN system option 136
%SYSFUNC function 186

T

TABLE statement 150, 186
TABULATE procedure 149–151
Taxonomy Codes 32–33, 32t, 36–38
Telecommunication Standard Implementation Guide 84t
temporary lodging, Place of Service codes for 208t
terminology 2–3, 3t
therapeutic class codes, on pharmacy claims 88–89
third-party administrator (TPA) 31
TIN, as provider ID 33t
TITLE statement 99
Trailer segment, on pharmacy claims 86t
Transaction Header segment, on pharmacy claims 85t
TRANSLATE function 187
Tribal 638 facilities, Place of Service codes for 207t
TRICARE 26
Type of Bill (TOB)
about 40–42
defined 200
on facility claims 30*t*
Type of Service (TOS), defined 200

U

unbundling, defined 200
underwriting, defined 200
UNION operator 171
upcoding, defined 200
UPIN, as provider ID 33t

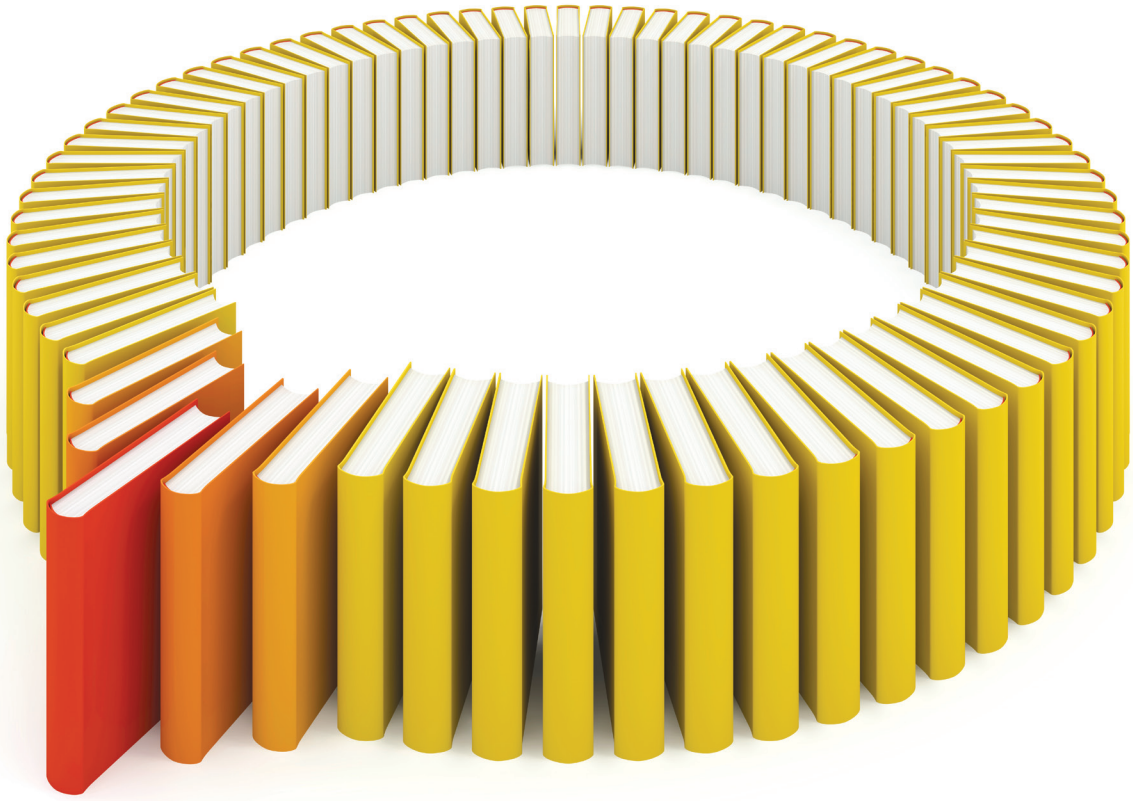
urgent care facilities, Place of Service codes for 208t
U.S. healthcare system 1
utility macros
about 181
Age-as-of-Calculation 181–182
identifying sparse variables 182–187
Utilization Management (UM), defined 200
Utilization Review (UR), defined 200

V

Value Codes 46t
Value Set Data (VSD), in HEDIS 156t
variability, HEDIS and 154
variables
See also specific variables
applying formats to SAS 19–21
formatting existing 19–20
identifying sparse 182–187
placing results of formats into new 20
variety, HEDIS and 154
VARNUM function 186
V-codes 106
velocity, HEDIS and 154
Version 10 (ICD-10) 105
visit, synonyms for 3t
volume, HEDIS and 154

W

walk-in retail health clinics, Place of Service codes for 208t
Washington Publishing Company 40
websites
All Payer Claims Database (APCD) 180
American Hospital Formulary Service (AHFs) 88
American Medical Association (AMA) 112
Booz Allen Hamilton 88
Centers for Medicare & Medicaid Services (CMS) 24, 105, 107, 123, 179
Department of Health and Human Services 24
First DataBank 87
government health IT 179
Healthcare Provider Taxonomy Code (HPTC) 32
industry algorithms 55
Kaiser Family Foundation 25
Medicare Physician Fee Schedule (MPFS) 63
Medi-Span 87
National Association of Boards of Pharmacy (NABP) 86
National Center for Health Statistics 107
National Committee for Quality Assurance (NCQA) 156
National Correct Coding Initiative (NCCI) 116
National Council for Prescription Drug Programs (NCPDP) 31, 84
National Drug Code Directory 87
National Uniform Billing Committee (NUBC) 40, 42, 43, 201
National Uniform Claim Committee (NUCC) 60, 202
Pharmacy Quality Alliance (PQA) 90
place of service 62
POS codes 31
provider specialty 62
Red Book 87
Regenstrief Institute 118
resources 200
Washington Publishing Company 40
World Health Organization (WHO) 104
WHEN statement 145
WHERE data set option 99, 140, 187
WHERE statement 76, 89, 100, 168
WORK library 16
Workers' Compensation segment, on pharmacy claims 85t
World Health Organization (WHO) 104–105



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